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Stonehouse et al.

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(54) **YEAST STRAINS EXHIBITING PROLONGED PERSISTENCE DURING A PLURALITY OF FERMENTATION CYCLES**

(71) Applicant: **Lallemand Hungary Liquidity Management LLC**, Budapest (HU)

(72) Inventors: **Emily Agnes Stonehouse**, Lebanon, NH (US); **John McBride**, Lebanon, NH (US); **Justin Van Rooyen**, Lebanon, NH (US); **Zach Losordo**, Lebanon, NH (US); **Petra Deane**, Lebanon, NH (US)

(73) Assignee: **Danstar Ferment AG**, Zug (CH)

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C12N 1/16 (2006.01)

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CPC **C12N 15/81** (2013.01); **C12N 1/165** (2021.05); **C12N 2510/02** (2013.01); **C12N 2511/00** (2013.01); **C12Y 101/01177** (2013.01)

(58) **Field of Classification Search**

CPC **C12N 15/81**; **C12N 2510/02**; **C12N 2511/00**; **C12N 1/18**; **C12N 15/1034**; **C12Y 101/01177**

USPC **435/254.2**
See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

9,988,652 B2	6/2018	McBride et al.
10,385,345 B2	8/2019	Brevnova et al.
11,332,728 B2	5/2022	Rice et al.
2013/0323822 A1	12/2013	Brevnova et al.
2018/0265853 A1	9/2018	Rice et al.
2020/0224209 A1	7/2020	Stonehouse et al.
2022/0127564 A1	4/2022	Argyros et al.

OTHER PUBLICATIONS

Kiselev L., (Structure, 2002, vol. 10: 8-9.*
Kwiatkowski et al., (Biochemistry 38:11643-11650, 1999.*
Wristlock et al., (Quarterly Reviews of Biophysics 2003, vol. 36 (3): 307-340.*
Davos et al., (Proteins: Structure, Function and Genetics, 2000, vol. 41: 98-107.*

Basso et al., "Ethanol Production in Brazil: The Industrial Process and Its Impact on Yeast Fermentation," *Biofuel Production—Recent Developments and Prospects*, Sep. 2011 (18 pages).

Brociner et al., "Thickening Flocculated Kaolinite Slurries in the Nozzle Discharge, Multi-Disc, Bowl Centrifuge," *Clay Minerals* 10:99, 1973 (14 pages).

Conrad et al., "Nutrient sensing and signaling in the yeast *Saccharomyces cerevisiae*," *FEMS Microbiol Rev* 38:254-299, Feb. 2014.

Danecek et al., "The variant call format and VCFtools," *Bioinformatics* 27(15):2156-2158, Jun. 2011.

Fisher et al., "Adaptive genome duplication affects patterns of molecular evolution in *Saccharomyces cerevisiae*," *PLoS Genet* 14(5):e 1007396, 2018 (15 pages).

Li, "A statistical framework for SNP calling, mutation discovery, association mapping and population genetical parameter estimation from sequencing data," *Bioinformatics* 27(21):2987-2993, Sep. 2011.

Li et al., "The Sequence Alignment/Map format and SAMtools," *Bioinformatics* 25(16):2078-2079, Jun. 2009.

Oud et al., "Genome duplication and mutations in *ACE2* cause multicellular, fast-sedimenting phenotypes in evolved *Saccharomyces cerevisiae*," *PNAS*, pp. E4223-E4231, Oct. 2013.

Purcell et al., "PLINK: A Tool Set for Whole-Genome Association and Population-Based Linkage Analyses," *The American Journal of Human Genetics* 81:559-575, Sep. 2007.

Scott et al., "The Influence of Polyploidy on the Evolution of Yeast Grown in a Sub-Optimal Carbon Source," *Mol. Biol. Evol.* 34(10): 2690-2703, Jul. 2017.

Selmecki et al., "Polyploidy can drive rapid adaptation in yeast," *Nature* 519(7543):349-352, Mar. 2015.

Zadrag-Tecza et al., "Cell Size Influences the Reproductive Potential and Total Lifespan of the *Saccharomyces cerevisiae* Yeast as Revealed by the Analysis of Polyploid Strains," *Oxidative Medicine and Cellular Longevity* 2018, Mar. 2018 (18 pages).

Zhang et al., "Effects of genome duplication on phenotypes and industrial applications of *Saccharomyces cerevisiae* strains," *App Microbiol Biotechnol* 101:5405-5414, 2017.

Basso et al., "Yeast selection for fuel ethanol production in Brazil," *FEMS Yeast Res* 8:1155-1163, 2008.

Basso et al., "Engineering topology and kinetics of sucrose metabolism in *Saccharomyces cerevisiae* for improved ethanol yield," *Metabolic Engineering* 13:694-703, 2011.

Bellon et al., "Introducing a New Breed of Wine Yeast: Interspecific Hybridisation between a Commercial *Saccharomyces cerevisiae* Wine Yeast and *Saccharomyces mikatae*," *PLoS ONE* 8(4):e62053, 2013, 14 pages.

Panchal et al., "Yeast Stability in the Brewing and Industrial Fermentation Ethanol Industries," *Critical Reviews in Biotechnology* 4:3:253-262, 1986.

Stanhill et al., "The Yeast Ras/Cyclic AMP Pathway Induces Invasive Growth by Suppressing the Cellular Stress Response," *Molecular and Cellular Biology* 19(11):7529-7538, 1999.

(Continued)

Primary Examiner — Robert B Mondesi

Assistant Examiner — Mohammad Y Meah

(74) *Attorney, Agent, or Firm* — Seed IP Law Group LLP

(57) **ABSTRACT**

The present disclosure provides yeasts, which can be recombinant yeast host cells, exhibiting prolonged persistence when submitted to a plurality of fermentation cycles. The yeasts exhibit at least one of the following phenotypic trait: a fast settling phenotype, a rugose phenotype, an improved invertase activity, triploidy, increased signaling in a RAS/cAMP/PKA pathway or combinations thereof.

23 Claims, 19 Drawing Sheets

Specification includes a Sequence Listing.

(56)

References Cited

OTHER PUBLICATIONS

Wang et al., "Increasing ethanol titer and yield in a gpd1 A gpd24 strain by simultaneous overexpression of *GLT1* and *STL1* in *Saccharomyces cerevisiae*," *Biotechnol Lett* 35:1859-1864, 2013.

Wu et al., "The transcription factor Ace2 and its paralog Swi5 regulate ethanol production during static fermentation through their targets Cts1 and Rps4a in *Saccharomyces cerevisiae*," *FEMS Yeast Research* 16:1-10, 2016.

Amorim et al., "Science and technology in the selection of yeasts for ethanol production," *Proceedings of the Symposium on Microorganisms in Agroenergy: from prospecting to bioprocesses*, 1 pg., Nov. 2013.

Entian et al., "25 Yeast Genetic Strain and Plasmid Collections," *Methods in Microbiology* 36:629-666, 2007.

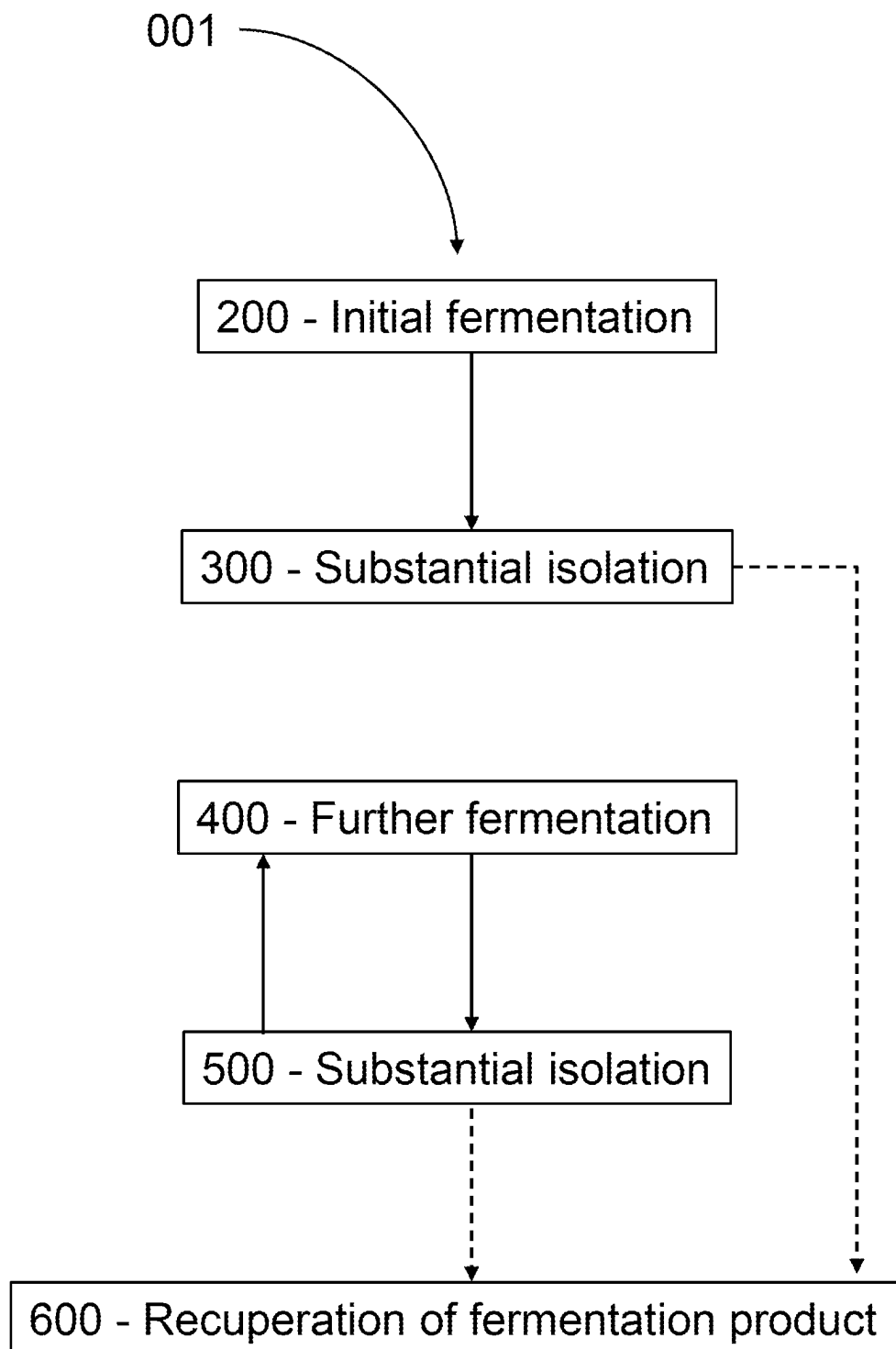
Figueirêdo et al., "Molecular analysis of flocculation and foam formation by yeasts used in the industrial production of fuel alcohol in Brazil," *Dissertation to the Postgraduate Program in Biotechnology of the Federal University of Santa Catarina*, 1 pg., 2008.

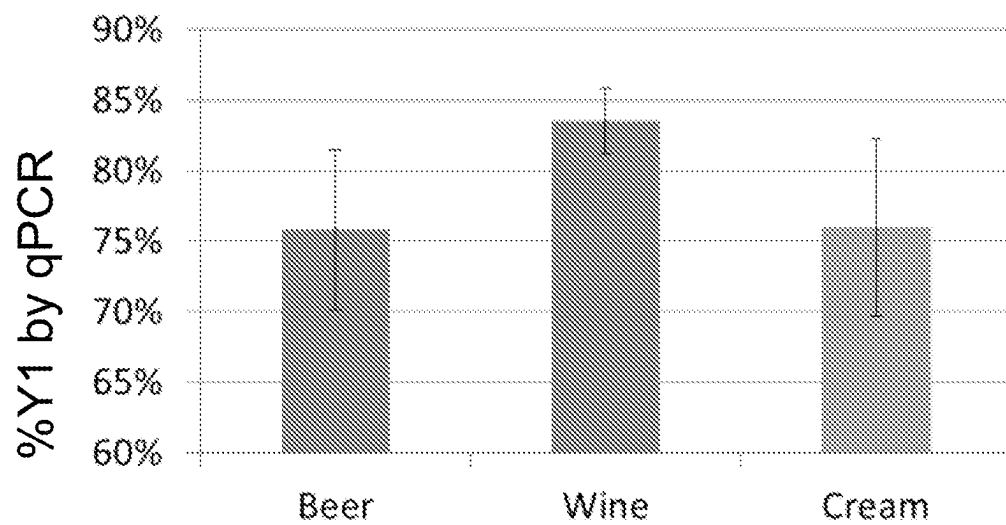
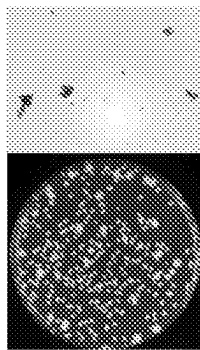
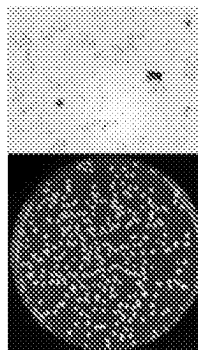
Figueirêdo et al., "Study of the influence of the FLO8 gene on phenotypes of *Saccharomyces cerevisiae* strains isolated in the fuel ethanol production process in Brazil," Thesis (PhD in Biotechnology)—Institute of Biomedical Sciences, University of São Paulo, São Paulo, 2012, 1 pg., Oct. 2012.

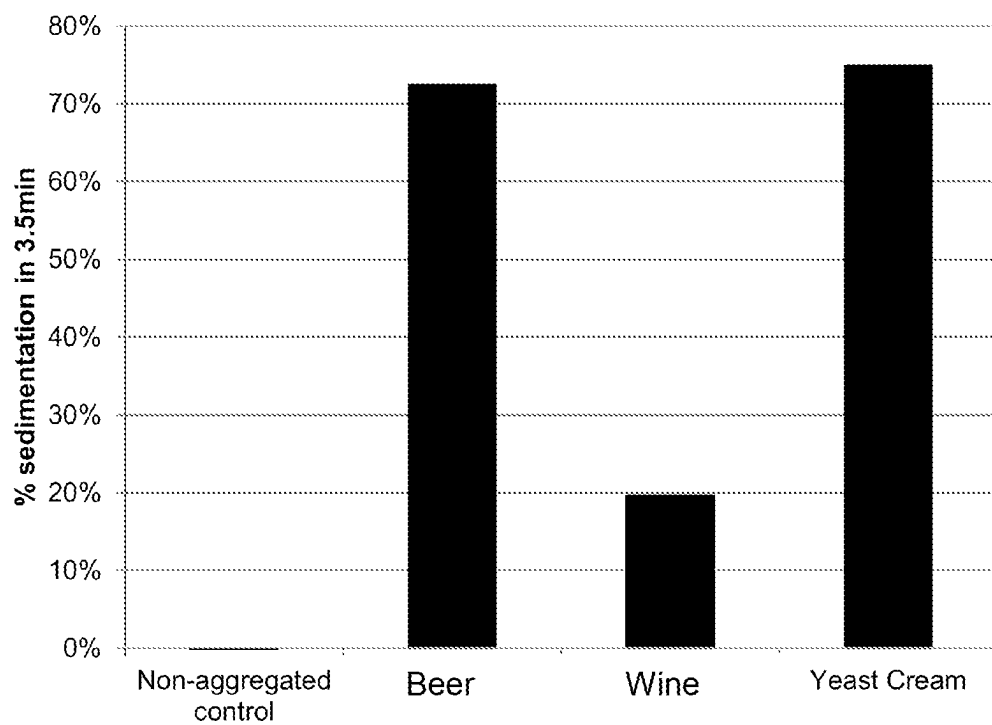
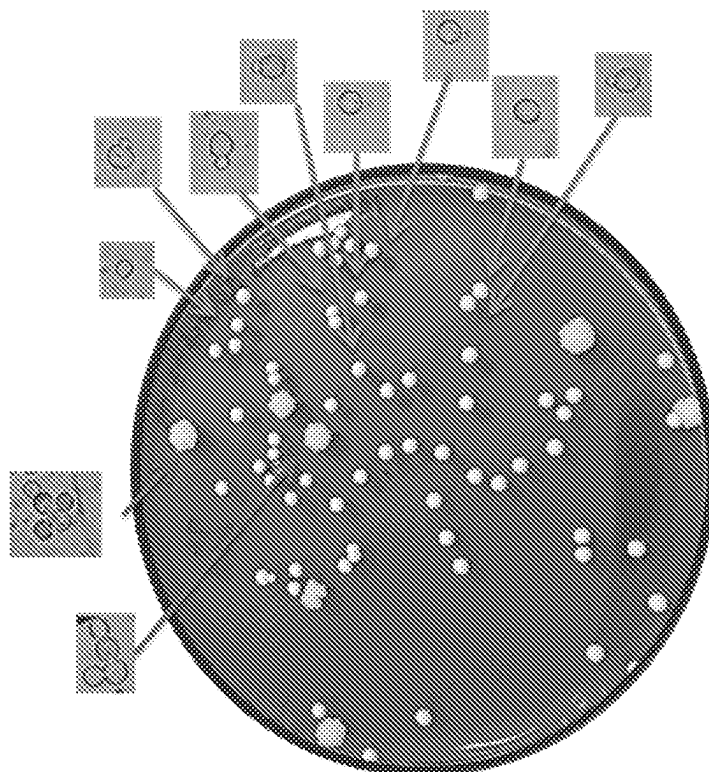
Nascimento et al., "Effects of the carbon source on the physiology and invertase activity of the yeast *Saccharomyces cerevisiae* FT858," *3 Biotech* 10:348, 9 pgs., 2020.

Zou et al., "Comparison of the Unfolded Protein Response in Cellobiose Utilization of Recombinant Angel-and W303-1A-Derived Yeast Expressing β -Glucosidase," *Frontiers in Bioengineering and Biotechnology* 10, 13 pgs., Mar. 2022.

* cited by examiner

**FIGURE 1**

**FIGURE 2A****FIGURE 2B****FIGURE 2C****FIGURE 2D**

**FIGURE 3****Figure 4A**

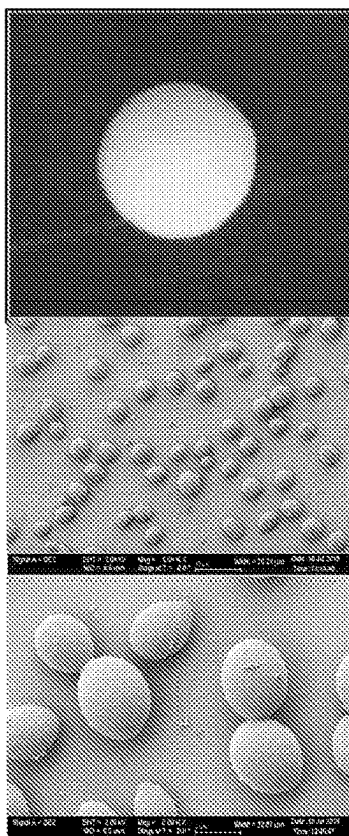


Figure 4B

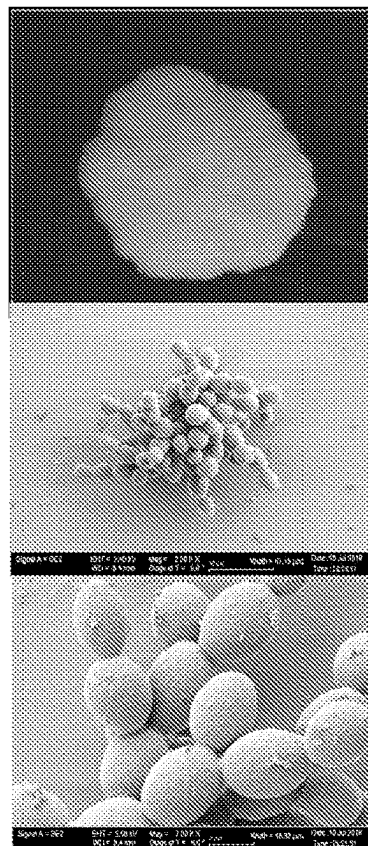


Figure 4C

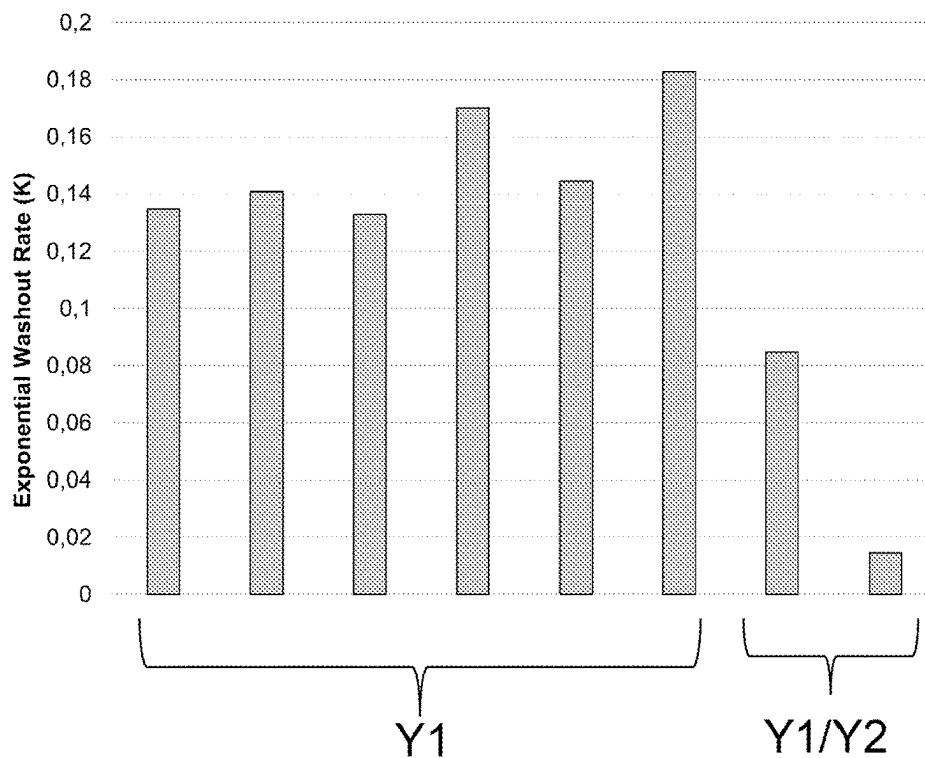


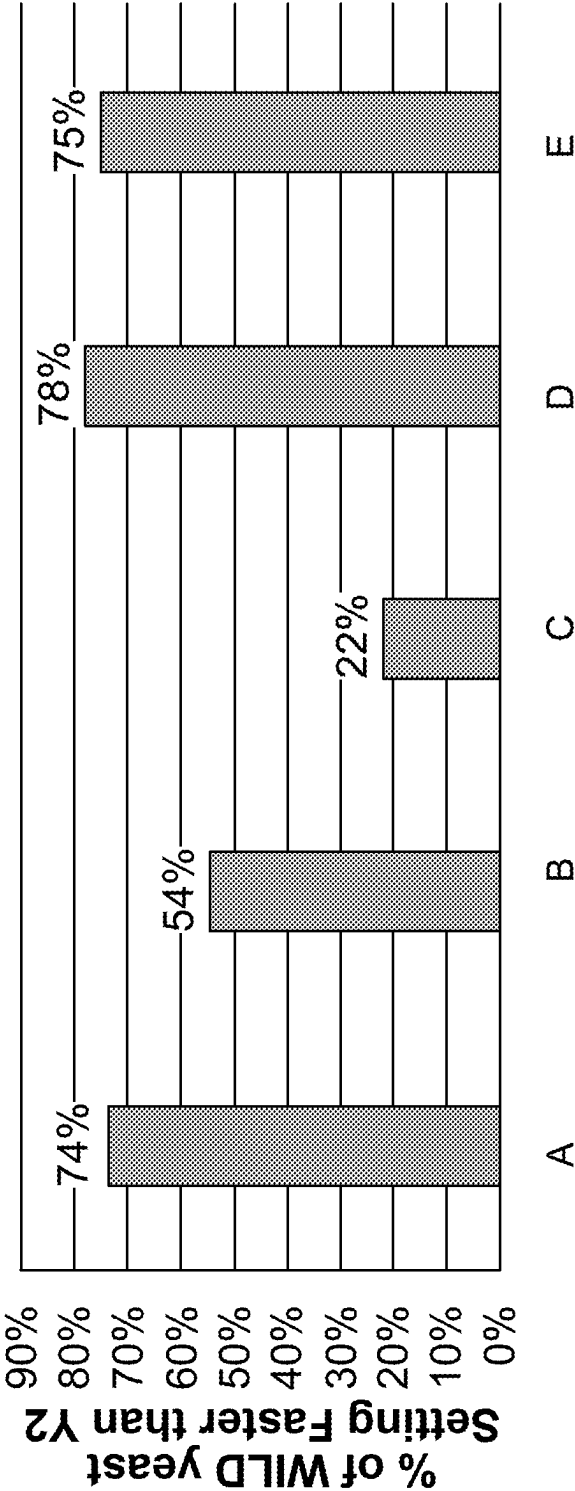
FIGURE 5

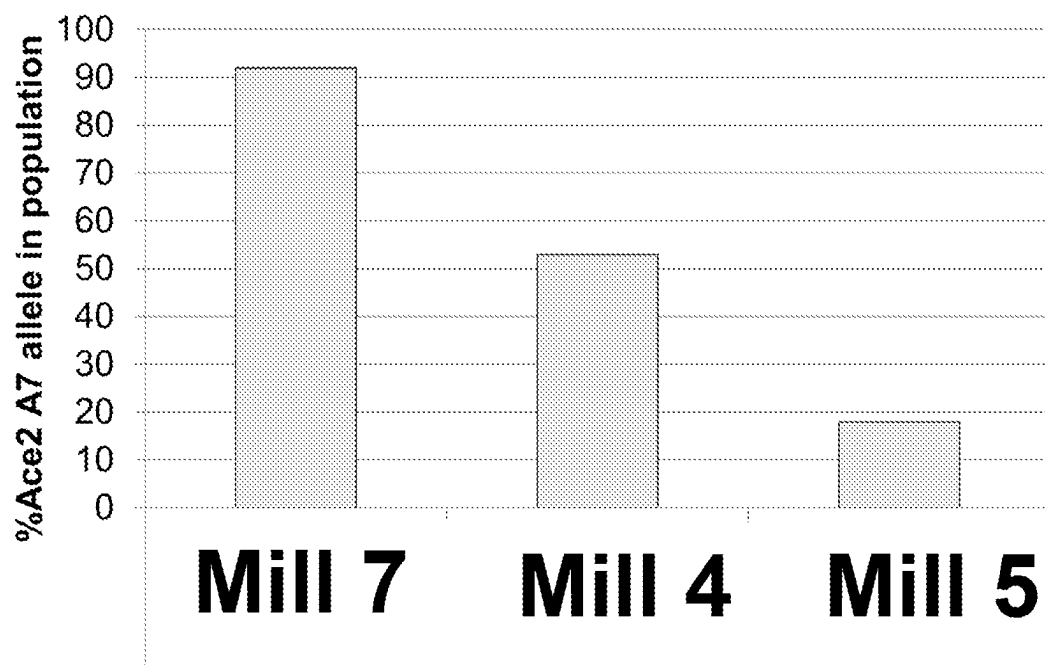
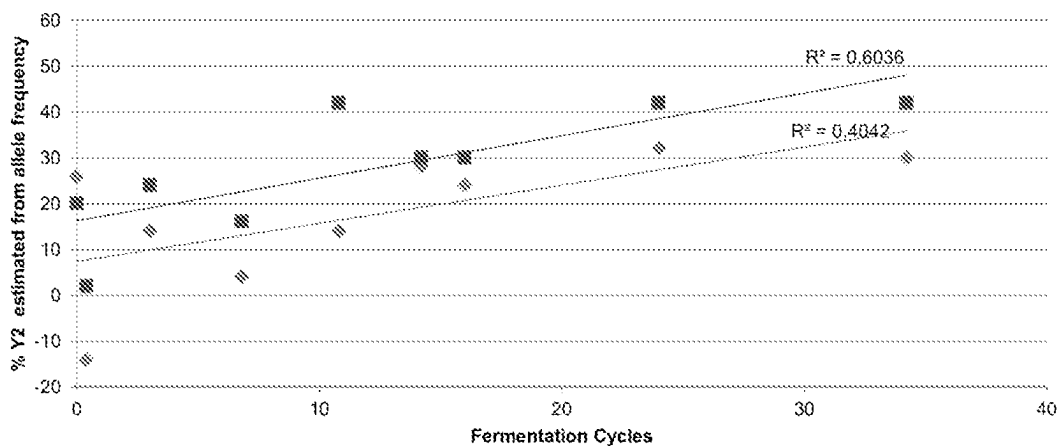
FIGURE 6

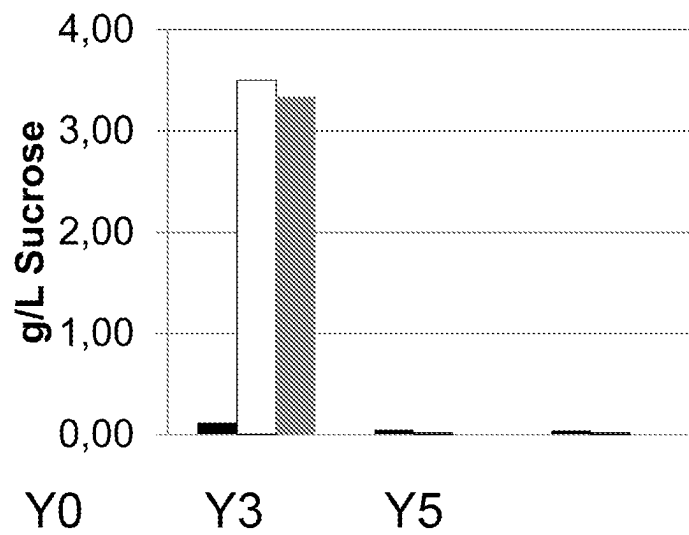
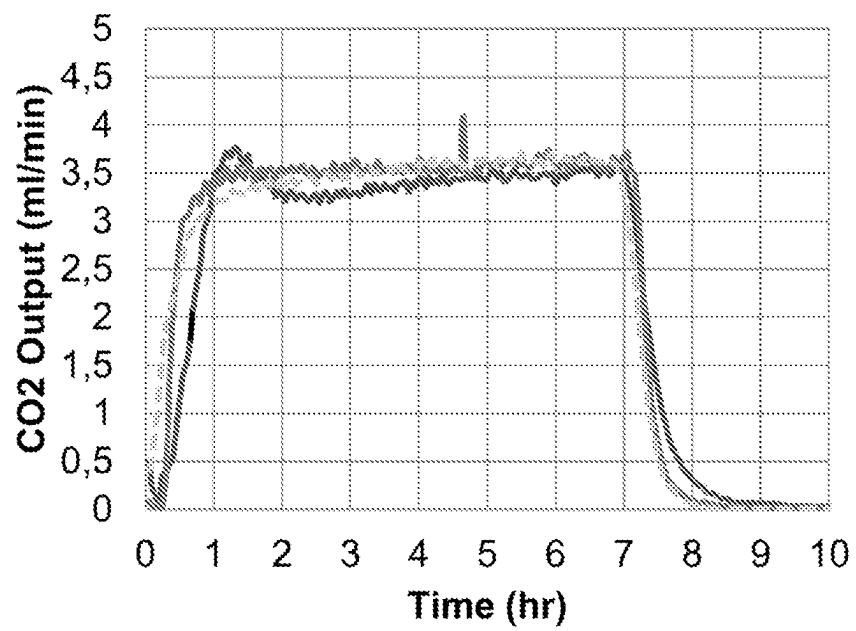
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	GTGATAAAAACAATAATGATNAAAAAATAGTACTGGTGATAACAT					
Identity 1. M10682-ACE2-A(8) Translation	GTGATAAAAACAATAATGATAAAAAAATAGTACTGGTGATAACAT					
	S D K N N N D K K N S T G D N I					
2. M10682-ACE2-A(7) Translation	GTGATAAAAACAATAATGAT-AAAAAATAGTACTGGTGATAACAT					
	S D K N N N D - K K I V L V I T					
Consensus Translation	1,140	1,150	1,160	1,170	1,180	
	ATTCCGTCGTTCGAAAAGACTTCCCCGGTGGGCTAAGTATCTCTCCAA					
Identity 1. M10682-ACE2-A(8) Translation	ATTCCGTCGTTCGAAAAGACTTCCCCGGTGGGCTAAGTATCTCTCCAA					
	F R L F E K T S P G G L S I S P					
2. M10682-ACE2-A(7) Translation	ATTCCGTCGTTCGAAAAGACTTCCCCGGTGGGCTAAGTATCTCTCCAA					
	Y S V C S K R L P R V G * V S L Q					



FIGURE 7



**FIGURE 8****FIGURE 9**

**FIGURE 10A****FIGURE 10B**

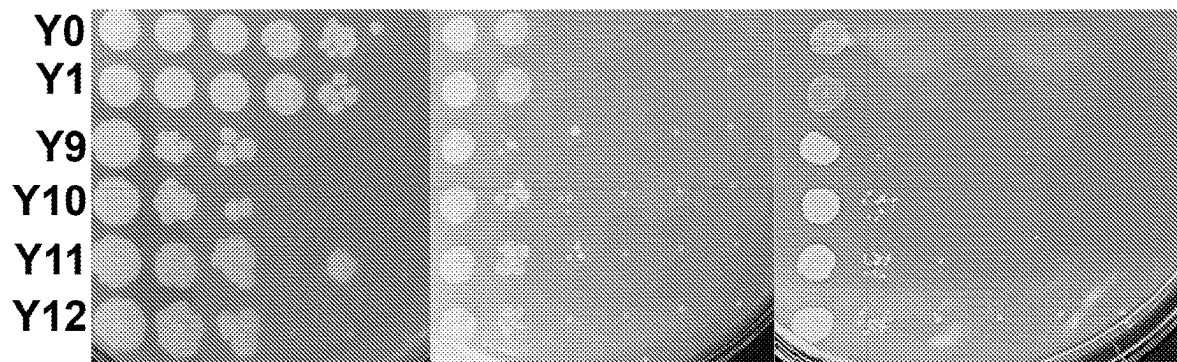


FIGURE 11A

FIGURE 11B

FIGURE 11C

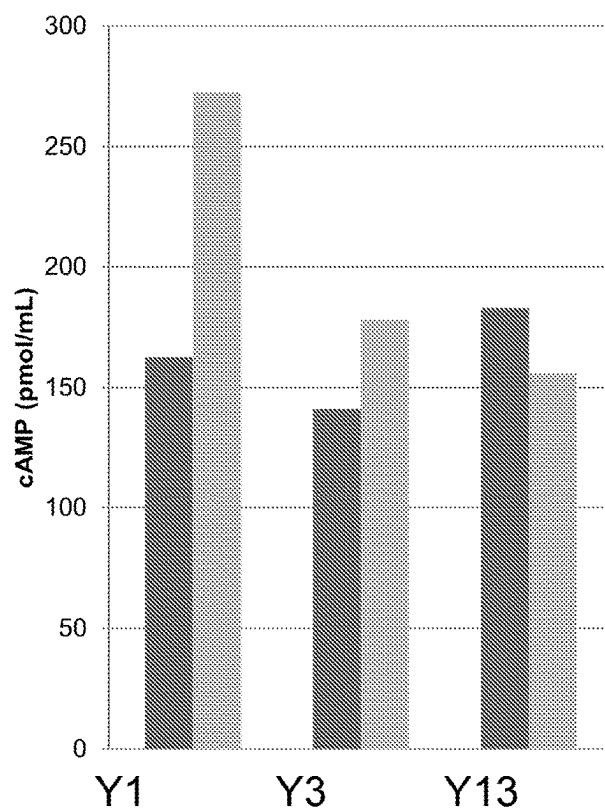
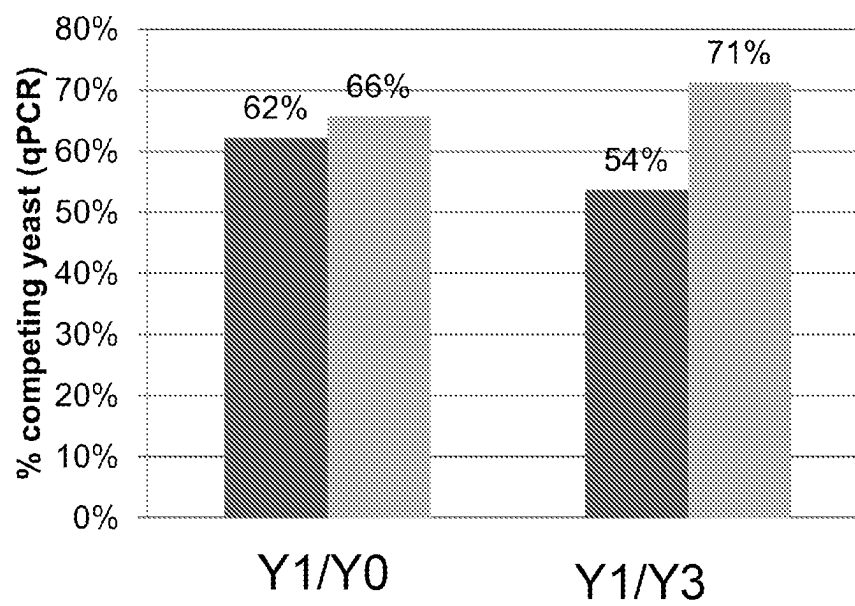


FIGURE 12

**FIGURE 13**

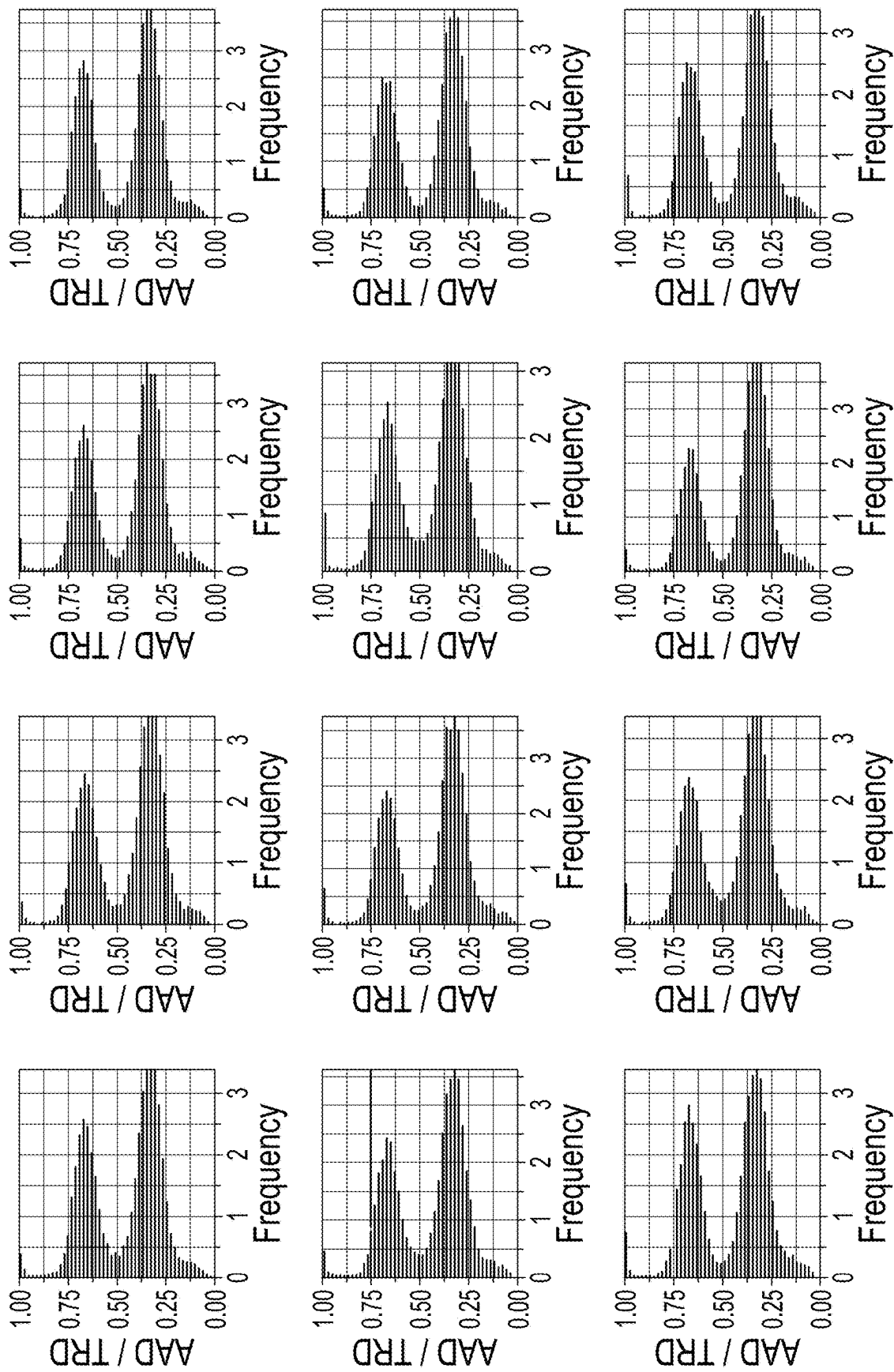


FIGURE 14A

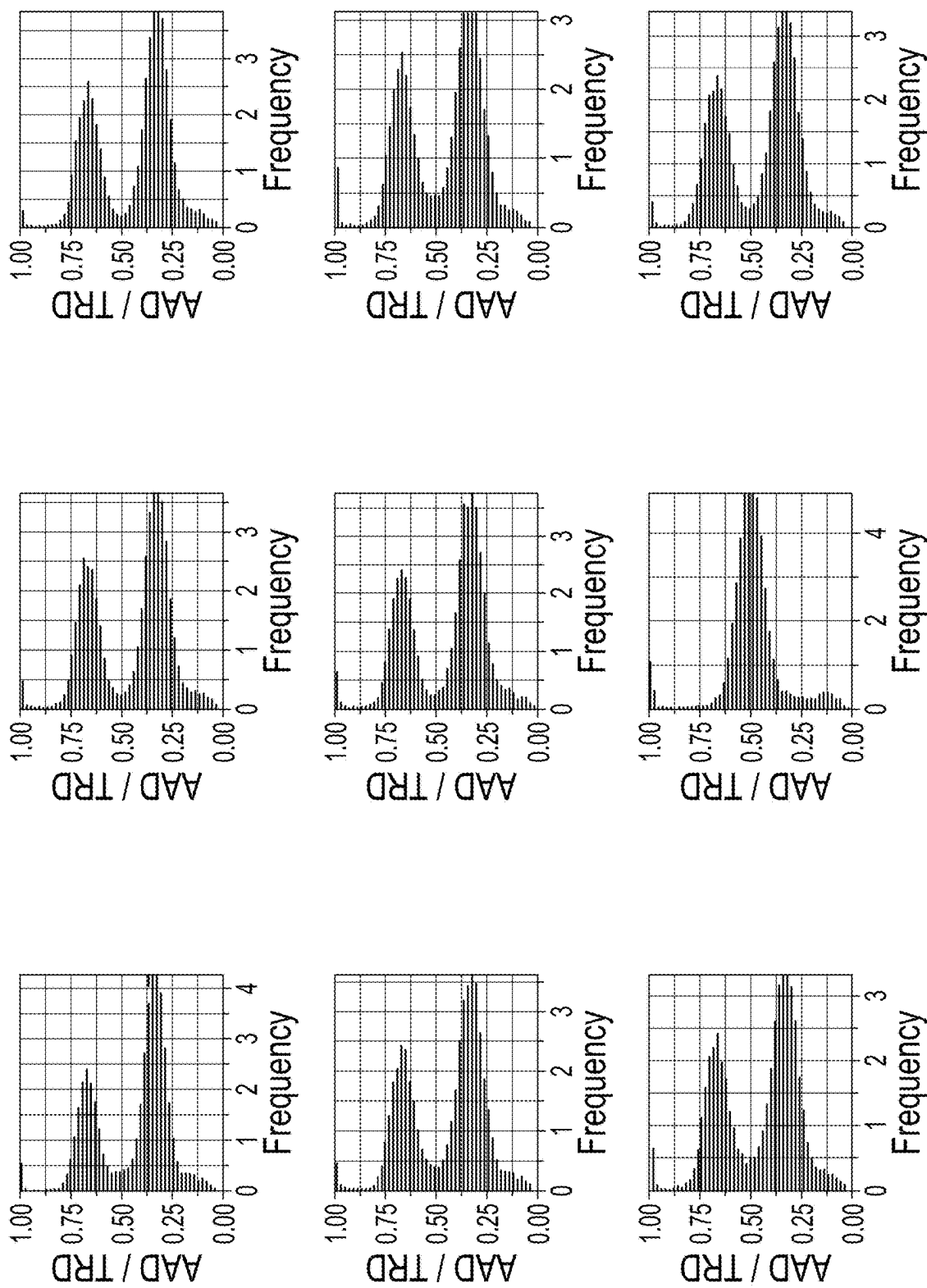


FIGURE 14B

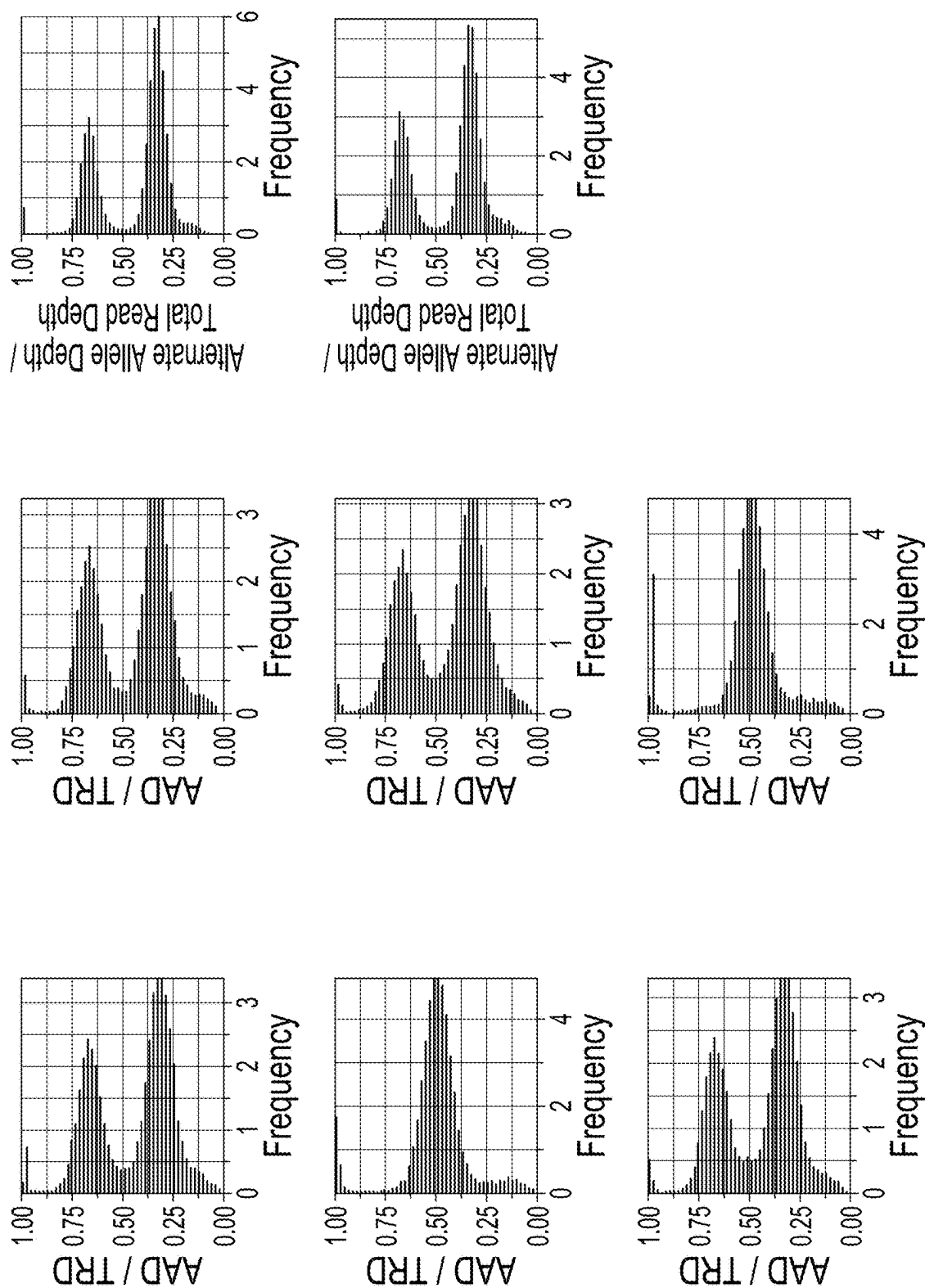


FIGURE 14C

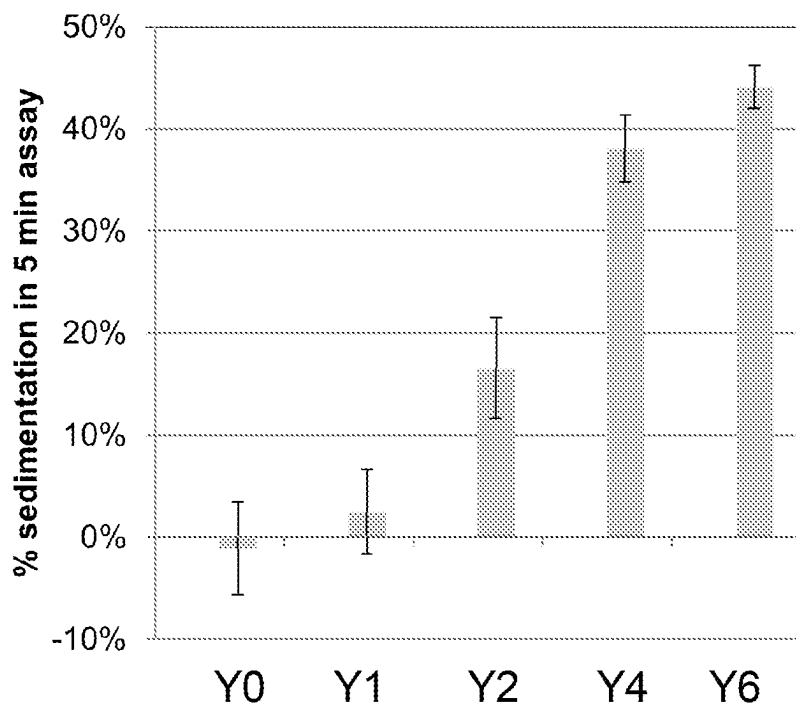
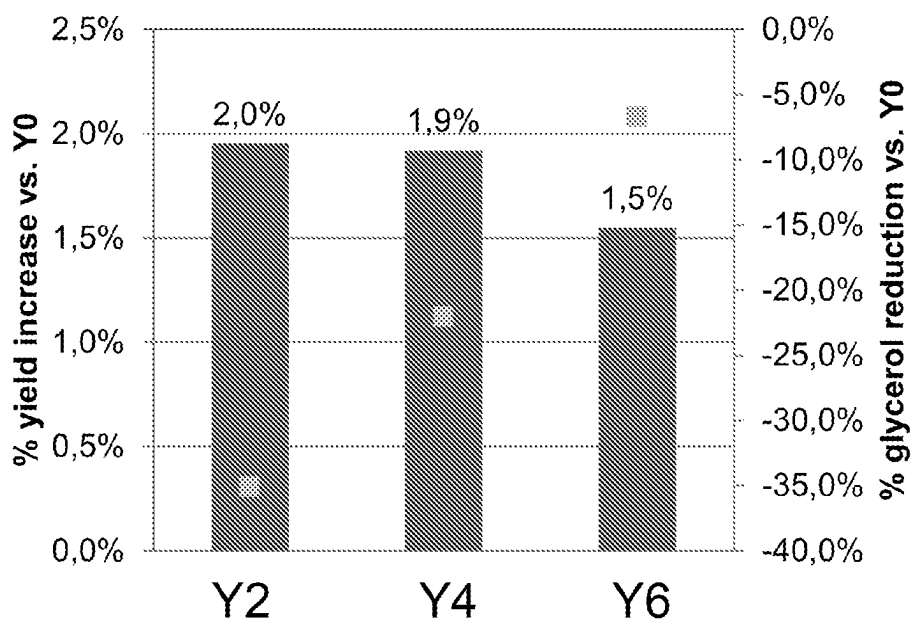
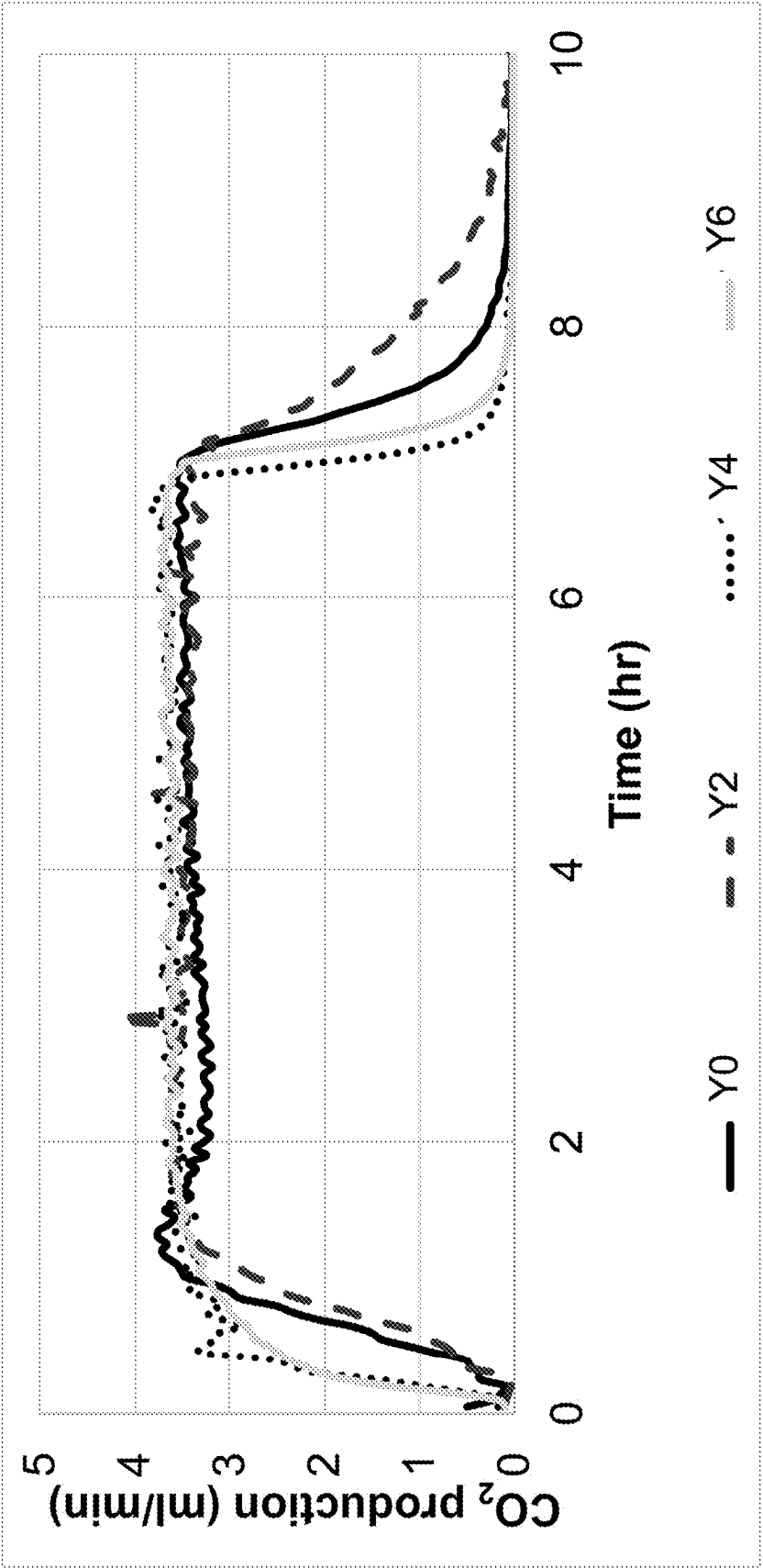
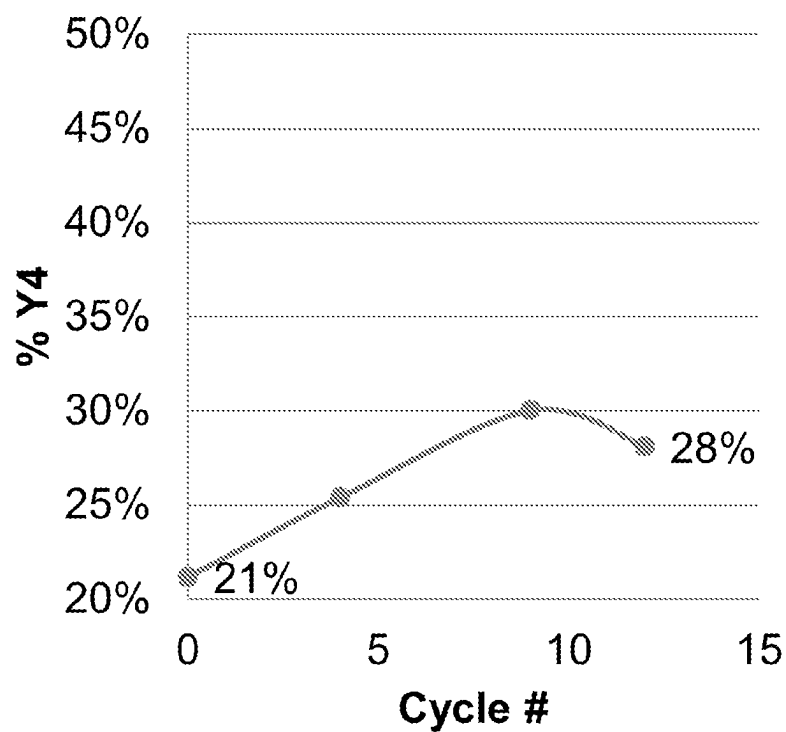
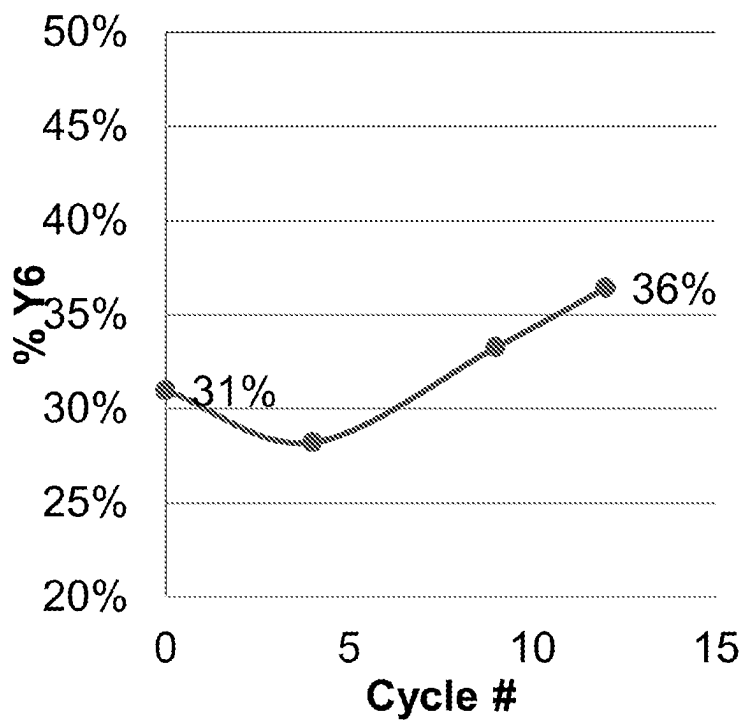
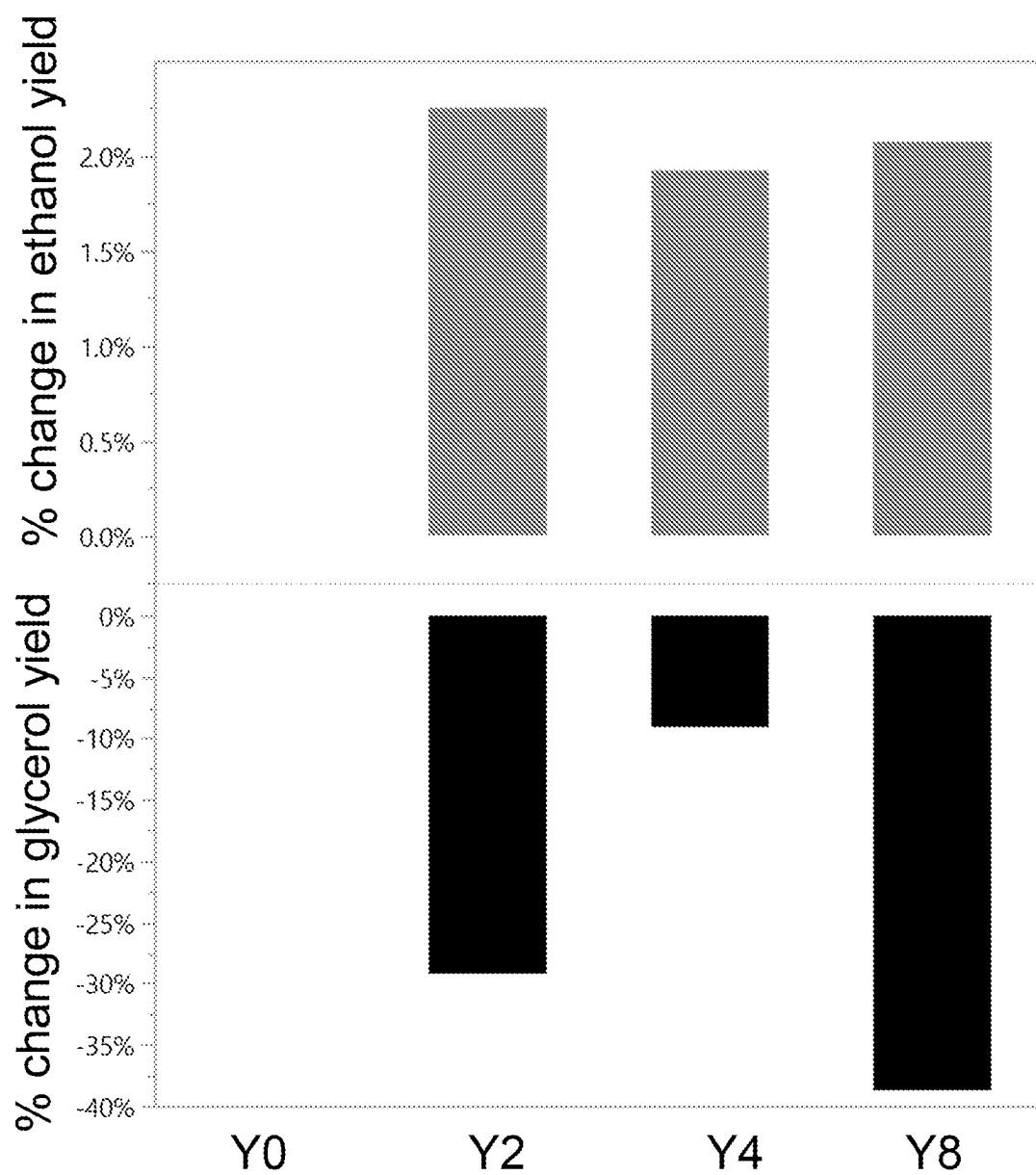
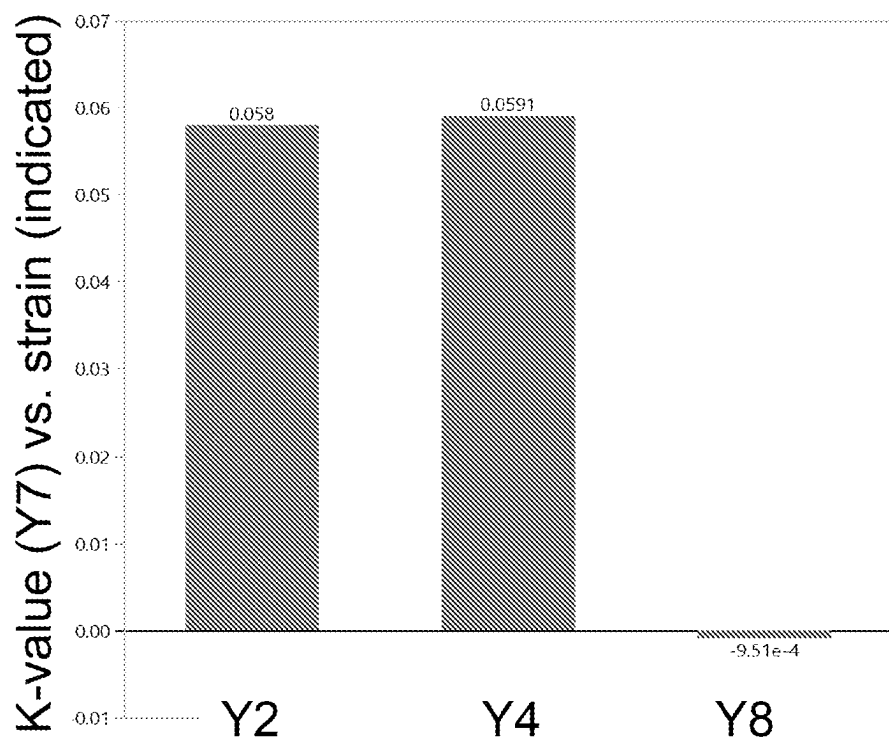
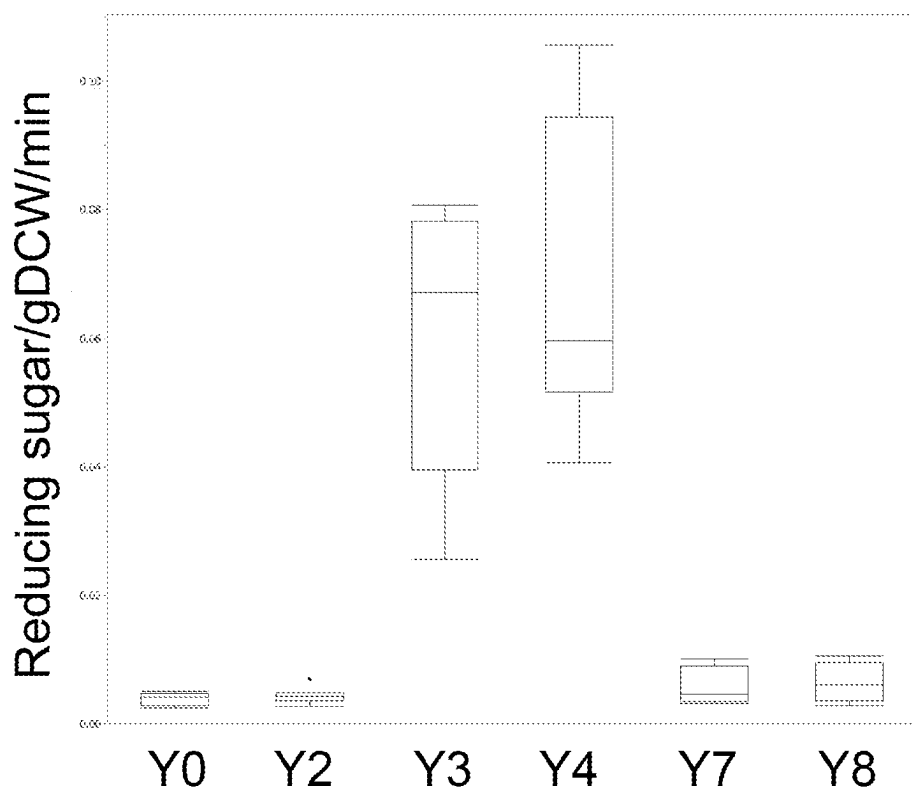
**FIGURE 15****FIGURE 16A**

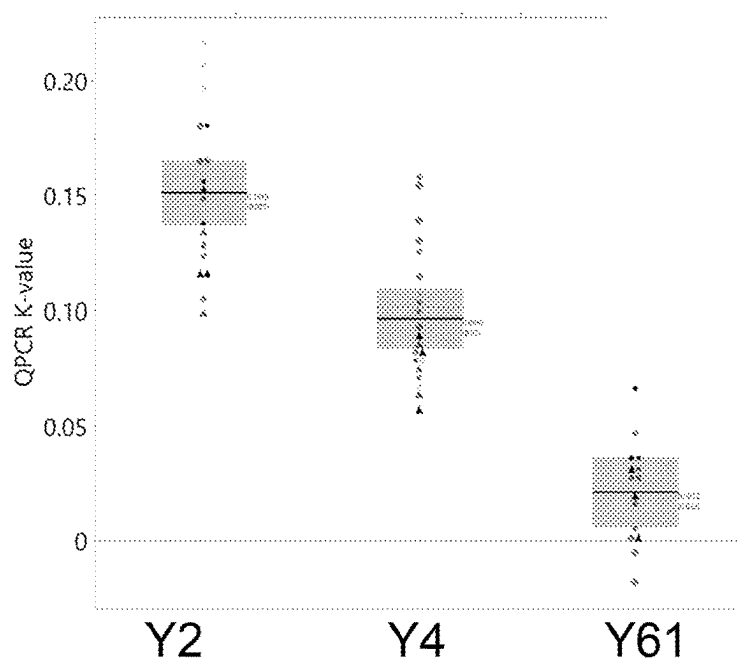
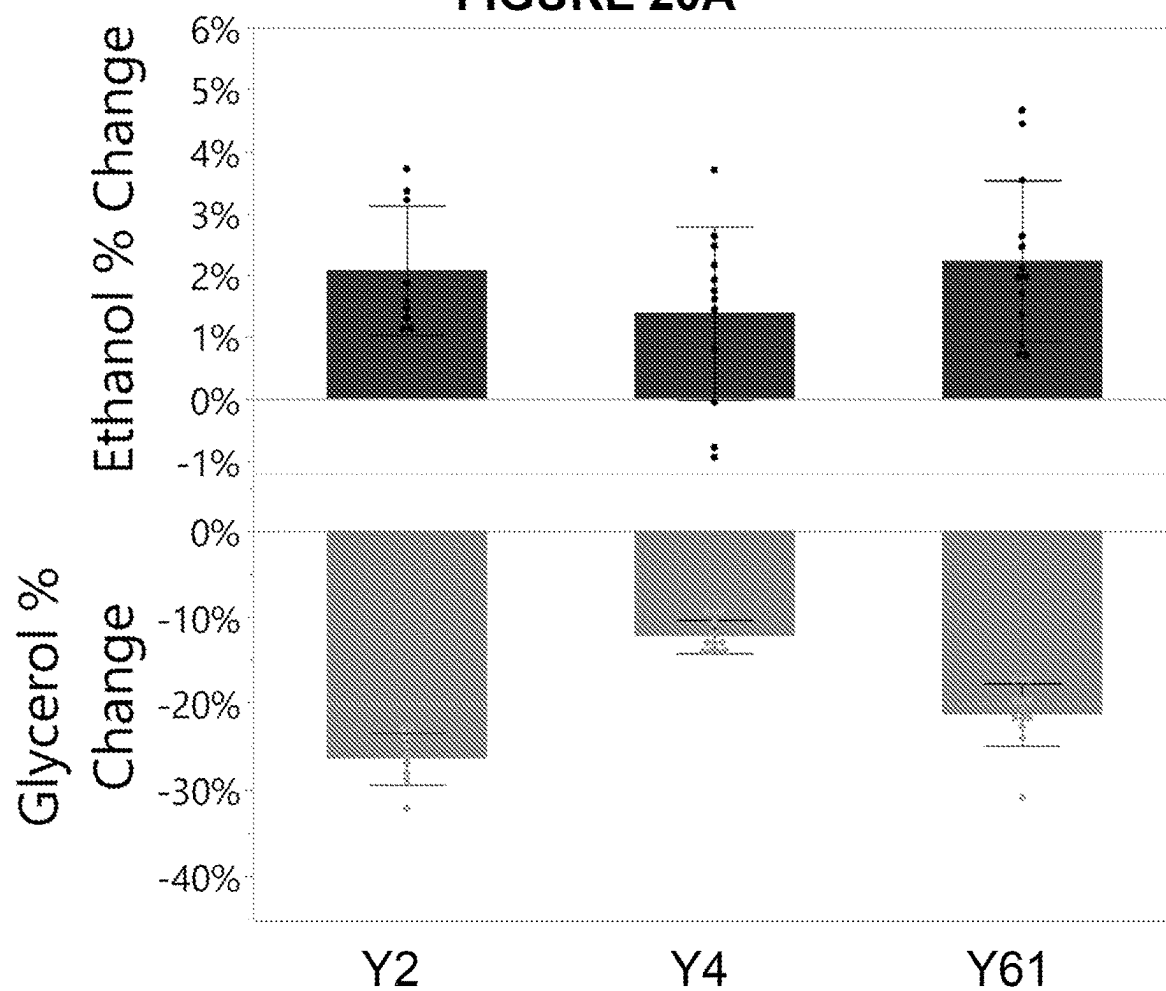
FIGURE 16B



**FIGURE 17A****FIGURE 17B**

**FIGURE 18A**

**FIGURE 18B****FIGURE 19**

**FIGURE 20A****FIGURE 20B**

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YEAST STRAINS EXHIBITING PROLONGED PERSISTENCE DURING A PLURALITY OF FERMENTATION CYCLES

CROSS-REFERENCE TO RELATED APPLICATION AND DOCUMENTS

This application claims priority from U.S. provisional application 63/211,831 filed on Jun. 17, 2021 and herewith incorporated in its entirety.

REFERENCE TO SEQUENCE LISTING SUBMITTED ELECTRONICALLY

The sequence listing associated with this application is provided in text format in lieu of a paper copy and is hereby incorporated by reference into the specification. The name of the text file containing the sequence listing is "580127_436_SEQ". The text file is 362,181 bytes, was created on Jun. 4, 2024, and is being submitted electronically.

TECHNOLOGICAL FIELD

The present disclosure concerns yeast host strains exhibiting prolonged persistence during a plurality of fermentation cycles in which the fermenting population is recycled.

BACKGROUND

Multiple fermentations cycles using a recycled biomass are susceptible to contaminations by wild yeasts. For example, in Brazilian fuel ethanol fermentations, the yeasts are pitched at the beginning of the sugarcane crushing season and are continually recycled for more than 200 days. The yeasts are recycled using continuous centrifugation and acid washing to improve productivity. Wild yeast contaminants are continually entering the fermentation since the fermentation substrates (e.g., sugarcane juice and molasses) are not sterilized. In addition, the predominate *Saccharomyces cerevisiae* yeast strains used in the Brazilian fuel ethanol industry are highly heterozygous and are known to have genomic rearrangements which creates challenges to the traditional molecular identification methods used to monitor yeast populations (such as, for example, microsatellite and inter-delta sequence amplification, random amplified polymorphic DNA (RAPD) or karyotyping by pulse-field gel electrophoresis (PFGE)).

Limiting contamination is important in continuous fermentations from both an economic and a processing perspective. Contaminating yeast have been associated with decreased ethanol yields, flocculation, and foaming. Greater than 95% of the contaminating yeasts are reported to be other *Saccharomyces* strains many of which have unfavorable fermentation characteristics and can lead to large productivity losses if allowed to proliferate. Less than 5% are non-*Saccharomyces* such as *Dekkera bruxellensis*, *Candida krusei* and *Schizosaccharomyces pombe*, but these strains can cause issues if left unchecked.

There is a need for limiting wild yeast contamination over numerous fermentation cycles using a recycled biomass or continuous fermentations so as to prolong the persistence of fermenting recombinant yeast host cells.

BRIEF SUMMARY

The present disclosure provides a yeast exhibiting prolonged persistence when submitted to a plurality of fermenta-

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tation cycles. The yeast exhibits at least one of the following phenotypic trait: a fast settling phenotype, a rugose phenotype, an improved invertase activity, triploidy, increased signaling in a RAS/cAMP/PKA pathway or combinations thereof.

In a first aspect, the present disclosure provides a recombinant yeast host cell capable of modulating the activity or the expression of a first polypeptide and/or a second polypeptide for increasing, when compared to a parental cell, the conversion of a biomass into a fermentation product and/or for reducing the conversion of the biomass into a fermentation by-product. The recombinant yeast host cell comprises of at least one of phenotypic trait providing persistence of the recombinant yeast host cell in a plurality of fermentation cycles. The at least one phenotypic trait is: a fast settling phenotype, a rugose phenotype, an improved invertase activity, triploidy, increased signaling in a RAS/cAMP/PKA pathway or combinations thereof. In an embodiment, the recombinant yeast host cell, after a total of 40 fermentation cycles, is present in a proportion to at least 99% in a fermenting population. In an embodiment, the fermentation product is an alcohol, such as, for example, ethanol. In another embodiment, the fermentation by-product is glycerol.

In another embodiment, the recombinant yeast host cell is capable of increasing the expression of the first polypeptide. In yet another embodiment, the first polypeptide is a heterologous polypeptide capable of being expressed (and in some embodiments is expressed) in the recombinant yeast host cell and the recombinant yeast host cell comprises a first heterologous nucleic acid encoding the first polypeptide. In yet a further embodiment, the first polypeptide is a sugar transporter-like protein (STL1). In still another embodiment, STL1 has the amino acid sequence of SEQ ID NO: 8, is a variant of the amino acid sequence of SEQ ID NO: 8 having glycerol transport activity or is a fragment of the amino acid sequence of SEQ ID NO: 8 having glycerol transport activity and/or wherein the first heterologous nucleic acid molecule comprises the nucleic acid sequence of SEQ ID NO: 7 or is a degenerate sequence of SEQ ID NO: 7 encoding SEQ ID NO: 8. In still a further embodiment, the recombinant yeast host cell is capable of decreasing the expression of the second polypeptide, wherein the second polypeptide is a native polypeptide. In an embodiment, the native gene encoding the second polypeptide is inactivated. In some embodiments, the second polypeptide has NAD-dependent glycerol-3-phosphate dehydrogenase activity, such as, for example, NAD-dependent glycerol-3-phosphate dehydrogenase is glycerol-3-phosphate 1 (GPD1) and/or NAD-dependent glycerol-3-phosphate is glycerol-3-phosphate 2 (GPD2).

In some embodiments, the recombinant yeast host cell exhibits the fast settling phenotype. In some further embodiments, at least 5% of a population consisting essentially of the recombinant yeast host cells is able to sediment by gravity after 5 minutes. In some additional embodiment, a population consisting essentially of the recombinant yeast host cells is able to sediment by gravity after 5 minutes in a proportion equal to or higher than 15% than a control population consisting essentially of control yeast cells lacking the fast settling phenotypic trait. In an embodiment, the control yeast cells are from a *Saccharomyces cerevisiae* PE-2 strain.

In some embodiments, the recombinant yeast host cell exhibits the rugose phenotype. In some further embodiments, at least 90% of a population consisting essentially of the recombinant yeast host cells, after exponential growth in

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a medium inoculated at low recombinant yeast host cell density, has at least two daughter cells attached. In some additional embodiments, the recombinant yeast host cell is capable of reducing the transcription factor activity of a Activator of CUP1 Expression (ACE2) polypeptide. In some further embodiments, the recombinant yeast host cell is capable of expressing (and in some embodiments expresses) a mutated ACE2 polypeptide, wherein the mutated ACE2 polypeptide has decreased activity when compared to a wild type ACE2 polypeptide. In yet some additional embodiments, the mutated ACE2 polypeptide is a variant or a fragment of the amino acid sequence of SEQ ID NO: 10.

In some embodiments, the recombinant yeast host cell exhibits the improved invertase activity phenotypic trait. In some further embodiments, a population consisting essentially of the recombinant yeast host cells is able to consume hydrolyze more than 0.05 gram of sucrose per gram of dry cell weight per minute and/or exhibits more than 1.0 time more invertase activity than a control population consisting essentially of control yeast cells lacking the improved invertase activity phenotypic trait. The invertase activity can be measured after exponential growth of the population diluted to a concentration of 9 mg/mL on a wet cell weight in a buffer and wherein the buffer comprises 40 g/L of sucrose, is at of pH 5 and at a temperature of 35° C. In an embodiment, the control yeast cells are from a *Saccharomyces cerevisiae* PE-2 strain. In an embodiment, the recombinant yeast host cell is capable of increasing the enzymatic activity of at least one polypeptide having invertase activity. In still another embodiment, the at least one polypeptide having invertase activity comprises SUC1, SUC2, SUC3, SUC4, SUC5, SUC6, SUC7 SUC8 or SUC9.

In some embodiments, the recombinant yeast host cell is a triplloid cell.

In some embodiments, the recombinant yeast host cell exhibits the increased signaling in the RAS/cAMP/PKA pathway phenotypic trait. In some additional embodiments, a population consisting essentially of the recombinant yeast host cells is able to exhibit a fold increase in the production of cAMP of equal to or less than 1.7 and/or a fold increase in the production of cAMP of less than 70% when compared a control population consisting essentially of control yeast cells lacking the increased signaling in the RAS/cAMP/PKA pathway. In some embodiments, the production of cAMP is measured in the population having been glucose depleted and 5 minutes after a glucose spike. In an embodiment, the control yeast cells are from a *Saccharomyces cerevisiae* PE-2 strain. In another embodiment, the recombinant yeast host cell is capable of expressing (and in some embodiments, expresses) a mutated polypeptide involved in the RAS/cAMP/PKA pathway. In some further embodiments, the mutated polypeptide involved in the RAS/cAMP/PKA pathway comprises a mutated RAS2 polypeptide having increased activity when compared to a wild-type RAS2 polypeptide. In yet additional embodiments, the mutated RAS2 polypeptide is a variant or a fragment of the amino acid sequence of SED ID NO: 19. In some further embodiments, the mutated polypeptide involved in the RAS/cAMP/PKA pathway comprises a mutated IRA2 polypeptide having a reduced inhibitory activity towards a wild-type RAS1 and/or a wild-type RAS2 polypeptide when compared to a wild-type IRA2 polypeptide. In still yet another embodiment, the mutated IRA2 polypeptide is a variant and/or a fragment of the amino acid sequence of SEQ ID NO: 22.

In some embodiments, the recombinant yeast host cell is from the genus *Saccharomyces* sp. or from the species *Saccharomyces cerevisiae*.

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According to a second aspect, the present disclosure provides a process for prolonging the persistence of a yeast in a fermenting population in a plurality of fermentation cycles. The plurality of fermentation cycles comprises an initial fermentation cycle and at least one further fermentation cycle. The initial fermentation cycle comprises: (i) contacting an initial fermentation medium comprising a fermentable carbohydrate with an initial fermenting population to obtain a fermented medium comprising a fermentation product and a fermenting population and (ii) substantially isolating the fermenting population from the fermented medium. Each further fermentation cycle comprises: (iii) contacting the fermented population obtained from a previous fermentation cycle with a further fermentation medium comprising the fermentable carbohydrate to obtain a further fermented medium and (iv) substantially isolating the fermenting population from the further fermented medium. In the process, the initial fermenting population consists essentially of persistent yeast cells having at least one of the phenotypic trait as defined herein.

In some embodiments, the persistent yeast cell exhibits the fast settling phenotype. In some further embodiments, at least 5% of a population consisting essentially of the persistent yeast cells is able to sediment by gravity after 5 minutes. In some additional embodiment, a population consisting essentially of the persistent yeast cells is able to sediment by gravity after 5 minutes in a proportion equal to or higher than 15% than a control population consisting essentially of control yeast cells lacking the fast settling phenotypic trait. In an embodiment, the control yeast cells are from a *Saccharomyces cerevisiae* PE-2 strain.

In some embodiments, the persistent yeast cell exhibits the rugose phenotype. In some further embodiments, at least 90% of a population consisting essentially of the persistent yeast cells, after exponential growth in a medium inoculated at low persistent yeast cell density, has at least two daughter cells attached. In some additional embodiments, the persistent yeast cell is capable of reducing the transcription factor activity of a Activator of CUP1 Expression (ACE2) polypeptide. In some further embodiments, the persistent yeast cell is capable of expressing (and in some embodiments expresses) a mutated ACE2 polypeptide, wherein the mutated ACE2 polypeptide has decreased activity when compared to a wild type ACE2 polypeptide. In yet some additional embodiments, the mutated ACE2 polypeptide is a variant or a fragment of the amino acid sequence of SEQ ID NO: 10.

In some embodiments, the persistent yeast cell exhibits the improved invertase activity phenotypic trait. In some further embodiments, a population consisting essentially of persistent yeast cells is able to consume hydrolyze more than 0.05 gram of sucrose per gram of dry cell weight per minute and/or exhibits more than 1.0 time more invertase activity than a control population consisting essentially of control yeast cells lacking the improved invertase activity phenotypic trait. The invertase activity can be measured after exponential growth of the population diluted to a concentration of 9 mg/mL on a wet cell weight in a buffer and wherein the buffer comprises 40 g/L of sucrose, is at of pH 5 and at a temperature of 35° C. In an embodiment, the control yeast cells are from a *Saccharomyces cerevisiae* PE-2 strain. In an embodiment, the persistent yeast cell is capable of increasing the enzymatic activity of at least one polypeptide having invertase activity. In still another embodiment, the at least one polypeptide having invertase activity comprises SUC1, SUC2, SUC3, SUC4, SUC5, SUC6, SUC7, SUC8 or SUC9.

In some embodiments, the persistent yeast cell is a triploid cell.

In some embodiments, the persistent yeast cell exhibits the increased signaling in the RAS/cAMP/PKA pathway phenotypic trait. In some additional embodiments, a population consisting essentially of the persistent yeast cells is able to exhibit a fold increase in the production of cAMP of equal to or less than 1.7 and/or a fold increase in the production of cAMP of less than 70% when compared to a control population consisting essentially of a control yeast cell lacking the increased signaling in the RAS/cAMP/PKA pathway. In some embodiments, the production of cAMP is measured in the population having been glucose depleted and 5 minutes after a glucose spike. In an embodiment, the control yeast cells are from a *Saccharomyces cerevisiae* PE-2 strain. In another embodiment, the persistent yeast cell is capable of expressing (and in some embodiments, expresses) a mutated polypeptide involved in the RAS/cAMP/PKA pathway. In some further embodiments, the mutated polypeptide involved in the RAS/cAMP/PKA pathway comprises a mutated RAS2 polypeptide having increased activity when compared to a wild-type RAS2 polypeptide. In yet additional embodiments, the mutated RAS2 polypeptide is a variant or a fragment of the amino acid sequence of SEQ ID NO: 19. In some further embodiments, the mutated polypeptide involved in the RAS/cAMP/PKA pathway comprises a mutated IRA2 polypeptide having a reduced inhibitory activity towards a wild-type RAS1 and/or a wild-type RAS2 polypeptide when compared to a wild-type IRA2 polypeptide. In still yet another embodiment, the mutated IRA2 polypeptide is a variant and/or a fragment of the amino acid sequence of SEQ ID NO: 22.

In some embodiments, the persistent yeast cell is from the genus *Saccharomyces* sp. or from the species *Saccharomyces cerevisiae*.

In some embodiments, the persistent yeast cells can be a recombinant yeast host cell as defined herein. In additional embodiments of the process, the persistent yeast cells, after a total of 40 fermentation cycles, are present in a proportion to at least 99% in the substantially isolated fermenting population. In an embodiment, the fermentation product is an alcohol, such as, for example, ethanol. In yet another embodiment, the fermentation by-product is glycerol. In still further embodiments, the initial and/or the further fermentation medium comprises sugarcane, a sugarcane derivative, molasses, a molasses derivative or a mixture thereof. In yet additional embodiments, the plurality of fermentation cycles comprises at least one continuous fermentation and/or at least one batch fermentation. In further embodiments, the substantially isolating step comprises centrifuging the fermented medium and/or the further fermented medium to substantially isolate the fermenting population. In yet some additional embodiments, each fermentation cycle further comprises acid washing the substantially isolated fermenting population prior to a further fermentation cycle. In some embodiments, the process comprises at least two or more fermentation cycles. In some embodiments, the process further comprises recuperating the fermentation product from the fermented medium and/or the further fermented medium.

According to a third aspect, the present disclosure provides a fermentation medium comprising the persistent yeast cell having the at least one phenotypic trait defined herein. In some embodiments, the persistent yeast cell can be the recombinant yeast host cell described herein.

According to a fourth aspect, the present disclosure provides a fermenting population comprising a proportion of at

least 99% of persistent yeast cells having the at least one phenotypic trait defined herein. In some embodiments, the persistent yeast cells can be the recombinant yeast host cells described herein. In some embodiments, the persistent yeast cells have been submitted a substantially isolating step and/or an acid-washing step.

According to a fifth aspect, the present disclosure provides a process for making a yeast composition. The process comprises propagating a persistent yeast cell having the at least one phenotypic trait defined herein in a propagation medium to obtain a propagated medium, propagated persistent yeast cells. In some embodiments, the process can also comprise substantially isolating the propagated persistent yeast cells or the propagated recombinant yeast host cells from the propagated medium to obtain the yeast composition. In an embodiment, the persistent yeast cells can be the recombinant yeast host cells described herein.

According to a sixth aspect, the present disclosure provides a yeast composition comprising propagated persistent yeast cells having the at least one phenotypic trait defined herein. In an embodiment, the persistent yeast cells are propagated recombinant yeast host cells described herein.

According to a seventh aspect, the present disclosure provides a process for making a persistent yeast cell. The process comprises submitting an initial fermenting population consisting essentially of an initial cell to a plurality of fermentation cycles as defined herein and substantially isolating at least one yeast cell from the fermenting population to obtain the persistent yeast cell. In an embodiment, the process further comprises introducing at least genetic modification for modulating the activity or the expression of a polypeptide for increasing, when compared to a parental cell, the conversion of a biomass into a fermentation product and/or for reducing the conversion of the biomass into a fermentation by-product.

BRIEF DESCRIPTION OF THE DRAWINGS

Having thus generally described the nature of the invention, reference will now be made to the accompanying drawings, showing by way of illustration, a preferred embodiment thereof, and in which:

FIG. 1 is an embodiment of the process of the present disclosure.

FIGS. 2A to 2D show the sampling of inputs and outputs of the centrifugation process. Microscopic and plating results from sampling of inputs and outputs from the centrifugation process. (FIG. 2A) provides the percentage of recombinant yeasts host cells (Y1 strain) (with respect to the total number of yeasts recuperated) in the beer (prior to centrifugation), the wine (the supernatant proceeding to distillation also referred to a "de-yeasted wine") and the cream (the pellet intended to be recycled in the process) as measured by qPCR. Microscopic (top panel) and macroscopic (bottom panel) of (FIG. 2B) the beer, (FIG. 2C) the wine, and (FIG. 2D) the cream.

FIG. 3 provides the results of the settling assay data of a non-aggregated control, a sample of the beer, of the wine and of the cream. Results are shown as the % of sedimentation in 3.5 min in function of the sample tested.

FIGS. 4A to 4C compares the characteristics of yeasts of the rugose and of the smooth phenotype. (FIG. 4A) Example of rugose (multicellular) and smooth (monocellular or budding) occurring on a sample of yeast taken from a commercial ethanol mill. (FIG. 4B) Light (top panel) and electron scanning electron microscopy (middle (scale bar 10 μ M) and lower panel (scale bar 3 μ M)) images of a smooth yeast

strain. (FIG. 4C) Light (top panel) and electron scanning microscopy (middle (scale bar 10 μ M) and lower panel (scale bar 3 μ M)) of a rugose yeast strain.

FIG. 5 provides the calculated exponential washout rates at eight commercial ethanol facilities. Strain(s) used to inoculate various facilities (each bar represents a different facility) is/are provided below each bar (Y1 alone, or a combination of Y1 and Y2).

FIG. 6 shows the alignment of the functional A8 and nonfunctional A7 alleles at ACE2 for Y1, indicating the translation frameshift and early stop codon introduced at amino acid residue 389 (identified with an arrow).

FIG. 7 provides the sedimentation of various isolates from commercial mills. Yeasts were isolated from five different mills and tested for their sedimentation in a liquid assay measuring the change in OD₆₀₀ in 5 minutes of settling. Results are shown as the percent of the wild yeast settling faster than strain Y2.

FIG. 8 provides the percentage of the A7 ACE2 mutation in three different commercial wild yeast populations (e.g., Mill 7, Mille 4 and Mill 5). Results are shown as the percentage of the A7 allele in the wild yeast population.

FIG. 9 shows the increase of the concentration of strain Y2 during a commercial implementation. Results are shown as the percentage of strain Y2 (estimated from the determination of allele frequency) in function of the number of fermentation cycle based on the estimation of the CDA1-2 allele (◆) or the ACE2-A7 allele (■).

FIGS. 10A and 10B show the results of strains fermented in a cell recycle process with acid wash at 33.5° C. for 10 hours with a 7 hour feed of industrially sourced must. (FIG. 10A) Sucrose consumption (measured by HPLC at time 0 (black bars), 1 (white bars) and 2 hrs (grey bars) after the start of feed in function of the yeast strains tested. (FIG. 10B) Fermentation kinetics monitored by CO₂ off gas over the course of the fermentation in function of the yeast strains tested (Y0=dark gray line; Y3=light grey line, Y5=stapled line).

FIGS. 11A to 11C shows the phenotypic characterization of wild yeast isolates (Y9, Y10, Y11 and Y12) compared to their parental strains Y0 and Y1. The strains or isolates were spot plated on (FIG. 11A) YPD medium, (FIG. 11B) YPS medium supplemented with 2-deoxyglucose and (FIG. 11C) YPD supplemented with rapamycin.

FIG. 12 compares the response of yeast produced cAMP in basal conditions (dark grey bars) and in response to a 100 mM glucose spike (light gray bars) for yeast strains Y1, Y3 and Y13. Results are shown as the cAMP produced (pmol/mL) in function of the experimental conditions and the strain tested.

FIG. 13 shows the population monitoring of co-cultures in a lab scale cell recycling process. Results are shown as the percentage of competing yeasts (as measure by qPCR) for two different yeast strain populations (Y1/Y0 and Y1/Y3) prior to (dark grey bars) and after three recycling cycles (light grey bars).

FIGS. 14A to 14C show the strain ploidy of wild yeast isolates from commercial cane ethanol facilities. Results are shown as the ratio of the alternate allele depth/total read region in function of frequency. Boxes highlight the triploid strains. Diploids are not boxed. Results are shown for the following strains (left to right, top to bottom). (FIG. 14A): Y14 (triploid), Y15 (triploid), Y16 (triploid), Y17 (triploid), Y18 (triploid), Y19 (triploid), Y20 (triploid), Y21 (triploid), Y22 (triploid), Y23 (triploid), Y24 (triploid), Y25 (triploid); (FIG. 14B): Y26 (triploid), Y27 (triploid), Y28 (triploid), Y29 (triploid), Y30 (triploid), Y31 (triploid), Y32 (triploid),

Y33 (diploid), Y34 (triploid); (FIG. 14C): Y35 (triploid), Y36 (triploid), Y37 (diploid), Y38 (triploid), Y39 (triploid), Y40 (triploid), and Y10 (diploid).

FIG. 15 compares the sedimentation rate of various yeast strains. Results are shown as the percentage of sedimentation in function of the yeast strain tested (Y0, Y1, Y2, Y4 or Y6).

FIGS. 16A and 16B provide the yield increase and kinetics of engineered yeasts with persistence traits. Y0, Y2, Y4 and Y6 were recycled for 13 cycles of acid treatment and fermentation on commercial must. (FIG. 16A) The percentage in yield increase (dark gray bars, left axis) and percentage glycerol reduction (light gray squares, right axis) are shown relative to the control strain Y0. (FIG. 16B) The fermentation kinetics were monitored by measuring CO₂ production during the 10 hour fermentation. Results are shown as CO₂ production (ml/min) in function of time (hours) and strain used (Y0: solid black line; Y2: dashed line; Y4: dotted line; Y6: solid light gray line).

FIGS. 17A and 17B provide the population tracking of yeast strains Y2 compared with (FIG. 17A) Y4 or (FIG. 17B) Y6 over 13 cycles of acid treatment and fermentation. Results are shown as the percentage of the yeast strain in function of the number of cycles.

FIGS. 18A and 18B provide the fermentation yields obtained with strains Y0, Y2, Y4 and Y0. (FIG. 18A) shows the % change in ethanol yield (top panel, grey bars) and in glycerol production (lower panel, black bars) in function of Y0. (FIG. 18B) shows the K value of each strain in function of contaminating strain Y7.

FIG. 19 provides the amount of reducing sugar per gram of dry cell weight (gDCW) per minute for yeast strains Y0, Y2, Y3, Y4, Y7 or Y8 on commercial must. The horizontal line within the box represents the median sample value. The ends of the box represent the 25th and 75th quantiles, also expressed as the 1st and 3rd quartile, respectively. The difference between the 1st and 3rd quartiles is referred to as the interquartile range. The whiskers extend from the ends of the box to the outermost data point that falls within the distances computed as follows: 1st quartile-1.5*(interquartile range) and 3rd quartile+1.5*(interquartile range). If the data points do not reach the computed ranges, then the whiskers are determined by the upper and lower data point values (not including outliers).

FIGS. 20A and 20B compare the performances of yeast strains Y2, Y4 and Y61 during multiple rounds of cocultures. (FIG. 20A) provides the calculated exponential washout rates of each tested strains. The origin of different commercial substrates is identified with different symbols. (FIG. 20B) provides the percentage in ethanol change (top panel) and in glycerol change (bottom panel) for each of the tested strains when compared to yeast strain Y0.

DETAILED DESCRIPTION

It was sought to obtain yeasts having a prolonged presence (e.g., to persist) over a plurality of fermentation cycles. In order to do so, Applicant has identified phenotypic traits which allowed a recombinant yeast host cell to persist during a the plurality of fermentation cycles. Recombinant yeast host cells lacking such phenotypic traits were rapidly selected out from the fermenting population.

The persistent yeast cells of the present disclosure can be submitted to a plurality of fermentation cycles and persist in the fermenting population that is being used throughout the fermentation cycles. The plurality of fermentation cycles in which the persistent yeast cells are submitted comprises at

least two distinct fermentation cycles: an initial fermentation cycle and one or more further fermentation cycles. In the initial fermentation cycle, a fermenting population consisting essentially of persistent yeast cells exhibiting the at least one phenotypic traits disclosed herein is contacted with a fermentation medium under conditions so as to obtain a fermentation product (and concurrently a fermented medium). As used in the present disclosure, "a fermenting population consisting essentially of the yeasts" refers to a population of cells which contains the yeasts having the at least one phenotypic traits and is substantially free of contaminating (wild) yeasts. In some embodiments, when the yeasts are recombinant yeast host cells, "a fermenting population consisting essentially of the recombinant yeast host" refers to a population of cells which contains the recombinant yeast host cells as described herein and is substantially free of contaminating (wild, non-genetically modified) yeasts. The fermenting population obtained at the end of this initial fermentation cycle is recycled for a further fermentation cycle (e.g., substantially isolated and used to inoculate a further fermentation medium). In this further fermentation cycle, no additional persistent yeast cells are added to the fermenting population. As such, the fermenting population used to inoculate the further fermentation medium consists essentially in the fermenting population substantially isolated in the initial fermentation cycle. It is recognized that the fermenting population used to inoculate the further fermentation medium can include contaminating wild yeasts which may have been introduced in the fermentation medium of the initial fermentation cycle. The inoculated further fermentation medium is then placed under conditions so as to obtain the fermented product and subsequently substantially isolate a (further) fermenting population (from a further fermented medium). The substantially isolated further fermenting population can be recycled and used to conduct one or more further fermentation cycle. It is understood that, in yet a further fermentation cycle, no additional persistent yeast cells are added to the further fermenting population. As such, the further fermenting population used to inoculate the yet further fermentation medium consists essentially in the fermenting population substantially isolated in the further fermentation cycle. It is recognized that the fermenting population used to inoculate the yet further fermentation medium can include contaminating wild yeasts which may have been introduced in the further fermentation medium of the further fermentation cycle. In an embodiment, a fermenting population obtained by using an initial fermenting population consisting essentially of the persistent yeast cells as described herein submitted to at least 40 fermentation cycles (total) comprises at least 90%, 99%, 99.9% or more persistent yeast cells having the at least one phenotype traits as described herein. In another embodiment, a fermenting population obtained by using an initial fermenting population consisting essentially of the recombinant persistent yeast host cells as described herein submitted to at least 40 fermentation cycles (total) comprises at least 90%, 99%, 99.9% or more of the recombinant persistent yeast host cells described herein.

An embodiment of a fermentation process **001** using a plurality of fermentation cycles is shown as FIG. 1. In the embodiments shown on FIG. 1, steps **200** and **300** refer to the initial fermentation cycle and steps **400** and **500** (which can be repeated) refer to the further fermentation cycle(s). In process **001**, at step **200**, an initial fermenting population is inoculated in a fermentation medium which is then submitted to an initial fermentation. The initial fermenting population that is added to the fermentation medium consists

essentially of the persistent yeast cells (which include, in some embodiments, the recombinant yeast host cells exhibiting the one or more phenotypic trait as described herein). It is understood that the fermentation medium of step **200** can initially include contaminating wild yeasts or can be contaminated during fermentation with wild yeasts. Once the initial fermentation has been completed (e.g., a fermentation product and a fermenting population have accumulated in the fermentation medium to provide a fermented fermentation medium), the resulting fermenting population, at step **300**, is substantially isolated from the fermented fermentation medium. As it will be explained below, the isolating step can include, without limitation, centrifuging the fermented fermentation medium and/or acid washing the substantially isolated fermenting population (not shown on FIG. 1). Once the initial fermentation cycle has been completed (at the conclusion of step **300**), the substantially isolated fermenting population is placed, at step **400**, into contact (e.g., used to inoculate) a further fermentation medium and allowed to perform a further fermentation. Once the further fermentation has been completed (e.g., a fermentation product and a further fermenting population have accumulated in the further fermentation medium to provide a further fermented fermentation medium), the resulting fermenting population, at step **500**, is substantially isolated from the fermented fermentation medium. As it will be explained below, the isolating step can include, without limitation, centrifuging the further fermented fermentation medium and/or acid washing the substantially isolated fermenting population (not shown on FIG. 1). In some embodiments, no additional persistent yeast cell exhibiting the one or more phenotypic trait, including the recombinant yeast host cell described herein, is added to the fermentation medium after step **200**, including during the one or more further fermentation cycles. However, in some embodiments, especially in the presence of contaminating microbes (such as bacteria and/or yeasts), it may be possible to add further persistent yeasts cells at the beginning of step **400** to perform the further fermentation. The substantially isolated fermenting population obtained at step **500** can be submitted to yet a further fermentation cycle at step **400**. In some embodiments, the process can also include, after steps **300** or **500**, recuperating, at step **600**, the fermentation product from the fermented fermentation medium or the further fermented fermentation medium. This can be used, for example, by distilling the fermented fermentation medium or the further fermented fermentation medium (not shown on FIG. 1).

In some embodiments, the persistent yeast cells of the present disclosure are recombinant yeast host cells capable of modulating the expression of one or more polypeptide for increasing, when compared to a non-genetically modified parental cell, the conversion of a biomass into a fermentation product during a fermentation and/or for reducing the production of a fermentation by-product during the fermentation. In one embodiment, the recombinant yeast host cells of the present disclosure include at least one genetic modification to increase or decrease the activity (and in some embodiments the expression) of one or more polypeptide involved in the conversion of a biomass into a fermentation product and optionally the reduction of a fermentation by-product. In some optional embodiments, the recombinant yeast host cells of the present disclosure can also include one or more further genetic modification for providing the at least one phenotypic traits disclosed herein.

When the genetic modification is aimed at increasing the activity of a specific targeted polypeptide (which may native

or heterologous) or the expression of a specific targeted gene (which may native or heterologous), the genetic modification can be made in one or multiple genetic locations. When the genetic modification is aimed at reducing or inhibiting the activity of a specific targeted polypeptide (which is native) or the expression of a specific targeted gene (which is native), the genetic modifications can be made in one or all copies of the targeted gene(s).

In the context of the present disclosure, the one or more genetic modifications are aimed at increasing, when compared to a parental cell, the conversion of a biomass into a fermentation product and/or at reducing the conversion of the biomass into a fermentation by-product in the recombinant yeast host cell. In an embodiment, when the one or more genetic modifications are aimed at increasing the conversion of a biomass (e.g., sugarcane or a biomass derived therefrom) into a fermentation product (e.g., an alcohol, for example, ethanol), an increase of at least 0.5, 0.6, 0.7, 0.8, 0.9, 1.0, 1.1, 1.2, 1.3, 1.4, 1.5, 1.6, 1.7, 1.8, 1.9, 2.0, 2.1, 2.2, 2.3, 2.4, 2.5% or more of ethanol change in the recombinant yeast host cell when compared to a parental cell can be observed. In an embodiment, the parental cell is the *Saccharomyces cerevisiae* PE-2 strain. In another embodiment, when the one or more genetic modifications are aimed at reducing the conversion of the biomass (e.g., sugarcane or a biomass derived therefrom) into a fermentation by-product (e.g., glycerol), a decrease of at least 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30% or more of glycerol change in the recombinant yeast host cell when compared to a parental cell can be observed. In an embodiment, the parental cell is the *Saccharomyces cerevisiae* PE-2 strain.

In the context of the present disclosure, the recombinant yeast host cells are qualified as being “genetically engineered”, e.g., they have been manipulated to either add at least one or more heterologous or exogenous nucleic acid residue and/or remove at least one endogenous (or native) nucleic acid residue. In some embodiments, the one or more nucleic acid residues that are added can be derived from a heterologous cell or the recombinant cell itself. In the latter scenario, the nucleic acid residue(s) is (are) added at a genomic location which is different than the native genomic location. The genetic manipulations did not occur in nature and are the results of in vitro manipulations of the native yeast.

In some embodiments, the genetic modification can be encoded on one or more heterologous molecules. In some embodiments, the heterologous nucleic acid molecule can encode one or more polypeptide (which may be additional copies of a native gene). In other embodiments, the heterologous nucleic acid molecules can encode a promoter or other regulatory sequence for upregulating or downregulating the expression of a native gene encoding a native polypeptide. In some embodiments, the heterologous nucleic acid molecules of the present disclosure can include a signal sequence to favor the secretion of the heterologous polypeptide or the native polypeptide.

The term “heterologous” when used in reference to a nucleic acid molecule (such as a promoter, a terminator or a coding sequence) or a protein/polypeptide refers to a nucleic acid molecule or a protein/polypeptide that is not natively found in the recombinant host cell. “Heterologous” also includes a native coding region/promoter/terminator, or portion thereof, that was removed from the source organism and subsequently reintroduced into the source organism in a form that is different from the corresponding native gene, e.g., not in its natural location in the organism’s genome.

The heterologous nucleic acid molecule is purposively introduced into the recombinant yeast host cell. For example, a heterologous element could be derived from a different strain of host cell, or from an organism of a different taxonomic group (e.g., different kingdom, phylum, class, order, family genus, or species, or any subgroup within one of these classifications). As used herein, the term “native” when used in inference to a gene, polypeptide, enzymatic activity, or pathway refers to an unmodified gene, polypeptide, enzymatic activity, or pathway originally found in the recombinant host cell. In some embodiments, heterologous polypeptides derived from a different strain of host cell, or from an organism of a different taxonomic group (e.g., different kingdom, phylum, class, order, family genus, or species, or any subgroup within one of these classifications) can be used in the context of the present disclosure.

The heterologous nucleic acid molecules of the present disclosure can comprise a coding region for the heterologous polypeptide. A DNA or RNA “coding region” is a DNA or RNA molecule (preferably a DNA molecule) which is transcribed and/or translated into a heterologous polypeptide in a cell in vitro or in vivo when placed under the control of appropriate regulatory sequences. “Suitable regulatory regions” refer to nucleic acid regions located upstream (5' non-coding sequences), within, or downstream (3' non-coding sequences) of a coding region, and which influence the transcription, RNA processing or stability, or translation of the associated coding region. Regulatory regions may include promoters, transcription terminators, translation leader sequences, RNA processing site, effector binding site and stem-loop structure. The boundaries of the coding region are determined by a start codon at the 5' (amino) terminus and a translation stop codon at the 3' (carboxyl) terminus. A coding region can include, but is not limited to, prokaryotic regions, cDNA from mRNA, genomic DNA molecules, synthetic DNA molecules, or RNA molecules. If the coding region is intended for expression in a eukaryotic cell (such as the recombinant yeast host cell of the present disclosure), a polyadenylation signal and transcription termination sequence will usually be located 3' to the coding region. In an embodiment, the coding region can be referred to as an open reading frame. “Open reading frame” is abbreviated ORF and means a length of nucleic acid, either DNA, cDNA or RNA, that comprises a translation start signal or initiation codon, such as an ATG or AUG, and a termination codon and can be potentially translated into a polypeptide sequence.

The heterologous nucleic acid molecules described herein can comprise transcriptional and/or translational control regions. “Transcriptional and translational control regions” are DNA regulatory regions, such as promoters, enhancers, terminators, and the like, that provide for the expression of a coding region in a recombinant host cell. In eukaryotic cells, polyadenylation signals are considered control regions.

In some embodiments, the heterologous nucleic acid molecules of the present disclosure include a coding sequence for a heterologous polypeptide, optionally in combination with a promoter and/or a terminator. In some embodiments, the heterologous nucleic acid molecules of the present disclosure include a nucleic acid sequence encoding a promoter for overexpressing a native gene encoding a native polypeptide. In the heterologous nucleic acid molecules of the present disclosure, the promoter and the terminator (when present) are operatively linked to the nucleic acid coding sequence of the heterologous or native polypeptide, e.g., they control the expression and the termi-

nation of expression of the nucleic acid sequence of the heterologous or the native polypeptide. The heterologous nucleic acid molecules of the present disclosure can also include a nucleic acid sequence coding for a signal sequence, e.g., a short peptide sequence for exporting the heterologous polypeptide outside the host cell. When present, the nucleic acid sequence coding for the signal sequence is directly located upstream and in frame of the nucleic acid sequence coding for the heterologous polypeptide.

In the persistent yeast cells described herein, the nucleic acid molecule coding for the promoter and the nucleic acid molecule coding for the heterologous or the native polypeptide are operatively linked to one another. In the context of the present disclosure, the expressions "operatively linked" or "operatively associated" refers to fact that the promoter is physically associated to the nucleotide acid molecule coding for the heterologous or the native polypeptide in a manner that allows, under certain conditions, for expression of the heterologous polypeptide from the nucleic acid molecule. In an embodiment, the promoter can be located upstream (5') of the nucleic acid sequence coding for the heterologous polypeptide. In still another embodiment, the promoter can be located downstream (3') of the nucleic acid sequence coding for the heterologous polypeptide. In the context of the present disclosure, one or more than one promoter can be included in the heterologous nucleic acid molecule. When more than one promoter is included in the heterologous nucleic acid molecule, each of the promoters is operatively linked to the nucleic acid sequence coding for the heterologous or native polypeptide. The promoters can be located, in view of the nucleic acid molecule coding for the heterologous or native polypeptide, upstream, downstream as well as both upstream and downstream.

The term "promoter" refers to a DNA fragment capable of controlling the expression of a coding sequence or functional RNA. The term "expression," as used herein, refers to the transcription and stable accumulation of sense mRNA from the heterologous nucleic acid molecule or the native gene described herein. Expression may also refer to translation of mRNA into a polypeptide. Promoters may be derived in their entirety from the promoter of a native gene, or be composed of different elements derived from different promoters found in nature, or even comprise synthetic DNA segments. It is understood by those skilled in the art that different promoters may direct the expression at different stages of development, or in response to different environmental or physiological conditions. Promoters which cause a gene to be expressed in most cells at most times at a substantial similar level are commonly referred to as "constitutive promoters". Promoters which cause a gene to be expressed during the propagation phase of a yeast cell are herein referred to as "propagation promoters". Propagation promoters include both constitutive and inducible promoters, such as, for example, glucose-regulated, molasses-regulated, stress-response promoters (including osmotic stress response promoters) and aerobic-regulated promoters. Promoters which cause a gene to be expressed during the fermentation phase of a yeast cell are herein referred to as "fermentation promoters". Fermentation promoters include both constitutive and inducible promoters such as, for example, anaerobic promoters. In the context of the present disclosure, a "glycolytic promoter" is a promoter (or a combination of promoters) allowing the expression (or, in some embodiments, the overexpression) of a gene operatively associated thereto when the recombinant microbial cell is in placed in glycolytic conditions. The glycolytic promoter can be a constitutive promoter or a glucose-

inducible promoter. Glycolytic promoters exclude glucose-repressible promoters. It is further recognized that since in most cases the exact boundaries of regulatory sequences have not been completely defined, DNA fragments of different lengths may have identical promoter activity. A promoter is generally bounded at its 3' terminus by the transcription initiation site and extends upstream (5' direction) to include the minimum number of bases or elements necessary to initiate transcription at levels detectable above background. Within the promoter will be found a transcription initiation site (conveniently defined for example, by mapping with nuclease S1), as well as protein binding domains (consensus sequences) responsible for the binding of the polymerase.

The promoter can be native or heterologous to the nucleic acid molecule encoding the native or the heterologous polypeptide. The promoter can be heterologous to the native gene encoding the native polypeptide to be overexpressed. The promoter can be heterologous or derived from a strain being from the same genus or species as the recombinant host cell. In an embodiment, the promoter is derived from the same genus or species of the yeast host cell and the heterologous polypeptide is derived from a different genus than the host cell. The promoter can be a single promoter or a combination of different promoters.

In the context of the present disclosure, the promoter controlling the expression of the heterologous polypeptide or the native polypeptide can be a constitutive promoter (such as, for example, *tef2p* (e.g., the promoter of the *tef2* gene), *cwp2p* (e.g., the promoter of the *cwp2* gene), *ssa1p* (e.g., the promoter of the *ssa1* gene), *eno1p* (e.g., the promoter of the *eno1* gene), *hxx1* (e.g., the promoter of the *hxx1* gene) and *pgk1p* (e.g., the promoter of the *pgk1* gene)). In some embodiment, the promoter is *tef2p* (e.g., the promoter of the *tef2* gene). In some embodiment, the promoter is *adh1p* (e.g., the promoter of the *adh1* gene). However, in some embodiments, it is preferable to limit the expression of the polypeptide. As such, the promoter controlling the expression of the heterologous polypeptide or the native polypeptide can be an inducible or modulated promoters such as, for example, a glucose-regulated promoter (e.g., the promoter of the *hxt7* gene (referred to as *hxt7p*)) or a sulfite-regulated promoter (e.g., the promoter of the *gpd2* gene (referred to as *gpd2p* or the promoter of the *fzf1* gene (referred to as the *fzf1p*)), the promoter of the *ssu1* gene (referred to as *ssulp*), the promoter of the *ssu1-r* gene (referred to as *ssu1-rp*). In an embodiment, the promoter is an anaerobic-regulated promoters, such as, for example *tdh1p* (e.g., the promoter of the *tdh1* gene), *pau5p* (e.g., the promoter of the *pau5* gene), *hor7p* (e.g., the promoter of the *hor7* gene), *adh1p* (e.g., the promoter of the *adh1* gene), *tdh2p* (e.g., the promoter of the *tdh2* gene), *tdh3p* (e.g., the promoter of the *tdh3* gene), *gpd1p* (e.g., the promoter of the *gdp1* gene), *cdc19p* (e.g., the promoter of the *cdc19* gene), *eno2p* (e.g., the promoter of the *eno2* gene), *pdc1p* (e.g., the promoter of the *pdc1* gene), *hxt3p* (e.g., the promoter of the *hxt3* gene), *dan1* (e.g., the promoter of the *dan1* gene) and *tpilp* (e.g., the promoter of the *tpi1* gene). One or more promoters can be used to allow the expression of each heterologous polypeptides in the recombinant yeast host cell.

Still in the context of the present disclosure, the promoter controlling the expression of the heterologous polypeptide or the native polypeptide can be a glycolytic promoter. For example, the glycolytic promoter can be a promoter (or a combination of promoters) from an alcohol dehydrogenase gene, a glucose-6-phosphate isomerase gene, a phosphofruct-

tokinase gene, an aldolase gene, a triosephosphate isomerase gene, a glyceraldehyde-3-phosphate dehydrogenase gene, a 3-phosphoglycerate kinase gene, a phosphoglycerate mutase, an enolase and/or a pyruvate kinase gene.

One or more promoters can be used to allow the expression of each heterologous/native polypeptides in the persistent yeast cell. In the context of the present disclosure, the expression “functional fragment of a promoter” when used in combination to a promoter refers to a shorter nucleic acid sequence than the native promoter which retain the ability to control the expression of the nucleic acid sequence encoding the heterologous polypeptide. Usually, functional fragments are either 5' and/or 3' truncation of one or more nucleic acid residue from the native promoter nucleic acid sequence.

The heterologous nucleic acid molecule of the present disclosure can be integrated in the chromosome(s) of the recombinant yeast host cell. The term “integrated” as used herein refers to genetic elements that are placed, through molecular biology techniques, into the chromosome of a host cell. In some embodiments, the heterologous nucleic acid molecule(s) is/are integrated at one or more neutral integration site. For example, genetic elements can be placed into the chromosomes of the host cell as opposed to in a vector such as a plasmid carried by the host cell. Methods for integrating genetic elements into the chromosome of a host cell are well known in the art and include homologous recombination. The heterologous nucleic acid molecule can be present in one or more copies in the recombinant yeast host cell's chromosome. Alternatively, the heterologous nucleic acid molecule can be independently replicating from the yeast's chromosome. In such embodiment, the nucleic acid molecule can be stable and self-replicating. The heterologous nucleic acid molecules can be present in one or more copies in the recombinant yeast host cell. For example, each heterologous nucleic acid molecules can be present in 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12 copies or more per chromosome.

In some embodiments, the heterologous nucleic acid molecules which can be introduced into the recombinant host cells are codon-optimized with respect to the intended recipient recombinant yeast host cell. As used herein the term “codon-optimized coding region” means a nucleic acid coding region that has been adapted for expression in the cells of a given organism by replacing at least one, or more than one, codons with one or more codons to optimize expression levels. In general, highly expressed genes in an organism are biased towards codons that are recognized by the most abundant tRNA species in that organism. One measure of this bias is the “codon adaptation index” or “CAI,” which measures the extent to which the codons used to encode each amino acid in a particular gene are those which occur most frequently in a reference set of highly expressed genes from an organism.

The heterologous nucleic acid molecules can be introduced in the yeast host cell using a vector. A “vector,” e.g., a “plasmid,” “cosmid” or “artificial chromosome” (such as, for example, a yeast artificial chromosome) refers to an extra chromosomal element and is usually in the form of a circular double-stranded DNA molecule. Such vectors may be autonomously replicating sequences, genome integrating sequences, phage or nucleotide sequences, linear, circular, or supercoiled, of a single- or double-stranded DNA or RNA, derived from any source, in which a number of nucleotide sequences have been joined or recombined into a unique construction which is capable of introducing a promoter fragment and DNA sequence for a selected gene product along with appropriate 3' untranslated sequence into a cell.

Phenotypic Traits

The persistent yeast cells of the present disclosure exhibit its one or more phenotypic trait which allows them to be present over more fermentation cycles (e.g., persist during a plurality of fermentation cycles) than a corresponding control cell which lacks the phenotypic trait(s). The persistent yeast cells can be selected for the presence of one or more phenotypic trait or could be genetically engineered to provide one or more phenotypic trait. In some embodiments, the persistent yeast cells can be selected for the presence of at least one phenotypic trait and can be genetically engineered for the another phenotypic trait.

As it will be explained into more details below, the recombinant yeast host cells of the present disclosure necessarily includes at least one genetic modification (for modulating the activity or the expression of a first and/or a second polypeptide). It is possible to select/engineer a parental yeast host cell which possess the one or more phenotypic traits and modify such parental yeast host cell to include the genetic modification(s) for modulating the activity or the expression of a first and/or a second polypeptide. It is also possible to first introduce in a first yeast host cell the genetic modification(s) for modulating the activity or the expression of a first and/or a second polypeptide and afterwards select/engineer for the at least one phenotypic traits to obtain the recombinant yeast host cells of the present disclosure.

In an embodiment, the persistent yeast cells of the present disclosure exhibit at least one of the following phenotypic trait: a fast settling phenotype, a rugose phenotype, an improved invertase activity, triploidy, or increased signaling in a RAS/cAMP/PKA pathway. In another embodiment, the persistent yeast cells of the present disclosure exhibits at least two of the following phenotypic trait: a fast settling phenotype, a rugose phenotype, an improved invertase activity, triploidy, or increased signaling in a RAS/cAMP/PKA pathway. In a further embodiment, the persistent yeast cells of the present disclosure exhibits at least three of the following phenotypic trait: a fast settling phenotype, a rugose phenotype, an improved invertase activity, triploidy, or increased signaling in a RAS/cAMP/PKA pathway. In yet another embodiment, the persistent yeast cells of the present disclosure exhibits at least four of the following phenotypic trait: a fast settling phenotype, a rugose phenotype, an improved invertase activity, triploidy, or increased signaling in a RAS/cAMP/PKA pathway. In one embodiment, the persistent yeast cells of the present disclosure exhibits the following phenotypic traits: a fast settling phenotype, a rugose phenotype, an improved invertase activity, triploidy, and increased signaling in a RAS/cAMP/PKA pathway.

The persistent yeast cells of the present disclosure can advantageously be used in a plurality of fermentation cycles for converting a biomass into a fermentation product (e.g., an alcohol, such as, for example, ethanol) and in some optional embodiments, for reducing the conversion of the biomass into a fermentation by-product (e.g., distinct from ethanol, such as, for example, glycerol). In some embodiments, when the persistent yeast cells are provided as an initial fermenting population in a plurality of fermentation cycles, after 40, 45, 50, 55, 60, 65, 70, 75, 80, 90 or more fermentation cycles, the persistent yeast cells remain present in a proportion of at least 90, 91, 92, 93, 94, 95, 96, 97, 98, 99.0, 99.1, 99.2, 99.3, 99.4, 99.5, 99.6, 99.7, 99.8, 99.9% or more in the resulting fermenting population (when measured as its contribution to the DNA of the resulting fermenting population).

Fast Settling Phenotype

In some embodiments, the persistent yeast cells of the present disclosure can exhibit a fast settling phenotype. Yeasts exhibiting the fast settling phenotype are able to be centrifuged more efficiently and are therefore positively selected to be present in the substantially isolated fermenting population (and therefore persist further during subsequent fermentation cycles). As used in the context of the present disclosure, a "fast settling phenotype" refers to the ability of the persistent yeast cell to settle more rapidly (either by gravity or during centrifugation) than a control yeast lacking the fast settling phenotype. The fast settling phenotype can be due, at least in part, with an increased ability of the persistent yeast cell to flocculate when compared to a control yeast lacking the fast settling phenotype. The fast settling phenotype can be due, at least in part, by an increase ability of the persistent yeast cell to form cell clumps when compared to a control yeast lacking the fast settling phenotype. The fast settling phenotype can be due, at least in part, by the presence of the rugose phenotype in the persistent yeast cell.

In one embodiment, the persistent yeast cell of the present disclosure has the ability to settle more rapidly than a control non-persistent yeast, e.g., the PE-2 strain. The commercially available strain PE-2 (described in Argueso et al., 2009 as well as Basso et al., 2011 and having the JAY291 genome in the *Saccharomyces* Genome Database (SGD)) lacks the fast settling phenotype. In the Example of the present disclosure, strain PE-2 is referred to as Y0.

In a specific embodiment, when a population consisting essentially of the persistent yeast cells of the present disclosure is provided in a relatively homogeneous distribution in a liquid medium, at least 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20% or more of the persistent yeast cells of the population will sediment by gravity after 5 minutes (when measured using optical density). In another specific embodiment, when a population consisting essentially of the persistent yeast cells of the present disclosure is provided in a relatively homogeneous distribution in a liquid medium, at least 5% of the persistent yeast cells of the population will sediment by gravity after 5 minutes (when measured using optical density). In still a further embodiment, a higher proportion (at least 15, 20, 25, 30, 35, 40, 45, 50, 60, 65, 70, 75, 80, 85, 90, 95 or 100%) of a population consisting essentially of persistent yeast cells of the present disclosure, when compared to a control population consisting essentially of control yeast cells, will sediment by gravity after 5 minutes (when measured using optical density). The control yeast cells lack the fast settling phenotype. In an embodiment, the control population consists essentially of the *Saccharomyces cerevisiae* PE-2 strain which is shown in the Example to lack the fast settling phenotype. In order to determine the proportion of the populations that is able to sediment by gravity, the populations are provided in a relatively homogeneous distribution in a liquid medium and the proportion can be determined by optical density.

The persistent yeast cell can be selected, from a population, for the fast settling phenotype. This can be done, for example, by submitting a population of yeasts to a plurality of fermentation cycles which includes centrifuging the fermenting population in between cycles and selecting the persistent yeast cells having the ability to settle more rapidly during the process. Alternatively or in combination, the yeasts can be genetically engineered to provide a fast settling phenotype. This can be done, for example, by introducing one or more genetic modification to change the phenotype of a yeast from being smooth to rugose.

In an embodiment, the persistent yeast cell exhibits the fast settling phenotype optionally in combination with at least one of the following additional phenotypic traits: a rugose phenotype, an improved invertase activity, triploidy or increased signaling in a RAS/cAMP/PKA pathway. In another embodiment, the persistent yeast cell exhibits the fast settling phenotype optionally in combination with at least two of the following additional phenotypic traits: a rugose phenotype, an improved invertase activity, triploidy or increased signaling in a RAS/cAMP/PKA pathway. In a further embodiment, the persistent yeast cell exhibits the fast settling phenotype optionally in combination with at least three of the following additional phenotypic traits: a rugose phenotype, an improved invertase activity, triploidy or increased signaling in a RAS/cAMP/PKA pathway. In a further embodiment, the persistent yeast cell exhibits the fast settling phenotype in combination with the following additional phenotypic traits: a rugose phenotype, an improved invertase activity, triploidy and increased signaling in a RAS/cAMP/PKA pathway.

Rugose Phenotype

In some embodiments, the persistent yeast cells of the present disclosure can exhibit the rugose phenotype. The rugose phenotype can be observed in cells after having been exponentially grown in a medium inoculated at low density. In an embodiment, low density refers to a density which would allow for 3 to 10 generations. In some embodiments, the low density refers to a density between about 0.01 and about 1 g/L of dry cell weight. The rugose phenotype is associated with the reduced ability of the persistent yeast cells to sever the septum between daughter cells. Yeasts exhibiting the rugose phenotype are able to form clumps in liquid medium which allows them to settle more rapidly (either by gravity or by centrifugation). Yeasts exhibiting the rugose phenotype are thus preferably selected to be present in the substantially isolated fermenting population (and therefore further persist in subsequent fermentation cycles). Colonies of persistent yeast cells exhibiting the rugose phenotype have irregular edges when cultured on a solid medium.

In a specific embodiment, when a population consisting essentially of the persistent yeast cells having the rugose phenotype is provided after exponential growth in a medium inoculated at low cell density, at least 90, 91, 92, 93, 94, 95, 96, 97, 98, 99% or more of the persistent yeast cells having at least two daughter cells attached. This assessment can be made, for example, by microscopic observation of the population after the exponential growth. In some embodiments, a population consisting essentially of persistent yeast cell exhibiting the rugose phenotype, after exponential growth in a medium inoculated at high cell density, may exhibit a lesser percentage of cells having at least two daughter cells attached. In some embodiments, the expression "high cell density" refers to a cell density of at least about 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29 or 30 g/L of dry cell weight. In further embodiments, the expression "high cell density" refers to a cell density between about 1 to about 30 g/L of dry cell weight.

In an embodiment, the persistent yeast cells exhibiting the rugose phenotype have a reduced transcription factor activity of an Activator of CUP1 Expression (ACE2) polypeptide (when compared to a control yeast expressing a wild-type ACE2 polypeptide, such as, for example, the control PE-2 strain). This reduction in activity can be caused, in some embodiments, by the presence of a genetic modification (e.g., insertion, deletion, substitution or indel) in the regu-

latory region (e.g., promoter) or the coding region of the gene encoding the ACE2 polypeptide. In a further embodiment, the persistent yeast cells of the present disclosure includes at least one genetic modification (when compared to the wild-type nucleic acid sequence of the ACE2 gene provided as SEQ ID NO: 9 or a degenerate sequence encoding the amino acid sequence of SEQ ID NO: 10) in one, two or all alleles encoding the ACE2 gene. Modifications of the ACE2 gene to provide a rugose phenotype are known (Oud et al., 2013) and can introduced in the recombinant yeast host cell of the present disclosure to provide a persistent yeast cell. In a specific example, the persistent yeast cells of the present disclosure can include a deletion of at least one or more nucleic acid residues (which, in some embodiments, may be a deletion at or in the 3' region of the ACE2 gene) when compared to the wild-type nucleic acid sequence of the ACE2 gene provided as SEQ ID NO: 9 or a degenerate sequence encoding SEQ ID NO: 10. In some embodiments, this genetic modification is provided in FIG. 8, Tables 4 or 5. The genetic modification in the ACE2 gene may be present in one, two or all alleles of the ACE2 gene. As such, the persistent yeast cells of the present disclosure may be heterozygous or homozygous with respect to the genetic modification in the ACE2 gene.

In yet another embodiment, the persistent yeast cells of the present disclosure can include at least one amino acid modification (when compared to the wild-type amino acid sequence of the ACE2 polypeptide provided as SEQ ID NO: 10) in a mutated ACE2 polypeptide. In a specific embodiment, the persistent yeast cells exhibiting the rugose phenotype expresses a fragment of the wild-type ACE2 polypeptide, such as, for example, a C-terminal truncation of the wild-type ACE2 polypeptide (when compared to the wild-type amino acid sequence of the ACE2 polypeptide provided as SEQ ID NO: 10). In still further embodiments, the persistent yeast cells of the present disclosure can express a mutated ACE2 polypeptide having the amino acid sequence of SEQ ID NO: 12, 13, 14, 15 or 16. In some embodiments, the persistent yeast cells exhibiting the rugose phenotype are capable of expressing a variant of a mutated ACE2 polypeptide having the amino acid sequence of SEQ ID NO: 12, 13, 14, 15 or 16 having reduced transcription factor activity (when compared to the wild-type ACE2 polypeptide). In some further embodiments, the persistent yeast cells exhibiting the rugose phenotype are capable of expressing a fragment of a mutated ACE2 polypeptide having the amino acid sequence of SEQ ID NO: 12, 13, 14, 15 or 16 having reduced transcription factor activity (when compared to the wild-type ACE2 polypeptide).

In a specific embodiment, the persistent yeast cells of the present disclosure can express a mutated ACE2 polypeptide having the amino acid sequence of SEQ ID NO: 12. In such embodiment, the persistent yeast cells of the present disclosure can include one or both mutated ACE2 alleles comprising the nucleic acid sequence of SEQ ID NO: 11 (or a degenerate sequence encoding SEQ ID NO: 12). In some embodiments, the mutated allele for the ACE2 polypeptide can be located on a heterologous nucleic acid molecule introduced in the recombinant yeast host cell.

The persistent yeast cells can be selected, from a population of yeast cells, for the rugose phenotype. This can be done, for example, by submitting a population of yeast cells to a plurality of fermentation cycles which includes centrifuging the fermenting population in between cycles and selecting the yeast cells having the ability to settle more rapidly during the process (which are believed to have the rugose phenotype). This can also be done, for example, by

culturing on a solid medium a population of yeast cells, and selecting the colonies exhibiting the rugose phenotype (either by visual observation or by microscopy for example). Alternatively or in combination, the yeast can be genetically engineered to provide a rugose phenotype. This can be done, for example, by introducing one or more genetic modification to change the phenotype of the yeast from being smooth to rugose and in some embodiments, for expressing a mutated ACE2 polypeptide.

In an embodiment, the persistent yeast cell exhibits the rugose phenotypic trait optionally in combination with at least one of the following additional phenotypic traits: a fast settling phenotype, an improved invertase activity, triploidy or increased signaling in a RAS/cAMP/PKA pathway. In an embodiment, the persistent yeast cell exhibits the rugose phenotypic trait optionally in combination with at least two of the following additional phenotypic traits: a fast settling phenotype, an improved invertase activity, triploidy or increased signaling in a RAS/cAMP/PKA pathway. In an embodiment, the persistent yeast cell exhibits the rugose phenotypic trait optionally in combination with at least three of the following additional phenotypic traits: a fast settling phenotype, an improved invertase activity, triploidy or increased signaling in a RAS/cAMP/PKA pathway. In an embodiment, the persistent yeast cell exhibits the rugose phenotypic trait in combination the following additional phenotypic traits: a fast settling phenotype, an improved invertase activity, triploidy and increased signaling in a RAS/cAMP/PKA pathway.

Improved Invertase Activity

In some embodiments, the persistent yeast cells of the present disclosure can exhibit improved invertase activity when compared to a control yeast cell lacking the phenotype. Yeasts exhibiting the improved invertase activity phenotype are able to hydrolyse more rapidly and/or at lower pH a carbohydrate source (e.g., a disaccharide (such as, for example, sucrose) or a trisaccharide (such as, for example, raffinose or kestose)) that is present in the biomass of the fermentation medium. In some embodiments, the persistent yeast cells of the present disclosure are used in fermentations in which the initial levels of fermentable glucose and fructose are relatively low (e.g., between 1 to 5 g/L) and as such their improved invertase activity can provide a selective advantage by accessing a complementary carbohydrate source (and consequently begin the fermentation process more rapidly).

In one embodiment, the persistent yeast cells of the present disclosure exhibit increased invertase activity when compared to the a control strain (e.g., such as the PE-2 strain) in comparable conditions. A population consisting essentially of the persistent yeast cells of the present disclosure is able to consume more than 1.0, 1.7, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11 times sucrose when compared to a population consisting essentially of control yeasts (e.g., such as the PE-2 strain) in comparable conditions. In order to make such comparison, both populations, after exponential growth, can be diluted, on a wet cell basis, to a concentration of 9 mg/mL in a buffer and sucrose consumption can be measured (by HPLC for example) at a specified time interval. In some embodiments, invertase activity is measured prior to saturation of the assay. In some specific embodiments, invertase activity is measured within 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14 or 15 minutes of the beginning of the assay. In some additional embodiments, invertase activity is measured 12 to 15 minutes after the beginning of the assay. Still in such embodiment, the buffer initially comprises 40 g/L of sucrose, is at of pH 5 and at a temperature of 35° C.

In a specific embodiment, a population consisting essentially of the persistent yeast cells of the present disclosure, after exponential growth, diluted to a concentration of 9 mg/mL, on a wet cell basis, in a buffer is able to consume more than 0.05 gram of sucrose per gram of dry cell weight per minute. In some additional embodiments, a population consisting essentially of the persistent yeast cells of the present disclosure, after exponential growth, diluted to a concentration of 9 mg/mL, on a wet cell basis, in a buffer is able to consume at least 0.06, 0.07, 0.08, 0.09, 0.10, 0.11, 0.12, 0.13, 0.14, 0.15, 0.16, 0.17, 0.18, 0.19, 0.20 or more gram of sucrose per gram of dry cell weight per minute. As indicated above, the invertase assay can use a buffer initially comprising 40 g/L of sucrose, at pH 5 and at a temperature of 35° C. In some embodiments, invertase activity is measured prior to saturation of the assay. In some specific embodiments, invertase activity is measured within 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14 or 15 minutes of the beginning of the assay. In some additional embodiments, invertase activity is measured 12 to 15 minutes after the beginning of the assay. Furthermore, in additional embodiments, the sugar consumption can be measured, for example, by HPLC.

The persistent yeast cells of the present disclosure can have increased enzymatic activity of at least one polypeptide having invertase activity. This increased enzymatic activity can be observed, in some embodiments, at low pH (for example, at a pH equal to or lower than 5.0, 4.9, 4.8, 4.7, 4.6, 4.5, 4.4, 4.3, 4.2, 4.1, 4.0 or lower). This increased invertase activity can be associated with a genetic modification in the regulatory region (e.g., promoter region) and/or the coding region of a gene encoding an invertase. This increased invertase activity can be associated with additional copies or duplications of a gene encoding an invertase. In yeasts, the following polypeptides are known to exhibit invertase activity: SUC1 (encoded by the SUC1 gene), SUC2 (encoded by the SUC2 gene), SUC3 (encoded by the SUC3 gene), SUC4 (encoded by the SUC4 gene), SUC5 (encoded by the SUC5 gene), SUC6 (encoded by the SUC6 gene), SUC7 (encoded by the SUC7 gene), SUC8 (encoded by the SUC8 gene) and SUC9 (encoded by the SUC9 gene). In some embodiments, the persistent yeast cells of the present disclosure have a genetic modification in one or more genes encoding an invertase. In further embodiments, the genetic modification (s) present in the persistent yeast cells can be associated with a modification in the amino acid sequence of the polypeptide having invertase activity (providing the modified polypeptide with increased invertase activity).

The persistent yeast cells can be selected, from a population of yeast cells, for the improved invertase activity phenotype. This can be done, for example, by submitting a population of yeast cells to a plurality of fermentation cycles which includes limiting the amount of fermentable sugars (like glucose and fructose) and selecting the yeast cells having the ability to digest unfermentable sugars more rapidly (like sucrose). Alternatively or in combination, the yeasts can be genetically engineered to provide an improved invertase activity phenotype. This can be done, for example, by introducing one or more genetic modification to increase the activity or the expression of one or more polypeptide having invertase activity.

In an embodiment, the persistent yeast cell exhibits the improved invertase activity phenotypic trait optionally in combination with at least one of the following additional phenotypic traits: a fast settling phenotype, a rugose, triploidy or increased signaling in a RAS/cAMP/PKA pathway. In an embodiment, the persistent yeast cell exhibits the

improved invertase activity phenotypic trait optionally in combination with at least two of the following additional phenotypic traits: a fast settling phenotype, a rugose, triploidy or increased signaling in a RAS/cAMP/PKA pathway.

In an embodiment, the persistent yeast cell exhibits the improved invertase activity phenotypic trait optionally in combination with at least three of the following additional phenotypic traits: a fast settling phenotype, a rugose, triploidy or increased signaling in a RAS/cAMP/PKA pathway. In an embodiment, the persistent yeast cell exhibits the improved invertase activity phenotypic trait in combination with the following additional phenotypic traits: a fast settling phenotype, a rugose, triploidy and increased signaling in a RAS/cAMP/PKA pathway.

15 Triploidy

In some embodiments, the persistent yeast cells of the present disclosure are triploids. Cell size (volume) is larger in higher ploidy strains and is associated with robustness to toxic compounds as well as increased adaptive fitness.

The persistent yeast cells can be selected, from a population of yeast cells, for a higher ploidy, like triploidy. This can be done, for example, by submitting a population of yeast cells to a plurality of fermentation cycles and selecting the yeast cells being triploids. Alternatively or in combination, the persistent yeast cells can be submitted to a process to provide triploidy. This can be done, for example, by using various mating techniques known in the art.

In an embodiment, the persistent yeast cell is a triploid optionally in combination with at least one of the following additional phenotypic traits: a fast settling phenotype, a rugose phenotype, improved invertase activity or increased signaling in a RAS/cAMP/PKA pathway. In an embodiment, the persistent yeast cell is a triploid optionally in combination with at least two of the following additional phenotypic traits: a fast settling phenotype, a rugose phenotype, improved invertase activity or increased signaling in a RAS/cAMP/PKA pathway. In an embodiment, the persistent yeast cell is a triploid optionally in combination with at least three of the following additional phenotypic traits: a fast settling phenotype, a rugose phenotype, improved invertase activity or increased signaling in a RAS/cAMP/PKA pathway. In an embodiment, the persistent yeast cell is a triploid optionally in combination with the following additional phenotypic traits: a fast settling phenotype, a rugose phenotype, improved invertase activity and increased signaling in a RAS/cAMP/PKA pathway.

Increased Signaling in a RAS/cAMP/PKA Pathway

In another embodiment, the persistent yeast cells exhibit increased signaling in a RAS/cAMP/PKA pathway when compared to control yeast cell lacking this phenotype (e.g., the PE-2 strain for example). Signaling via the RAS/cAMP/PKA pathway is important for nutrient signaling in yeasts and is associated to responses to glucose, nitrogen and phosphate levels. Increasing signaling in the RAS/cAMP/PKA pathway is associated with glucose derepression which eventually leads to better glucose and nitrogen uptake even in the presence of low levels of glucose (e.g., between 1 to 5 g/L) or other nutrients.

As used in the context of the present disclosure a persistent yeast cell having "increased signaling activity in the RAS/cAMP/PKA pathway" exhibits an increase in biological activity in one or more protein in the RAS/cAMP/PKA pathway. Persistent yeast cells exhibiting the increased signaling activity in the RAS/cAMP/PKA pathway phenotypic trait can show a decrease in the production of cAMP after a glucose spike, when compared to corresponding control yeasts lacking the increased signaling activity in the RAS/

cAMP/PKA pathway phenotype. Alternatively or in combination, persistent yeast cells exhibiting the increased signaling activity in the RAS/cAMP/PKA pathway phenotypic trait, can show an increase in the production of cAMP in a basal medium, when compared to corresponding control yeasts lacking the increased signaling activity in the RAS/cAMP/PKA pathway phenotype. In some embodiments, the control yeasts lacking the increased signaling activity in the RAS/cAMP/PKA pathway phenotype are from the *Saccharomyces cerevisiae* PE-2 strain. As it is known in the art, the increase in cAMP caused by this biological pathway causes the dissociation of the PKA protein into the BCY1 protein and the TPK1-3 protein. This increase in RAS/cAMP/PKA signaling is preferably observed during fermentation (e.g., for example, in anaerobic conditions) and, in some embodiments, is not observed during propagation (e.g., for example, in glucose-limited aerobic conditions).

In one embodiment, the persistent yeast cells of the present disclosure exhibit increased signaling in the RAS/cAMP/PKA pathway when compared to a control strain lacking the increased signaling activity in the RAS/cAMP/PKA pathway phenotype (e.g., the PE-2 strain for example) in comparable conditions. In an embodiment, a population consisting essentially of the persistent yeast cells exhibits a fold increase in cAMP production less than 70, 65, 60, 55, 50, 45, 40, 35, 30, 25% or less when compared to the fold increase in cAMP production in a population consisting essentially of control yeast cells lacking the increased signaling activity in the RAS/cAMP/PKA pathway phenotype (e.g., the PE-2 strain) in comparable conditions. In order to make such comparison, both populations can be glucose depleted and submitted to a glucose spike. In addition, the cAMP production can be measured before and after a predetermined time (e.g., 5 min) of a glucose spike.

In a specific embodiment, a population consisting essentially of the persistent yeast cells having been glucose depleted and submitted to a glucose spike exhibits a fold increase in cAMP production equal to or less than 1.7, 1.6, 1.5, 1.4, 1.3 or less. In order to make such determination, the cAMP production can be measured before and after a predetermined time (e.g., 5 min) of a glucose spike.

In order to achieve such increase in RAS/cAMP/PKA signaling, the expression and/or activity of one or more polypeptide of the RAS/cAMP/PKA pathway can be increased (when compared to a corresponding control yeast lacking the phenotype, such as, for example, the PE-2 strain). The one or more polypeptides whose expression or biological activity can be increased include, but are not limited to a CDC25 polypeptide (a membrane bound guanine nucleotide exchange factor capable of activating a RAS1 polypeptide and/or a RAS2 polypeptide), a SDC25 polypeptide (a Ras guanine nucleotide exchange factor capable of activating the RAS1 polypeptide and/or the RAS2 polypeptide), a RAS1 polypeptide (GTPase whose activity increase the activity of the Cyr1 polypeptide) and/or a RAS2 polypeptide (a GTPase whose activity increases the activity of the Cyr1).

In an embodiment, the RAS2 polypeptide expression or its biological activity is increased to cause an increase in the signaling activity of the RAS/cAMP/PKA pathway in the persistent yeast cell. In such embodiment, the persistent yeast cell can include a mutation in the RAS2 polypeptide (herein referred to as a mutated RAS2 polypeptide) which increases its biological activity. For example, the mutated RAS2 polypeptide can be a variant or a fragment of the wild-type RAS2 polypeptide resulting in an increase in the biological activity of the RAS2 polypeptide. The RAS2

polypeptide is a GTPase and as such its biological activity includes binding to GTP and hydrolyzing GTP into GDP. As such, in the context of the present disclosure, a mutated RAS2 polypeptide having increased (biological) activity can exhibit a higher binding affinity for GTP, a higher GTP hydrolyzing activity or both, when compared to the wild-type RAS2 polypeptide. In an embodiment, the mutated RAS2 polypeptide can have one or more amino acid substitutions with respect to the amino acid sequence of the wild-type RAS2 polypeptide.

For example, the mutated RAS2 polypeptide can have an amino acid substitution at a residue corresponding to location 66 of SEQ ID NO: 19 (or a corresponding residue in another wild-type RAS2 polypeptide). In an embodiment, the amino acid substitution of the mutated RAS2 polypeptide is limited to the residue located at position 66 of SEQ ID NO: 19 (or a corresponding residue in another wild-type RAS2 polypeptide). In the wild-type RAS2 polypeptide of *S. cerevisiae* (SEQ ID NO: 19), the amino acid residue at location 66 is an alanine residue. In an embodiment, the mutated RAS2 polypeptide does not have an alanine residue located at position 66 of SEQ ID NO: 19 (or at a corresponding position in another wild-type RAS2 polypeptide), but instead has an histidine, an isoleucine, an arginine, a leucine, an asparagine, a lysine, an aspartic acid, a methionine, a cysteine, a phenylalanine, a glutamic acid, a threonine, a glutamine, a tryptophan, a glycine, a valine, a proline, a serine or a tyrosine residue. In an embodiment, the mutated RAS2 polypeptide has, at position 66 of SEQ ID NO: 19 (or at a corresponding position in another wild-type RAS2 polypeptide) does not have an aliphatic amino acid residue, such as, for example, a glycine, a valine, a leucine or an isoleucine residue. In still another embodiment, the mutated RAS2 polypeptide has, at position 66 of SEQ ID NO: 19 (or at a corresponding position in another wild-type RAS2 polypeptide) a hydroxyl or sulfur/selenium-containing amino acid, such as, for example, a serine, a cysteine, a threonine or a methionine residue. In yet another embodiment, the mutated RAS2 polypeptide has, at position 66 of SEQ ID NO: 19 (or at a corresponding position in another wild-type RAS2 polypeptide) a threonine residue. In still a further embodiment, the mutated RAS2 polypeptide has the amino acid sequence of SEQ ID NO: 17 and can be encoded by a nucleic acid molecule having a nucleic acid molecule having the sequence of SEQ ID NO: 18 or a degenerate sequence of SEQ ID NO: 18 encoding SEQ ID NO: 17.

In another example, the mutated RAS2 polypeptide can have an amino acid substitution at a residue corresponding to location 19 of SEQ ID NO: 19 (or a corresponding residue in another wild-type RAS2 polypeptide). In the wild-type RAS2 polypeptide of *S. cerevisiae* (SEQ ID NO: 19), the amino acid residue at location 19 is a glycine residue. In an embodiment, the mutated RAS2 polypeptide does not have a glycine residue located at position 19 of SEQ ID NO: 19 (or at a corresponding position in another wild-type RAS2 polypeptide), but instead has an histidine, an isoleucine, an arginine, a leucine, an asparagine, a lysine, an aspartic acid, a methionine, a cysteine, a phenylalanine, a glutamic acid, a threonine, a glutamine, a tryptophan, an alanine, a valine, a proline, a serine or a tyrosine residue. In yet another embodiment, the mutated RAS2 polypeptide has, at position 19 of SEQ ID NO: 19 (or at a corresponding position in another wild-type RAS2 polypeptide) a valine residue.

In another example, the mutated RAS2 polypeptide can have an amino acid substitution at a residue corresponding to location 77 of SEQ ID NO: 19 (or a corresponding residue in another wild-type RAS2 polypeptide). In the wild-type

RAS2 polypeptide of *S. cerevisiae* (SEQ ID NO: 19), the amino acid residue at location 77 is a glutamine residue. In an embodiment, the mutated RAS2 polypeptide does not have a glutamine residue located at position 77 of SEQ ID NO: 19 (or at a corresponding position in another wild-type RAS2 polypeptide), but instead has an histidine, an isoleucine, an arginine, a leucine, an asparagine, a lysine, an aspartic acid, a methionine, a cysteine, a phenylalanine, a glutamic acid, a threonine, an alanine, a tryptophan, a glycine, a valine, a proline, a serine or a tyrosine residue. In yet another embodiment, the mutated RAS2 polypeptide has, at position 77 of SEQ ID NO: 19 (or at a corresponding position in another wild-type RAS2 polypeptide) a lysine residue.

In another example, the mutated RAS2 polypeptide can have an amino acid substitution at a residue corresponding to location 112 of SEQ ID NO: 19 (or a corresponding residue in another wild-type RAS2 polypeptide). In the wild-type RAS2 polypeptide of *S. cerevisiae* (SEQ ID NO: 19), the amino acid residue at location 112 is an aspartic acid residue. In an embodiment, the mutated RAS2 polypeptide does not have an aspartic acid residue located at position 112 of SEQ ID NO: 19 (or at a corresponding position in another wild-type RAS2 polypeptide), but instead has an histidine, an isoleucine, an arginine, a leucine, an asparagine, a lysine, a glutamine, a methionine, a cysteine, a phenylalanine, a glutamic acid, a threonine, an alanine, a tryptophan, a glycine, a valine, a proline, a serine or a tyrosine residue. In yet another embodiment, the mutated RAS2 polypeptide has, at position 112 of SEQ ID NO: 19 (or at a corresponding position in another wild-type RAS2 polypeptide) a tyrosine residue.

In an embodiment, the RAS1 polypeptide expression or biological activity is increased to cause an increase in the signaling activity of the RAS/cAMP/PKA pathway in the persistent yeast cell. In such embodiment, the persistent yeast cell includes a mutation in the RAS1 polypeptide (herein referred to as a mutated RAS1 polypeptide) which increases its biological activity. For example, the mutated RAS1 polypeptide can be a variant or a fragment of the wild-type RAS1 polypeptide resulting in an increase in the biological activity of the RAS1 polypeptide. The RAS1 polypeptide is a GTPase and as such its biological activity include binding to GTP and hydrolyzing GTP into GDP. As such, in the context of the present disclosure, a mutated RAS1 polypeptide having increased (biological) activity can exhibit a higher binding affinity for GTP, a higher GTP hydrolyzing activity or both, when compared to the wild-type RAS1 polypeptide. In an embodiment, the mutated RAS1 polypeptide can have an amino acid substitution. For example, the mutated RAS1 polypeptide can have an amino acid substitution at a residue corresponding to location 66 of SEQ ID NO: 26 (or at a corresponding residue in another wild-type RAS1 polypeptide). In an embodiment, the amino acid substitution of the mutated RAS1 polypeptide is limited to the residue located at position 66 of SEQ ID NO: 26 (or a corresponding residue in another wild-type RAS1 polypeptide). In the wild-type RAS1 polypeptide of *S. cerevisiae* (SEQ ID NO: 26), the amino acid residue at location 66 is an alanine residue. In an embodiment, the mutated RAS1 polypeptide does not have an alanine residue located at position 66 of SEQ ID NO: 26 (or at a corresponding position in another wild-type RAS1 polypeptide), but instead has an histidine, an isoleucine, an arginine, a leucine, an asparagine, a lysine, an aspartic acid, a methionine, a cysteine, a phenylalanine, a glutamic acid, a threonine, a glutamine, a tryptophan, a glycine, a valine, a proline, a

serine or a tyrosine residue. In an embodiment, the mutated RAS1 polypeptide has, at position 66 of SEQ ID NO: 26 (or at a corresponding position in another wild-type RAS1 polypeptide) does not have an aliphatic amino acid residue, such as, for example, a glycine, a valine, a leucine or an isoleucine residue. In still another embodiment, the mutated RAS1 polypeptide has, at position 66 of SEQ ID NO: 26 (or at a corresponding position in another wild-type RAS1 polypeptide) a hydroxyl or sulfur/selenium-containing amino acid, such as, for example, a serine, a cysteine, a threonine or a methionine residue. In yet another embodiment, the mutated RAS1 polypeptide has, at position 66 of SEQ ID NO: 26 (or at a corresponding position in another wild-type RAS1 polypeptide) a threonine residue. In still a further embodiment, the mutated RAS1 polypeptide has the amino acid sequence of SEQ ID NO: 27.

In another example, the expression and/or activity of one or more polypeptide of the RAS/cAMP/PKA pathway can be decreased (when compared to a corresponding control yeast lacking the phenotype, such as the PE-2 strain) to achieve an increase in the signaling activity RAS/cAMP/PKA pathway. The one or more polypeptide whose expression or biological activity can be decreased include, but is not limited to, a IRA1 polypeptide (a GTPase-activating polypeptide whose activity decreases the activity of the wild-type Ras1 polypeptide and/or the wild-type RAS2 polypeptide) and/or an IRA2 polypeptide (a GTPase-activating polypeptide whose activity decreases the activity of the wild-type RAS1 polypeptide and/or the wild-type RAS2 polypeptide).

As indicated above, in an embodiment, the expression and/or activity of the IRA2 polypeptide can be decreased to achieve an increase in the signaling activity in the RAS/cAMP/PKA pathway in the persistent yeast cell. In an embodiment, the persistent yeast cell can include a mutation in the IRA2 polypeptide (herein referred to as a mutated IRA2 polypeptide) which decreases its biological activity. As it is known in the art, the IRA2 polypeptide converts the wild-type RAS1 polypeptide or the wild-type RAS2 polypeptide from their GTP-bound to their GDP-bound inactive form. The biological activity of the IRA2 polypeptide includes binding to the wild-type RAS1 polypeptide and to the wild-type RAS2 polypeptide. As such, in the context of the present disclosure, a mutated IRA2 polypeptide having decreased (biological activity) can exhibit a lower binding affinity for the wild-type RAS1 polypeptide, the wild-type RAS2 polypeptide or both, when compared to the wild-type IRA2. For example, the mutated IRA2 polypeptide can be a variant or a fragment of the wild-type IRA2 polypeptide (which can have, in some embodiments, the amino acid sequence of SEQ ID NO: 22 or be encoded by a nucleic acid molecule having the nucleic acid sequence of SEQ ID NO: 28 or be a degenerate sequence encoding the amino acid sequence of SEQ ID NO: 22) exhibiting in an increase in the biological activity of the wild-type RAS1 polypeptide and/or the wild-type RAS2 polypeptide. In an embodiment, the consensus sequence of the mutated IRA2 polypeptide can have the amino acid sequence of SEQ ID NO: 23. In yet a further embodiment, the mutated IRA2 polypeptide can have a mutation at the amino acid residue located at position 2440 of the wild-type IRA2 polypeptide of SEQ ID NO 22 (or a corresponding position in another wild-type IRA2 polypeptide). This mutation at position 2440 can cause the substitution of a glutamic acid residue to a lysine residue. In a specific embodiment, the mutated IRA2 polypeptide can have the amino acid sequence of SEQ ID NO: 20 and/or be encoded by a nucleic acid molecule having the nucleic acid

sequence of SEQ ID NO: 21 or a degenerate sequence of SEQ ID NO: 21 encoding SEQ ID NO: 20. In a specific embodiment, the mutated IRA2 polypeptide can be a truncated IRA2 polypeptide encoded by a nucleic acid molecule or a gene which includes a frame-shift mutation. The mutated fragment of the IRA2 polypeptide can have the amino acid sequence of SEQ ID NO: 24.

In an embodiment, the expression and/or activity of the IRA1 polypeptide can be decreased to achieve an increase in the signaling activity in the RAS/cAMP/PKA pathway in the persistent yeast cell. In an embodiment, the persistent yeast cell expresses a mutation in the IRA1 polypeptide (herein referred to as a mutated IRA1 polypeptide) which decreases its biological activity. For example, the mutated IRA1 polypeptide can be a variant or a fragment of the wild-type IRA1 polypeptide resulting in an increase in the biological activity of the wild-type RAS1 polypeptide and/or the wild-type RAS2 polypeptide. The IRA1 polypeptide converts the wild-type RAS1 polypeptide or the wild-type RAS2 polypeptide from their GTP-bound to their GDP-bound inactive form. The biological activity of the IRA1 polypeptide includes binding to the wild-type RAS1 polypeptide and/or to the wild-type RAS2 polypeptide. As such, in the context of the present disclosure, a mutated IRA1 polypeptide having decreased (biological activity) can exhibit a lower binding affinity for the wild-type RAS1 polypeptide, the wild-type RAS2 polypeptide or both. In a specific embodiment, the mutated IRA1 polypeptide can be a truncated IRA1 polypeptide encoded by a nucleic acid molecule or a gene which includes a frame-shift mutation.

In yet another example, the expression and/or activity of one or more polypeptide of the RAS/cAMP/PKA pathway can be increased and the expression and/or activity of one or more polypeptide of the RAS/cAMP/PKA pathway can be decreased both in comparison with a corresponding control yeast lacking the phenotype (like the PE-2 strain for example) to achieve an increase in the signaling activity RAS/cAMP/PKA pathway.

In order to achieve such increase in RAS/cAMP/PKA signaling, it is also possible to regulate the activity of one or more polypeptide of the RAS/cAMP/PKA signaling pathway at the post-transcriptional level. For example, it is possible to genetically modify the recombinant yeast host cell to allow for the glucose-induced polypeptide turnover of one or more polypeptides in the RAS/cAMP/PKA signaling pathway (e.g., the IRA1 polypeptide and/or the IRA2 polypeptide for example).

The persistent yeast cell can be selected, from a population of yeast cells, for the increased signaling in the RAS/cAMP/PKA pathway. This can be done, for example, by submitting a population of yeast cells to a plurality of fermentation cycles in limited nutrient availability and selecting the yeast cells exhibiting increased signaling in the RAS/cAMP/PKA pathway. Alternatively or in combination, the persistent yeast cell can be genetically engineered to provide increased signaling in the RAS/cAMP/PKA pathway. This can be done, for example, by introducing one or more genetic modification to introduce one or more genetic modification in the polypeptides involved in the RAS/cAMP/PKA pathway.

In an embodiment, the persistent yeast cell exhibits the increased signaling in a RAS/cAMP/PKA pathway phenotypic trait optionally in combination with at least one of the following additional phenotypic traits: a fast settling phenotype, a rugose phenotype, an improved invertase activity or triploidy. In an embodiment, the persistent yeast cell exhibits the increased signaling in a RAS/cAMP/PKA path-

way phenotypic trait optionally in combination with at least two of the following additional phenotypic traits: a fast settling phenotype, a rugose phenotype, an improved invertase activity or triploidy. In an embodiment, the persistent yeast cell exhibits the increased signaling in a RAS/cAMP/PKA pathway phenotypic trait optionally in combination with at least three of the following additional phenotypic traits: a fast settling phenotype, a rugose phenotype, an improved invertase activity or triploidy. In an embodiment, the persistent yeast cell exhibits the increased signaling in a RAS/cAMP/PKA pathway phenotypic trait in combination with the following additional phenotypic traits: a fast settling phenotype, a rugose phenotype, an improved invertase activity and triploidy.

15 Recombinant Yeast Host Cells

The recombinant yeast host cells of the present disclosure are capable of modulating the activity or the expression of at least one polypeptide for increasing, with compared to a parental cell, the conversion of a biomass into a fermentation product and optionally for reducing the production of a fermentation by-product. As used in the context of the present application, a "non-genetically modified parental cell" refer to the parental cell that was modified to provide the recombinant yeast host cell. The parental host cell does not include the one or more genetic modification that are present in the recombinant yeast host cell renders or augments the capacity of the latter capable to increase the conversion of a biomass into a fermentation product and optionally reducing the production of a fermentation by-product. The parental host cell can be a non-genetically modified cell. Alternatively, the parental host cell can include one or more genetic modifications that are unrelated to the conversion of a biomass into a fermentation product or in the production of a fermentation by-product. In some embodiments, the parental host cell can be a persistent yeast cell and include the one or more phenotypic traits described herein.

The recombinant yeast host cells of the present disclosure are intended to be used in a plurality of fermentation cycles as described herein. In each fermentation cycles, the recombinant yeast host cells of the present disclosure is involved in fermentation, e.g. the conversion of a biomass into a fermentation product (which can be an alcohol, such as, for example, ethanol). In one embodiment, the polypeptide(s) whose activity or expression is modulated in the recombinant yeast host cells can be directly involved in converting the biomass into the fermentation product. In another embodiment, the polypeptide(s) whose activity or expression is modulated in the recombinant yeast host cells can be indirectly involved in converting the biomass into the fermentation product by reducing or limiting the production of a fermentation by-product (such as, for example, glycerol) and ultimately increasing the fermentation yield (e.g., the yield of the fermentation product).

During yeast metabolism, a major by-product of biomass fermentation is glycerol. Glycerol is produced in microorganisms, such as yeasts, in response to a redox or osmotic stress. The glycerol produced is then exported from the cell where it is considered waste. While the production of glycerol is important to protect microorganisms from various stressors, it also tends to decrease ethanol yields, especially when the microorganisms are growing or encountering osmotic stress.

In yeasts, glycerol is a required metabolic end-product of ethanol fermentation allowing the yeast to balance its redox state and regenerate NAD⁺ used as a cofactor during glycolysis. During anaerobic growth on carbohydrates, produc-

tion of ethanol and carbon dioxide is redox neutral, while the reactions that create cell biomass and associated carbon dioxide are more oxidized relative to carbohydrates. The production of glycerol, which is more reduced relative to carbohydrates, functions as an electron sink to off-set cell biomass formation, so that overall redox neutrality is conserved. This is essential from a theoretical consideration of conservation of mass, and in practice strains unable to produce glycerol are unable to grow under anaerobic conditions. As glycerol is a byproduct with low value, it can be an undesirable by-product of fermentation. There is a strong commercial incentive to reduce glycerol as a by-product during the production of fuels and chemicals, as reduction typically results in an increased yield of the desired compound.

In an embodiment, the recombinant yeast host cells of the present disclosure are capable of increasing the activity or the expression of a first polypeptide involved in the conversion of a biomass into a fermentation product and/or the reduction in the production of a fermentation by-product. As such, the recombinant yeast host cells of the present disclosure can include one or more genetic modification for increasing the activity or the expression of a first polypeptide. In an embodiment, the genetic modification can be located in a regulatory region (such as a promoter region) of a native gene encoding a native (first) polypeptide so as to increase the expression (and ultimately the activity) of the native (first) polypeptide). Alternatively or in combination, the genetic modification can be the introduction of one or more (first) heterologous nucleic acid molecules encoding a heterologous (first) polypeptide in the recombinant yeast host cells so as to increase or provide the recombinant yeast host cell with increased activity of the first polypeptide. In embodiments in which the genetic modification is intended to cause the expression of a secreted first polypeptide, the first nucleic acid heterologous nucleic acid molecule can also include a portion encoding a signal sequence operatively associated with another portion encoding the secreted polypeptide. The heterologous nucleic acid molecules that may be present in the recombinant yeast host cells can be integrated at the same or different integration sites.

In an embodiment, the first polypeptide is a saccharolytic enzyme (or a polypeptide having saccharolytic activity) and as such the recombinant yeast host cell comprises one or more first genetic modification for overexpressing a native saccharolytic or expressing heterologous saccharolytic enzymes. As used in the context of the present disclosure, a "saccharolytic enzyme" can be any enzyme involved in carbohydrate digestion, metabolism and/or hydrolysis, including amylases, cellulases, hemicellulases, cellulolytic and amylolytic accessory enzymes, inulinases, levanases, and pentose sugar utilizing enzymes. amylolytic enzyme. In an embodiment, the saccharolytic enzyme is an amylolytic enzyme. As used herein, the expression "amylolytic enzyme" refers to a class of enzymes capable of hydrolyzing starch or hydrolyzed starch. Amylolytic enzymes include, but are not limited to alpha-amylases (EC 3.2.1.1, sometimes referred to fungal alpha-amylase, see below), maltogenic amylase (EC 3.2.1.133), glucoamylase (EC 3.2.1.3), glucan 1,4-alpha-maltotetrahydrolase (EC 3.2.1.60), pullulanase (EC 3.2.1.41), iso-amylase (EC 3.2.1.68) and amyloamylase (EC 2.4.1.25).

In an embodiment, the one or more amylolytic enzymes can be, a maltogenic alpha-amylase from *Geobacillus stearothermophilus*, a glucoamylase from *Saccharomycopsis fibuligera*, a glucan 1,4-alpha-maltotetrahydrolase from *Pseudomonas saccharophila*, a pullulanase from *Bacillus*

naganoensis, a pullulanase from *Bacillus acidopullulyticus*, an iso-amylase from *Pseudomonas amyloclavata*, and/or amyloamylase from *Thermus thermophilus*. Some amylolytic enzymes have been described in US20220127564 and are incorporated herein by reference.

For example, the heterologous alpha-amylase can be from a *Rhizomucor* sp., such as, for example, from *Rhizomucor pusillus*. In an embodiment, the heterologous alpha-amylase corresponds to Uniprot M9T189. For example, the heterologous alpha-amylase can be from a *Aspergillus* sp., such as, for example, from *Aspergillus luchuensis*. In an embodiment, the heterologous alpha-amylase corresponds to Uniprot A0A146F6W4 or to GenBank Accession Number GAT21778. In an embodiment, the heterologous alpha-amylase corresponds to Uniprot 013296 or to GenBank Accession Number BAA22993. For example, the heterologous alpha-amylase can be from *Aspergillus oryzae*. In an embodiment, the heterologous alpha-amylase corresponds to Uniprot Q2UIS5 or to GenBank Accession Number XP_001820542. For example, the heterologous alpha-amylase can be from *Aspergillus niger*. In an embodiment, the heterologous alpha-amylase corresponds to Uniprot A2QTS4 or to GenBank Accession Number XP_001393626. In an embodiment, the heterologous alpha-amylase corresponds to Uniprot A2R6F9 or to GenBank Accession Number XP_001397301. In an embodiment, the heterologous alpha-amylase corresponds to GenBank Accession Number XP_001395328. In an embodiment, the heterologous alpha-amylase corresponds to Uniprot A0A370BQ30 or to GenBank Accession Number RDH15462. For example, the heterologous alpha-amylase can be from *Aspergillus fischeri*. In an embodiment, the heterologous alpha-amylase corresponds to Uniprot A1CYB1 or to GenBank Accession Number XP_001265628. For example, the heterologous alpha-amylase can be from a *Homo* sp., such as, for example, from *Homo sapiens*. In an embodiment, the heterologous alpha-amylase corresponds to GenBank Accession Number 1B2Y_A.

For example, the yeast cell can bear one or more genetic modifications allowing for the production of a heterologous glucoamylase. Many microbes produce an amylase to degrade extracellular starches. In addition to cleaving the last $\alpha(1-4)$ glycosidic linkages at the non-reducing end of amylose and amylopectin, yielding glucose, γ -amylase will cleave $\alpha(1-6)$ glycosidic linkages. The heterologous glucoamylase can be derived from any organism. In an embodiment, the heterologous polypeptide is derived from a γ -amylase, such as, for example, the glucoamylase of *Saccharomycopsis fibuligera* (e.g., encoded by the glu 0111 gene). Examples of recombinant yeast cells bearing such genetic modifications and expressing saccharolytic enzymes are described in US20130323822 as well as in US20180265853 and are both herewith incorporated in its entirety.

In another embodiment, the first polypeptide can be involved in the transport of a fermentation by-product, like glycerol, in the recombinant yeast cell. In an embodiment, the first polypeptide is a glycerol transporter and can, in some further embodiments, be responsible for the import of glycerol in the recombinant yeast host cells. In a specific embodiment, the first polypeptide is a sugar transporter-like protein (STL1). As such, the recombinant yeast host cells comprise one or more first genetic modification for overexpressing a native STL1 or expressing a heterologous STL1. By increasing the activity or expression of the STL1 poly-

peptide, it is possible to control, to some extent, glycerol synthesis and ultimately increase the fermentation (ethanol) yield.

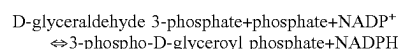
The STL1 polypeptide is natively expressed in yeasts and fungi, therefore the heterologous polypeptide functioning to import glycerol can be derived from yeasts and fungi. STL1 genes encoding the STL1 polypeptide include, but are not limited to, Gene ID: 852149 (encoded by SEQ ID NO: 7 and shown in SEQ ID NO: 8), *Candida albicans*, *Kluyveromyces lactis* Gene ID: 2896463 (SEQ ID NO: 67), *Eremothecium gossypii* Gene ID: 4620396 (SEQ ID NO: 36), *Eremothecium sincaudum* Gene ID: 28724161 (SEQ ID NO: 37), *Torulaspora delbrueckii* Gene ID: 11505245 (SEQ ID NO: 57), *Lachancea thermotolerans* Gene ID: 8290820 (SEQ ID NO: 60), *Phialophora attinorum* Gene ID: 28742143 (SEQ ID NO: 47), *Penicillium digitatum* Gene ID: 26229435 (SEQ ID NO: 46), *Aspergillus oryzae* Gene ID: 5997623 (SEQ ID NO: 61), *Aspergillus fumigatus* Gene ID: 3504696 (SEQ ID NO: 32), *Talaromyces atroroseus* Gene ID: 31007540 (SEQ ID NO: 53), *Rasamsonia emersonii* Gene ID: 25315795 (SEQ ID NO: 68), *Aspergillus terreus* Gene ID: 4322759 (SEQ ID NO: 33), *Penicillium rubens* Gene ID: 8310605 (SEQ ID NO: 58), *Alternaria alternata* Gene ID: 29120952 (SEQ ID NO: 31), *Paraphaeosphaeria sporulosa* Gene ID: 28767590 (SEQ ID NO: 45), *Pyrenophora tritici-repentis* Gene ID: 6350281 (SEQ ID NO: 49), *Metarhizium robertsii* Gene ID: 19259252 (SEQ ID NO: 41), *Isaria fumosorosea* Gene ID: 30023973 (SEQ ID NO: 39), *Cordyceps militaris* Gene ID: 18171218 (SEQ ID NO: 34), *Pochonia chlamydosporia* Gene ID: 28856912 (SEQ ID NO: 48), *Metarhizium majus* Gene ID: 26274087 (SEQ ID NO: 40), *Neofusicoccum parvum* Gene ID: 19029314 (SEQ ID NO: 63), *Diplodia corticola* Gene ID: 31017281 (SEQ ID NO: 35), *Verticillium dahliae* Gene ID: 20711921 (SEQ ID NO: 56), *Verticillium alfalfa* Gene ID: 9537052 (SEQ ID NO: 55), *Paracoccidioides lutzii* Gene ID: 9094964 (SEQ ID NO: 44), *Trichophyton rubrum* Gene ID: 10373998 (SEQ ID NO: 59), *Nannizzia gypsea* Gene ID: 10032882 (SEQ ID NO: 42), *Trichophyton verrucosum* Gene ID: 9577427 (SEQ ID NO: 62), *Trichophyton benhamiae* Gene ID: 9523991 (SEQ ID NO: 54), *Pyricularia oryzae* Gene ID: 2678012 (SEQ ID NO: 50), *Gaeumannomyces tritici* Gene ID: 20349750 (SEQ ID NO: 38), *Phaeoacremonium minimum* Gene ID: 19329524 (SEQ ID NO: 65), *Eutypa lata* Gene ID: 19232829 (SEQ ID NO: 64), *Scedosporium apiospermum* Gene ID: 27721841 (SEQ ID NO: 51), *Aureobasidium namibiae* Gene ID: 25414329 (SEQ ID NO: 66), *Sphaerulina musiva* Gene ID: 27905328 (SEQ ID NO: 52) as well as *Pachysolen tannophilus* GenBank Accession Numbers JQ481633 (SEQ ID NO: 69) and JQ481634 (SEQ ID NO: 43).

The first polypeptide can be encoded by STL1 gene as indicated herein or a STL1 gene ortholog or paralog. The heterologous polypeptide functioning to import glycerol can be a STL1 polypeptide as defined herein, a variant of the STL1 polypeptide and/or a fragment of the STL1 polypeptide. In addition, when more than one copy of the first heterologous nucleic acid molecule encoding STL1 is included in the recombinant yeast host cell, the plurality of first heterologous nucleic acid molecules encoding the STL1 polypeptide could be the same or different, integrated at the same or different integration sites.

In a specific embodiment, the recombinant yeast host cells of the present disclosure is capable of expressing a STL1 polypeptide having the amino acid sequence of SEQ ID NO: 8, a variant of the amino acid sequence of SEQ ID NO: 8 having glycerol transport activity or a fragment of the amino

acid sequence of SEQ ID NO: 8 having glycerol transporter activity. In another specific embodiment, the recombinant yeast host cells of the present disclosure comprises a heterologous nucleic acid molecule comprising the nucleic acid sequence of SEQ ID NO: 7, corresponds to a degenerate sequence of the nucleic acid sequence of SEQ ID NO: 7 (encoding SEQ ID NO: 8) or encodes the variant or the fragment of the amino acid sequence of SEQ ID NO: 8. The heterologous nucleic acid molecule encoding the STL1 polypeptide, variant or fragment can be present in one copy in the recombinant yeast host cell. The heterologous nucleic acid molecule encoding the STL1 polypeptide, variant or fragment can be present in two copies in the recombinant yeast host cell. The heterologous nucleic acid molecule encoding the STL1 polypeptide, variant or fragment can be present in three copies in the recombinant yeast host cell. The heterologous nucleic acid molecule encoding the STL1 polypeptide, variant or fragment can be present in four copies in the recombinant yeast host cell. The heterologous nucleic acid molecule encoding the STL1 polypeptide, variant or fragment can be present in five copies in the recombinant yeast host cell. The heterologous nucleic acid molecule encoding the STL1 polypeptide, variant or fragment can be present in six copies in the recombinant yeast host cell. The heterologous nucleic acid molecule encoding the STL1 polypeptide, variant or fragment can be present in seven copies in the recombinant yeast host cell. The heterologous nucleic acid molecule encoding the STL1 polypeptide, variant or fragment can be present in eight copies in the recombinant yeast host cell. The heterologous nucleic acid molecule encoding the STL1 polypeptide, variant or fragment can be present in nine copies in the recombinant yeast host cell. The heterologous nucleic acid molecule encoding the STL1 polypeptide, variant or fragment can be present in ten copies in the recombinant yeast host cell. The heterologous nucleic acid molecule encoding the STL1 polypeptide, variant or fragment can be present in eleven copies in the recombinant yeast host cell. The heterologous nucleic acid molecule encoding the STL1 polypeptide, variant or fragment can be present in twelve copies in the recombinant yeast host cell.

In an embodiment, the first polypeptide is a heterologous glyceraldehyde-3-phosphate dehydrogenase. Glyceraldehyde-3-phosphate dehydrogenases allow the catalysis of the reaction of glyceraldehyde-3-phosphate to 3-phosphoglycerate in glycolysis, using NADP⁺ as a cofactor. In some embodiments, regeneration of NADPH and/or NADH by way a glycolytic pathway using glyceraldehyde-3-phosphate also improves ethanol production and reduces glycerol production. The glyceraldehyde-3-phosphate dehydrogenase is a non-phosphorylating glyceraldehyde-3-phosphate dehydrogenase, e.g., it is incapable of mediating a phosphorylation reaction. In some embodiments, the glyceraldehyde-3-phosphate dehydrogenase is of enzyme commission (EC) class 1.2.1, however it excludes the enzymes capable of mediating a phosphorylating reaction. The glyceraldehyde-3-phosphate dehydrogenase of the present disclosure specifically exclude enzymes capable of directly using or generating of 3-phospho-D-glyceroyl phosphate, such as enzymes of EC 1.2.1.13. Enzymes of EC 1.2.1.13 catalyze the following reaction:



In one embodiment, the glyceraldehyde-3-phosphate dehydrogenase is NADP⁺ dependent (EC 1.2.1.9) and allows

the conversion of NADP⁺ to NADPH. Enzymes of EC1.2.1.9 can only use NADP⁺ as a cofactor.

In one embodiment, the glyceraldehyde-3-phosphate dehydrogenase is bifunctional NADP⁺/NAD⁺ dependent (EC1.2.1.90) and allows the conversion of NADP⁺ to NADPH and/or NAD⁺ to NAD⁺. Enzymes of EC1.2.1.90 can use NADP⁺ or NAD⁺ as a cofactor. In some embodiments, glyceraldehyde-3-phosphate dehydrogenase uses NADP⁺ and/or NAD⁺ as a cofactor. In one embodiment, the glyceraldehyde-3-phosphate dehydrogenase is encoded by a GAPN gene. In one embodiment, the glyceraldehyde-3-phosphate dehydrogenase is GAPN. In some embodiments, the recombinant yeast host cell includes two first genetic modifications and is capable of expressing STL1 and GAPN.

In some embodiments, the glyceraldehyde-3-phosphate dehydrogenase can be derived from a bacteria, for example, from the genus *Streptococcus* and, in some instances, from the species *Streptococcus mutans*. The glyceraldehyde-3-phosphate dehydrogenase can be encoded by the GAPN gene from *Streptococcus mutans*, or a GAPN gene ortholog, or a GAPN gene paralog. In an embodiment, the GAPN gene comprises the nucleic acid sequence of SEQ ID NO: 30, is a variant of the nucleic acid sequence of SEQ ID NO: 30 (including but not limited to a degenerate variant of SEQ ID NO: 30 encoding the amino acid sequence of SEQ ID NO: 29) or is a fragment of the nucleic acid sequence of SEQ ID NO: 30. In an embodiment, the GAPN has the amino acid sequence of SEQ ID NO: 29, is a variant of the amino acid sequence of SEQ ID NO: 29 or is a fragment of SEQ ID NO: 29.

In some embodiments, the glyceraldehyde-3-phosphate dehydrogenase can be derived from a bacteria, for example, from the genus *Lactobacillus* and, in some instances, from the species *Lactobacillus delbrueckii*. The glyceraldehyde-3-phosphate dehydrogenase can be encoded by the GAPN gene from *Lactobacillus delbrueckii*, or a GAPN gene ortholog, or a GAPN gene paralog.

In some embodiments, the glyceraldehyde-3-phosphate dehydrogenase can be derived from a bacteria, for example, from the genus *Streptococcus* and, in some instances, from the species *Streptococcus thermophilus*. The glyceraldehyde-3-phosphate dehydrogenase can be encoded by the GAPN gene from *Streptococcus thermophilus*, or a GAPN gene ortholog, or a GAPN gene paralog.

In some embodiments, the glyceraldehyde-3-phosphate dehydrogenase can be derived from a bacteria, for example, from the genus *Streptococcus* and, in some instances, from the species *Streptococcus macacae*. The glyceraldehyde-3-phosphate dehydrogenase can be encoded by the GAPN gene from *Streptococcus macacae*, or a GAPN gene ortholog, or a GAPN gene paralog.

In some embodiments, the glyceraldehyde-3-phosphate dehydrogenase can be derived from a bacteria, for example, from the genus *Streptococcus* and, in some instances, from the species *Streptococcus hyointestinalis*. The glyceraldehyde-3-phosphate dehydrogenase can be encoded by the GAPN gene from *Streptococcus hyointestinalis*, or a GAPN gene ortholog, or a GAPN gene paralog.

In some embodiments, the glyceraldehyde-3-phosphate dehydrogenase can be derived from a bacteria, for example, from the genus *Streptococcus* and, in some instances, from the species *Streptococcus urinalis*. The glyceraldehyde-3-phosphate dehydrogenase can be encoded by the GAPN gene from *Streptococcus urinalis*, or a GAPN gene ortholog, or a GAPN gene paralog.

In some embodiments, the glyceraldehyde-3-phosphate dehydrogenase can be derived from a bacteria, for example,

from the genus *Streptococcus* and, in some instances, from the species *Streptococcus canis*. The glyceraldehyde-3-phosphate dehydrogenase can be encoded by the GAPN gene from *Streptococcus canis*, or a GAPN gene ortholog, or a GAPN gene paralog.

In some embodiments, the glyceraldehyde-3-phosphate dehydrogenase can be derived from a bacteria, for example, from the genus *Streptococcus* and, in some instances, from the species *Streptococcus thoraltensis*. The glyceraldehyde-3-phosphate dehydrogenase can be encoded by the GAPN gene from *Streptococcus thoraltensis*, or a GAPN gene ortholog, or a GAPN gene paralog.

In some embodiments, the glyceraldehyde-3-phosphate dehydrogenase can be derived from a bacteria, for example, from the genus *Streptococcus* and, in some instances, from the species *Streptococcus dysgalactiae*. The glyceraldehyde-3-phosphate dehydrogenase can be encoded by the GAPN gene from *Streptococcus dysgalactiae*, or a GAPN gene ortholog, or a GAPN gene paralog.

In some embodiments, the glyceraldehyde-3-phosphate dehydrogenase can be derived from a bacteria, for example, from the genus *Streptococcus* and, in some instances, from the species *Streptococcus pyogenes*. The glyceraldehyde-3-phosphate dehydrogenase can be encoded by the GAPN gene from *Streptococcus pyogenes*, or a GAPN gene ortholog, or a GAPN gene paralog.

In some embodiments, the glyceraldehyde-3-phosphate dehydrogenase can be derived from a bacteria, for example, from the genus *Streptococcus* and, in some instances, from the species *Streptococcus ictaluri*. The glyceraldehyde-3-phosphate dehydrogenase can be encoded by the GAPN gene from *Streptococcus ictaluri*, or a GAPN gene ortholog, or a GAPN gene paralog.

In some embodiments, the glyceraldehyde-3-phosphate dehydrogenase can be derived from a bacteria, for example, from the genus *Clostridium* and, in some instances, from the species *Clostridium perfringens*. The glyceraldehyde-3-phosphate dehydrogenase can be encoded by the GAPN gene from *Clostridium perfringens*, or a GAPN gene ortholog, or a GAPN gene paralog.

In some embodiments, the glyceraldehyde-3-phosphate dehydrogenase can be derived from a bacteria, for example, from the genus *Clostridium* and, in some instances, from the species *Clostridium chromiireducens*. The glyceraldehyde-3-phosphate dehydrogenase can be encoded by the GAPN gene from *Clostridium chromiireducens*, or a GAPN gene ortholog, or a GAPN gene paralog.

In some embodiments, the glyceraldehyde-3-phosphate dehydrogenase can be derived from a bacteria, for example, from the genus *Clostridium* and, in some instances, from the species *Clostridium botulinum*. The glyceraldehyde-3-phosphate dehydrogenase can be encoded by the GAPN gene from *Clostridium botulinum*, or a GAPN gene ortholog, or a GAPN gene paralog.

In some embodiments, the glyceraldehyde-3-phosphate dehydrogenase can be derived from a bacteria, for example, from the genus *Bacillus* and, in some instances, from the species *Bacillus cereus*. The glyceraldehyde-3-phosphate dehydrogenase can be encoded by the GAPN gene from *Bacillus cereus*, or a GAPN gene ortholog, or a GAPN gene paralog.

In some embodiments, the glyceraldehyde-3-phosphate dehydrogenase can be derived from a bacteria, for example, from the genus *Bacillus* and, in some instances, from the species *Bacillus anthracis*. The glyceraldehyde-3-phosphate

dehydrogenase can be encoded by the GAPN gene from *Bacillus anthracis*, or a GAPN gene ortholog, or a GAPN gene paralog.

In some embodiments, the glyceraldehyde-3-phosphate dehydrogenase can be derived from a bacteria, for example, from the genus *Bacillus* and, in some instances, from the species *Bacillus thuringiensis*. The glyceraldehyde-3-phosphate dehydrogenase can be encoded by the GAPN gene from *Bacillus thuringiensis*, or a GAPN gene ortholog, or a GAPN gene paralog.

In some embodiments, the glyceraldehyde-3-phosphate dehydrogenase can be derived from a bacteria, for example, from the genus *Pyrococcus* and, in some instances, from the species *Pyrococcus furiosus*. The glyceraldehyde-3-phosphate dehydrogenase can be encoded by the GAPN gene from *Pyrococcus furiosus*, or a GAPN gene ortholog, or a GAPN gene paralog.

Embodiments of glyceraldehyde-3-phosphate dehydrogenase can also be derived, without limitation, from the following (the number in brackets correspond to the Gene ID number): *Triticum aestivum* (543435); *Streptococcus mutans* (1028095); *Streptococcus agalactiae* (1013627); *Streptococcus pyogenes* (901445); *Clostridioides difficile* (4913365); *Mycoplasma mycoides* subsp. *mycoides* SC str. (2744894); *Streptococcus pneumoniae* (933338); *Streptococcus sanguinis* (4807521); *Acinetobacter pittii* (11638070); *Clostridium botulinum* A str. (5185508); [*Bacillus thuringiensis*] serovar *konkukian* str. (2857794); *Bacillus anthracis* str. Ames (1088724); *Phaeodactylum tricornutum* (7199937); *Emiliania huxleyi* (17251102); *Zea mays* (542583); *Helianthus annuus* (110928814); *Streptomyces coelicolor* (1101118); *Burkholderia pseudomallei* (U.S. Pat. Nos. 3,097,058, 3,095,849); variants thereof as well as fragments thereof.

Additional embodiments of glyceraldehyde-3-phosphate dehydrogenase can also be derived, without limitation, from the following (the number in brackets correspond to the Pubmed Accession number): *Streptococcus macacae* (WP_003081126.1), *Streptococcus hyointestinalis* (WP_115269374.1), *Streptococcus urinalis* (WP_006739074.1), *Streptococcus canis* (WP_003044111.1), *Streptococcus pluranimalium* (WP_104967491.1), *Streptococcus equi* (WP_012678132.1), *Streptococcus thoraltensis* (WP_018380938.1), *Streptococcus dysgalactiae* (WP_138125971.1), *Streptococcus halotolerans* (WP_062707672.1), *Streptococcus pyogenes* (WP_136058687.1), *Streptococcus ictaluri* (WP_008090774.1), *Clostridium perfringens* (WP_142691612.1), *Clostridium chromiireducens* (WP_079442081.1), *Clostridium botulinum* (WP_012422907.1), *Bacillus cereus* (WP_000213623.1), *Bacillus anthracis* (WP_098340670.1), *Bacillus thuringiensis* (WP_087951472.1), *Pyrococcus furiosus* (WP_011013013.1) as well as variants thereof and fragments thereof.

Embodiments of glyceraldehyde-3-phosphate dehydrogenase as well as yeasts expressing same are disclosed in US patent application 20210380989, incorporated herewith in its entirety.

In an embodiment, the recombinant yeast host cells of the present disclosure are capable of decreasing the activity or the expression of a second polypeptide involved in the conversion of a biomass into a fermentation product and/or the reduction in the production of a fermentation by-product. In still another embodiment, the recombinant yeast host cells is capable of reducing the production of the fermentation

by-product, such as, for example, glycerol. The recombinant yeast host cells of the present disclosure can include one or more genetic modification for decreasing the activity or the expression of a second polypeptide. In an embodiment, the genetic modification can be located in a regulatory region (such as a promoter region) of a native gene encoding a native (second) polypeptide and/or in the coding region of a native gene encoding a native (second) polypeptide. Alternatively or in combination, the genetic modification can be the introduction of one or more (second) heterologous nucleic acid molecules in the recombinant yeast host cells so as to inactivate, at least in part or totally, the expression of the second native polypeptide. In such embodiment, the second heterologous nucleic acid molecule(s) can be placed in the open-reading frame of the native gene to be inactivated. In still another embodiment, the second heterologous nucleic acid molecule(s) can be replace the open-reading frame of the native gene to be inactivated. The second genetic modification can be made in at least one (an in some embodiments in all) allele(s) of a copy of the native gene to be inactivated.

In still a further embodiment, the recombinant yeast host cells of the present disclosure are capable of reducing the activity or the expression of a (first) native NAD-dependent glycerol-3-phosphate dehydrogenase (GPD) polypeptide. Recombinant yeast host cells having decreased GPD activity have been described in US20200224209 which is herewith incorporated in its entirety.

Most mammalian cells express two different glycerol-3-phosphate dehydrogenases (GPDs) which are necessary for glycerol production and they are expressed in response to different cellular signals: the GPD1 and the GPD2 polypeptides. Both polypeptides share 75% amino acid identity and, while they catalyze the same reaction, the differences in their amino acid sequence make them more efficient enzymes under the environmental conditions that induce their expression. GPD2 is known to be unable to fully substitute for GPD1 in the production of osmotically induced glycerol production suggesting that this enzyme has lower activity than GPD1 under osmotic stress.

In an embodiment, the recombinant yeast host cells bear a genetic modification in at least one of the *gpd1* gene (encoding the GPD1 polypeptide), the *gpd2* gene (encoding the GPD2 polypeptide), the *gpp1* gene (encoding the GPP1 polypeptide) or the *gpp2* gene (encoding the GPP2 polypeptide). In another embodiment, the recombinant yeast host cells bear a genetic modification in at least two of the *gpd1* gene (encoding the GPD1 polypeptide), the *gpd2* gene (encoding the GPD2 polypeptide), the *gpp1* gene (encoding the GPP1 polypeptide) or the *gpp2* gene (encoding the GPP2 polypeptide). Examples of recombinant yeast cells bearing such genetic modification(s) leading to the reduction in the production of one or more native enzymes that function to produce glycerol are described in WO 2012/138942. In some embodiments, the recombinant yeast host cells have a genetic modification (such as a genetic deletion or insertion) only in one enzyme that functions to produce glycerol, in the *gpd2* gene, which would cause the yeast cell to have a knocked-out *gpd2* gene. In some embodiments, the recombinant yeast host cell can have a genetic modification in the *gpd1* gene and the *gpd2* gene resulting is a recombinant yeast host cell being knock-out for the *gpd1* gene and the *gpd2* gene. In some specific embodiments, the yeast cell can be a knock-out for the *gpd1* gene and have duplicate copies of the *gpd2* gene (in some embodiments, under the control of the *gpd1* promoter). In yet another embodiment, the

recombinant yeast host cells do not bear such genetic modification and includes its native genes coding for the GPP/GDP polypeptides.

The recombinant yeast host cells of the present disclosure can include a genetic modification to inhibit (at least partially or totally) the expression of a first NAD-dependent glycerol-3-phosphate dehydrogenase. This first NAD-dependent glycerol-3-phosphate dehydrogenase can be a NAD-dependent glycerol-3-phosphate 1 (GPD1) polypeptide or a GPD1 gene ortholog or paralog. The second genetic modification can include a deletion, deletion or substitution of one or more of a nucleic acid residue(s) in a gene (or a gene ortholog) encoding the GPD1 polypeptide (particularly in the gene's coding sequence) which would cause a reduction in the activity of the GPD1 polypeptide. In an embodiment, the second genetic modification can include the deletion of all of the coding sequence of a gene (or a gene ortholog) encoding the GPD1 polypeptide. Alternatively or in combination, the recombinant yeast host cell can express a heterologous GPD1 polypeptide variant or fragment having a reduced activity when compared to the native GPD1 polypeptide.

The GPD1 polypeptide is natively expressed in yeasts, fungi, mammalian and plant cells. GPD1 genes encoding the GPD1 polypeptide include, but are not limited to *Saccharomyces cerevisiae* Gene ID: 851539, *Schizosaccharomyces pombe* Gene ID: 2540547, *Schizosaccharomyces pombe* Gene ID: 2540455, *Neurospora crassa* Gene ID: 3873099, *Candida albicans* Gene ID: 3643924, *Scheffersomyces stipitis* Gene ID: 4840320, *Spathaspora passalidarum* Gene ID: 18874668, *Trichoderma reesei* Gene ID: 18482691, *Nectria haematococca* Gene ID: 9668637, *Candida dubliniensis* Gene ID: 8046432, *Chlamydomonas reinhardtii* Gene ID: 5716580, *Brassica napus* Gene ID: 106365675, *Chlorella variabilis* Gene ID: 17355036, *Brassica napus* Gene ID: 106352802, *Mus musculus* Gene ID: 14555, *Homo sapiens* Gene ID: 2819, *Rattus norvegicus* Gene ID: 60666, *Sus scrofa* Gene ID: 100153250, *Gallus gallus* Gene ID: 426881, *Bos taurus* Gene ID: 525042, *Xenopus tropicalis* Gene ID: 448519, *Pan troglodytes* Gene ID: 741054, *Canis lupus familiaris* Gene ID: 607942, *Callorhinchus milii* Gene ID: 103188923, *Columba livia* Gene ID: 102088900, *Macaca fascicularis* Gene ID: 101865501, *Myotis brandtii* Gene ID: 102257341, *Heterocephalus glaber* Gene ID: 101702723, *Nannospalax galili* Gene ID: 103746543, *Mus-tela putorius furo* Gene ID: 101681348, *Callithrix jacchus* Gene ID: 100414900, *Labrus bergylta* Gene ID: 109980872, *Monopterus albus* Gene ID: 109969143, *Castor canadensis* Gene ID: 109695417, *Paralichthys olivaceus* Gene ID: 109635348, *Bos indicus* Gene ID: 109559120, *Hippocampus comes* Gene ID: 109507993, *Rhinolophus sinicus* Gene ID: 109443801, *Hipposideros armiger* Gene ID: 109393253, *Crocodylus porosus* Gene ID: 109324424, *Gavialis gangeticus* Gene ID: 109293349, *Panthera pardus* Gene ID: 109249099, *Cyprinus carpio* Gene ID: 109094445, *Scleropages formosus* Gene ID: 108931403, *Nanorana parkeri* Gene ID: 108789981, *Rhinopithecus bieti* Gene ID: 108543924, *Lepidothrix coronata* Gene ID: 108509436, *Pygocentrus nattereri* Gene ID: 108444060, *Manis javanica* Gene ID: 108406536, *Cebus capucinus imitator* Gene ID: 108316082, *Ictalurus punctatus* Gene ID: 108255083, *Kryptolebias marmoratus* Gene ID: 108231479, *Miniopterus natalensis* Gene ID: 107528262, *Rousettus aegyptiacus* Gene ID: 107514265, *Coturnix japonica* Gene ID: 107325705, *Protobothrops mucrosquamatus* Gene ID: 107302714, *Parus major* Gene ID: 107215690, *Marmota marmota* Gene ID: 107148619, *Gekko japonicus*

Gene ID: 107122513, *Cyprinodon variegatus* Gene ID: 107101128, *Acinonyx jubatus* Gene ID: 106969233, *Poecilia latipinna* Gene ID: 106959529, *Poecilia mexicana* Gene ID: 106929022, *Calidris pugnax* Gene ID: 106891167, *Sturnus vulgaris* Gene ID: 106863139, *Equus asinus* Gene ID: 106845052, *Thamnophis sirtalis* Gene ID: 106545289, *Apteryx australis mantelli* Gene ID: 106499434, *Anser cygnoides domesticus* Gene ID: 106047703, *Dipodomys ordii* Gene ID: 105987539, *Clupea harengus* Gene ID: 105897935, *Microcebus murinus* Gene ID: 105869862, *Pro-pithecus coquereli* Gene ID: 105818148, *Aotus nancymaae* Gene ID: 105709449, *Cercocebus atys* Gene ID: 105580359, *Mandrillus leucophaeus* Gene ID: 105527974, *Colobus angolensis palliatus* Gene ID: 105507602, *Macaca nemestrina* Gene ID: 105492851, *Aquila chrysaetos canadensis* Gene ID: 105414064, *Pteropus vampyrus* Gene ID: 105297559, *Camelus dromedarius* Gene ID: 105097186, *Camelus bactrianus* Gene ID: 105076223, *Esox lucius* Gene ID: 105016698, *Bison bison bison* Gene ID: 105001494, *Notothenia coriiceps* Gene ID: 104967388, *Larimichthys crocea* Gene ID: 104928374, *Fukomys damarensis* Gene ID: 04861981, *Haliaeetus leucocephalus* Gene ID: 104831135, *Corvus cornix cornix* Gene ID: 104683744, *Rhinopithecus roxellana* Gene ID: 104679694, *Balearica regulorum gibbericeps* Gene ID: 104630128, *Tinamus guttatus* Gene ID: 104575187, *Mesitornis unicolor* Gene ID: 104539793, *Anstrostromus carolinensis* Gene ID: 104532747, *Buceros rhinoceros silvestris* Gene ID: 104501599, *Chaetura pelagica* Gene ID: 104385595, *Leptosomus discolor* Gene ID: 104353902, *Opisthocomus hoazin* Gene ID: 104326607, *Charadrius vociferus* Gene ID: 104284804, *Struthio camelus australis* Gene ID: 104144034, *Egretta garzetta* Gene ID: 104132778, *Cuculus canorus* Gene ID: 104055090, *Nipponia nippon* Gene ID: 104011969, *Pygoscelis adeliae* Gene ID: 103914601, *Aptenodytes forsteri* Gene ID: 103894920, *Serinus canaria* Gene ID: 103823858, *Manacus vitellinus* Gene ID: 103760593, *Ursus maritimus* Gene ID: 103675473, *Corvus brachyrhynchos* Gene ID: 103613218, *Galeopterus variegatus* Gene ID: 103598969, *Equus przewalskii* Gene ID: 103546083, *Calypte anna* Gene ID: 103536440, *Poecilia reticulata* Gene ID: 103464660, *Cynoglossus semilaevis* Gene ID: 103386748, *Stegastes partitus* Gene ID: 103355454, *Eptesicus fuscus* Gene ID: 103285288, *Chlorocebus sabaes* Gene ID: 103238296, *Oryzomys afer afer* Gene ID: 103194426, *Poecilia formosa* Gene ID: 103134553, *Erinaceus europaeus* Gene ID: 103118279, *Lipotes vexillifer* Gene ID: 103087725, *Python bivittatus* Gene ID: 103049416, *Astyanax mexicanus* Gene ID: 103021315, *Balaenoptera acutorostrata scammoni* Gene ID: 103006680, *Physeter catodon* Gene ID: 102996836, *Panthera tigris altaica* Gene ID: 102961238, *Chelonia mydas* Gene ID: 102939076, *Peromyscus maniculatus bairdii* Gene ID: 102922332, *Pteropus alecto* Gene ID: 102880604, *Elephantulus edwardii* Gene ID: 102844587, *Chrysocloris asiatica* Gene ID: 102825902, *Myotis davidii* Gene ID: 102754955, *Leptonyctotes weddellii* Gene ID: 102730427, *Lepisosteus oculatus* Gene ID: 102692130, *Alligator mississippiensis* Gene ID: 102576126, *Vicugna pacos* Gene ID: 102542115, *Camelus ferus* Gene ID: 102507052, *Tupaia chinensis* Gene ID: 102482961, *Pelodiscus sinensis* Gene ID: 102446147, *Myotis lucifugus* Gene ID: 102420239, *Bubalus bubalis* Gene ID: 102395827, *Alligator sinensis* Gene ID: 102383307, *Latimeria chalumnae* Gene ID: 102345318, *Pantholops hodgsonii* Gene ID: 102326635, *Haplochromis burtoni* Gene ID: 102295539, *Bos mutus* Gene ID: 102267392, *Xiphophorus maculatus* Gene ID:

102228568, *Pundamilia nyererei* Gene ID: 102192578, *Capra hircus* Gene ID: 102171407, *Pseudopodoces humilis* Gene ID: 102106269, *Zonotrichia albicollis* Gene ID: 102070144, *Falco cherrug* Gene ID: 102047785, *Geospiza fortis* Gene ID: 102037409, *Chinchilla lanigera* Gene ID: 102014610, *Microtus ochrogaster* Gene ID: 101990242, *Ictidomys tridecemlineatus* Gene ID: 101955193, *Chrysemys picta* Gene ID: 101939497, *Falco peregrinus* Gene ID: 101911770, *Mesocricetus auratus* Gene ID: 101824509, *Ficedula albicollis* Gene ID: 101814000, *Anas platyrhynchos* Gene ID: 101789855, *Echinops telfairi* Gene ID: 101641551, *Condylura cristata* Gene ID: 101622847, *Jaculus jaculus* Gene ID: 101609219, *Octodon degus* Gene ID: 101563150, *Sorex araneus* Gene ID: 101556310, *Ochotona princeps* Gene ID: 101532015, *Maylandia zebra* Gene ID: 101478751, *Dasyurus novemcinctus* Gene ID: 101446993, *Odobenus rosmarus divergens* Gene ID: 101385499, *Tursiops truncatus* Gene ID: 101318662, *Orcinus orca* Gene ID: 101284095, *Oryzias latipes* Gene ID: 101154943, *Gorilla gorilla* Gene ID: 101131184, *Ovis aries* Gene ID: 101119894, *Felis catus* Gene ID: 101086577, *Takifugu rubripes* Gene ID: 101079539, *Saimiri boliviensis* Gene ID: 101030263, *Papio anubis* Gene ID: 101004942, *Pan paniscus* Gene ID: 100981359, *Otolemur garnettii* Gene ID: 100946205, *Sarcophilus harrisii* Gene ID: 100928054, *Cricetulus griseus* Gene ID: 100772179, *Cavia porcellus* Gene ID: 100720368, *Oreochromis niloticus* Gene ID: 100712149, *Loxodonta africana* Gene ID: 100660074, *Nomascus leucogenys* Gene ID: 100594138, *Anolis carolinensis* Gene ID: 100552972, *Meleagris gallopavo* Gene ID: 100542199, *Ailuropoda melanoleuca* Gene ID: 100473892, *Oryctolagus cuniculus* Gene ID: 100339469, *Taeniopygia guttata* Gene ID: 100225600, *Pongo abelii* Gene ID: 100172201, *Ornithorhynchus anatinus* Gene ID: 100085954, *Equus caballus* Gene ID: 100052204, *Mus musculus* Gene ID: 100198, *Xenopus laevis* Gene ID: 399227, *Danio rerio* Gene ID: 325181, *Danio rerio* Gene ID: 406615, *Melopsittacus undulatus* Gene ID: 101872435, *Ceratotherium simum simum* Gene ID: 101408813, *Trichechus manatus latirostris* Gene ID: 101359849 and *Takifugu rubripes* Gene ID: 101071719). In the present disclosure, the recombinant yeast cell can reduce or inhibit the expression of a GPD1 gene (or a GPD1 gene ortholog) encoding a GPD1 polypeptide, variant or fragment.

The recombinant yeast host cells of the present disclosure can include a genetic modification to inhibit (at least partially or totally) the expression of a second NAD-dependent glycerol-3-phosphate dehydrogenase. This second NAD-dependent glycerol-3-phosphate dehydrogenase can be a NAD-dependent glycerol-3-phosphate 2 (GPD2) polypeptide or a GPD2 gene ortholog or paralog. The genetic modification can include a deletion, deletion or substitution of one or more of a nucleic acid residue(s) in a gene (or a gene ortholog) encoding the GPD2 polypeptide (particularly in the gene's coding sequence) which would cause a reduction in the activity of the GPD2 polypeptide. In an embodiment, the second genetic modification can include the deletion of all of the coding sequence of a gene (or a gene ortholog) encoding the GPD2 polypeptide. Alternatively or in combination, the recombinant yeast host cell can express a heterologous GPD2 polypeptide variant or fragment having a reduced activity when compared to the native GPD2 polypeptide.

In some embodiments, the recombinant yeast host cells of the present disclosure, while having reduced activity or expression in a first NAD-dependent glycerol-3-phosphate (e.g., GPD1), can express a second NAD-dependent gly-

cerol-3-phosphate dehydrogenase exhibiting less enzymatic activity than the first NAD-dependent glycerol-3-phosphate (e.g., GPD2). For example, the recombinant yeast host cells of the present disclosure, while having a reduced GPD1 activity or express, is capable of expressing a heterologous NAD-dependent glycerol-3-phosphate dehydrogenase 2 (GPD2) polypeptide (which exhibits less enzymatic activity than GPD1). As such, the second genetic modification can include modifying the recombinant host cells to express a heterologous NAD-dependent glycerol-3-phosphate dehydrogenase 2 (GPD2) polypeptide. This can be done, for example, by expressing a heterologous nucleic acid encoding the heterologous GPD2 polypeptide using an osmotic promoter (such as, for example, the promoter of the GPD1 gene). The second heterologous nucleic acid molecule can, in some additional embodiments, replace the open-reading frame of at least one copy of the native GPD1 gene. The second heterologous nucleic acid molecule can, in some embodiments, replace the open-reading frame of all copies of the native GPD1 gene. In some embodiments, at least a single native copy of the gene (or the gene ortholog) encoding the GPD2 polypeptide be under the control of the native GPD2 promoter.

The GPD2 polypeptide is expressed in bacteria, yeasts, fungi, mammalian and plant cells. GPD2 genes encoding the GPD2 polypeptide include, but are not limited to *Mus musculus* Gene ID: 14571, *Homo sapiens* Gene ID: 2820, *Saccharomyces cerevisiae* Gene ID: 854095, *Rattus norvegicus* Gene ID: 25062, *Schizosaccharomyces pombe* Gene ID: 2541502, *Mus musculus* Gene ID: 14380, *Danio rerio* Gene ID: 751628, *Caenorhabditis elegans* Gene ID: 3565504, *Mesocricetus auratus* Gene ID: 101825992, *Xenopus tropicalis* Gene ID: 779615, *Macaca mulatta* Gene ID: 697192, *Bos taurus* Gene ID: 504948, *Canis lupus familiaris* Gene ID: 478755, *Cavia porcellus* Gene ID: 100721200, *Gallus gallus* Gene ID: 424321, *Pan troglodytes* Gene ID: 459670, *Oryctolagus cuniculus* Gene ID: 100101571, *Candida albicans* Gene ID: 3644563, *Xenopus laevis* Gene ID: 444438, *Macaca fascicularis* Gene ID: 102127260, *Ailuropoda melanoleuca* Gene ID: 100482626, *Cricetulus griseus* Gene ID: 100766128, *Heterocephalus glaber* Gene ID: 101715967, *Scheffersomyces stipitis* Gene ID: 4838862, *Ictalurus punctatus* Gene ID: 108273160, *Mustela putorius furo* Gene ID: 101681209, *Nannospalax galili* Gene ID: 103741048, *Callithrix jacchus* Gene ID: 100409379, *Lates calcarifer* Gene ID: 108873068, *Nothobranchius furzeri* Gene ID: 07384696, *Acanthisitta chloris* Gene ID: 103808746, *Acinonyx jubatus* Gene ID: 106978985, *Alligator mississippiensis* Gene ID: 102562563, *Alligator sinensis* Gene ID: 102380394, *Anas platyrhynchos*, *Anolis carolinensis* Gene ID: 100551888, *Anser cygnoides domesticus* Gene ID: 106043902, *Aotus nancymaeae* Gene ID: 105719012, *Apaloderma vittatum* Gene ID: 104281080, *Aptenodytes forsteri* Gene ID: 103893867, *Apteryx australis mantelli* Gene ID: 106486554, *Aquila chrysaetos canadensis* Gene ID: 105412526, *Astyanax mexicanus* Gene ID: 103029081, *Austrofundulus limnaeus* Gene ID: 106535816, *Balaenoptera acutorostrata scammoni* Gene ID: 103019768, *Balearica regulorum gibbericeps*, *Bison bison* Gene ID: 104988636, *Bos indicus* Gene ID: 109567519, *Bos mutus* Gene ID: 102277350, *Bubalus bubalis* Gene ID: 102404879, *Buceros rhinoceros silvestris* Gene ID: 104497001, *Calidris pugnax* Gene ID: 106902763, *Callorhinchus milii* Gene ID: 103176409, *Calypte anna* Gene ID: 103535222, *Camelus bactrianus* Gene ID: 105081921, *Camelus dromedarius* Gene ID: 105093713, *Camelus ferus* Gene ID: 102519983, *Capra hircus* Gene ID:

102176370, *Cariama cristata* Gene ID: 104154548, *Castor canadensis* Gene ID: 109700730, *Cebus capucinus imitator* Gene ID: 108316996, *Cercocebus atys* Gene ID: 105576003, *Chaetura pelagica* Gene ID: 104391744, *Charadrius vociferus* Gene ID: 104286830, *Chelonia mydas* Gene ID: 102930483, *Chinchilla lanigera* Gene ID: 102017931, *Chlamydotis macqueenii* Gene ID: 104476789, *Chlorocebus sabaues* Gene ID: 103217126, *Chrysemys picta* Gene ID: 101939831, *Chrysochloris asiatica* Gene ID: 102831540, *Clupea harengus* Gene ID: 105902648, *Colius striatus* Gene ID: 104549356, *Colobus angolensis palliatus* Gene ID: 105516852, *Columba livia* Gene ID: 102090265, *Condylura cristata* Gene ID: 101619970, *Corvus brachyrhynchos*, *Coturnix japonica* Gene ID: 107316969, *Crocodylus porosus* Gene ID: 109322895, *Cuculus canorus* Gene ID: 104056187, *Cynoglossus semilaevis* Gene ID: 103389593, *Dasypris novemcinctus* Gene ID: 101428842, *Dipodomys ordii* Gene ID: 105996090, *Echinops telfairi* Gene ID: 101656272, *Egretta garzetta* Gene ID: 104135263, *Elephantulus edwardii* Gene ID: 102858276, *Eptesicus fuscus* Gene ID: 103283396, *Equus asinus* Gene ID: 106841969, *Equus caballus* Gene ID: 100050747, *Equus przewalskii* Gene ID: 103558835, *Erinaceus europaeus* Gene ID: 103114599, *Eurypyga helias* Gene ID: 104502666, *Falco cherrug* Gene ID: 102054715, *Falco peregrinus* Gene ID: 101912742, *Felis catus* Gene ID: 101089953, *Ficedula albicollis* Gene ID: 101816901, *Fukomys damarensis* Gene ID: 104850054, *Fundulus heteroclitus* Gene ID: 105936523, *Galeopterus variegatus* Gene ID: 103586331, *Gavia stellata* Gene ID: 104250365, *Gavialis gangeticus* Gene ID: 109301301, *Gekko japonicus* Gene ID: 107110762, *Geospiza fortis* Gene ID: 102042095, *Gorilla gorilla* Gene ID: 101150526, *Haliaeetus albicilla* Gene ID: 104323154, *Haliaeetus leucocephalus* Gene ID: 104829038, *Haplochromis burtoni* Gene ID: 102309478, *Hippocampus comes* Gene ID: 109528375, *Hipposideros armiger* Gene ID: 109379867, *Ictidomys tridecemlineatus* Gene ID: 101965668, *Jaculus jaculus* Gene ID: 101616184, *Kryptolebias marmoratus* Gene ID: 108251075, *Labrus bergylta* Gene ID: 109984158, *Larimichthys crocea* Gene ID: 104929094, *Latimeria chalumnae* Gene ID: 102361446, *Lepidothrix coronata* Gene ID: 108501660, *Lepisosteus oculatus* Gene ID: 102691231, *Leptonychotes weddellii* Gene ID: 102739068, *Leptosomus discolor* Gene ID: 104340644, *Lipotes vexillifer* Gene ID: 103074004, *Loxodonta africana* Gene ID: 100654953, *Macaca nemestrina* Gene ID: 105493221, *Manacus vitellinus* Gene ID: 103757091, *Mandrillus leucophaeus* Gene ID: 105548063, *Manis javanica* Gene ID: 108392571, *Marmota marmota* Gene ID: 107136866, *Maylandia zebra* Gene ID: 101487556, *Mesitornis unicolor* Gene ID: 104545943, *Microcebus murinus* Gene ID: 105859136, *Microtus ochrogaster* Gene ID: 101999389, *Miniopterus natalensis* Gene ID: 107525674, *Monodelphis domestica* Gene ID: 100014779, *Monopterus albus* Gene ID: 109957085, *Myotis brandtii* Gene ID: 102239648, *Myotis davidii* Gene ID: 102770109, *Myotis lucifugus* Gene ID: 102438522, *Nanorana parkeri* Gene ID: 108784354, *Nestor notabilis* Gene ID: 104399051, *Nipponia nippon* Gene ID: 104012349, *Nomascus leucogenys* Gene ID: 100590527, *Notothenia coriiceps* Gene ID: 104964156, *Ochotona princeps* Gene ID: 101530736, *Octodon degus* Gene ID: 101591628, *Odobenus rosmarus divergens* Gene ID: 101385453, *Oncorhynchus kisutch* Gene ID: 109870627, *Opisthocomus hoazin* Gene ID: 104338567, *Orcinus orca* Gene ID: 101287409, *Oreochromis niloticus* Gene ID: 100694147, *Ornithorhynchus anatinus* Gene ID: 100081433, *Orycteropus afer* Gene ID: 103197834, *Oryzias latipes* Gene ID: 101167020, *Otolemurgarnettii* Gene ID: 100966064, *Ovis aries* Gene ID: 443090, *Pan paniscus* Gene ID: 100970779, *Panthera pardus* Gene ID: 109271431, *Panthera tigris altaica* Gene ID: 102957949, *Pantholops hodgsonii* Gene ID: 102323478, *Papio anubis* Gene ID: 101002517, *Paralichthys olivaceus* Gene ID: 109631046, *Pelodiscus sinensis* Gene ID: 102454304, *Peromyscus maniculatus bairdii* Gene ID: 102924185, *Phaethon lepturus* Gene ID: 104624271, *Phalacrocorax carbo* Gene ID: 104049388, *Physeter catodon* Gene ID: 102978831, *Picoides pubescens* Gene ID: 104296936, *Poecilia latipinna* Gene ID: 106958025, *Poecilia mexicana* Gene ID: 106920534, *Poecilia reticulata* Gene ID: 103473778, *Pongo abelii* Gene ID: 100452414, *Propithecus coquereli* Gene ID: 105807399, *Protobothrops mucrosquamatus* Gene ID: 107289584, *Pseudodocodes humilis* Gene ID: 102109711, *Pterocles gutturalis* Gene ID: 104461236, *Pteropus alecto* Gene ID: 102879110, *Pteropus vampyrus* Gene ID: 105291402, *Pundamilia nyererei* Gene ID: 102200268, *Pygocentrus nattereri* Gene ID: 108411786, *Pygoscelis adeliae* Gene ID: 103925329, *Python bivittatus* Gene ID: 103059167, *Rhincodon typus* Gene ID: 109920450, *Rhinolophus sinicus* Gene ID: 109445137, *Rhinopithecus bieti* Gene ID: 108538766, *Rhinopithecus roxellana* Gene ID: 104654108, *Rousettus aegyptiacus* Gene ID: 107513424, *Saimiri boliviensis* Gene ID: 101027702, *Salmo salar* Gene ID: 106581822, *Sarcophilus harrisii* Gene ID: 100927498, *Scleropages formosus* Gene ID: 108927961, *Serinus canaria* Gene ID: 103814246, *Sinocyclocheilus grahami* Gene ID: 107555436, *Sorex araneus* Gene ID: 101543025, *Stegastes partitus* Gene ID: 103360018, *Struthio camelus australis* Gene ID: 104138752, *Sturnus vulgaris* Gene ID: 106861926, *Sugiyamaella lignohabitans* Gene ID: 30033324, *Sus scrofa* Gene ID: 397348, *Taeniopygia guttata* Gene ID: 100222867, *Takifugu rubripes* Gene ID: 101062218, *Tarsius syrichta* Gene ID: 103254049, *Tauraco erythrolophus* Gene ID: 104378162, *Thamnomphs sirtalis* Gene ID: 106538827, *Tinamus guttatus* Gene ID: 104572349, *Tupaia chinensis* Gene ID: 102471148, *Tursiops truncatus* Gene ID: 101330605, *Ursus maritimus* Gene ID: 103659477, *Vicugna pacos* Gene ID: 102533941, *Xiphophorus maculatus* Gene ID: 102225536, *Zonotrichia albicollis* Gene ID: 102073261, *Ciona intestinalis* Gene ID: 100183886, *Meleagris gallopavo* Gene ID: 100546408, *Trichechus manatus latirostris* Gene ID: 101355771, *Ceratotherium simum simum* Gene ID: 101400784, *Melopittacus undulatus* Gene ID: 101871704, *Esox lucius* Gene ID: 10502249 and *Pygocentrus nattereri* Gene ID: 108411786. In an embodiment, the GPD2 polypeptide is encoded by *Saccharomyces cerevisiae* Gene ID: 854095. In some embodiments, the GPD2 polypeptide has the amino acid sequence of SEQ ID NO: 6, is a variant of the amino acid sequence of SEQ ID NO: 6 having NAD-dependent glycerol-3-phosphate activity or is a fragment of the amino acid sequence of SEQ ID NO: 6 having NAD-dependent glycerol-3-phosphate activity. In some embodiments, the second heterologous nucleic acid molecule comprises the nucleic acid sequence of SEQ ID NO: 5, is a degenerate sequence of SEQ ID NO: 5 (encoding SEQ ID NO: 6) or encodes a variant or a fragment of the amino acid sequence of SEQ ID NO: 6.

In some embodiments, this overall reduction in GPD activity can be observed in high osmotic conditions. As used herein, the expression “high osmotic conditions” refers to the presence of a high osmotic pressure, usually caused by an increase in the solute concentration in the environment surrounding the recombinant yeast host cell. In yeasts, “high

osmotic conditions” are associated with an upregulation of the HOG pathway, a concentration of sugars higher than about 50 g/L and/or equivalent to at least 1 g/L of a salt (such as NaCl). This decrease in NAD-dependent glycerol-3-phosphate activity can be observed with respect to the same recombinant yeast cell in normal or low osmotic conditions or with respect to a recombinant yeast host cell lacking the one or more second genetic modification. As also used in the present disclosure, the expression “normal or low osmotic conditions” refers to conditions that are not associated with high osmotic pressure.

The second heterologous nucleic acid molecules of the present disclosure can include or be operatively associated with an osmotic promoter. In the context of the present disclosure, an “osmotic promoter” can be a promoter (or a combination of promoters) allowing the expression (or, in some embodiments, the overexpression) of a gene when the recombinant yeast host cell is placed in high osmotic conditions but refraining the expression (or, in some embodiments, the overexpression) of a gene when the recombinant yeast host cell is placed in normal or low osmotic conditions. In this embodiment, the osmotic promoter can be an inducible promoter. Osmotic promoters are usually associated with genes in the HOG1 pathway and promoters controlling the expression of genes which are upregulated in the HOG1 pathway can be used in the recombinant yeast host cell of the present disclosure. Enzymes in the HOG1 pathway whose expression is upregulated in high osmotic conditions include, but are not limited to, a NAD-dependent glycerol-3-phosphate dehydrogenase 1 gene, a dihydroxyacetone kinase gene and a trehalose-phosphatase gene. As such, in the context of the present disclosure, the osmotic promoter can be a promoter (or a combination of promoters) from a NAD-dependent glycerol-3-phosphate dehydrogenase 1 gene, a dihydroxyacetone kinase gene and/or a trehalose-phosphatase gene. In *Saccharomyces cerevisiae*, enzymes in the HOG1 pathway whose expression is upregulated in the presence of high osmotic conditions include, but are not limited to, a GPD1 gene, a DAK1 gene and a TPS2 gene. As such, in the context of the present disclosure, the osmotic promoter can be a promoter (or a combination of promoters) from a GPD1 gene (referred to as the GPD1 promoter or *gpd1p*), a DAK1 gene (referred to as the DAK1 promoter or *dak1p*) and/or a TPS2 gene (referred to as the TPS2 promoter or *tps2p*).

An “osmotic promoter” can also be a constitutive promoter which allows the expression of coding sequences operatively associated thereto during osmotic conditions. In some embodiments, it is preferred that the constitutive promoter be a “low” constitutive promoter. Exemplary “low” constitutive promoters could be associated with the expression of housekeeping genes, and, for example, can include the promoter of the *CYC1* gene. In some embodiment, the osmotic promoter is not a high constitutive promoter.

In yet another embodiment, the recombinant yeast host cells of the present disclosure are capable of reducing the activity or the expression of a second polypeptide capable of exporting glycerol from the recombinant yeast host cell. Exemplary polypeptides capable of functioning to export glycerol include aquaporins as well as glycerol facilitators. The *fdp1* support (FPS1) polypeptide (encoded by Gene ID 850683 in *Saccharomyces cerevisiae*) is a glycerol facilitator capable of importing glycerol. As such, the polypeptide capable of functioning to export glycerol can be a FPS1 polypeptide or a polypeptide encoded by a FPS1 gene ortholog. The FPS1 polypeptide can be derived, for

example, from *Saccharomyces cerevisiae* or a corresponding ortholog found in *Pachysolen tannophilus*, *Komagataella pastoris*, *Yarrowia lipolytica* and/or *Cyberlindnera jadinii*.

In the present disclosure, it is possible to express a variant of a first polypeptide or of a second polypeptide in the recombinant yeast host cells. A variant comprises at least one amino acid difference (substitution or addition) when compared to the amino acid sequence of the wild type polypeptide and still exhibits the biological activity of the wild type polypeptide (e.g., for a STL1 variant, glycerol transport activity; for a glucoamylase, a starch-degrading activity; for a GPD2 variant, a NAD-dependent glycerol-3-phosphate dehydrogenase activity, etc.). In an embodiment, the variant polypeptide exhibits at least 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99% of the activity of the wild-type polypeptide. The variants also have at least 70%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99% identity when compared to the wild-type polypeptide over its entire length. The term “percent identity”, as known in the art, is a relationship between two or more polypeptide sequences, as determined by comparing the sequences. The level of identity can be determined conventionally using known computer programs. Identity can be readily calculated by known methods, including but not limited to those described in: Computational Molecular Biology (Lesk, A. M., ed.) Oxford University Press, NY (1988); Biocomputing: Informatics and Genome Projects (Smith, D. W., ed.) Academic Press, NY (1993); Computer Analysis of Sequence Data, Part I (Griffin, A. M., and Griffin, H. G., eds.) Humana Press, NJ (1994); Sequence Analysis in Molecular Biology (von Heinje, G., ed.) Academic Press (1987); and Sequence Analysis Primer (Gribskov, M. and Devereux, J., eds.) Stockton Press, NY (1991). Preferred methods to determine identity are designed to give the best match between the sequences tested. Methods to determine identity and similarity are codified in publicly available computer programs. Sequence alignments and percent identity calculations may be performed using the Megalign program of the LASERGENE bioinformatics computing suite (DNASTAR Inc., Madison, Wis.). Multiple alignments of the sequences disclosed herein were performed using the Clustal method of alignment (Higgins and Sharp (1989) CABIOS. 5:151-153) with the default parameters (GAP PENALTY=10, GAP LENGTH PEN ALT Y=10). Default parameters for pairwise alignments using the Clustal method were KTUPLB 1, GAP PENALTY=3, WINDOW=5 and DIAGONALS SAVED=5.

The variant polypeptides described herein may be (i) one in which one or more of the amino acid residues are substituted with a conserved or non-conserved amino acid residue (preferably a conserved amino acid residue) and such substituted amino acid residue may or may not be one encoded by the genetic code, or (ii) one in which one or more of the amino acid residues includes a substituent group, or (iii) one in which the mature polypeptide is fused with another compound, such as a compound to increase the half-life of the polypeptide (for example, polyethylene glycol), or (iv) one in which the additional amino acids are fused to the mature polypeptide for purification of the polypeptide. Conservative substitutions typically include the substitution of one amino acid for another with similar characteristics, e.g., substitutions within the following groups: valine, glycine; glycine, alanine; valine, isoleucine, leucine; aspartic acid, glutamic acid; asparagine, glutamine;

serine, threonine; lysine, arginine; and phenylalanine, tyrosine. Other conservative amino acid substitutions are known in the art and are included herein. Non-conservative substitutions, such as replacing a basic amino acid with a hydrophobic one, are also well-known in the art.

A variant polypeptide can also be a conservative variant or an allelic variant. As used herein, a conservative variant refers to alterations in the amino acid sequence that do not adversely affect the biological functions of the variant polypeptide. A substitution, insertion or deletion is said to adversely affect the polypeptide when the altered sequence prevents or disrupts a biological function associated with the variant polypeptide. For example, the overall charge, structure or hydrophobic-hydrophilic properties of the polypeptide can be altered without adversely affecting a biological activity. Accordingly, the amino acid sequence can be altered, for example to render the variant polypeptide more hydrophobic or hydrophilic, without adversely affecting its biological activity.

The present disclosure also provide fragments of the first and/or the second polypeptide described herein. A fragment comprises at least one less amino acid residue when compared to the amino acid sequence of the wild type polypeptide or variant and still possess the biological activity of the full-length wild type polypeptide. In an embodiment, the fragment exhibits at least 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99% activity when compared to the full-length wild type polypeptide or variant. The fragments can also have at least 70%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99% identity when compared to the wild type polypeptide or the variant. The fragment can be, for example, a truncation of one or more amino acid residues at the amino-terminus, the carboxy terminus or both termini of the wild type polypeptide or variant. Alternatively or in combination, the fragment can be generated from removing one or more internal amino acid residues.

Persistent Yeast Compositions and Processes for Propagating Persistent Yeast Cells

The present disclosure provides compositions comprising the persistent yeast cells as well as processes for making such compositions. Broadly, the process for making the yeast composition comprises two steps: a first step of propagating the persistent yeast cells and a second step of formulating the persistent yeast cells into a composition. It is understood that the process for making the fermented product (described herein below in more details) can include some propagation of the persistent yeast cells but mainly concerns converting the biomass into the fermentation product. The propagation step of the process for making the persistent yeast cell composition minimizes the conversion of the biomass into the fermentation product and concerns mainly concerns maximizing cellular division of the persistent yeast cells.

The propagation can be performed by sampling the fermentation medium of an initial and/or a further fermentation (obtained, for example, at steps 200 or 400 described in FIG. 1) or the substantially isolated persistent yeast host cells (obtained, for example, at steps 300 and/or 500 of FIG. 1). The sample comprises persistent yeast cells. In some embodiments, the sample of the fermentation medium and/or the substantially isolated persistent yeast host cell can be placed into contact directly with a propagation medium allowing propagation in order to be propagated. In such

embodiment, the propagation medium can be a fresh medium and/or also allow fermentation. The propagation medium can include, for example, molasses, cane juice, one or more nutrient and/or one or more antibiotic. In additional embodiments, the sample of the fermentation medium and/or the substantially isolated persistent yeast host cell can be placed directly in a fermentor or, in further embodiments, in a small vessel (such as, for example, a shake flask) to scale up the propagation prior to fermentation. In some embodiments, the sample of the fermentation medium and/or the substantially isolated persistent yeast host cell can be placed into contact with a solid medium (e.g., an agar plate for example) prior to propagation. Prior to being placed in a medium allowing for propagation, the sampled persistent yeast cells can be diluted or washed (with water for example) and/or concentrated (with centrifugation or filtration for example). The sampled persistent yeast cells can, prior to or during propagation, be supplemented with one or more nutrient or one or more antibiotic to maintain or prolong viability. The sampled persistent yeast cells can, prior to propagation, be stored. After having been propagated, the sampled persistent yeast cells can be diluted or washed (with water for example) and/or concentrated (with centrifugation or filtration for example).

The propagation can be conducted according to a traditional baker's yeast production process with the persistent yeast cells as described herein. The propagation step can be a continuous propagation, a batch propagation or a fed-batch propagation. The propagation medium intended to be inoculated with the persistent yeast cells can comprise a carbon source (such as, for example, molasses, sucrose, glucose, dextrose syrup, ethanol, corn, glycerol, corn steep liquor and/or a lignocellulosic biomass), a nitrogen source (such as, for example, ammonia or another inorganic source of nitrogen) and a phosphorous source (such as, for example, phosphoric acid or another inorganic source of phosphorous). The propagation medium can further comprise additional micronutrients such as vitamins and/or minerals to support the propagation of the persistent yeast cells. In some embodiments, the propagation medium can comprise molasses or be derived from molasses.

In the propagation process, the persistent yeast cells are placed in a propagation medium which can, in some embodiments, allow for a specific growth rate of 0.25, 0.24, 0.23, 0.22, 0.21, 0.20, 0.19, 0.18, 0.17, 0.16 or 0.15 h⁻¹ or less. In order to limit the growth rate of the persistent yeast cells, in some embodiments, the process can further comprise controlling the addition of nutrients, such as carbohydrates, during the propagation step. Limiting the growth rate of the persistent yeast cells during propagation can be achieved by maintaining the concentration of carbohydrates below 0.1, 0.01, 0.001 or 0.0001 weight % with respect to the volume of the fermentation medium. Controlling the concentration of carbohydrates of the propagation medium can be done by various means known in the art and can involve sampling the propagation medium at various intervals, determining the carbohydrate concentration, fermentation product (e.g., alcohol) concentration and/or gas (e.g., CO₂) concentration in those samples and adding or refraining from adding, if necessary, additional carbohydrates in the propagation medium to accelerate or decelerate the growth of the persistent yeast cells. In some embodiments, the process provides for adding nitrogen and/or phosphorous to match/support the growth rate of the persistent yeast cells.

The propagation process is preferably conducted under high aeration conditions. For example, in some embodiments, the process can include controlling the aeration of the

propagation medium (which is contained in a vessel of a specific volume) to achieve a specific aeration rate, for example, of at least 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1.0, 1.1, 1.2 or 1.3 air volume/vessel volume/minute.

The propagation process can be conducted at a specific pH and/or a specific temperature which may be optimal for the propagation of the persistent yeast cells. As such, in embodiments in which the persistent yeast cell is from the genus *Saccharomyces*, the process can comprise controlling the pH of the propagation medium to between about 3.0 to about 6.0, about 3.5 to about 5.5 or about 4.0 to about 5.5. In a specific embodiment, the pH is controlled at about 4.5. In another example, in embodiments in which the persistent yeast cell is from the genus *Saccharomyces*, the process can comprise controlling the temperature of the propagation medium between about about 20° C. to about 40° C., about 25° C. to about 30° C. or about 30° C. to about 35° C. In a specific embodiment, the temperature is controlled at between about about 30° C. to about 35° C. (32° C. for example).

At the end of the propagation step, a propagated medium comprising propagated persistent yeast cells (which can be propagated recombinant yeast host cells) is obtained. In some embodiments, a specific concentration of the propagated persistent yeast cells can be sought or achieved in the propagated medium. In some embodiments, the concentration of the propagated persistent yeast cells is at least about 0.1, 0.15, 0.2, 0.25, 0.3, 0.35, 0.4, 0.45, 0.5, 1.0, 1.5, 2.0, 2.5, 3.0, 3.5, 4.0, 4.5, 5.0 or more weight % with respect to the volume of the propagated medium. In a specific embodiment in which the persistent yeast cells are propagated using a fed-batch process, the concentration of the propagated persistent yeast cells is at least about 0.25 weight % with respect to the volume of the propagated medium.

In the formulating step, the mixture obtained after propagation (comprising the propagated persistent yeast cells) is modified to provide a yeast composition. In an embodiment of the formulating step, at least one component of the mixture obtained after propagation is removed from the propagated medium to provide the yeast composition. This component can be, without limitation, water, amino acids, peptides and proteins, nucleic acid residues and nucleic acid molecules, cellular debris, fermentation products, etc. In an embodiment, the formulating step comprises substantially isolating the propagated persistent yeast cells (e.g., the biomass) from the components of the propagated medium. As used in the context of the of the formulating step, the expression "substantially isolating" refers to the removal of the majority of the components of the propagated medium from the propagated persistent yeast cells. In some embodiments, "substantially isolating" refers to concentrating the propagated persistent yeast cells to at least 5, 10, 15, 20, 25, 30, 35, 45% or more when compared to the concentration of the persistent yeast cells prior to the substantial isolation. In order to provide the yeast composition, the propagated yeasts can be centrifuged (and the resulting cellular pellet comprising the propagated persistent yeast cells can optionally be washed), filtered and/or dried (optionally using a vacuum-drying technique). The formulation step can, in some embodiments, preserve the viability (at least in part) of the recombinant yeast host cells. As such, the yeast composition can be provided in an active or a semi-active form. The yeast composition can be provided in a liquid, semi-solid or dry form. In an embodiment, the composition can be provided in the form of a cream yeast.

Processes for Prolonging Persistence

[illegible]

In an embodiment, after a total 40 fermentation cycles (comprising both the initial fermentation cycle and 39 further fermentation cycles) according to the present process, the persistent yeast cells of the present disclosure are present in a proportion of at least 90% in the fermenting population (when measured with respect to its DNA contribution of the total fermenting population). In an embodiment, after a total 40 fermentation cycles (comprising both the initial fermentation cycle and 39 further fermentation cycles) according to the present process, the persistent yeast cells of the present disclosure are present in a proportion of at least 91% in the fermenting population (when measured with respect to its DNA contribution of the total fermenting population). In an embodiment, after a total 40 fermentation cycles (comprising both the initial fermentation cycle and 39 further fermentation cycles) according to the present process, the persistent yeast cells of the present disclosure are present in a proportion of at least 92% in the fermenting population (when measured with respect to its DNA contribution of the total fermenting population). In an embodiment, after a total 40 fermentation cycles (comprising both the initial fermentation cycle and 39 further fermentation cycles) according to the present process, the persistent yeast cells of the present disclosure are present in a proportion of at least 93% in the fermenting population (when measured with respect to its DNA contribution of the total fermenting population). In an embodiment, after a total 40 fermentation cycles (comprising both the initial fermentation cycle and 39 further fermentation cycles) according to the present process, the persistent yeast cells of the present disclosure are present in a proportion of at least 94% in the fermenting population (when measured with respect to its DNA contribution of the total fermenting population). In an embodiment, after a total 40 fermentation cycles (comprising both the initial fermentation cycle and 39 further fermentation cycles) according to the present process, the persistent yeast cells of the present disclosure are present in a proportion of at least 95% in the fermenting population (when measured with respect to its DNA contribution of the total fermenting population). In an embodiment, after a total 40 fermentation cycles (comprising both the initial fermentation cycle and 39 further fermentation cycles) according to the present process, the persistent yeast cells of the present disclosure are present in a proportion of at least 96% in the fermenting population (when measured with respect to its DNA contribution of the total fermenting population).

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tation cycle and 49 further fermentation cycles) according to the present process, the persistent yeast cells of the present disclosure are present in a proportion of at least 99.9% in the fermenting population (when measured with respect to its DNA contribution of the total fermenting population).

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In an embodiment, after a total 70 fermentation cycles (comprising both the initial fermentation cycle and 69

In an embodiment, after a total 80 fermentation cycles (comprising both the initial fermentation cycle and 79 further fermentation cycles) according to the present process, the persistent yeast cells of the present disclosure are present in a proportion of at least 90% in the fermenting population (when measured with respect to its DNA contribution of the total fermenting population). In an embodiment, after a total 80 fermentation cycles (comprising both the initial fermentation cycle and 79 further fermentation

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In an embodiment, after a total 110 fermentation cycles (comprising both the initial fermentation cycle and 109 further fermentation cycles) according to the present process, the persistent yeast cells of the present disclosure are present in a proportion of at least 90% in the fermenting population (when measured with respect to its DNA contribution of the total fermenting population). In an embodiment, after a total 110 fermentation cycles (comprising both the initial fermentation cycle and 109 further fermentation cycles) according to the present process, the persistent yeast cells of the present disclosure are present in a proportion of at least 91% in the fermenting population (when measured with respect to its DNA contribution of the total fermenting population). In an embodiment, after a total 110 fermentation cycles (comprising both the initial fermentation cycle and 109 further fermentation cycles) according to the present process, the persistent yeast cells of the present disclosure are present in a proportion of at least 92% in the fermenting population (when measured with respect to its DNA contribution of the total fermenting population). In an embodiment, after a total 110 fermentation cycles (comprising both the initial fermentation cycle and 109 further fermentation cycles) according to the present process, the persistent yeast cells of the present disclosure are present in a proportion of at least 93% in the fermenting population (when measured with respect to its DNA contribution of the total fermenting population). In an embodiment, after a total 110 fermentation cycles (comprising both the initial fermentation cycle and 109 further fermentation cycles) of the total fermenting

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In an embodiment, after a total 120 fermentation cycles (comprising both the initial fermentation cycle and 119 further fermentation cycles) according to the present process, the persistent yeast cells of the present disclosure are present in a proportion of at least 90% in the fermenting population (when measured with respect to its DNA contribution of the total fermenting population). In an embodiment, after a total 120 fermentation cycles (comprising both the initial fermentation cycle and 119 further fermentation cycles) according to the present process, the persistent yeast cells of the present disclosure are present in a proportion of at least 91% in the fermenting population (when measured with respect to its DNA contribution of the total fermenting population). In an embodiment, after a total 120 fermentation cycles (comprising both the initial fermentation cycle and 119 further fermentation cycles) according to the present process, the persistent yeast cells of the present disclosure are present in a proportion of at least 92% in the fermenting population (when measured with respect to its DNA contribution of the total fermenting population). In an embodiment, after a total 120 fermentation cycles (comprising both the initial fermentation cycle and 119 further fermentation cycles) according to the present process, the persistent yeast cells of the present disclosure are present in a proportion of at least 93% in the fermenting population (when measured with respect to its DNA contribution of the total fermenting population). In an embodiment, after a total 120 fermentation cycles (comprising both the initial fermentation cycle and 119 further fermentation cycles) according to the present process, the persistent yeast cells of the present disclosure are present in a proportion of at least 94% in the fermenting population (when measured with respect to its DNA contribution of the total fermenting population). In an embodiment, after a total 120 fermentation

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In an embodiment, after a total 140 fermentation cycles (comprising both the initial fermentation cycle and 139 further fermentation cycles) according to the present process, the persistent yeast cells of the present disclosure are present in a proportion of at least 90% in the fermenting population (when measured with respect to its DNA contribution of the total fermenting population). In an embodiment, after a total 140 fermentation cycles (comprising both the initial fermentation cycle and 139 further fermentation cycles) according to the present process, the persistent yeast cells of the present disclosure are present in a proportion of at least 91% in the fermenting population (when measured with respect to its DNA contribution of the total fermenting population). In an embodiment, after a total 140 fermentation cycles (comprising both the initial fermentation cycle and 139 further fermentation cycles) according to the present process, the persistent yeast cells of the present disclosure are present in a proportion of at least 92% in the fermenting population (when measured with respect to its DNA contribution of the total fermenting population). In an embodiment, after a total 140 fermentation cycles (comprising both the initial fermentation cycle and 139 further fermentation cycles of the total fermenting population) according to the present process, the persistent yeast cells of the present disclosure are present in a proportion of at least 93% in the fermenting population (when measured with respect to its DNA contribution of the total fermenting population). In an embodiment, after a total 140 fermentation cycles (comprising both the initial fermentation cycle and 139 further fermentation cycles of the total fermenting population) according to the present process, the persistent yeast cells of the present disclosure are present in a proportion of at least 94% in the fermenting population (when measured with respect to its DNA contribution of the total fermenting population). In an embodiment, after a total 140 fermentation cycles (comprising both the initial fermentation cycle and 139 further fermentation cycles of the total fermenting population) according to the present process, the persistent yeast cells of the present disclosure are present in a proportion of at least 95% in the fermenting population (when measured with respect to its DNA contribution of the total fermenting population). In an embodiment, after a total 140 fermentation cycles (comprising both the initial fermentation cycle and 139 further fermentation cycles of the total fermenting population) according to the present process, the

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In an embodiment, after a total 150 fermentation cycles (comprising both the initial fermentation cycle and 149 further fermentation cycles) according to the present process, the persistent yeast cells of the present disclosure are present in a proportion of at least 90% in the fermenting population (when measured with respect to its DNA contribution of the total fermenting population). In an embodiment, after a total 150 fermentation cycles (comprising both the initial fermentation cycle and 149 further fermentation cycles) according to the present process, the persistent yeast cells of the present disclosure are present in a proportion of at least 91% in the fermenting population (when measured with respect to its DNA contribution of the total fermenting population). In an embodiment, after a total 150 fermentation cycles (comprising both the initial fermentation cycle and 149 further fermentation cycles) according to the present process, the persistent yeast cells of the present disclosure are present in a proportion of at least 92% in the fermenting population (when measured with respect to its DNA contribution of the total fermenting population). In an embodiment, after a total 150 fermentation cycles (comprising both the initial fermentation cycle and 149 further fermentation cycles of the total fermenting population) according to the present process, the persistent yeast cells of the present disclosure are present in a proportion of at least 93% in the fermenting population (when measured with respect to its DNA contribution of the total fermenting population). In an embodiment, after a total 150 fermentation cycles (comprising both the initial fermentation cycle and 149 further fermentation cycles of the total fermenting population) according to the present process, the persistent yeast cells of the present disclosure are present in a proportion of at least 94% in the fermenting population (when measured with respect to its DNA contribution of the total fermenting population). In an embodiment, after a total 150 fermentation cycles (comprising both the initial fermentation cycle and 149 further fermentation cycles of the total fermenting population) according to the present process, the persistent yeast cells of the present disclosure are present in a proportion of at least 95% in the fermenting population (when measured with respect to its DNA contribution of the total fermenting population). In an embodiment, after a total 150 fermentation cycles (comprising both the initial fermentation cycle and 149 further fermentation cycles of the total fermenting population) according to the present process, the persistent yeast cells of the present disclosure are present in a proportion of at least 96% in the fermenting population (when measured with respect to its DNA contribution of the total fermenting population). In an embodiment, after a total

population). In an embodiment, after a total 160 fermentation cycles (comprising both the initial fermentation cycle and 159 further fermentation cycles) according to the present process, the persistent yeast cells of the present disclosure are present in a proportion of at least 99.9% in the fermenting population (when measured with respect to its DNA contribution of the total fermenting population).

[illegible]

fermenting population (when measured with respect to its DNA contribution of the total fermenting population).

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In an embodiment, after a total 190 fermentation cycles (comprising both the initial fermentation cycle and 189 further fermentation cycles) according to the present pro-

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In an embodiment, after a total 200 fermentation cycles (comprising both the initial fermentation cycle and 199 further fermentation cycles) according to the present process, the persistent yeast cells of the present disclosure are present in a proportion of at least 90% in the fermenting population (when measured with respect to its DNA contribution of the total fermenting population). In an embodiment, after a total 200 fermentation cycles (comprising both

total fermenting population). In an embodiment, after a total 200 fermentation cycles (comprising both the initial fermentation cycle and 199 further fermentation cycles) according to the present process, the persistent yeast cells of the present disclosure are present in a proportion of at least 99.2% in the fermenting population (when measured with respect to its DNA contribution of the total fermenting population). In an embodiment, after a total 200 fermentation cycles (comprising both the initial fermentation cycle and 199 further fermentation cycles) according to the present process, the persistent yeast cells of the present disclosure are present in a proportion of at least 99.3% in the fermenting population (when measured with respect to its DNA contribution of the total fermenting population). In an embodiment, after a total 200 fermentation cycles (comprising both the initial fermentation cycle and 199 further fermentation cycles) according to the present process, the persistent yeast cells of the present disclosure are present in a proportion of at least 99.4% in the fermenting population (when measured with respect to its DNA contribution of the total fermenting population). In an embodiment, after a total 200 fermentation cycles (comprising both the initial fermentation cycle and 199 further fermentation cycles) according to the present process, the persistent yeast cells of the present disclosure are present in a proportion of at least 99.5% in the fermenting population (when measured with respect to its DNA contribution of the total fermenting population). In an embodiment, after a total 200 fermentation cycles (comprising both the initial fermentation cycle and 199 further fermentation cycles) according to the present process, the persistent yeast cells of the present disclosure are present in a proportion of at least 99.6% in the fermenting population (when measured with respect to its DNA contribution of the total fermenting population). In an embodiment, after a total 200 fermentation cycles (comprising both the initial fermentation cycle and 199 further fermentation cycles) according to the present process, the persistent yeast cells of the present disclosure are present in a proportion of at least 99.7% in the fermenting population (when measured with respect to its DNA contribution of the total fermenting population). In an embodiment, after a total 200 fermentation cycles (comprising both the initial fermentation cycle and 199 further fermentation cycles) according to the present process, the persistent yeast cells of the present disclosure are present in a proportion of at least 99.8% in the fermenting population (when measured with respect to its DNA contribution of the total fermenting population). In an embodiment, after a total 200 fermentation cycles (comprising both the initial fermentation cycle and 199 further fermentation cycles) according to the present process, the persistent yeast cells of the present disclosure are present in a proportion of at least 99.9% in the fermenting population (when measured with respect to its DNA contribution of the total fermenting population).

The initial and further fermentation medium comprises or is derived from a biomass. The biomass that can be fermented with the recombinant host cell described herein includes any type of biomass known in the art and described herein. For example, the biomass can include, but is not limited to, starch, sugar and lignocellulosic materials. Starch materials can include, but are not limited to, mashes such as corn, wheat, rye, barley, rice, or milo. Sugar materials can include, but are not limited to, sugar beets, artichoke tubers, sweet sorghum, molasses or sugarcane. The terms "lignocellulosic material", "lignocellulosic substrate" and "cellulosic biomass" mean any type of biomass comprising cellulose, hemicellulose, lignin, or combinations thereof, such

as but not limited to woody biomass, forage grasses, herbaceous energy crops, non-woody-plant biomass, agricultural wastes and/or agricultural residues, forestry residues and/or forestry wastes, paper-production sludge and/or waste paper sludge, waste-water-treatment sludge, municipal solid waste, corn fiber from wet and dry mill corn ethanol plants and sugar-processing residues. The terms "hemicellulosics", "hemicellulosic portions" and "hemicellulosic fractions" mean the non-lignin, non-cellulose elements of lignocellulosic material, such as but not limited to hemicellulose (i.e., comprising xyloglucan, xylan, glucuronoxylan, arabinoxylan, mannan, glucomannan and galactoglucomannan), pectins (e.g., homogalacturonans, rhamnogalacturonan I and II, and xylogalacturonan) and proteoglycans (e.g., arabinogalactan-protein, extensin, and pro line-rich proteins). In a further embodiment, the initial and/or the further fermentation medium comprises sucrose as the main fermentable carbohydrate. In one embodiment, the initial and/or the further fermentation medium comprises or is derived from sugarcane, molasses, derivatives thereof as well as mixtures thereof.

In a non-limiting example, the lignocellulosic material can include, but is not limited to, woody biomass, such as recycled wood pulp fiber, sawdust, hardwood, softwood, and combinations thereof; grasses, such as switch grass, cord grass, rye grass, reed canary grass, *miscanthus*, or a combination thereof; sugar-processing residues, such as but not limited to sugar cane bagasse; agricultural wastes, such as but not limited to rice straw, rice hulls, barley straw, corn cobs, cereal straw, wheat straw, canola straw, oat straw, oat hulls, and corn fiber; stover, such as but not limited to soybean stover, corn stover; succulents, such as but not limited to, agave; and forestry wastes, such as but not limited to, recycled wood pulp fiber, sawdust, hardwood (e.g., poplar, oak, maple, birch, willow), softwood, or any combination thereof. Lignocellulosic material may comprise one species of fiber; alternatively, lignocellulosic material may comprise a mixture of fibers that originate from different lignocellulosic materials. Other lignocellulosic materials are agricultural wastes, such as cereal straws, including wheat straw, barley straw, canola straw and oat straw; corn fiber; stovers, such as corn stover and soybean stover; grasses, such as switch grass, reed canary grass, cord grass, and *miscanthus*; or combinations thereof.

Substrates for cellulose activity assays can be divided into two categories, soluble and insoluble, based on their solubility in water. Soluble substrates include celloextrins or derivatives, carboxymethyl cellulose (CMC), or hydroxyethyl cellulose (HEC). Insoluble substrates include crystalline cellulose, microcrystalline cellulose (Avicel), amorphous cellulose, such as phosphoric acid swollen cellulose (PASC), dyed or fluorescent cellulose, and pretreated lignocellulosic biomass. These substrates are generally highly ordered cellulosic material and thus only sparingly soluble.

It will be appreciated that suitable lignocellulosic material may be any feedstock that contains soluble and/or insoluble cellulose, where the insoluble cellulose may be in a crystalline or non-crystalline form. In various embodiments, the lignocellulosic biomass comprises, for example, wood, corn, corn stover, sawdust, bark, molasses, sugarcane, leaves, agricultural and forestry residues, grasses such as switchgrass, ruminant digestion products, municipal wastes, paper mill effluent, newspaper, cardboard or combinations thereof.

Paper sludge is also a viable feedstock for lactate or acetate production. Paper sludge is solid residue arising from pulping and paper-making, and is typically removed from process wastewater in a primary clarifier. The cost of

disposing of wet sludge is a significant incentive to convert the material for other uses, such as conversion to ethanol. Processes provided by the present invention are widely applicable. Moreover, the saccharification and/or fermentation products may be used to produce ethanol or higher value added chemicals, such as organic acids, aromatics, esters, acetone and polymer intermediates.

In the process described herein, it is possible to add an exogenous source (e.g., to dose) of an enzyme to facilitate saccharification or improve fermentation yield. As such, the process can comprise including one or more dose(s) of one or more enzyme(s). The exogenous enzyme that can be used can include, without limitation, an alpha-amylase, a glucoamylase, a protease, a phytase, a pullulanase, a cellulase, a hemi-cellulase such as a xylanase, a trehalase, or any combination thereof. The exogenous enzyme can be provided, in some embodiments, in a purified form and/or provided as part of a cocktail.

The production of ethanol can be performed, for example, at temperatures of at least about 30° C., about 31° C., about 32° C., about 33° C., about 34° C., about 35° C., about 36° C., about 37° C., about 38° C., about 39° C., about 40° C., about 41° C., about 42° C., about 43° C., about 44° C., about 45° C., about 46° C., about 47° C., about 48° C., about 49° C., or about 50° C. In some embodiments, the production of ethanol from cellulose can be performed, for example, at temperatures above about 30° C., about 31° C., about 32° C., about 33° C., about 34° C., about 35° C., about 36° C., about 37° C., about 38° C., about 39° C., about 40° C., about 41° C., about 42° C., or about 43° C., or about 44° C., or about 45° C., or about 50° C. In some embodiments, the persistent yeast can produce ethanol from cellulose at temperatures from about 30° C. to 60° C., about 30° C. to 55° C., about 30° C. to 50° C., about 40° C. to 60° C., about 40° C. to 55° C. or about 40° C. to 50° C.

In the processes described herein, at the end of the fermentation, the fermenting population is substantially isolated from the fermented fermentation medium. As used in the context of the present disclosure, the expression "substantially isolating" refers to the removal of the majority of the components of the fermented fermentation medium from the fermenting population. In some embodiments, "substantially isolating" refers to concentrating the fermenting population to at least 5, 10, 15, 20, 25, 30, 35, 45% or more when compared to the concentration of the fermenting population prior to the substantially isolation. In order to substantially isolate to fermenting population, the fermented fermentation medium can be centrifuged. Cell separation and recovery in the fuel ethanol process is carried out using stacked-disk, nozzle discharge type centrifuges (see Brociner et al.). In these machines, the feed-broth from the end of fermentation, often referred to in the process as "vinho bruto" or "beer" is introduced into the top of the machine, circulates to the bottom, and is then forced upward through a set of rotating disks. The rotation of these disks imparts a centrifugal force on the total feed, and particles. Yeast cells and other solids are forced downward and to the side of the machine. The cells then exit through nozzles at the outer edge of the machine creating a concentrated yeast cream. Clarified liquid, often called "vinho," "vinho delevurado" or "wine" exits the machine out the top.

Optionally the substantially isolated fermenting population can be washed. In a specific embodiment, the substantially isolated fermenting population can be submitted to an acid washing step. In the acid washing step, an acid or an acidic solution is put into contact with the fermenting population. In some embodiments, the acid or the acidic

solution has a pH of between 2.0 and 2.2. In some embodiments, the contact between the substantially isolated fermenting population and the acid/acidic solution is maintained so as to reduce the contaminating bacterial population that may be present. For example, the contact between the substantially isolated fermenting population and the acid or the acidic solution can last at least 30, 40, 50, 60, 70, 80, 90, 100, 110, 120 minutes or more. In certain embodiments, the acid is sulphuric acid and/or the acidic solution comprises sulphuric acid. After the acid washing step, the pH of the acid washed fermenting population can be adjusted prior to the further fermentation cycle.

In some embodiments, methods of producing ethanol can comprise contacting the fermentation substrate with a persistent yeast described herein and additionally contacting the substrate with externally produced enzymes which can be provided in a purified form. Exemplary externally produced enzymes include, but are not limited to starch degrading enzymes, dextran degrading enzymes, phytase, protease, cellulases and/or xylose isomerase. Specific externally produced (and optionally purified) enzymes include, but are not limited to, trehalases, glucoamylases, alpha-amylases, alpha-glucosidases, glucanases (endo/exo), pullulanases, phytases and/or proteases.

In some embodiments, the methods comprise producing ethanol at a particular rate. For example, in some embodiments, ethanol is produced at a rate of at least about 0.1 mg per hour per liter, at least about 0.25 mg per hour per liter, at least about 0.5 mg per hour per liter, at least about 0.75 mg per hour per liter, at least about 1.0 mg per hour per liter, at least about 2.0 mg per hour per liter, at least about 5.0 mg per hour per liter, at least about 10 mg per hour per liter, at least about 15 mg per hour per liter, at least about 20.0 mg per hour per liter, at least about 25 mg per hour per liter, at least about 30 mg per hour per liter, at least about 50 mg per hour per liter, at least about 100 mg per hour per liter, at least about 200 mg per hour per liter, at least about 300 mg per hour per liter, at least about 400 mg per hour per liter, at least about 500 mg per hour per liter, at least about 600 mg per hour per liter, at least about 700 mg per hour per liter, at least about 800 mg per hour per liter, at least about 900 mg per hour per liter, at least about 1 g per hour per liter, at least about 1.5 g per hour per liter, at least about 2 g per hour per liter, at least about 2.5 g per hour per liter, at least about 3 g per hour per liter, at least about 3.5 g per hour per liter, at least about 4 g per hour per liter, at least about 4.5 g per hour per liter, at least about 5 g per hour per liter, at least about 5.5 g per hour per liter, at least about 6 g per hour per liter, at least about 6.5 g per hour per liter, at least about 7 g per hour per liter, at least about 7.5 g per hour per liter, at least about 8 g per hour per liter, at least about 8.5 g per hour per liter, at least about 9 g per hour per liter, at least about 9.5 g per hour per liter, at least about 10 g per hour per liter, at least about 10.5 g per hour per liter, at least about 11 g per hour per liter, at least about 11.5 g per hour per liter, at least about 12 g per hour per liter, at least about 12.5 g per hour per liter, at least about 13 g per hour per liter, at least about 13.5 g per hour per liter, at least about 14 g per hour per liter, at least about 14.5 g per hour per liter or at least about 15 g per hour per liter.

In some embodiments, the persistent yeast cells can produce ethanol at a rate of at least about 0.1 mg per hour per liter, at least about 0.25 mg per hour per liter, at least about 0.5 mg per hour per liter, at least about 0.75 mg per hour per liter, at least about 1.0 mg per hour per liter, at least about 2.0 mg per hour per liter, at least about 5.0 mg per hour per liter, at least about 10 mg per hour per liter, at least about

15 mg per hour per liter, at least about 20.0 mg per hour per liter, at least about 25 mg per hour per liter, at least about 30 mg per hour per liter, at least about 50 mg per hour per liter, at least about 100 mg per hour per liter, at least about 200 mg per hour per liter, at least about 300 mg per hour per liter, at least about 400 mg per hour per liter, at least about 500 mg per hour per liter, at least about 600 mg per hour per liter, at least about 700 mg per hour per liter, at least about 800 mg per hour per liter, at least about 900 mg per hour per liter, at least about 1 g per hour per liter, at least about 1.5 g per hour per liter, at least about 2 g per hour per liter, at least about 2.5 g per hour per liter, at least about 3 g per hour per liter, at least about 3.5 g per hour per liter, at least about 4 g per hour per liter, at least about 4.5 g per hour per liter, at least about 5 g per hour per liter, at least about 5.5 g per hour per liter, at least about 6 g per hour per liter, at least about 6.5 g per hour per liter, at least about 7 g per hour per liter, at least about 7.5 g per hour per liter, at least about 8 g per hour per liter, at least about 8.5 g per hour per liter, at least about 9 g per hour per liter, at least about 9.5 g per hour per liter, at least about 10 g per hour per liter, at least about 10.5 g per hour per liter, at least about 11 g per hour per liter, at least about 11.5 g per hour per liter, at least about 12 g per hour per liter, at least about 12.5 g per hour per liter, at least about 13 g per hour per liter, at least about 13.5 g per hour per liter, at least about 14 g per hour per liter, at least about 14.5 g per hour per liter, at least about 15 g per hour per liter or more than a control strain (e.g., a wild-type strain, such as, for example, strain PE-2) and grown under the same conditions. In some embodiments, the ethanol can be produced in the absence of any externally added cellulases.

Ethanol production can be measured using any method known in the art. For example, the quantity of ethanol in fermentation samples can be assessed using HPLC analysis. Many ethanol assay kits are commercially available that use, for example, alcohol oxidase enzyme based assays.

Processes for Screening/Generating Persistent Yeast Cells

The present disclosure also provides for a process for determining if a test yeast is considered persistent or not. The process comprises conducting a plurality of fermentation cycles using, in the initial fermenting population a test yeast and determining, after a specific number of fermentation cycles, if the test yeast is present in the fermented medium and if so, the proportion of the test yeast in the fermentation medium. The test yeast has a detectable feature which is absent from other wild yeasts which may contaminate the fermentation medium. If the test yeast is present, after a total of 40, 50, 60, 70, 80, 90, 100, 110, 120, 130, 140, 150, 160, 170, 180, 190, 200 fermentation cycles or more, in a proportion at least 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, 99.1%, 99.2%, 99.3%, 99.4%, 99.5%, 99.6%, 99.7%, 99.8%, 99.9% or more in the fermenting population, the test yeast is considered to be persistent. If the test yeast is present, after a total of 40, 50, 60, 70, 80, 90, 100, 110, 120, 130, 140, 150, 160, 170, 180, 190, 200 fermentation cycles or more, in a proportion lower than 90% in the fermenting population, the test yeast is not considered to be persistent. The process can include, prior to performing the plurality of fermentation cycles, providing a test yeast as a recombinant yeast host cell capable of modulating the activity or the expression of a first polypeptide and/or a second polypeptide for increasing, when compared to a parental cell, the conversion of a biomass into a fermentation product and/or for reducing the conversion of the biomass into a fermentation by-product as described herein. In some embodiments, the process can include introducing one or more genetic modification, for example those presented

above, to modulate the activity or the expression of the first polypeptide and/or the second polypeptide in the test yeast which is considered to be persistent or intended to be screened for persistence phenotypes. The process can include, in some embodiments, determining if the test yeast is persistent by determining the presence or the absence of at least one or any combination of phenotypic traits. A test yeast comprising at least one or any combination of the phenotypic traits described herein is considered to be persistent.

The present disclosure also provides for a process for generating a persistent yeast. The process comprises conducting a plurality of fermentation cycles using, in the initial fermenting population comprising an initial yeast or a combination of initial yeasts and conducting a specific number of fermentation cycles. The initial yeast(s) may or may not be considered to be persistent. In some embodiments, it is not known if the initial yeast(s) is persistent or not. The fermenting population obtained after a total of 40, 50, 60, 70, 80, 90, 100, 110, 120, 130, 140, 150, 160, 170, 180, 190, 200 fermentation cycles or more is considered to include persistent yeast cells. The persistent yeast cells may or may not correspond to the initial yeast(s) initially inoculated in the fermentation medium. The process can include, in some embodiments, determining if the fermenting population obtained after a total of at least 40 fermentation cycles is persistent by determining the presence or the absence of the at least one phenotypic traits in one or more cells of the fermenting population. A yeast cell obtained (and in some embodiments substantially isolated) from the fermenting population after at least 40 fermentation cycles exhibiting at least one or any combination of the phenotypic traits described herein is considered to be persistent. The persistent yeast cells obtained by this process can, in some embodiments, be genetically modified to have the ability of modulating the activity or the expression of a first polypeptide and/or a second polypeptide for increasing, when compared to a parental cell, the conversion of a biomass into a fermentation product and/or for reducing the conversion of the biomass into a fermentation by-product as described herein.

The present invention will be more readily understood by referring to the following examples which are given to illustrate the invention rather than to limit its scope.

EXAMPLE

Material and Methods

Gene sequencing. DNA was extracted from particular fermentation cycles throughout the implementation test and Illumina sequenced (2x126 bp read data; short reads of all DNA present).

Quantitative PCR (qPCR) assay. The amount of the recombinant yeast strain was measured by a quantitative polymerase chain reaction (qPCR) technique that quantifies the amount of the IME1 gene present in a sample compared to the amount of the ALG9 gene. Since the recombinant yeast strain does not have the IME1 gene (because it includes at least one genetic modification disrupting the IME1 gene), the amount of quantified IME1 DNA relative to the amount of ALG9 DNA increases as the amount of contaminating DNA (from wild yeasts) increases. The absolute quantification method was used to analyze the samples by comparison to a standard curve for each gene of interest. The qPCR reactions for the unknowns contains the following components, genomic DNA (loaded at 0.05 ng/μl), SsoAdvanced™ Universal SYBR® Green Supermix (Bio-

rad #172-5271), and primer pairs specific to either the IME1 or the ALG9 gene at 0.5 uM final concentration. The IME1 gene was amplified with primer X33463 (cacgtgccttagaagatgg, SEQ ID NO: 1) and primer X33464 (gttctgcagctgagatgagg, SEQ ID NO: 2). The ALG9 gene was amplified with primer X34382 (gtttaatccggctggttccat, SEQ ID NO: 3) and primer X34383 (TAGACCCAGTGGACAGATAGCG, SEQ ID NO: 4). The amplification was performed in a BioRad CFX96 Touch Real-Time PCR Detection System using 2-step PCR cycling conditions with a melting temperature of 60° C. for 30 seconds for a total of 40 cycles. Finally, the starting quantity (SQ) of the sample DNA was determined by comparing the cycle of quantitation (Cq) for the unknown to the Cq of the known DNA quantities in the standard curve for each primer set. The calculation (IME1 SQ/ALG9 SQ*100) determined the percentage of wild type DNA in the population and was used to estimate the presence of the recombinant yeast strain.

Scanning electron microscopy. Scanning electron microscopy (SEM) was used to study the morphological shape of yeast cells. The cells were fixed in glutaraldehyde, post-fixed in OsO_4 , dried and sputter-coated with gold/palladium (Au/Pd) for viewing. The specimen were fixed by immersing them in 2.5%-4% glutaraldehyde in 0.1 M sodium cacodylate or phosphate buffer solution (pH 6.8-7.4 as appropriate) and incubating for at least two hours. The specimen were washed by immersing them 3 times with the 0.1 M cacodylate or phosphate buffer solution for 10 minutes each to remove excess fixative. The specimen were then submitted to a post-fixation treatment by immersing them in a 1% OsO_4 in buffer at 4° C. and further washed in distilled or de-ionized water to remove all traces of fixative and buffer solutions. The specimens were dehydrated by immersing them for 10 minutes each in 25%, 50%, 70%, 85%, 95%, and three 100% ethanol solutions to remove all traces of water. The specimens were submitted to a critical point dry or three washes in HMDS for 10 minutes each followed by air drying. The specimens were mounted and a sputter coat a thin layer (~5 nm) of AuPd was applied.

Commercial fed batch fermentation process. A detailed description of the fermentation process with yeast recycle with its various forms can be found in Basso et al. (2001). Briefly, non-sterilized sugarcane juice and/or molasses with a typical concentration of between 18% and 22% total sugars was fermented at a high yeast inoculum (10 to 14% wet weight per volume) to achieve fast (8 to 12 hours) fermentations. The sugar concentration can vary over the course of the sugarcane harvest season, and is typically lowest at the start and end of the season. The yeast was repeatedly recycled over the crop season (~200 days). After each cycle of fermentation, yeast cells were separated from the whole fermentation broth by centrifugation, with some of the yeast being lost with the wine from the centrifuge and proceeding to distillation, and the retained yeast remaining as a concentrated yeast cream. The yeast cream was then subjected to a treatment with sulphuric acid where water and acid are added until the yeast cream reaches a pH of between 3.5 to 1.8 (although in some cases no acid treatment is used). This treatment lasted 1 to 2 hours and was used to decrease bacterial contamination and break up flocculation of the yeast cells. In some cases antimicrobials were added to the yeast cream prior to fermentation as well. After the treatment was complete, the yeast cell cream was repitched to a fermentation tank and fresh substrate (molasses, sugarcane juice, or a mixture) was then added to the cells and fermentation is started again. Fermentation temperatures varied based on the configuration of the industrial facility, but were

generally targeted to between 32 and 35° C. In many cases, the temperature cannot be controlled in this range and can reach as high as 40° C. or higher. Fermentation vessels were typically only agitated by the CO_2 generated during fermentation escaping the broth and by the action of a pump around loop with a heat exchanger to remove excess heat.

Laboratory scale fed batch cell recycle fermentation process. Lab scale fermentations were set up to mimic the industrial scale process. They were carried out using 50 mL vessels filled with yeast cream either from propagation, or from a previous fermentation, at a level to reproduce standard yeast concentrations in fermentations recycling yeasts (~10% wet cell mass). This yeast cream was subjected to acid treatment under the conditions provided above in the section "Commercial fed batch fermentation process". A feeding system was used to provide a feed of substrate or "must" (sugar cane must sourced from operating facilities), again at rates and concentrations dictated by average conditions occurring in commercial facilities. This feed stream was provided via a syringe pump to each reactor. Fermentations were held under temperature controlled conditions (generally 32 to 35° C.) and gently agitated, and were allowed to proceed until the evolution of CO_2 falls below a minimum threshold. Once complete, samples were taken for analysis by HPLC to compare the production of ethanol, glycerol, organic acids, and other compounds. Yeast cultures were run as either pure cultures or as mixtures of yeast strains as indicated in the experimental description or figure legends. The amounts of different strains present after a particular number of cycles was determined using a qPCR technique specific to the strain of interest.

Batch laboratory scale fermentation. Anaerobic batch fermentations were run in 60 mL pressure bottles with 20 mL of media (commercially sourced must or defined laboratory media). Fermentations were held under temperature controlled conditions and gently agitated, and were allowed to proceed until the evolution of CO_2 falls below a minimum threshold. Once complete, samples were taken for analysis by HPLC to compare the production of ethanol, glycerol, organic acids, and other compounds. Aerobic cultures were similarly set-up but with a permeable cap. Cultures of yeast strains Y2, Y4 and Y8 were pitched in monoculture or in co-culture with ~15% of the wild type strain Y7. The fermentations were monitored by HPLC and yeast populations were monitored by qPCR.

Suspension assay. Cells were suspended evenly at time zero, taking a sample from the top of the suspension and measuring the sample's optical density at 600 nm. The sample was then incubated for a designated amount of time (3.5 or 5 min). After the incubation, another sample was obtained from the top of the suspension and the optical density is again measured at 600 nm. The percentage of sedimentation corresponds to the percentage change in optical density between the two samples. Samples in this assay included a control, which was a non-clustered yeast strain freshly grown up on media, as well as samples taken directly from the centrifugation process.

Wash out rate determination. Washout rates were determined by measuring the amount of non-engineered strain present in the fermentation system of the facility by carrying out qPCR of composite samples (as described above). Then, overall exponential washout rates (K value) were calculated using the formula

$$K = \ln \left(\frac{C_{\text{FINAL}}}{C_{\text{INITIAL}}} \right) / (\text{CYCLE}_{\text{FINAL}} - \text{CYCLE}_{\text{INITIAL}})$$

where C_{FINAL} =the fraction of contaminant yeast DNA at the final cycle measured; $C_{INITIAL}$ =the fraction of contaminant yeast DNA at the first cycle measured; $CYCLE_{FINAL}$ =the number of the final cycle measured; $CYCLE_{INITIAL}$ =the number of the first cycle measured.

Genome-wide association tests. The software Plink (Purcell et al. 2007) can conduct a genome-wide scan for alleles that significantly associate with a phenotype of interest. Following ploidy inference, the filtered variant callset of the appropriate ploidy were converted to Plink format using BCFtools (Li et al. 2009; Li, 2011) to create a chromosome-map file, and VCFtools (Danecek et al., 2011) to execute the conversion. Each strain was coded as being either smooth (0) or rugose (1) and conducted the association test for SNPs and indels independently, log-transforming the resulting p-values to plot the relative strength of the genotype-phenotype association across each position. In conventional genome-wide association studies (GWAS) involving hundreds to thousands of samples, p-values $< 1 \times 10^{-5}$ are considered suggestive and p-values $< 5 \times 10^{-8}$ are considered significant. In the present case, comparatively few samples (N=25), even perfect associations may not yield conventionally significant p-values depending on the level of noise

in the genome-wide background, but the results can be used to narrow down the field of candidate causal mutations by quantifying the relative strength of the association across a genome-wide panel of variants.

Invertase assay. Cells were grown overnight in aerobic or anaerobic culture, or harvested at the end of one or more fermentation cycles in the fed-batch cell recycle system. Cultures were diluted to 9 mg/mL (wet weight basis, ~2 g/L dry weight basis) in a citrate-phosphate buffer pH 5 with 40 g/L sucrose and incubated at 35° C. for 12 to 15 minutes. The mixture is then incubated with 3,5-dinitrosalicylic acid (DNS). DNS reacts with reducing sugars and to form 3-amino-5-nitrosalicylic acid, which absorbs light at 540 nm. Glucose and fructose released by the yeast invertase activity is quantified by spectrophotometry at 540 nm. Dry weight measurements of cell samples are used to calculate the amount of dry cells added in each reaction and the sugar release per gram of dry cells loaded per time.

Response to cAMP. Cell were grown in YPD for 48 hours until they were glucose depleted followed by a spike of 100 mM glucose to the cells. Cells were snap frozen before glucose addition and after 5 minutes of incubation with the glucose. Cells were thawed, lysed and the amount of cAMP was then measured by a standard kit.

TABLE 1

Description of the strains that were made and/or characterizes. GPD2 had the amino acid sequence of SEQ ID NO: 5 was encoded by the nucleic acid sequence of SEQ ID NO: 6. STL1 had the amino acid sequence of SEQ ID NO: 7 was encoded by the nucleic acid sequence of SEQ ID NO: 8. SmGAPN had the amino acid sequence of SEQ ID NO: 29 and was encoded by the nucleic acid sequence of SEQ ID NO: 30. The presence of the a plurality of acronyms in a strain (STL1-STL1 for example) refers that two copies of a gene encoding for the polypeptide referred to by the acronym has been inserted at each integration site.

Genetically modified strain phenotype	Parental strain phenotype	Genes overexpresses in the genetically modified strain
Y1: smooth	M710: smooth	gpd1Δ::gpd2 fcy1Δ::STL1-STL1 ime1Δ::STL1-STL1
Y2: rugose	Isolate from Y1: smooth	gpd1Δ::gpd2 fcy1Δ::STL1-STL1 ime1Δ::STL1-STL1
Y3: rugose, fast settling, high invertase activity, triploid, hyperactivated Ras/cAMP	NA - wild-type	NA - wild-type
Y4: rugose, fast settling, high invertase activity, triploid, hyperactivated Ras/cAMP	Y3	fcy1Δ::STL1-STL1 ime1Δ::STL1-STL1
Y5: rugose, fast settling, high invertase activity, triploid, hyperactivated Ras/cAMP	NA - wild-type	NA - wild-type
Y6: rugose, fast settling, high invertase activity, triploid, hyperactivated Ras/cAMP	Y5	fcy1Δ::STL1-STL1 ime1Δ::STL1-STL1
Y7: rugose, fast settling, triploid, hyperactivated Ras/cAMP	NA - wild-type	NA - wild-type
Y8: rugose, fast settling, triploid, hyperactivated Ras/cAMP	Y7	gpd1Δ::gpd2 fcy1Δ::STL1-STL1

TABLE 1-continued

Description of the strains that were made and/or characterizes. GPD2 had the amino acid sequence of SEQ ID NO: 5 was encoded by the nucleic acid sequence of SEQ ID NO: 6. STL1 had the amino acid sequence of SEQ ID NO: 7 was encoded by the nucleic acid sequence of SEQ ID NO: 8. SmGAPN had the amino acid sequence of SEQ ID NO: 29 and was encoded by the nucleic acid sequence of SEQ ID NO: 30. The presence of the a plurality of acronyms in a strain (STL1-STL1 for example) refers that two copies of a gene encoding for the polypeptide referred to by the acronym has been inserted at each integration site.

Genetically modified strain phenotype	Parental strain phenotype	Genes overexpresses in the genetically modified strain
Y61	Y3: rugose, fast settling, high invertase activity, triploid, hyperactivated Ras/cAMP	gpd1Δ::gpd2-STL1-STL1
Y62	Y3: rugose, fast settling, high invertase activity, triploid, hyperactivated Ras/cAMP	fcy1Δ::SmGAPN-SmGAPN
Y63	Y3: rugose, fast settling, high invertase activity, triploid, hyperactivated Ras/cAMP	zwf1Δ::SmGAPN-SmGAPN
Y64	Y6: rugose, fast settling, high invertase activity, triploid, hyperactivated Ras/cAMP	fur1Δ::SmGAPN-SmGAPN fcy1Δ::STL1-STL1 ime1Δ::STL1-STL1
Y65	Y6	fcy1Δ::STL1-STL1 ime1Δ::STL1-STL1 fur1Δ::SmGAPN-SmGAPN zwf1Δ::SmGAPN-SmGAPN
Y66	Y6	fcy1Δ::STL1-STL1 ime1Δ::STL1-STL1

Fast Settling and Rugose Phenotype

Strain Y1 was submitted to a commercial fermentation. The input and output streams of yeast strain Y1 from a centrifuge obtained at a commercial ethanol mill were examined. FIG. 2A shows that the percentage of Y1 DNA is higher in the wine stream as compared to the beer stream, indicating that the yeasts exiting the fermentation process and heading to the distillation process is enriched for strain Y1 compared to wild yeast strains. In addition, the Y1 strain percentage of total yeast DNA in the yeast cream being recycled to the process is the same or somewhat lower than what is found in the beer. These results suggest that yeast strain Y1 is being selected against during the process (most likely during the centrifuge step).

The Y1 strain is a smooth colony on a plate and does not present the rugose morphology. Under the microscope, the Y1 strain forms single or doublet cells similar to what is depicted in FIGS. 4A and 4B.

Microscopic analysis showed that the feed to the centrifuge, the beer, was composed of a mix of particle of different sizes (FIG. 2B), whereas the wine was composed of mostly small particles (FIG. 2C). The cream yeast, which is usually retained and sent back to the fermentation process, was composed of mostly larger particles (FIG. 2D). FIGS. 2B to 2D show the presence of "rugose" type colonies found in these streams. The feed to the centrifuge (beer) was composed of cells that form a mixture of rugose and smooth colonies (FIG. 2B). The wine was composed of cells that

form exclusively smooth colonies (FIG. 2C). The yeast cream, which was retained, showed a mixture of cells forming smooth and rugose colonies, but was enriched for those cells that form rugose colonies relative to the beer (FIG. 2D). This data suggests that the centrifuge was preferentially selecting for cells that form rugose colonies to be retained in the process. Conversely, it showed that the centrifuge selected against those cells that form smooth colonies.

The settling velocity of the various yeast samples was determined. FIG. 3 shows the measurements of the rate of settling by gravity in a suspension after incubation for 3.5 minutes. The control, non-aggregated strain, does not settle at all under the conditions used for the assay. The input to the centrifuge (the beer) settles rapidly, as does the yeast cream output that will be recycled to the process. The wine showed much less/much slower settling of the particles it contained. This supports the principle that the centrifuge was acting to retain those particles that settle more rapidly.

Colonies that appear rugose on a growth plate form clusters of daughter yeast cells still connected to their mother (FIGS. 4A and 4C), while those that are smooth show the typical single cell or the budding morphology typical of *S. cerevisiae* (FIGS. 4A and 4B). The rugose/smooth phenotype of various commercial wild yeast populations was determined and is provided in Table 2.

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TABLE 2

Colony morphology of yeast isolated from various commercial cane fuel ethanol mills	
Sample	% Rugose
Mill 1	83%
Mill 2	91%
Mill 3 2017	90%
Mill 3 2018	100%
Mill 4	100%
Mill 4 End of season	100%
Mill 5	100%
Mill 5 End of season	100%
Mill 6	100%
Mill 6 End of season	100%
Mill 7	80%
Mill 8	98%
All Mills	95%

Six commercial scale fermentations of the Y1 strain and two commercial scale fermentation of a 75:25 (weight) mixture of the Y1 and Y2 strains were conducted and the wash-out rates were determined. As shown in FIG. 5, when the rugose strain, Y2, was present, the exponential rate of washout slowed considerably. In one case (facility 7), it decreased by >40%, and in another (facility 8) it decreased by >90%.

Various isolates from the Y1 strain (like the Y2 strain) from the commercial process were found to be rugose after five months in a commercial fed batch fermentation process. Genome sequencing of various rugose isolates was performed to understand how yeasts can change from smooth to rugose during commercial implementations. Out of 53 936 SNPs and 7 273 indels present in the panel of 25 sequenced strains, only 1 SNP and 1 indel (both on a section of Y1 scaffold 6 corresponding to chromosome XII; data not shown) were associated with the rugose phenotype via genome-wide association test ($p=1\times 10^{-6}$; data not shown). Both variants on scaffold 6 were nonsynonymous changes that predicted the rugose phenotype: a T→C variant in Choline Kinase (CK11) converting phenylalanine (TTT) to serine (TCT) at scaffold position 858 413, and a deletion at scaffold position 854 924 causing a translation frameshift in the Activator of CUP1 Expression (ACE2). These two candidate variants may be in linkage disequilibrium due to their close proximity (3 489 bp) and both fall in a distinct region of lost heterozygosity among rugose strains (data not shown). Given the prior functional evidence for ACE2, it is reasonable to assume that the ACE2 deletion was the strongest candidate causal mutation for the rugose phenotype, and the association at the CK11 SNP is a consequence of proximity.

All the characterized rugose strains were homozygous for a single base pair deletion at base pair 1,112 in the poly-A region of ACE2 (genotype A(7):A(7), SEQ ID NO: 11), causing a translation frameshift and introducing an early stop codon at amino acid residue 389, 21 residues downstream of the deletion (FIG. 6). In comparison, as shown in Table 3, all smooth mill inoculants and isolates were heterozygous for the deletion (retaining one functional copy of ACE2, genotype A(7):A(8)), including Y1 itself, and the wild type parental strain PE-2. In contrast to the smooth industrial mill isolates, neither of two smooth isolates from a laboratory experiment possessed the deletion (homozygous genotype A(8):A(8)).

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TABLE 3

Source, phenotype, and ACE2 genotype for smooth and rugose strains sequenced for comparative genomics. A(8) is encoded by the gene having the nucleic acid sequence of SEQ ID NO: 9 and has the amino acid sequence of SEQ ID NO: 10. A(7) is encoded by the gene having the nucleic acid sequence of SEQ ID NO: 11 and has the amino acid sequence of SEQ ID: 12			
Source	Phenotype	Strain designation	ACE2 genotype
Wild type parental strain	Smooth	Y0	A(7):A(8)
Industrial mill inoculants	Smooth	Y1	A(7):A(8)
	Smooth	Y41	A(7):A(8)
	Smooth	Y42	A(7):A(8)
	Smooth	Y43	A(8):A(8)
Lab experiment isolates	Smooth	Y44	A(8):A(8)
	Rugose	Y45	A(7):A(7)
	Rugose	Y46	A(7):A(7)
	Smooth	Y47	A(7):A(8)
Surviving mill isolates	Smooth	Y48	A(7):A(8)
	Smooth	Y49	A(7):A(8)
	Smooth	Y50	A(7):A(8)
	Rugose	Y2	A(7):A(7)
	Rugose	Y51	A(7):A(7)
	Rugose	Y52	A(7):A(7)
	Rugose	Y53	A(7):A(7)
	Rugose	Y54	A(7):A(7)
	Rugose	Y55	A(7):A(7)
	Rugose	Y56	A(7):A(7)
	Rugose	Y57	A(7):A(7)
	Rugose	Y58	A(7):A(7)
Contaminant mill isolates	Rugose	Y59	A(7):A(7)
	Rugose	Y60	A(7):A(7)
	Rugose	Y3	A(7):A(7)
	Rugose	Y39	A(7):A(7)

As indicated above, the evaluation of wild contaminating yeasts in the fed batch fermentation process showed that >90% of colonies analyzed were rugose (Table 3) suggesting that the rugose morphology provides an advantage during the fermentation process.

In addition, it was noted that the majority of yeasts in the mills were also extremely fast settling (e.g., fast sedimentation) when isolated and tested in a liquid settling assay (FIG. 7). The settling rate was maintained in the yeast even over 100 generations of passaging in rich glucose media showing that this is a stable feature of these mill yeasts (data not shown). Ninety (90) yeasts from various mills were tested in sedimentation and while all were fast settling, as expected due to their rugose phenotype, the majority at most mills settling even faster than the Y2 rugose natural isolate (data not shown).

Additionally, when the wild population was analyzed for mutations in the ACE2 gene, it was found that there was a high percentage of the truncated Ace2, 7 A poly A tract, allele in the mill populations (FIG. 8).

Other mutations in the ACE2 gene were also identified amongst the wild yeasts that, similar to the A8 to A7, cause a premature stop codon. These mutations (presented in Table 4) can be seen in the wild population as shown in Table 5.

TABLE 4

Mutations observed in the ACE2 gene			
Mutation location in genome	Nucleotide location in Ace2	Effect of mutation on ACE2 polypeptide	Amino acid sequence
385, 423-385, 424: A6 > A7 insertion	549	Stop codon position at 169	MDNVVDPWYINPSGFAKDTQDEEYVQHHDNVNPTIPPPDN YILNNENDDGLDNLGMDYYNIDDLTQELRDLDDIPLVPS PKTGDGSSDKKNIDRTWNLGDENNKVSHYSKKSMSHSHKRG LSGTAIFGFLGHNKTLSSISLQQSILNMSKDPQPMELINE LGNHNTVKK (SEQ ID NO: 13)
385, 379: A4 > A3 deletion	551	Stop codon at position 186	MDNVVDPWYINPSGFAKDTQDEEYVQHHDNVNPTIPPPDN YILNNENDDGLDNLGMDYYNIDDLTQELRDLDDIPLVPS PKTGDGSSDKKNIDRTWNLGDENNKVSHYSKKSMSHSHKRG LSGTAIFGFLGHNKTLSSISLQQSILNMSKDPQPMELINE LGNHNTVKNNNDPFDHIRENDGEIAI (SEQ ID NO: 14)
385, 358: T4 > T3 deletion	572	Stop codon at position 191	MDNVVDPWYINPSGFAKDTQDEEYVQHHDNVNPTIPPPDN YILNNENDDGLDNLGMDYYNIDDLTQELRDLDDIPLVPS PKTGDGSSDKKNIDRTWNLGDENNKVSHYSKKSMSHSHKRG LSGTAIFGFLGHNKTLSSISLQQSILNMSKDPQPMELINE LGNHNTVKNNNDPFDHIRENDGENSYLSQVC (SEQ ID NO: 15)
384, 818: A8 > A7 deletion	1112	Stop codon position 389	MDNVVDPWYINPSGFAKDTQDEEYVQHHDNVNPTIPPPDN YILNNENDDGLDNLGMDYYNIDDLTQELRDLDDIPLVPS PKTGDGSSDKKNIDRTWNLGDENNKVSHYSKKSMSHSHKRG LSGTAIFGFLGHNKTLSSISLQQSILNMSKDPQPMELINE LGNHNTVKNNNDPFDHIRENDGENSYLSQVLLKQQEELRI ALEKQKEVNEKLEKQLRDNQIQEKLKRVLEEQEVAQKL VSGATNSNSKPGSPVILKTPAMQNGRMKDNAIIVTTNSAN GGYQFPPTLISPRMSNTSINGSPSRKYHRQRYPNKSPES NGLNLFSSNSGYLRDSELLSFSPQNYNLNLDGLTYNDHNN TSDKNNNDKKIVLVITYSVCSKRLPRVG (SEQ ID NO: 16)

TABLE 5

Percentage of ACE2 mutants observed in wild populations. A = % of contaminating cells, B = % of products, C = % of Y1, D = % of Y2, E = % ACE2 A7 @384,818, F = % ACE2 T3 @385,358, G = % ACE2 A3 @385,379, H = % ACE2 A7 @385,423								
Sample	A	B	C	D	E	F	G	H
1	0	100	100	0	49			
2	0	100	100	0	52			
3	0	100	100	0	52			
4	0	100	100	0	30			
5	0	100	100	0	57			
6	0	100	100	0	49			
7	0	100	100	0	47			
8	0	100	100	0	50			
9	0	100	100	0	62			
10	1	99	99	0	46			
11	1	99	99	0	50			
12	43	57	57	0	40			
13	94	6	6	0	64			
14	94	6	6	0	60			
15	19	81	81	0	46			
16	58	42	42	0	29			
17	0	100	74	26	60			
18	0	100	100	0	51			
19	0	100	86	14	62			
20	0	100	96	4	58			
21	0	100	86	14	71			
22	0	100	72	28	65			
23	0	100	76	24	65			
24	0	100	67	32	71	0	0	0
25	3	97	64	33	71	0	0	0
26	33	67	40	27	41	0	11	14
27	85	15	11	5	12	0	21	51
28	94	6	6	0	3	0	28	70
29	88	12	12	0	92			
30	99	1	1	0	35	0	10	0

TABLE 5-continued

Percentage of ACE2 mutants observed in wild populations. A = % of contaminating cells, B = % of products, C = % of Y1, D = % of Y2, E = % ACE2 A7 @384,818, F = % ACE2 T3 @385,358, G = % ACE2 A3 @385,379, H = % ACE2 A7 @385,423								
Sample	A	B	C	D	E	F	G	H
31	0	100	78	22	64	0	0	0
32	0	100	64	36	68			
33	0	100	96	4	64			
34	0	100	66	34	76			
35	0	100	100	0	72			
36	0	100	80	20	54			
37	0	100	76	24	66			
38	0	100	80	20	66			
39	0	100	84	16	62			
40	0	100	74	26	70			
41	0	100	74	26	61			
42	0	100	80	20	70			
43	0	100	72	28	72			
44	0	100	92	8	71			
45	0	100	80	20	68	0	0	0
46	0	100	66	34	64	0	0	0
47	0	100	74	26	59	0	0	0
48	0	100	68	32	63	0	0	0
49	1	99	64	35	68	0	0	0
50	97	3	3	0	53			
51	99	1	1	0	55	23	28	0
52	96	4	4	0	18			
53	98	2	2	0	20	0	85	0
54	89	11	4	7	40			
56	99	1	1	0	16	0	82	0
57	7	93	90	3	39			
58	97	3	3	0	40			

The Y1 strain was run in co-inoculation with Y2 (at a 75:25 ratio) at commercial scale to determine if selection of the Y2 strain occurred in a commercial process. DNA was extracted from particular fermentation cycles throughout the implementation test and Illumina sequenced (2×126 bp read data; short reads of all DNA present). Read data was processed and aligned against the Y0 (e.g., PE-2) reference genome, computing read coverage across the genome and calling variants from each alignment. The relative abundance was determined for Y1 vs. Y2 in co-pitched fermentations. Tracking of two alleles that are present at higher levels in the Y2 background (cda1-2 A:A and Ace2 A7:A7) allowed to estimate if Y2 was increasing relative to Y1. During the implementation the level of Y2 as reported by the CDA1-2 A and Ace2 A7 alleles increase by ~2-fold over the 30 cycles of the commercial test (FIG. 9).

Invertase Activity

The acid treatment in the fed batch fermentation process is typically run at pH 2-3. After acid treatment sugarcane must is fed to the fermentation over 3-8 hours which buffers the pH of the fermentation. A typical fermentation starts at pH 2-3 and ends with a pH of 4-5.5 depending on the buffering capacity of the fed must. Sucrose is the predominant sugar in must and needs to be hydrolyzed by yeast expressed invertase to glucose and fructose. The typical pH optimum for *S. cerevisiae* invertase ranges over 3.5-5. It was determined if strains with increased invertase activity in the process could be more competitive. Without wishing to be bound to theory, increased invertase activity could provide an advantage during the first few hours of feeding when the pH is low coming out of acid treatment. The improved invertase activity could be derived from expressing an enzyme that has higher activity at lower pH, expressing and/or secreting higher level of an invertase (which may be more acid stable).

Invertase activity of 75 yeast isolates from a commercial mill was measured on YPD and commercial must after 48 hours of anaerobic growth. Invertase activity was measured by mixing cells and supernatant with a 40 g/L sucrose solution at pH 5 and incubated at 32° C. for 12 minutes and reducing sugar was determined using DNS and monitored by OD₅₄₀. In Table 6, the average fold increase in invertase activity vs. Y0 is ~12 fold higher on either YPD or must.

TABLE 6

Invertase activity of 75 isolates from a commercial mill compared to commercial strain Y0.	
fold difference in invertase activity from Y0	
YPD anaerobic	13.8
Must anaerobic	12.0

The invertase activity was also measured for two mill isolates on a variety of substrates after 48 h of anaerobic growth compared to yeast samples taken from the end of 15 cycles of acid treatment and fermentation. The mill isolated yeasts had invertase activity on all media tested but drastically higher activity when the activity is tested on cells directly out of the fed-batch process (Table 7).

TABLE 7

Fold change in invertase activity of engineered wild yeast vs. Y0 on various media							
	YPD	YPS	YPF	Commercial Must 1	Commercial Must 2	Commercial Must 3	15 cycles in cell recycle system
Y4	2.0	2.6	2.6	1.8	2.2	1.7	11.8
Y6	2.3	2.5	2.9	1.9	2.5	1.7	7.0

This higher level of invertase was reflected in the faster hydrolysis of sucrose of strains Y3 and Y5 during a fed-batch sucrose fermentation compared to the Y0 strain (FIG. 10A). Faster sucrose hydrolysis allowed strains Y3 and Y5 to ferment faster than strain Y0 as seen by the faster production of ethanol (FIG. 10B).

The rate of invertase activity associated with strains Y0, Y2, Y3, Y4, Y7 and Y8 (as the amount of reducing sugar/dry cell weight per minute) was determined on commercial must and is shown on FIG. 19 as a box and whisker plot obtained using the JMP software.

The rate of invertase activity associated with Y61 was determined to be equal to or higher than its parental strain Y3.

RAS/cAMP/PKA Activity

Strains showing normal RAS/cAMP/PKA activity (Y0 and Y1) and strains showing hyperactivated RAS/cAMP activity (Y9, Y10, Y11 and Y12) were tested on media containing 2-deoxyglucose and rapamycin. As shown in FIGS. 11B and 11C, the strains exhibiting hyperactivated Ras/cAMP activity were less sensitive to these compounds (which perturb glucose and nitrogen sensing in yeast). Without wishing to be bound to theory, these results suggests that the strains may have differential regulation of the RAS/cAMP/PKA pathway which provide them with a growth.

The ability of strains Y0, Y3 and Y13 to modulate their cAMP production was measured before and after (5 min) a 100 mM glucose spike. As shown in Table 8, strains Y3 and Y13 were not able to produce as much cAMP as strain Y0 (which does not exhibit an increase in the signaling on the RAS/cAMP/PKA pathway).

TABLE 8

cAMP production following glucose spike				
Strain	cAMP levels in basal medium	cAMP levels 5 min after a 100 mM glucose spike	Fold increase in cAMP level at 5 min following glucose spike	% change in fold increase compared to Y0
Y0	162.592	272.416	1.7	N. A.
Y3	141.108	178.044	1.3	-25%
Y13	182.816	155.808	0.9	-49%

The RAS/cAMP/PKA activity was specifically measured for recombinant strain Y1 as well as wild yeast strains Y3 and Y13. Strain Y1 showed almost 2 fold increase in cAMP production upon glucose spike showing a typical upregulation of the cAMP/PKA pathway (FIG. 12). Two commercial isolates, Y3 and Y13 showed no cAMP spike after the addition of glucose. This suggests that the cAMP pathway is hyperactive even at a basal state in strains Y3 and Y13.

Yeast cream samples were taken from various commercial mills (from which the Y3 and Y13 isolates were obtained) and Illumina sequencing was performed on the mixed commercial samples. The sequence of genes involved in the cAMP/PKA pathway were analyzed and a homozygous IRA2 Lys:Lys 2440/3079 at position was identified that differed from the beginning of the season where the strains were homozygous Glu:Glu (Table 9) The prevalence of this mutation in various mills suggested an adaptive advantage. Consistent with the IRA2 Lys mutation in these strain leading to derepression cAMP/PKA pathway.

TABLE 9

IRA2 mutations observed in wild populations. A = % of contaminating cells, B = % of products, C = % of Y1, D = % of Y2, E = % cells expressing the CDA1/2 A allele and F = % of cells expressing an homozygous Lys:Lys mutation at positions 2440 and 3079. The sample presented in this table correspond to the samples presented in Table 5.						
Sample	A	B	C	D	E	F
1	0	100	100	0	53	0
2	0	100	100	0	42	0
3	0	100	100	0	57	0
4	0	100	100	0	45	0
5	0	100	100	0	47	0
6	0	100	100	0	56	0
7	0	100	100	0	51	0
8	0	100	100	0	51	0
9	0	100	100	0	56	0
10	1	99	99	0	47	0
11	1	99	99	0	65	0
12	43	57	57	0	23	21
13	94	6	6	0	9	61
14	94	6	6	0	0	66
15	19	81	81	0	0	17
16	58	42	42	0	0	48
17	0	100	74	26	63	0
18	0	100	100	0	43	0
19	0	100	86	14	57	0
20	0	100	96	4	52	0
21	0	100	86	14	57	0
22	0	100	72	28	64	0
23	0	100	76	24	62	0
24	0	100	67	32	66	0
25	3	97	64	33	65	0
26	33	67	40	27	47	33
27	85	15	11	5	10	79
28	94	6	6	0	0	93
29	88	12	12	0	0	96
30	99	1	1	0	0	68
31	0	100	78	22	61	0
32	0	100	64	36	68	0
33	0	100	96	4	52	0
34	0	100	66	34	67	0
35	0	100	100	0	47	0
36	0	100	80	20	60	0
37	0	100	76	24	62	0
38	0	100	80	20	60	0
39	0	100	84	16	58	0
40	0	100	74	26	63	0
41	0	100	74	26	63	0
42	0	100	80	20	60	0
43	0	100	72	28	64	0
44	0	100	92	8	54	0
45	0	100	80	20	60	0
46	0	100	66	34	67	0
47	0	100	74	26	63	0
48	0	100	68	32	66	0
49	1	99	64	35	67	0
50	97	3	3	0	0	100
51	99	1	1	0	0	100
52	96	4	4	0	0	78
53	98	2	2	0	0	89
54	89	11	4	7	9	49
55	99	1	1	0	0	65
56	7	93	90	3	48	0
57	97	3	3	0	0	0

Strain Y1 was admixed with strain Y0 or a commercial isolate bearing a IRA Lys mutation, Y3 (ratio is provided on FIG. 13). As shown on FIG. 13, isolate Y3 rapidly outcompeted Y1 in three cycles of the fed-batch lab scale system rising from 54% to 71% of the population demonstrating the competitive advantage of the wild yeast.

Triploid Phenotype

Various commercial and contaminating yeasts from mills were isolated and sequenced by Illumina sequencing. While each of the available commercial strains were found to be highly heterozygous diploid (data not shown), the majority

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of the strains isolated from the fermentation process in the end of the season were found to be highly heterozygous triploids (3n) (FIG. 14).

Recombinant Yeast Expressing Heterologous STL1

Strains Y4, Y6 and Y61 were genetically engineered to express the STL1 polypeptide. Briefly, two parental yeast strains (Y3 and Y5) isolated from the fed batch process with the dominant features of rugose colony formation, fast settling, high invertase activity post-acid treatment, triploid and features of cAMP/PKA hyperactivation were selected. These parental strains were then engineered with additional copies of STL1 to reduce glycerol and increase ethanol yields (see Table 1). After engineering, the strains were tested and demonstrated that they maintained the characteristics that they were chosen for fast settling (FIG. 15 for Y4 and Y6), hyperactivation of the RAS/cAMP pathway (data not shown) and high invertase activity (data not shown).

The performance of strains Y4 and Y6 was then monitored in a fed-batch high cell density fermentation with acid recycle for 13 rounds of fermentation. Over the 13 cycles, the yield of ethanol and glycerol were compared to the conventional strain Y0 and achieved a 1.9% yield increase for Y4 and a 1.5% yield increase for Y6 (FIG. 16A). These yield were close to the corresponding Y0-based strain (Y2) that is engineered and achieved a 1.9% yield increase over Y0. In addition, faster kinetics was maintained for strains Y4 and Y6 after engineering (FIG. 16B).

The strains Y4 and Y6 were then mixed in co-culture with Y2 to determine if they had growth advantage in co-culture and could displace the Y2 strain. A mixture of 20% Y4 to 80% Y2 and 30% Y6 to 70% Y2 were co-fermented in the fed-batch system with acid recycle. Populations were monitored using the qPCR method described above. Over the course of the 13 cycles of fermentation, both Y4 and Y6 increased within the population at an average of about 0.5% per cycle demonstrating their competitive fitness and growth advantage in this type of fermentation process (FIGS. 17A and 17B).

The performance of strain Y61 was monitored in co-culture fermentation experiments with strain Y3 on seven different commercial substrates and compared to the performances of strains Y2 or Y4 also in co-culture fermentation experiments with Y3. Populations were monitored using the qPCR method described above. Ethanol and glycerol levels were determined using HPLC. Over the various fermentations, the washout rate of strains Y4 and Y61 was lower than the washout rate of strain Y2 (FIG. 20A). These results indicated that the strains built in the Y3 background have improved competitive fitness during fermentation compared to the Y2 strain which was built in the Y0 background. In addition, strains Y2, Y4 and Y61 exhibited an increase in the percent in ethanol change as well as a decrease in the percent in glycerol changed when compared to strain Y0 (FIG. 20B). These results indicated that the strains with the persistent phenotypic traits maintained fermentation performances during multiple rounds of fermentation even though they included genetic modifications and expressed the heterologous STL1 gene.

Recombinant Yeast Having Inactivated GPD1

Strains Y0, Y2, Y4 and Y8 were pitched in monoculture or in co-culture with ~15% of the wild type strain Y7. As described in Table 1 above, strain Y8 contained an additional modification that is not present in Y4, a replacement of the native GPD1 gene with GPD2 ($\Delta gpd1::gpd2$) in addition to the overexpression of STL1 ($\Delta fcy1::stl1$). This additional modification lead to a 2% yield increase over Y0 as well as higher levels of glycerol reduction (>35% vs. Y0, FIG.

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18A). In addition, strain Y8 showed a high levels of persistence over the recycle testing when in co-culture with Y7 (FIG. 18B).

Improved Persistence

Several strains were submitted to commercial fermentations on cane must in which they were recycled and submitted to several consecutive cycles of fermentations. It was then determined, using quantitative PCR, the number of cycles at which the strain (or the combination of strains) represented 99%, 90% or 50% of the fermenting population. The results are presented at Table 10.

TABLE 10

Number of cycles at which the strain (or the combination of strains) represented 99% ("number of cycles to 99%"), 90% ("number of cycles to 90%") or 50% ("number of cycles to 50%") of total the fermenting population in function of each mill.				
Mill Code	Strain	Number of cycles to 99%	Number of cycles to 90%	Number of cycles to 50%
Mill #1	Y1	50	53	72
Mill #2	Y1	29	45	53
Mill #3	Y1	41	48	53
Mill #4	Y1	26	30	38
Mill #5	Y1	29	38	42
Mill #6	Y2	100	120	NA
Mill #7	Y2	23	35	39
Mill #8	Y2	42	58	80
Mill #9	Y2	47	59	69
Mill #10	Y2	64	73	90
Mill #11	Y2	48	57	75
Mill #12	Y2	37	47	58
Mill #13	Y2	96	115	140
Mill #14	Y2	60	65	82
Mill #15	Y2	53	55	65
Mill #16	Y2	56	60	70
Mill #17	Y2	30	40	46
Mill #18	Y2	60	72	83
Mill #19	Y2	63	75	94
Mill #20	Y4	124	160	170
Mill #21	Y4	120	133	163
Mill #22	Y4	160	190	200
Mill #23	Y4	225	250	270
Mill #24	Y2 & Y4	75	100	116
Mill #25	Y2 & Y4	110	135	160
Mill #26	Y2 & Y4	88	100	120
Mill #27	Y2 & Y4	42	70	97

While the invention has been described in connection with specific embodiments thereof, it will be understood that the scope of the claims should not be limited by the preferred embodiments set forth in the examples, but should be given the broadest interpretation consistent with the description as a whole.

REFERENCES

- Luiz Carlos Basso, Thiago Olitta Basso and Saul Nitsche Rocha (Sep. 15, 2011). Ethanol Production in Brazil: The Industrial Process and Its Impact on Yeast Fermentation, Biofuel Production—Recent Developments and Prospects, Marco Aurelio dos Santos Bernardes, IntechOpen, DOI: 10.5772/17047.
- Brociner R. E., Vollans E. C. (1973) Thickening flocculated kaolinite slurries in the nozzle discharge, multi disc, bowl centrifuge. Clay Minerals 10 (99).
- Conrad M, Schothorst J, Kankipati H N, Van Zeebroeck G, Rubio-Teixeira M, Thevelein J M. Nutrient sensing and signaling in the yeast *Saccharomyces cerevisiae*. FEMS Microbiol Rev. 2014; 38(2):254-299.

- Danecek, Petr, Adam Auton, Goncalo Abecasis, Cornelis A. Albers, Eric Banks, Mark A. DePristo, Robert E. Handsaker, et al. "The Variant Call Format and VCFtools." *Bioinformatics* (Oxford, England) 27, no. 15 (Aug. 1, 2011): 2156-58.
- Fisher K J, Buskirk S W, Vignogna R C, Marad D A, Lang G I. Adaptive genome duplication affects patterns of molecular evolution in *Saccharomyces cerevisiae*. *PLoS Genet.* 2018; 14(5):e1007396. Published 2018 May 25.
- Li, Heng. "A Statistical Framework for SNP Calling, Mutation Discovery, Association Mapping and Population Genetical Parameter Estimation from Sequencing Data." *Bioinformatics* 27, no. 21 (Nov. 1, 2011): 2987-93.
- Li, Heng, Bob Handsaker, Alec Wysoker, Tim Fennell, Jue Ruan, Nils Homer, Gabor Marth, Goncalo Abecasis, Richard Durbin, and 1000 Genome Project Data Processing Subgroup. "The Sequence Alignment/Map Format and SAMtools." *Bioinformatics* (Oxford, England) 25, no. 16 (Aug. 15, 2009): 2078-79.
- Oud B, Guadalupe-Medina V, Nijkamp J F, de Ridder D, Pronk J T, van Maris A J, Daran J M. Genome duplication and mutations in ACE2 cause multicellular, fast-sedimenting phenotypes in evolved *Saccharomyces cerevisiae*. *Proc Natl Acad Sci USA.* 2013 Nov. 5; 110(45):

- E4223-31. doi: 10.1073/pnas.1305949110. Epub 2013 Oct. 21. PMID: 24145419; PMCID: PMC3831460.
- Purcell, Shaun, Benjamin Neale, Kathe Todd-Brown, Lori Thomas, Manuel A. R. Ferreira, David Bender, Julian Maller, "PLINK: A Tool Set for Whole-Genome Association and Population-Based Linkage Analyses." *American Journal of Human Genetics* 81, no. 3 (September 2007): 559-75.
- Scott A L, Richmond P A, Dowell R D, Selmecki A M. The Influence of Polyploidy on the Evolution of Yeast Grown in a Sub-Optimal Carbon Source. *Mol Biol Evol.* 2017; 34(10):2690-2703.
- Selmecki A M, Maruvka Y E, Richmond P A, et al. Polyploidy can drive rapid adaptation in yeast. *Nature.* 2015; 519(7543):349-352.
- Zadrag-Tecza R, Kwolek-Mirek M, Alabrudzińska M, Skoneczna A. Cell Size Influences the Reproductive Potential and Total Lifespan of the *Saccharomyces cerevisiae* Yeast as Revealed by the Analysis of Polyploid Strains. *Oxid Med Cell Longev.* 2018; 2018:1898421. Published 2018 Mar. 20.
- Zhang, K., Fang, Y., Gao, K. et al. Effects of genome duplication on phenotypes and industrial applications of *Saccharomyces cerevisiae* strains. *Appl Microbiol Biotechnol* 101, 5405-5414 (2017).

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Leu Gln Arg Ala Leu Ile Ala Ala Ser Thr Gln Phe Phe Gln Gln Phe	290	295	300
Thr Gly Cys Asn Ala Ala Ile Tyr Tyr Ser Thr Val Leu Phe Asn Lys	305	310	315
Thr Ile Lys Leu Asp Tyr Arg Leu Ser Met Ile Ile Gly Gly Val Phe	325	330	335
Ala Thr Ile Tyr Ala Leu Ser Thr Ile Gly Ser Phe Phe Leu Ile Glu	340	345	350
Lys Leu Gly Arg Arg Lys Leu Phe Leu Leu Gly Ala Thr Gly Gln Ala	355	360	365
Val Ser Phe Thr Ile Thr Phe Ala Cys Leu Val Lys Glu Asn Lys Glu	370	375	380
Asn Ala Arg Gly Ala Ala Val Gly Leu Phe Leu Phe Ile Thr Phe Phe	385	390	395
Gly Leu Ser Leu Leu Ser Leu Pro Trp Ile Tyr Pro Pro Glu Ile Ala	405	410	415
Ser Met Lys Val Arg Ala Ser Thr Asn Ala Phe Ser Thr Cys Thr Asn	420	425	430
Trp Leu Cys Asn Phe Ala Val Val Met Phe Thr Pro Ile Phe Ile Gly	435	440	445
Gln Ser Gly Trp Gly Cys Tyr Leu Phe Phe Ala Val Met Asn Tyr Leu	450	455	460
Tyr Ile Pro Val Ile Phe Phe Phe Tyr Pro Glu Thr Ala Gly Arg Ser	465	470	475
Leu Glu Glu Ile Asp Ile Ile Phe Ala Lys Ala Tyr Glu Asp Gly Thr	485	490	495

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Gln Pro Trp Arg Val Ala Asn His Leu Pro Lys Leu Ser Leu Gln Glu
 500 505 510

Val Glu Asp His Ala Asn Ala Leu Gly Ser Tyr Asp Asp Glu Met Glu
 515 520 525

Lys Glu Asp Phe Gly Glu Asp Arg Val Glu Asp Thr Tyr Asn Gln Ile
 530 535 540

Asn Gly Asp Asn Ser Ser Ser Ser Ser Asn Ile Lys Asn Glu Asp Thr
 545 550 555 560

Val Asn Asp Lys Ala Asn Phe Glu Gly
 565

<210> SEQ ID NO 9

<211> LENGTH: 2313

<212> TYPE: DNA

<213> ORGANISM: *Saccharomyces cerevisiae*

<400> SEQUENCE: 9

atggataacg tttagatgcc gtggtatata aatccctcag gcttcgcgaa agacactcaa	60
gatgaggagt atgttcaaca tcatgataat gtcaatccta ccataccccc acccgacaat	120
tatatTTTga ataatgaaaa cgatgatggc ctcgataact tgtaggtat ggactactat	180
aacatcgatg acctgttgac tcaagagtta agagatctgg atattccttt agtgccttct	240
cctaagacgg gcgatggttc ttctgataaa aagaatattg atagaacttg gaaccttggt	300
gatgaaaaca acaaagtctc ccactatagc aaaaaatcaa tgccctcaca caagagaggt	360
ctaagtggca cagcgatatt tggatttctc ggccataata agacattgag tatttccagt	420
ttacagcaat ccattctaaa tatgtctaaa gatccgcaac ccattggaact cataaatgaa	480
ttgggtaatc ataatacggg aaaaaataac aatgatgact ttgaccatat aagggaatat	540
gatggtgaaa atagctatTTT gagccaagtt ttgttgaaac agcaggagga gttaagaatt	600
gctcttgaaa aacaaaagga agtgaacgaa aaattggaga agcagttgag agacaatcaa	660
atacagcaag aaaagtTgcg taaagtatta gaagagcaag aagaggTggc gcagaagttg	720
gtttctgggg ctacaaaTtc taattccaaa cctggatctc cagtaatact aaagacacct	780
gccaatgaaa acggtagaat gaaagataat gctataatcg tcacaacgaa ctctgcaaat	840
ggcgatatac aatttctctc tccgacgtta atatcgctc ggatgtcaaa tacttcaata	900
aatggttcac catccaggaa ataccatagg caacgatata caaataaaaag ccagaaagt	960
aatggattga acctTTTTT ctctaacagt ggttatttga gagattctga actgctttca	1020
ttttctccac aaaattataa tttaaacttg gacggcttga cttataatga ccataataac	1080
accagtgata aaaacaataa tgataaaaaa aatagtactg gtgataacat attccgtctg	1140
ttcgaaaaga cttccccggg tgggctaagt atctctccaa ggataaatgg aaatagtTtg	1200
agatcgccct tcctcgtcgg cacagataaa agcagggatg atcgatatgc tgctggcacg	1260
ttcacgccta gaacacagtt gtcacctatc cacaagaaaa gggaatccgt agtttccacg	1320
gtctcgacaa tatcacaact gcaggatgac actgaaccca tccacatgcg aaataaccag	1380
aaaccaacat taagaaatgc aaacgcttta gcgatcatcaa gtgtactacc tcctattcct	1440
ggttccagca ataactctc aattaagaat tctttgccac aaaaacatgt atttcaacat	1500
actcccgTca aagctccacc aaagaacgga agtaacctag ctccgcttct aaatgcaccg	1560
gatttaacag atcatcagtt agaaattaag acaccatac gaaataacag tcaactgtgaa	1620
gtggaaagct atccgcaagt accacctgtc acacatgata ttcacaaaaag ccccactttg	1680

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catagtacgt ctcctttacc agatgaata atacctagga ctacgccaat gaaaataacc 1740
aagaaaccaa ctactctgcc tccgggtacc attgaccagt acgtcaagga actaccgcac 1800
aaactattcg agtgcttata ccctaactgt aacaaagtat tcaagcgtag atacaacata 1860
aggctgcata ttcagacaca ttgcaagat agaccgtatt catgcgactt tcccggttgc 1920
accaaggcgt ttgttcgcaa tcatgattta ataagacaca aaatctccca taatgccaa 1980
aaatacatct gccatgcgg aaagagattt aataggagg atgctcta ggtgcataga 2040
agtcggatga ttgcaccgg cggaagaaa ttagaacatt cgatcaacaa gaaacttaca 2100
tctcccaaaa aaagcctgct tgacagcccg catgacacaa gtcccgtaaa agaaactatc 2160
gcccgggata aagatgggag cgtcctaata aaaatggagg aacagctgcg agatgatatg 2220
cgcaaacatg gattactgga tccaccccca tccacagcag cgcacgagca aaactcgaac 2280
cgcacccctt caaacgaaac tgatgctctc tga 2313

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<210> SEQ ID NO 10

<211> LENGTH: 770

<212> TYPE: PRT

<213> ORGANISM: *Saccharomyces cerevisiae*

<400> SEQUENCE: 10

```

Met Asp Asn Val Val Asp Pro Trp Tyr Ile Asn Pro Ser Gly Phe Ala
1           5           10          15

Lys Asp Thr Gln Asp Glu Glu Tyr Val Gln His His Asp Asn Val Asn
20          25          30

Pro Thr Ile Pro Pro Pro Asp Asn Tyr Ile Leu Asn Asn Glu Asn Asp
35          40          45

Asp Gly Leu Asp Asn Leu Leu Gly Met Asp Tyr Tyr Asn Ile Asp Asp
50          55          60

Leu Leu Thr Gln Glu Leu Arg Asp Leu Asp Ile Pro Leu Val Pro Ser
65          70          75          80

Pro Lys Thr Gly Asp Gly Ser Ser Asp Lys Lys Asn Ile Asp Arg Thr
85          90          95

Trp Asn Leu Gly Asp Glu Asn Asn Lys Val Ser His Tyr Ser Lys Lys
100         105         110

Ser Met Ser Ser His Lys Arg Gly Leu Ser Gly Thr Ala Ile Phe Gly
115         120         125

Phe Leu Gly His Asn Lys Thr Leu Ser Ile Ser Ser Leu Gln Gln Ser
130         135         140

Ile Leu Asn Met Ser Lys Asp Pro Gln Pro Met Glu Leu Ile Asn Glu
145         150         155         160

Leu Gly Asn His Asn Thr Val Lys Asn Asn Asn Asp Asp Phe Asp His
165         170         175

Ile Arg Glu Asn Asp Gly Glu Asn Ser Tyr Leu Ser Gln Val Leu Leu
180         185         190

Lys Gln Gln Glu Glu Leu Arg Ile Ala Leu Glu Lys Gln Lys Glu Val
195         200         205

Asn Glu Lys Leu Glu Lys Gln Leu Arg Asp Asn Gln Ile Gln Gln Glu
210         215         220

Lys Leu Arg Lys Val Leu Glu Glu Gln Glu Glu Val Ala Gln Lys Leu
225         230         235         240

Val Ser Gly Ala Thr Asn Ser Asn Ser Lys Pro Gly Ser Pro Val Ile
245         250         255

Leu Lys Thr Pro Ala Met Gln Asn Gly Arg Met Lys Asp Asn Ala Ile

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260					265					270				
Ile	Val	Thr	Thr	Asn	Ser	Ala	Asn	Gly	Gly	Tyr	Gln	Phe	Pro	Pro
	275							280				285		
Thr	Leu	Ile	Ser	Pro	Arg	Met	Ser	Asn	Thr	Ser	Ile	Asn	Gly	Ser
	290					295					300			Pro
Ser	Arg	Lys	Tyr	His	Arg	Gln	Arg	Tyr	Pro	Asn	Lys	Ser	Pro	Glu
305					310					315				320
Asn	Gly	Leu	Asn	Leu	Phe	Ser	Ser	Asn	Ser	Gly	Tyr	Leu	Arg	Asp
			325						330					335
Glu	Leu	Leu	Ser	Phe	Ser	Pro	Gln	Asn	Tyr	Asn	Leu	Asn	Leu	Asp
			340					345					350	Gly
Leu	Thr	Tyr	Asn	Asp	His	Asn	Asn	Thr	Ser	Asp	Lys	Asn	Asn	Asp
		355					360					365		
Lys	Lys	Asn	Ser	Thr	Gly	Asp	Asn	Ile	Phe	Arg	Leu	Phe	Glu	Lys
	370					375					380			Thr
Ser	Pro	Gly	Gly	Leu	Ser	Ile	Ser	Pro	Arg	Ile	Asn	Gly	Asn	Ser
385					390					395				400
Arg	Ser	Pro	Phe	Leu	Val	Gly	Thr	Asp	Lys	Ser	Arg	Asp	Asp	Arg
			405						410					415
Ala	Ala	Gly	Thr	Phe	Thr	Pro	Arg	Thr	Gln	Leu	Ser	Pro	Ile	His
			420					425					430	Lys
Lys	Arg	Glu	Ser	Val	Val	Ser	Thr	Val	Ser	Thr	Ile	Ser	Gln	Leu
	435						440					445		Gln
Asp	Asp	Thr	Glu	Pro	Ile	His	Met	Arg	Asn	Thr	Gln	Asn	Pro	Thr
450						455					460			Leu
Arg	Asn	Ala	Asn	Ala	Leu	Ala	Ser	Ser	Ser	Val	Leu	Pro	Pro	Ile
465					470					475				480
Gly	Ser	Ser	Asn	Asn	Thr	Pro	Ile	Lys	Asn	Ser	Leu	Pro	Gln	Lys
			485						490					495
Val	Phe	Gln	His	Thr	Pro	Val	Lys	Ala	Pro	Pro	Lys	Asn	Gly	Ser
			500					505					510	Asn
Leu	Ala	Pro	Leu	Leu	Asn	Ala	Pro	Asp	Leu	Thr	Asp	His	Gln	Leu
	515						520					525		Glu
Ile	Lys	Thr	Pro	Ile	Arg	Asn	Asn	Ser	His	Cys	Glu	Val	Glu	Ser
	530					535					540			Tyr
Pro	Gln	Val	Pro	Pro	Val	Thr	His	Asp	Ile	His	Lys	Ser	Pro	Thr
545					550					555				560
His	Ser	Thr	Ser	Pro	Leu	Pro	Asp	Glu	Ile	Ile	Pro	Arg	Thr	Thr
				565					570					575
Met	Lys	Ile	Thr	Lys	Lys	Pro	Thr	Thr	Leu	Pro	Pro	Gly	Thr	Ile
			580					585					590	Asp
Gln	Tyr	Val	Lys	Glu	Leu	Pro	Asp	Lys	Leu	Phe	Glu	Cys	Leu	Tyr
	595						600					605		Pro
Asn	Cys	Asn	Lys	Val	Phe	Lys	Arg	Arg	Tyr	Asn	Ile	Arg	Ser	His
	610					615					620			Ile
Gln	Thr	His	Leu	Gln	Asp	Arg	Pro	Tyr	Ser	Cys	Asp	Phe	Pro	Gly
625					630					635				640
Thr	Lys	Ala	Phe	Val	Arg	Asn	His	Asp	Leu	Ile	Arg	His	Lys	Ile
				645					650					655
His	Asn	Ala	Lys	Lys	Tyr	Ile	Cys	Pro	Cys	Gly	Lys	Arg	Phe	Asn
			660					665					670	Arg
Glu	Asp	Ala	Leu	Met	Val	His	Arg	Ser	Arg	Met	Ile	Cys	Thr	Gly
	675						680					685		Gly

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Lys Lys Leu Glu His Ser Ile Asn Lys Lys Leu Thr Ser Pro Lys Lys
 690 695 700
 Ser Leu Leu Asp Ser Pro His Asp Thr Ser Pro Val Lys Glu Thr Ile
 705 710 715 720
 Ala Arg Asp Lys Asp Gly Ser Val Leu Met Lys Met Glu Glu Gln Leu
 725 730 735
 Arg Asp Asp Met Arg Lys His Gly Leu Leu Asp Pro Pro Pro Ser Thr
 740 745 750
 Ala Ala His Glu Gln Asn Ser Asn Arg Thr Leu Ser Asn Glu Thr Asp
 755 760 765
 Ala Leu
 770

<210> SEQ ID NO 11
 <211> LENGTH: 2312
 <212> TYPE: DNA
 <213> ORGANISM: *Saccharomyces cerevisiae*

<400> SEQUENCE: 11

atggataacg ttgtagatcc gtggtatata aatccctcag gcttcgcgaa agacactcaa	60
gatgaggagt atgttcaaca tcattgataat gtcaatccta ccataccccc acccgacaat	120
tatatatttga ataataaaaa cgatgatggc ctgcataact tgtaggtat ggactactat	180
aacatcgatg acctgttgac tcaagagtta agagatctgg atattccttt agtgccttct	240
cctaagacgg gcgatggttc tcttgataaa aagaatattg atagaacttg gaaccttggt	300
gatgaaaaca acaagtctc ccactatagc aaaaaatcaa tgccttcaca caagagaggt	360
ctaagtggca cagcgatatt tggatttctc ggccataata agacattgag tatttccagt	420
ttacagcaat ccattctaaa tatgtctaaa gatccgcaac ccatggaact cataaatgaa	480
ttgggtaatc ataatacggc aaaaaataac aatgatgact ttgaccatat aagggaaaat	540
gatggtgaaa atagctatct gagccaagtt ttgttgaaac agcaggagga gttaagaatt	600
gctcttgaaa aacaaaagga agtgaacgaa aaattggaga agcagttgag agacaatcaa	660
atacagcaag aaaagttgag taaagtatta gaagagcaag aagaggtggc gcagaagttg	720
gtttctgggg ctacaaattc taattccaaa cctggatctc cagtaatact aaagacacct	780
gccatgcaaa acggtagaat gaaagataat gctataatcg tcacaacgaa ctctgcaaat	840
ggcgatatac aatttctctc tccgacgta atactgcctc ggatgtcaaa tacttcaata	900
aatggttcac catccaggaa ataccatagg caacgatata caaataaaaag cccagaaaagt	960
aatggattga accttttttc ctctaacagt ggttatttga gagattctga actgctttca	1020
ttttctccac aaaattataa tttaaacttg gacggcttga cttataatga ccataataac	1080
accagtgata aaaacaataa tgataaaaaa atagtactgg tgataacata ttccgtctgt	1140
tcgaaaagac ttccccgggt gggctaagta tctctccaag gataaatgga aatagtttga	1200
gatcgccctt cctcgtcggc acagataaaa gcagggatga tcgatatgct gctggcacgt	1260
tcacgcctag aacacagttg tcacctatcc acaagaaaag ggaatccgta gtttccacgg	1320
tctcgacaat atcacaactg caggatgaca ctgaacccat ccacatgcga aatacccaga	1380
acccaacatt aagaaatgca aacgctttag cgtcatcaag tgtactacct cctattctg	1440
gttccagcaa taacactcca attaagaatt ctttgccaca aaaacatgta tttcaacata	1500
ctcccgtaaa agctccacca aagaacggaa gtaacctagc tccgcttcta aatgcaccgg	1560

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atttaacaga tcatcagtta gaaattaaga caccatacag aaataacagt cactgtgaag 1620
tggaagcta tccgaagta ccacctgtca cacatgatat tcacaaaagc cccactttgc 1680
atagtacgtc tcctttacca gatgaaataa tacctaggac tacgccaatg aaaataacca 1740
agaaaccaac tactctgcct ccgggtacca ttgaccagta cgtcaaggaa ctacccgaca 1800
aactattcga gtgcttatac cctaactgta acaaagtatt caagcgtaga tacaacataa 1860
ggtcgcataat tcagacacat ttgcaagata gaccgtattc atgcgacttt cccggttgca 1920
ccaaggcggtt tgttcgcaat catgatttaa taagacacaa aatctcccat aatgccaaga 1980
aatacatctg cccatgcgga aagagattta atagggagga tgctctaata gtgcatagaa 2040
gtcggatgat ttgcaccggc ggtaagaaat tagaacattc gatcaacaag aaacttacat 2100
ctcccaaaaa aagcctgctt gacagccgc atgacacaag tcccgtaaaa gaaactatcg 2160
cccgggataa agatgggagc gtccaatga aaatggagga acagctgcga gatgatatgc 2220
gcaaacatgg attactggat ccaccccat ccacagcagc gcacgagcaa aactgaacc 2280
gcacccttc aaacgaaact gatgctctct ga 2312

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<210> SEQ ID NO 12

<211> LENGTH: 387

<212> TYPE: PRT

<213> ORGANISM: *Saccharomyces cerevisiae*

<400> SEQUENCE: 12

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Met Asp Asn Val Val Asp Pro Trp Tyr Ile Asn Pro Ser Gly Phe Ala
1           5           10          15
Lys Asp Thr Gln Asp Glu Glu Tyr Val Gln His His Asp Asn Val Asn
20          25          30
Pro Thr Ile Pro Pro Pro Asp Asn Tyr Ile Leu Asn Asn Glu Asn Asp
35          40          45
Asp Gly Leu Asp Asn Leu Leu Gly Met Asp Tyr Tyr Asn Ile Asp Asp
50          55          60
Leu Leu Thr Gln Glu Leu Arg Asp Leu Asp Ile Pro Leu Val Pro Ser
65          70          75          80
Pro Lys Thr Gly Asp Gly Ser Ser Asp Lys Lys Asn Ile Asp Arg Thr
85          90          95
Trp Asn Leu Gly Asp Glu Asn Asn Lys Val Ser His Tyr Ser Lys Lys
100         105         110
Ser Met Ser Ser His Lys Arg Gly Leu Ser Gly Thr Ala Ile Phe Gly
115         120         125
Phe Leu Gly His Asn Lys Thr Leu Ser Ile Ser Ser Leu Gln Gln Ser
130         135         140
Ile Leu Asn Met Ser Lys Asp Pro Gln Pro Met Glu Leu Ile Asn Glu
145         150         155         160
Leu Gly Asn His Asn Thr Val Lys Asn Asn Asn Asp Asp Phe Asp His
165         170         175
Ile Arg Glu Asn Asp Gly Glu Asn Ser Tyr Leu Ser Gln Val Leu Leu
180         185         190
Lys Gln Gln Glu Glu Leu Arg Ile Ala Leu Glu Lys Gln Lys Glu Val
195         200         205
Asn Glu Lys Leu Glu Lys Gln Leu Arg Asp Asn Gln Ile Gln Gln Glu
210         215         220
Lys Leu Arg Lys Val Leu Glu Glu Gln Glu Glu Val Ala Gln Lys Leu
225         230         235         240

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Val	Ser	Gly	Ala	Thr	Asn	Ser	Asn	Ser	Lys	Pro	Gly	Ser	Pro	Val	Ile
				245					250					255	
Leu	Lys	Thr	Pro	Ala	Met	Gln	Asn	Gly	Arg	Met	Lys	Asp	Asn	Ala	Ile
			260					265					270		
Ile	Val	Thr	Thr	Asn	Ser	Ala	Asn	Gly	Gly	Tyr	Gln	Phe	Pro	Pro	Pro
		275					280					285			
Thr	Leu	Ile	Ser	Pro	Arg	Met	Ser	Asn	Thr	Ser	Ile	Asn	Gly	Ser	Pro
	290					295					300				
Ser	Arg	Lys	Tyr	His	Arg	Gln	Arg	Tyr	Pro	Asn	Lys	Ser	Pro	Glu	Ser
305					310					315					320
Asn	Gly	Leu	Asn	Leu	Phe	Ser	Ser	Asn	Ser	Gly	Tyr	Leu	Arg	Asp	Ser
			325						330					335	
Glu	Leu	Leu	Ser	Phe	Ser	Pro	Gln	Asn	Tyr	Asn	Leu	Asn	Leu	Asp	Gly
			340					345					350		
Leu	Thr	Tyr	Asn	Asp	His	Asn	Asn	Thr	Ser	Asp	Lys	Asn	Asn	Asn	Asp
		355					360					365			
Lys	Lys	Ile	Val	Leu	Val	Ile	Thr	Tyr	Ser	Val	Cys	Ser	Lys	Arg	Leu
	370					375					380				
Pro	Arg	Val													
385															

<210> SEQ ID NO 13
 <211> LENGTH: 169
 <212> TYPE: PRT
 <213> ORGANISM: *Saccharomyces cerevisiae*

<400> SEQUENCE: 13

Met	Asp	Asn	Val	Val	Asp	Pro	Trp	Tyr	Ile	Asn	Pro	Ser	Gly	Phe	Ala
1			5						10					15	
Lys	Asp	Thr	Gln	Asp	Glu	Glu	Tyr	Val	Gln	His	His	Asp	Asn	Val	Asn
		20					25						30		
Pro	Thr	Ile	Pro	Pro	Pro	Asp	Asn	Tyr	Ile	Leu	Asn	Asn	Glu	Asn	Asp
	35					40					45				
Asp	Gly	Leu	Asp	Asn	Leu	Leu	Gly	Met	Asp	Tyr	Tyr	Asn	Ile	Asp	Asp
	50				55					60					
Leu	Leu	Thr	Gln	Glu	Leu	Arg	Asp	Leu	Asp	Ile	Pro	Leu	Val	Pro	Ser
65				70					75					80	
Pro	Lys	Thr	Gly	Asp	Gly	Ser	Ser	Asp	Lys	Lys	Asn	Ile	Asp	Arg	Thr
		85						90						95	
Trp	Asn	Leu	Gly	Asp	Glu	Asn	Asn	Lys	Val	Ser	His	Tyr	Ser	Lys	Lys
		100						105					110		
Ser	Met	Ser	Ser	His	Lys	Arg	Gly	Leu	Ser	Gly	Thr	Ala	Ile	Phe	Gly
	115					120						125			
Phe	Leu	Gly	His	Asn	Lys	Thr	Leu	Ser	Ile	Ser	Ser	Leu	Gln	Gln	Ser
	130				135					140					
Ile	Leu	Asn	Met	Ser	Lys	Asp	Pro	Gln	Pro	Met	Glu	Leu	Ile	Asn	Glu
145				150						155				160	
Leu	Gly	Asn	His	Asn	Thr	Val	Lys	Lys							
				165											

<210> SEQ ID NO 14
 <211> LENGTH: 186
 <212> TYPE: PRT
 <213> ORGANISM: *Saccharomyces cerevisiae*

<400> SEQUENCE: 14

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Met Asp Asn Val Val Asp Pro Trp Tyr Ile Asn Pro Ser Gly Phe Ala
1      5      10      15
Lys Asp Thr Gln Asp Glu Glu Tyr Val Gln His His Asp Asn Val Asn
20      25      30
Pro Thr Ile Pro Pro Pro Asp Asn Tyr Ile Leu Asn Asn Glu Asn Asp
35      40      45
Asp Gly Leu Asp Asn Leu Leu Gly Met Asp Tyr Tyr Asn Ile Asp Asp
50      55      60
Leu Leu Thr Gln Glu Leu Arg Asp Leu Asp Ile Pro Leu Val Pro Ser
65      70      75      80
Pro Lys Thr Gly Asp Gly Ser Ser Asp Lys Lys Asn Ile Asp Arg Thr
85      90      95
Trp Asn Leu Gly Asp Glu Asn Asn Lys Val Ser His Tyr Ser Lys Lys
100     105     110
Ser Met Ser Ser His Lys Arg Gly Leu Ser Gly Thr Ala Ile Phe Gly
115     120     125
Phe Leu Gly His Asn Lys Thr Leu Ser Ile Ser Ser Leu Gln Gln Ser
130     135     140
Ile Leu Asn Met Ser Lys Asp Pro Gln Pro Met Glu Leu Ile Asn Glu
145     150     155     160
Leu Gly Asn His Asn Thr Val Lys Asn Asn Asn Asp Asp Phe Asp His
165     170     175
Ile Arg Glu Asn Asp Gly Glu Ile Ala Ile
180     185

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<210> SEQ ID NO 15

<211> LENGTH: 191

<212> TYPE: PRT

<213> ORGANISM: *Saccharomyces cerevisiae*

<400> SEQUENCE: 15

```

Met Asp Asn Val Val Asp Pro Trp Tyr Ile Asn Pro Ser Gly Phe Ala
1      5      10      15
Lys Asp Thr Gln Asp Glu Glu Tyr Val Gln His His Asp Asn Val Asn
20      25      30
Pro Thr Ile Pro Pro Pro Asp Asn Tyr Ile Leu Asn Asn Glu Asn Asp
35      40      45
Asp Gly Leu Asp Asn Leu Leu Gly Met Asp Tyr Tyr Asn Ile Asp Asp
50      55      60
Leu Leu Thr Gln Glu Leu Arg Asp Leu Asp Ile Pro Leu Val Pro Ser
65      70      75      80
Pro Lys Thr Gly Asp Gly Ser Ser Asp Lys Lys Asn Ile Asp Arg Thr
85      90      95
Trp Asn Leu Gly Asp Glu Asn Asn Lys Val Ser His Tyr Ser Lys Lys
100     105     110
Ser Met Ser Ser His Lys Arg Gly Leu Ser Gly Thr Ala Ile Phe Gly
115     120     125
Phe Leu Gly His Asn Lys Thr Leu Ser Ile Ser Ser Leu Gln Gln Ser
130     135     140
Ile Leu Asn Met Ser Lys Asp Pro Gln Pro Met Glu Leu Ile Asn Glu
145     150     155     160
Leu Gly Asn His Asn Thr Val Lys Asn Asn Asn Asp Asp Phe Asp His
165     170     175
Ile Arg Glu Asn Asp Gly Glu Asn Ser Tyr Leu Ser Gln Val Cys
180     185     190

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<210> SEQ ID NO 16
<211> LENGTH: 388
<212> TYPE: PRT
<213> ORGANISM: Saccharomyces cerevisiae

<400> SEQUENCE: 16

Met Asp Asn Val Val Asp Pro Trp Tyr Ile Asn Pro Ser Gly Phe Ala
1          5          10          15

Lys Asp Thr Gln Asp Glu Glu Tyr Val Gln His His Asp Asn Val Asn
20          25          30

Pro Thr Ile Pro Pro Pro Asp Asn Tyr Ile Leu Asn Asn Glu Asn Asp
35          40          45

Asp Gly Leu Asp Asn Leu Leu Gly Met Asp Tyr Tyr Asn Ile Asp Asp
50          55          60

Leu Leu Thr Gln Glu Leu Arg Asp Leu Asp Ile Pro Leu Val Pro Ser
65          70          75          80

Pro Lys Thr Gly Asp Gly Ser Ser Asp Lys Lys Asn Ile Asp Arg Thr
85          90          95

Trp Asn Leu Gly Asp Glu Asn Asn Lys Val Ser His Tyr Ser Lys Lys
100         105         110

Ser Met Ser Ser His Lys Arg Gly Leu Ser Gly Thr Ala Ile Phe Gly
115         120         125

Phe Leu Gly His Asn Lys Thr Leu Ser Ile Ser Ser Leu Gln Gln Ser
130         135         140

Ile Leu Asn Met Ser Lys Asp Pro Gln Pro Met Glu Leu Ile Asn Glu
145         150         155         160

Leu Gly Asn His Asn Thr Val Lys Asn Asn Asn Asp Asp Phe Asp His
165         170         175

Ile Arg Glu Asn Asp Gly Glu Asn Ser Tyr Leu Ser Gln Val Leu Leu
180         185         190

Lys Gln Gln Glu Glu Leu Arg Ile Ala Leu Glu Lys Gln Lys Glu Val
195         200         205

Asn Glu Lys Leu Glu Lys Gln Leu Arg Asp Asn Gln Ile Gln Gln Glu
210         215         220

Lys Leu Arg Lys Val Leu Glu Glu Gln Glu Glu Val Ala Gln Lys Leu
225         230         235         240

Val Ser Gly Ala Thr Asn Ser Asn Ser Lys Pro Gly Ser Pro Val Ile
245         250         255

Leu Lys Thr Pro Ala Met Gln Asn Gly Arg Met Lys Asp Asn Ala Ile
260         265         270

Ile Val Thr Thr Asn Ser Ala Asn Gly Gly Tyr Gln Phe Pro Pro Pro
275         280         285

Thr Leu Ile Ser Pro Arg Met Ser Asn Thr Ser Ile Asn Gly Ser Pro
290         295         300

Ser Arg Lys Tyr His Arg Gln Arg Tyr Pro Asn Lys Ser Pro Glu Ser
305         310         315         320

Asn Gly Leu Asn Leu Phe Ser Ser Asn Ser Gly Tyr Leu Arg Asp Ser
325         330         335

Glu Leu Leu Ser Phe Ser Pro Gln Asn Tyr Asn Leu Asn Leu Asp Gly
340         345         350

Leu Thr Tyr Asn Asp His Asn Asn Thr Ser Asp Lys Asn Asn Asn Asp
355         360         365

Lys Lys Ile Val Leu Val Ile Thr Tyr Ser Val Cys Ser Lys Arg Leu

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370	375	380
Pro Arg Val Gly		
385		
<210> SEQ ID NO 17		
<211> LENGTH: 322		
<212> TYPE: PRT		
<213> ORGANISM: <i>Saccharomyces cerevisiae</i>		
<400> SEQUENCE: 17		
Met Pro Leu Asn Lys Ser Asn Ile Arg Glu Tyr Lys Leu Val Val Val		
1 5 10 15		
Gly Gly Gly Gly Val Gly Lys Ser Ala Leu Thr Ile Gln Leu Thr Gln		
20 25 30		
Ser His Phe Val Asp Glu Tyr Asp Pro Thr Ile Glu Asp Ser Tyr Arg		
35 40 45		
Lys Gln Val Val Ile Asp Asp Glu Val Ser Ile Leu Asp Ile Leu Asp		
50 55 60		
Thr Thr Gly Gln Glu Glu Tyr Ser Ala Met Arg Glu Gln Tyr Met Arg		
65 70 75 80		
Asn Gly Glu Gly Phe Leu Leu Val Tyr Ser Ile Thr Ser Lys Ser Ser		
85 90 95		
Leu Asp Glu Leu Met Thr Tyr Tyr Gln Gln Ile Leu Arg Val Lys Asp		
100 105 110		
Thr Asp Tyr Val Pro Ile Val Val Val Gly Asn Lys Ser Asp Leu Glu		
115 120 125		
Asn Glu Lys Gln Val Ser Tyr Gln Asp Gly Leu Asn Met Ala Lys Gln		
130 135 140		
Met Asn Ala Pro Phe Leu Glu Thr Ser Ala Lys Gln Ala Ile Asn Val		
145 150 155 160		
Glu Glu Ala Phe Tyr Thr Leu Ala Arg Leu Val Arg Asp Glu Gly Gly		
165 170 175		
Lys Tyr Asn Lys Thr Leu Thr Glu Asn Asp Asn Ser Lys Gln Thr Ser		
180 185 190		
Gln Asp Thr Lys Gly Ser Gly Ala Asn Ser Val Pro Arg Asn Ser Gly		
195 200 205		
Gly His Arg Lys Met Ser Asn Ala Ala Asn Gly Lys Asn Val Asn Ser		
210 215 220		
Ser Thr Thr Val Val Asn Ala Arg Asn Ala Ser Ile Glu Ser Lys Thr		
225 230 235 240		
Gly Leu Ala Gly Asn Gln Ala Thr Asn Gly Lys Thr Gln Thr Asp Arg		
245 250 255		
Thr Asn Ile Asp Asn Ser Thr Gly Gln Ala Gly Gln Ala Asn Ala Gln		
260 265 270		
Ser Ala Asn Thr Val Asn Asn Arg Val Asn Asn Asn Ser Lys Ala Gly		
275 280 285		
Gln Val Ser Asn Ala Lys Gln Ala Arg Lys Gln Gln Ala Ala Pro Gly		
290 295 300		
Gly Asn Thr Ser Glu Ala Ser Lys Ser Gly Ser Gly Gly Cys Cys Ile		
305 310 315 320		
Ile Ser		

<210> SEQ ID NO 18

<211> LENGTH: 969

<212> TYPE: DNA

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<213> ORGANISM: *Saccharomyces cerevisiae*
 <220> FEATURE:
 <221> NAME/KEY: misc_feature
 <222> LOCATION: (198)..(198)
 <223> OTHER INFORMATION: n is a, c, g, or t

<400> SEQUENCE: 18

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atgcctttga acaagtcgaa cataagagag tacaagctag tcgtcgttgg tgggtggtggt      60
gttggtaaat ctgctttgac catacaattg acccaatcgc actttgtaga tgaatacgat      120
cccacaattg aggattcata caggaagcaa gtggtgattg atgatgaagt gtctatatattg      180
gacatttttg atactacnng gcaggaagaa tactctgcta tgagggaaca atacatgcgc      240
aacggcgaag gattcctatt ggtttactct ataacatcca agtcgtctct tgatgagctt      300
atgacttact atcaacagat attgagagtc aaagataccg actatgttcc aattgtggtt      360
gttggttaaca aatctgattt agaaaacgaa aaacaggctt cttaccagga cgggttgaac      420
atggcaaagc aaatgaagcg tctttctctg gagacatctg ctaagcaagc aatcaacgtg      480
gaagaggcgt tttacactct agcacgttta gttagagacg aaggcggcaa gtacaacaag      540
actttgacgg aaaatgacaa ctccaagcaa acttctcaag atacaaaagg gagcggtgcc      600
aactctgtgc ctgaaatag cggtgggccac aggaagatga gcaatgctgc caacggtaaa      660
aatgtgaaca gtagcacaac tgcgtgtaat gccaggaatg caagcataga gagtaagaca      720
gggttgccag gcaaccaggc gacaaatggt aagacacaaa ctgatcgcac caatatagac      780
aattccacgg gccaaactgg tcaggccaac gctcaaagcg ctaatacggg taataatcgt      840
gtaaataata atagtaaggc cggccaagtt tcaaagtcta aacaggctag gaagcagcaa      900
gtgcacccg gcggtaacac cagtgaagcc tccaagagcg gatcgggtgg ctgttgattt      960
ataagttaa
  
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<210> SEQ ID NO 19
 <211> LENGTH: 322
 <212> TYPE: PRT
 <213> ORGANISM: *Saccharomyces cerevisiae*

<400> SEQUENCE: 19

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Met Pro Leu Asn Lys Ser Asn Ile Arg Glu Tyr Lys Leu Val Val Val
 1          5          10          15
Gly Gly Gly Gly Val Gly Lys Ser Ala Leu Thr Ile Gln Leu Thr Gln
 20          25          30
Ser His Phe Val Asp Glu Tyr Asp Pro Thr Ile Glu Asp Ser Tyr Arg
 35          40          45
Lys Gln Val Val Ile Asp Asp Glu Val Ser Ile Leu Asp Ile Leu Asp
 50          55          60
Thr Ala Gly Gln Glu Glu Tyr Ser Ala Met Arg Glu Gln Tyr Met Arg
 65          70          75          80
Asn Gly Glu Gly Phe Leu Leu Val Tyr Ser Ile Thr Ser Lys Ser Ser
 85          90          95
Leu Asp Glu Leu Met Thr Tyr Tyr Gln Gln Ile Leu Arg Val Lys Asp
100          105          110
Thr Asp Tyr Val Pro Ile Val Val Val Gly Asn Lys Ser Asp Leu Glu
115          120          125
Asn Glu Lys Gln Val Ser Tyr Gln Asp Gly Leu Asn Met Ala Lys Gln
130          135          140
Met Asn Ala Pro Phe Leu Glu Thr Ser Ala Lys Gln Ala Ile Asn Val
145          150          155          160
  
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Glu Glu Ala Phe Tyr Thr Leu Ala Arg Leu Val Arg Asp Glu Gly Gly
 165 170 175
 Lys Tyr Asn Lys Thr Leu Thr Glu Asn Asp Asn Ser Lys Gln Thr Ser
 180 185 190
 Gln Asp Thr Lys Gly Ser Gly Ala Asn Ser Val Pro Arg Asn Ser Gly
 195 200 205
 Gly His Arg Lys Met Ser Asn Ala Ala Asn Gly Lys Asn Val Asn Ser
 210 215 220
 Ser Thr Thr Val Val Asn Ala Arg Asn Ala Ser Ile Glu Ser Lys Thr
 225 230 235 240
 Gly Leu Ala Gly Asn Gln Ala Thr Asn Gly Lys Thr Gln Thr Asp Arg
 245 250 255
 Thr Asn Ile Asp Asn Ser Thr Gly Gln Ala Gly Gln Ala Asn Ala Gln
 260 265 270
 Ser Ala Asn Thr Val Asn Asn Arg Val Asn Asn Asn Ser Lys Ala Gly
 275 280 285
 Gln Val Ser Asn Ala Lys Gln Ala Arg Lys Gln Gln Ala Ala Pro Gly
 290 295 300
 Gly Asn Thr Ser Glu Ala Ser Lys Ser Gly Ser Gly Gly Cys Cys Ile
 305 310 315 320
 Ile Ser

<210> SEQ ID NO 20
 <211> LENGTH: 3079
 <212> TYPE: PRT
 <213> ORGANISM: *Saccharomyces cerevisiae*

<400> SEQUENCE: 20

Met Ser Gln Pro Thr Lys Asn Lys Lys Lys Glu His Gly Thr Asp Ser
 1 5 10 15
 Lys Ser Ser Arg Met Thr Arg Thr Leu Val Asn His Ile Leu Phe Glu
 20 25 30
 Arg Ile Leu Pro Ile Leu Pro Val Glu Ser Asn Leu Ser Thr Tyr Ser
 35 40 45
 Glu Val Glu Glu Tyr Ser Ser Phe Ile Ser Cys Arg Ser Val Leu Ile
 50 55 60
 Asn Val Thr Val Ser Arg Asp Ala Asn Ala Met Val Glu Gly Thr Leu
 65 70 75 80
 Glu Leu Ile Glu Ser Leu Leu Gln Gly His Glu Ile Ile Ser Asp Lys
 85 90 95
 Gly Ser Ser Asp Val Ile Glu Ser Ile Leu Ile Ile Leu Arg Leu Leu
 100 105 110
 Ser Asp Ala Leu Glu Tyr Asn Trp Gln Asn Gln Glu Ser Leu His Tyr
 115 120 125
 Asn Asp Ile Ser Thr His Val Glu His Asp Gln Glu Gln Lys Tyr Arg
 130 135 140
 Pro Lys Leu His Asn Ile Leu Pro Asp Tyr Ser Ser Thr His Ser Asn
 145 150 155 160
 Gly Asn Lys His Phe Phe His Gln Ser Lys Pro Gln Ala Leu Ile Pro
 165 170 175
 Glu Leu Ala Ser Lys Leu Leu Glu Ser Cys Ala Lys Leu Lys Phe Asn
 180 185 190
 Thr Arg Thr Leu Gln Ile Leu Gln Asn Met Ile Ser His Val His Gly
 195 200 205

Asn	Ile	Val	Thr	Thr	Leu	Ser	Ser	Ser	Ile	Leu	Pro	Arg	His	Lys	Ser
210						215					220				
Tyr	Leu	Thr	Arg	His	Asn	His	Pro	Ser	His	Cys	Lys	Met	Ile	Asp	Ser
225					230					235					240
Thr	Leu	Gly	His	Ile	Leu	Arg	Phe	Val	Ala	Ala	Ser	Asn	Pro	Ser	Glu
				245					250					255	
Tyr	Phe	Glu	Phe	Ile	Arg	Lys	Ser	Val	Gln	Val	Pro	Val	Thr	Gln	Thr
			260					265					270		
His	Thr	His	Ser	His	Ser	His	Ser	His	Ser	Leu	Pro	Ser	Ser	Val	Tyr
	275						280					285			
Asn	Ser	Ile	Val	Pro	His	Phe	Asp	Leu	Phe	Ser	Phe	Ile	His	Leu	Ser
290						295					300				
Lys	Asp	Asn	Phe	Lys	Lys	Tyr	Leu	Glu	Leu	Ile	Lys	Asn	Leu	Ser	Val
305					310					315					320
Thr	Leu	Arg	Lys	Thr	Ile	Tyr	His	Cys	Leu	Leu	Leu	His	Tyr	Ser	Ala
				325					330					335	
Lys	Ala	Ile	Met	Phe	Trp	Ile	Met	Thr	Arg	Pro	Ala	Glu	Tyr	Tyr	Glu
			340					345					350		
Leu	Phe	Asn	Leu	Leu	Lys	Asp	Asn	Asn	Asn	Glu	His	Ser	Lys	Ser	Leu
	355					360						365			
Asn	Thr	Leu	Asn	His	Thr	Leu	Phe	Glu	Glu	Ile	His	Ser	Thr	Phe	Asn
	370					375					380				
Val	Asn	Ser	Met	Ile	Thr	Thr	Asn	Gln	Asn	Ala	His	Gln	Gly	Ser	Ser
385					390					395					400
Ser	Pro	Ser	Ser	Ser	Ser	Pro	Ser	Ser	Pro	Pro	Ser	Ser	Ser	Ser	Ser
				405					410					415	
Asp	Asn	Asn	Asn	Gln	Asn	Ile	Ile	Ala	Lys	Ser	Leu	Ser	Arg	Gln	Leu
				420				425					430		
Ser	His	His	Gln	Ser	Tyr	Ile	Gln	Gln	Gln	Ser	Glu	Arg	Lys	Leu	His
	435					440						445			
Ser	Ser	Trp	Thr	Thr	Asn	Ser	Gln	Ser	Ser	Thr	Ser	Leu	Ser	Ser	Ser
	450				455						460				
Thr	Ser	Asp	Ser	Thr	Thr	Thr	Asp	Phe	Ser	Thr	His	Thr	Gln	Pro	Gly
465					470					475					480
Glu	Tyr	Asp	Pro	Ser	Leu	Pro	Asp	Thr	Pro	Thr	Met	Ser	Asn	Ile	Thr
				485					490					495	
Ile	Ser	Ala	Ser	Ser	Leu	Leu	Ser	Gln	Thr	Pro	Thr	Pro	Thr	Thr	Gln
			500					505					510		
Leu	Gln	Gln	Arg	Leu	Asn	Ser	Ala	Ala	Ala	Ala	Ala	Ala	Ala	Ala	Ala
	515						520					525			
Ser	Pro	Ser	Asn	Ser	Thr	Pro	Thr	Gly	Tyr	Thr	Ala	Glu	Gln	Gln	Ser
	530					535					540				
Arg	Ala	Ser	Tyr	Asp	Ala	His	Lys	Thr	Gly	His	Thr	Gly	Lys	Asp	Tyr
545					550					555					560
Asp	Glu	His	Phe												

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Pro	Lys	Glu	Arg	Thr	Ser	Ser	Phe	Asn	Asn	Thr	Ala	His	Glu	Thr	Gly	
625					630					635					640	
Ser	Glu	Lys	Thr	Ser	Gly	Ile	Lys	His	Ile	Thr	Gln	Gly	Leu	Lys	Lys	
				645					650					655		
Leu	Thr	Ser	Leu	Pro	Ser	Ser	Thr	Lys	Lys	Thr	Val	Lys	Phe	Met	Lys	
			660					665					670			
Met	Leu	Leu	Arg	Asn	Leu	Ile	Gly	Asn	Gln	Ala	Val	Ser	Asp	Val	Ala	
	675						680					685				
Leu	Leu	Asp	Thr	Met	Arg	Ala	Leu	Leu	Ser	Phe	Phe	Thr	Met	Thr	Ser	
	690					695					700					
Ala	Val	Phe	Leu	Val	Asp	Arg	Asn	Leu	Pro	Ser	Val	Leu	Phe	Ala	Lys	
705					710					715					720	
Arg	Leu	Ile	Pro	Ile	Met	Gly	Thr	Asn	Leu	Ser	Val	Gly	Gln	Asp	Trp	
				725					730					735		
Asn	Ser	Lys	Ile	Asn	Asn	Ser	Leu	Met	Val	Cys	Leu	Lys	Lys	Asn	Ser	
			740					745					750			
Thr	Thr	Phe	Val	Gln	Leu	Gln	Leu	Ile	Phe	Phe	Ser	Ser	Ala	Ile	Gln	
		755					760						765			
Phe	Asp	His	Glu	Leu	Leu	Leu	Ala	Arg	Leu	Ser	Ile	Asp	Thr	Met	Ala	
	770					775					780					
Asn	Asn	Leu	Asn	Met	Gln	Lys	Leu	Cys	Leu	Tyr	Thr	Glu	Gly	Phe	Arg	
785					790					795					800	
Ile	Phe	Phe	Asp	Ile	Pro	Ser	Lys	Lys	Glu	Leu	Arg	Lys	Ala	Ile	Ala	
				805					810					815		
Val	Lys	Ile	Ser	Lys	Phe	Phe	Lys	Thr	Leu	Phe	Ser	Ile	Ile	Ala	Asp	
			820					825					830			
Ile	Leu	Leu	Gln	Glu	Phe	Pro	Tyr	Phe	Asp	Glu	Gln	Ile	Thr	Asp	Ile	
	835						840					845				
Val	Ala	Ser	Ile	Leu	Asp	Gly	Thr	Ile	Ile	Asn	Glu	Tyr	Gly	Thr	Lys	
	850					855					860					
Lys	His	Phe	Lys	Gly	Ser	Ser	Pro	Ser	Leu	Cys	Ser	Thr	Thr	Arg	Ser	
865					870					875					880	
Arg	Ser	Gly	Ser	Thr	Ser	Gln	Ser	Ser	Met	Thr	Pro	Val	Ser	Pro	Leu	
				885					890					895		
Gly	Leu	Asp	Thr	Asp	Ile	Cys	Pro	Met	Asn	Thr	Leu	Ser	Leu	Val	Gly	
		900						905					910			
Ser	Ser	Thr	Ser	Arg	Asn	Ser	Asp	Asn	Val	Asn	Ser	Leu	Asn	Ser	Ser	
		915					920					925				
Pro	Lys	Asn	Leu	Ser	Ser	Asp	Pro	Tyr	Leu	Ser	His	Leu	Val	Ala	Pro	
	930					935					940					
Arg	Ala	Arg	His	Ala	Leu	Gly	Gly	Pro	Ser	Ser	Ile	Ile	Arg	Asn	Lys	
945					950					955					960	
Ile	Pro	Thr	Thr	Leu	Thr	Ser	Pro	Pro	Gly	Thr	Glu	Lys	Ser	Ser	Pro	
				965					970					975		
Val	Gln	Arg	Pro	Gln	Thr	Glu	Ser	Ile	Ser	Ala	Thr	Pro	Met	Ala	Ile	
			980					985					990			
Thr	Asn	Ser	Thr	Pro	Leu	Ser	Ser	Ala	Ala	Phe	Gly	Ile	Arg	Ser	Pro	
		995					1000					1005				
Leu	Gln	Lys	Ile	Arg	Thr	Arg	Arg	Tyr	Ser	Asp	Glu	Ser	Leu	Gly		
	1010					1015					1020					
Lys	Phe	Met	Lys	Ser	Thr	Asn	Asn	Tyr	Ile	Gln	Glu	His	Leu	Ile		
	1025					1030					1035					
Pro	Lys	Asp	Leu	Asn	Glu	Ala	Thr	Leu	Gln	Asp	Ala	Arg	Arg	Ile		

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1040	1045	1050
Met Ile Asn Ile Phe Ser Ile Phe Lys Arg Pro Asn Ser Tyr Phe		
1055	1060	1065
Ile Ile Pro His Asn Ile Asn Ser Asn Leu Gln Trp Val Ser Gln		
1070	1075	1080
Asp Phe Arg Asn Ile Met Lys Pro Ile Phe Val Ala Ile Val Ser		
1085	1090	1095
Pro Asp Val Asp Leu Gln Asn Thr Ala Gln Ser Phe Met Asp Thr		
1100	1105	1110
Leu Leu Ser Asn Val Ile Thr Tyr Gly Glu Ser Asp Glu Asn Ile		
1115	1120	1125
Ser Ile Glu Gly Tyr His Leu Leu Cys Ser Tyr Thr Val Thr Leu		
1130	1135	1140
Phe Ala Met Gly Leu Phe Asp Leu Lys Ile Asn Asn Glu Lys Arg		
1145	1150	1155
Gln Ile Leu Leu Asp Ile Thr Val Lys Phe Met Lys Val Arg Ser		
1160	1165	1170
His Leu Ala Gly Ile Ala Glu Ala Ser His His Met Glu Tyr Ile		
1175	1180	1185
Ser Asp Ser Glu Lys Leu Thr Phe Pro Leu Ile Met Gly Thr Val		
1190	1195	1200
Gly Arg Ala Leu Phe Val Ser Leu Tyr Ser Ser Gln Gln Lys Ile		
1205	1210	1215
Glu Lys Thr Leu Lys Ile Ala Tyr Thr Glu Tyr Leu Ser Ala Ile		
1220	1225	1230
Asn Phe His Glu Arg Asn Ile Asp Asp Ala Asp Lys Thr Trp Val		
1235	1240	1245
His Asn Ile Glu Phe Val Glu Ala Met Cys His Asp Asn Tyr Thr		
1250	1255	1260
Thr Ser Gly Ser Ile Ala Phe Gln Arg Arg Thr Arg Asn Asn Ile		
1265	1270	1275
Leu Arg Phe Ala Thr Ile Pro Asn Ala Ile Leu Leu Asp Ser Met		
1280	1285	1290
Arg Met Ile Tyr Lys Lys Trp His Thr Tyr Thr His Ser Lys Ser		
1295	1300	1305
Leu Glu Lys Gln Glu Arg Asn Asp Phe Arg Asn Phe Ala Gly Ile		
1310	1315	1320
Leu Ala Ser Leu Ser Gly Ile Leu Phe Ile Asn Lys Lys Ile Leu		
1325	1330	1335
Gln Glu Met Tyr Pro Tyr Leu Leu Asp Thr Val Ser Glu Leu Lys		
1340	1345	1350
Lys Asn Ile Asp Ser Phe Ile Ser Lys Gln Cys Gln Trp Leu Asn		
1355	1360	1365
Tyr Pro Asp Leu Leu Thr Arg Glu Asn Ser Arg Asp Ile Leu Ser		
1370	1375	1380
Val Glu Leu His Pro Leu Ser Phe Asn Leu Leu Phe Asn Asn Leu		
1385	1390	1395
Arg Leu Lys Leu Lys Glu Leu Ala Cys Ser Asp Leu Ser Ile Pro		
1400	1405	1410
Glu Asn Glu Ser Ser Tyr Val Leu Leu Glu Gln Ile Ile Lys Met		
1415	1420	1425
Leu Arg Thr Ile Leu Gly Arg Asp Asp Asp Asn Tyr Val Met Met		
1430	1435	1440

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Leu Phe	Ser Thr	Glu Ile	Val	Asp Leu	Ile Asp	Leu	Leu Thr	Asp
1445			1450			1455		
Glu Ile	Lys Lys	Ile Pro	Ala	Tyr Cys	Pro Lys	Tyr	Leu Lys	Ala
1460			1465			1470		
Ile Ile	Gln Met	Thr Lys	Met	Phe Ser	Ala Leu	Gln	His Ser	Glu
1475			1480			1485		
Val Asn	Leu Gly	Val Lys	Asn	His Phe	His Val	Lys	Asn Lys	Trp
1490			1495			1500		
Leu Arg	Gln Val	Thr Asp	Trp	Phe Gln	Val Ser	Ile	Ala Arg	Glu
1505			1510			1515		
Tyr Asp	Phe Glu	Asn Leu	Ser	Lys Pro	Leu Lys	Glu	Met Asp	Leu
1520			1525			1530		
Val Lys	Arg Asp	Met Asp	Ile	Leu Tyr	Ile Asp	Thr	Ala Ile	Glu
1535			1540			1545		
Ala Ser	Thr Ala	Ile Ala	Tyr	Leu Thr	Arg His	Thr	Phe Leu	Glu
1550			1555			1560		
Ile Pro	Pro Ala	Ala Ser	Asp	Pro Glu	Leu Ser	Arg	Ser Arg	Ser
1565			1570			1575		
Val Ile	Phe Gly	Phe Tyr	Phe	Asn Ile	Leu Met	Lys	Gly Leu	Glu
1580			1585			1590		
Lys Ser	Ser Asp	Arg Asp	Asn	Tyr Pro	Val Phe	Leu	Arg His	Lys
1595			1600			1605		
Met Ser	Val Leu	Asn Asp	Asn	Val Ile	Leu Ser	Leu	Thr Asn	Leu
1610			1615			1620		
Ser Asn	Thr Asn	Val Asp	Ala	Ser Leu	Gln Phe	Thr	Leu Pro	Met
1625			1630			1635		
Gly Tyr	Ser Gly	Asn Arg	Asn	Ile Arg	Asn Ala	Phe	Leu Glu	Val
1640			1645			1650		
Phe Ile	Asn Ile	Val Thr	Asn	Tyr Arg	Thr Tyr	Thr	Ala Lys	Thr
1655			1660			1665		
Asp Leu	Gly Lys	Leu Glu	Ala	Ala Asp	Lys Phe	Leu	Arg Tyr	Thr
1670			1675			1680		
Ile Glu	His Pro	Gln Leu	Ser	Ser Phe	Gly Ala	Ala	Val Cys	Pro
1685			1690			1695		
Ala Ser	Asp Ile	Asp Ala	Tyr	Ala Ala	Gly Leu	Ile	Asn Ala	Phe
1700			1705			1710		
Glu Thr	Arg Asn	Ala Thr	His	Ile Val	Val Ser	Gln	Leu Ile	Lys
1715			1720			1725		
Asn Glu	Ile Glu	Asn Ser	Ser	Arg Pro	Thr Asp	Ile	Leu Arg	Arg
1730			1735			1740		
Asn Ser	Cys Ala	Thr Arg	Ser	Leu Ser	Met Leu	Ala	Arg Ser	Lys
1745			1750			1755		
Gly Ser	Glu Tyr	Leu Ile	Arg	Thr Leu	Gln Pro	Leu	Leu Lys	Lys
1760			1765			1770		
Ile Ile	Gln Asn	Arg Asp	Phe	Phe Glu	Ile Glu	Lys	Leu Lys	Pro
1775			1780			1785		
Glu Asp	Ser Asp	Ala Glu	Arg	Gln Ile	Glu Leu	Phe	Val Lys	Tyr
1790			1795			1800		
Met Asn	Glu Leu	Leu Glu	Ser	Ile Ser	Asn Ser	Val	Ser Tyr	Phe
1805			1810			1815		
Pro Pro	Pro Leu	Phe Tyr	Ile	Cys Gln	Asn Ile	Tyr	Lys Val	Ala
1820			1825			1830		

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Cys 1835	Glu	Lys	Phe	Pro	Asp	His 1840	Ala	Ile	Ile	Ala	Ala 1845	Gly	Ser	Phe
Val 1850	Phe	Leu	Arg	Phe	Phe	Cys 1855	Pro	Ala	Leu	Val	Ser 1860	Pro	Asp	Ser
Glu 1865	Asn	Ile	Ile	Asp	Ile	Ser 1870	His	Leu	Ser	Glu	Lys 1875	Arg	Thr	Phe
Ile 1880	Ser	Leu	Ala	Lys	Val	Ile 1885	Gln	Asn	Ile	Ala	Asn 1890	Gly	Ser	Glu
Asn 1895	Phe	Ser	Arg	Trp	Pro	Ala 1900	Leu	Cys	Ser	Gln	Lys 1905	Asp	Phe	Leu
Lys 1910	Glu	Cys	Ser	Asp	Arg	Ile 1915	Phe	Arg	Phe	Leu	Ala 1920	Glu	Leu	Cys
Arg 1925	Thr	Asp	Arg	Thr	Ile	Asp 1930	Ile	Gln	Val	Arg	Thr 1935	Asp	Pro	Thr
Pro 1940	Ile	Ala	Phe	Asp	Tyr	Gln 1945	Phe	Leu	His	Ser	Phe 1950	Val	Tyr	Leu
Tyr 1955	Gly	Leu	Glu	Val	Arg	Arg 1960	Asn	Val	Leu	Asn	Glu 1965	Ala	Lys	His
Asp 1970	Asp	Gly	Asp	Ile	Asp	Gly 1975	Asp	Asp	Phe	Tyr	Lys 1980	Thr	Thr	Phe
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accattccta	aaattttgtt	ggagtggttg	agttcggata	acgaaaatgc	aaacattacc	8820

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ctctatcaag gtgcgatact gttcaaatgt tcagttacgg atgaaccaag tagatttagg 8880
tttgcgttga ttattaggca tctattgaca aagaaccca tttgtgcatt gcgtttttac 8940
agtgttattc gtaacgaaat aagaaaaata tcagcatttg agcaaaacttc ggattgtgtt 9000
ccacttgctt tcgatatttt aaacttatta gtgacgcatt cagagtctaa ttccttagaa 9060
aaacttcacg aagaatccat tgaacgtctg accaaaagag gtttatcaat tgtgacttct 9120
tctggatatat ttgcgaagaa ttccgacatg atgataccgt tagatgtaaa acctgaagat 9180
atctatgaac gtaagagaat aatgacaatg attttatcaa ggatgtcatg ttctgcttag 9240

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<210> SEQ ID NO 22

<211> LENGTH: 3079

<212> TYPE: PRT

<213> ORGANISM: *Saccharomyces cerevisiae*

<400> SEQUENCE: 22

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Met Ser Gln Pro Thr Lys Asn Lys Lys Lys Glu His Gly Thr Asp Ser
1          5          10          15
Lys Ser Ser Arg Met Thr Arg Thr Leu Val Asn His Ile Leu Phe Glu
20        25        30
Arg Ile Leu Pro Ile Leu Pro Val Glu Ser Asn Leu Ser Thr Tyr Ser
35        40        45
Glu Val Glu Glu Tyr Ser Ser Phe Ile Ser Cys Arg Ser Val Leu Ile
50        55        60
Asn Val Thr Val Ser Arg Asp Ala Asn Ala Met Val Glu Gly Thr Leu
65        70        75        80
Glu Leu Ile Glu Ser Leu Leu Gln Gly His Glu Ile Ile Ser Asp Lys
85        90        95
Gly Ser Ser Asp Val Ile Glu Ser Ile Leu Ile Ile Leu Arg Leu Leu
100       105       110
Ser Asp Ala Leu Glu Tyr Asn Trp Gln Asn Gln Glu Ser Leu His Tyr
115       120       125
Asn Asp Ile Ser Thr His Val Glu His Asp Gln Glu Gln Lys Tyr Arg
130       135       140
Pro Lys Leu His Asn Ile Leu Pro Asp Tyr Ser Ser Thr His Ser Asn
145       150       155       160
Gly Asn Lys His Phe Phe His Gln Ser Lys Pro Gln Ala Leu Ile Pro
165       170       175
Glu Leu Ala Ser Lys Leu Leu Glu Ser Cys Ala Lys Leu Lys Phe Asn
180       185       190
Thr Arg Thr Leu Gln Ile Leu Gln Ser Met Ile Ser His Val His Gly
195       200       205
Asn Ile Leu Thr Thr Leu Ser Ser Ser Ile Leu Pro Arg His Lys Ser
210       215       220
Tyr Leu Thr Arg His Asn His Pro Ser His Cys Lys Met Ile Asp Ser
225       230       235       240
Thr Leu Gly His Ile Leu Arg Phe Val Ala Ala Ser Asn Pro Ser Glu
245       250       255
Tyr Phe Glu Phe Ile Arg Lys Ser Val Gln Val Pro Val Thr Gln Thr
260       265       270
His Thr His Ser His Ser His Ser His Ser Leu Pro Ser Ser Val Tyr
275       280       285
Asn Ser Ile Val Pro His Phe Asp Leu Phe Ser Phe Ile His Leu Ser
290       295       300

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Lys	Asp	Asn	Phe	Lys	Lys	Tyr	Leu	Glu	Leu	Ile	Lys	Asn	Leu	Ser	Val	305	310	315	320
Thr	Leu	Arg	Lys	Thr	Ile	Tyr	His	Cys	Leu	Leu	Leu	His	Tyr	Ser	Ala	325	330	335	
Lys	Ala	Ile	Met	Phe	Trp	Ile	Met	Thr	Arg	Pro	Ala	Glu	Tyr	Tyr	Glu	340	345	350	
Leu	Phe	Asn	Leu	Leu	Lys	Asp	Asn	Asn	Asn	Glu	His	Ser	Lys	Ser	Leu	355	360	365	
Asn	Thr	Leu	Asn	His	Thr	Leu	Phe	Glu	Glu	Ile	His	Ser	Thr	Phe	Asn	370	375	380	
Val	Asn	Ser	Met	Ile	Thr	Thr	Asn	Gln	Asn	Val	His	Gln	Gly	Ser	Ser	385	390	395	400
Ser	Pro	Ser	Ser	Ser	Ser	Pro	Ser	Ser	Pro	Pro	Ser	Ser	Ser	Ser	Ser	405	410	415	
Asp	Asn	Asn	Asn	Gln	Asn	Ile	Ile	Ala	Lys	Ser	Leu	Ser	Arg	Gln	His	420	425	430	
Ser	His	His	Gln	Ser	Tyr	Ile	Gln	Gln	Gln	Ser	Glu	Arg	Lys	Leu	His	435	440	445	
Ser	Ser	Trp	Thr	Thr	Asn	Ser	Gln	Ser	Ser	Thr	Ser	Leu	Ser	Ser	Ser	450	455	460	
Thr	Ser	Asn	Ser	Thr	Thr	Thr	Asp	Phe	Ser	Thr	His	Thr	Gln	Pro	Gly	465	470	475	480
Glu	Tyr	Asp	Pro	Ser	Leu	Pro	Asp	Thr	Pro	Thr	Met	Ser	Asn	Ile	Thr	485	490	495	
Ile	Ser	Ala	Ser	Ser	Leu	Leu	Ser	Gln	Thr	Pro	Thr	Pro	Thr	Thr	Gln	500	505	510	
Leu	Gln	Gln	Arg	Leu	Asn	Ser	Ala	Ala	Ala	Ala	Ala	Ala	Ala	Ala	Ala	515	520	525	
Ser	Pro	Ser	Asn	Ser	Thr	Pro	Thr	Gly	Tyr	Thr	Ala	Glu	Gln	Gln	Ser	530	535	540	
Arg	Ala	Ser	Tyr	Asp	Ala	His	Lys	Thr	Gly	His	Thr	Gly	Lys	Asp	Tyr	545	550	555	560
Asp	Glu	His	Phe	Leu	Ser	Ile	Thr	Arg	Leu	Asp	Asn	Val	Leu	Glu	Leu	565	570	575	
Tyr	Thr	His	Phe	Asp	Asp	Thr	Glu	Val	Leu	Pro	His	Thr	Ser	Val	Leu	580	585	590	
Lys	Phe	Leu	Thr	Thr	Leu	Thr	Met	Phe	Asp	Ile	Asp	Leu	Phe	Asn	Glu	595	600	605	
Leu	Asn	Ala	Thr	Ser	Phe	Lys	Tyr	Ile	Pro	Asp	Cys	Thr	Met	His	Arg	610	615	620	
Pro	Lys	Glu	Arg	Thr	Ser	Ser	Phe	Asn	Asn	Thr	Ala	His	Glu	Thr	Gly	625	630	635	640
Ser	Glu	Lys	Thr	Ser	Gly	Ile	Lys	His	Ile	Thr	Gln	Gly	Leu	Lys	Lys	645	650	655	
Leu	Thr	Ser	Leu	Pro	Ser	Ser	Thr	Lys	Lys	Thr	Val	Lys	Phe	Met	Lys	660	665	670	
Met	Leu	Leu	Arg	Asn	Leu	Ile	Gly	Asn	Gln	Ala	Val	Ser	Asp	Val	Ala	675	680	685	
Leu	Leu	Asp	Thr	Met	Arg	Ala	Leu	Leu	Ser	Phe	Phe	Thr	Met	Thr	Ser	690	695	700	
Ala	Val	Phe	Leu	Val	Asp	Arg	Asn	Leu	Pro	Ser	Val	Leu	Phe	Ala	Lys	705	710	715	720
Arg	Leu	Ile	Pro	Ile	Met	Gly	Thr	Asn	Leu	Ser	Val	Gly	Gln	Asp	Trp				

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725										730					735				
Asn	Ser	Lys	Ile	Asn	Asn	Ser	Leu	Met	Val	Cys	Leu	Lys	Lys	Asn	Ser				
			740					745					750						
Thr	Thr	Phe	Val	Gln	Leu	Gln	Leu	Ile	Phe	Phe	Ser	Ser	Ala	Ile	Gln				
		755					760					765							
Phe	Asp	His	Glu	Leu	Leu	Leu	Ala	Arg	Leu	Ser	Ile	Asp	Thr	Met	Ala				
	770					775					780								
Asn	Asn	Leu	Asn	Met	Gln	Lys	Leu	Cys	Leu	Tyr	Thr	Glu	Gly	Phe	Arg				
785					790					795					800				
Ile	Phe	Phe	Asp	Ile	Pro	Ser	Lys	Lys	Glu	Leu	Arg	Lys	Ala	Ile	Ala				
			805						810					815					
Val	Lys	Ile	Ser	Lys	Phe	Phe	Lys	Thr	Leu	Phe	Ser	Ile	Ile	Ala	Asp				
			820					825					830						
Ile	Leu	Leu	Gln	Glu	Phe	Pro	Tyr	Phe	Asp	Glu	Gln	Ile	Thr	Asp	Ile				
		835					840					845							
Val	Ala	Ser	Ile	Leu	Asp	Gly	Thr	Ile	Ile	Asn	Glu	Tyr	Gly	Thr	Lys				
	850					855					860								
Lys	His	Phe	Lys	Gly	Ser	Ser	Pro	Ser	Leu	Cys	Ser	Thr	Thr	Arg	Ser				
865					870					875					880				
Arg	Ser	Gly	Ser	Thr	Ser	Gln	Ser	Ser	Met	Thr	Pro	Val	Ser	Pro	Leu				
				885					890					895					
Gly	Leu	Asp	Thr	Asp	Ile	Cys	Pro	Met	Asn	Thr	Leu	Ser	Leu	Val	Gly				
			900					905					910						
Ser	Ser	Thr	Ser	Arg	Asn	Ser	Asp	Asn	Val	Asn	Ser	Leu	Asn	Ser	Ser				
		915					920					925							
Pro	Lys	Asn	Leu	Ser	Ser	Asp	Pro	Tyr	Leu	Ser	His	Leu	Val	Ala	Pro				
	930					935					940								
Arg	Ala	Arg	His	Ala	Leu	Gly	Gly	Pro	Ser	Ser	Ile	Ile	Arg	Asn	Lys				
945					950					955					960				
Ile	Pro	Thr	Thr	Leu	Thr	Ser	Pro	Pro	Gly	Thr	Glu	Lys	Ser	Ser	Pro				
				965					970					975					
Val	Gln	Arg	Pro	Gln	Thr	Glu	Ser	Ile	Ser	Ala	Thr	Pro	Met	Ala	Ile				
			980					985					990						
Thr	Asn	Ser	Thr	Pro	Leu	Ser	Ser	Ala	Ala	Phe	Gly	Ile	Arg	Ser	Pro				
		995					1000					1005							
Leu	Gln	Lys	Ile	Arg	Thr	Arg	Arg	Tyr	Ser	Asp	Glu	Ser	Leu	Gly					
	1010					1015					1020								
Lys	Phe	Met	Lys	Ser	Thr	Asn	Asn	Tyr	Ile	Gln	Glu	His	Leu	Ile					
	1025					1030					1035								
Pro	Lys	Asp	Leu	Asn	Glu	Ala	Thr	Leu	Gln	Asp	Ala	Arg	Arg	Ile					
	1040					1045					1050								
Met	Ile	Asn	Ile	Phe	Ser	Ile	Phe	Lys	Arg	Pro	Asn	Ser	Tyr	Phe					
	1055					1060					1065								
Ile	Ile	Pro	His	Asn	Ile	Asn	Ser	Asn	Leu	Gln	Trp	Val	Ser	Gln					
	1070					1075					1080								
Asp	Phe	Arg	Asn	Ile	Met	Lys	Pro	Ile	Phe	Val	Ala	Ile	Val	Ser					
	1085					1090					1095								
Pro	Asp	Val	Asp	Leu	Gln	Asn	Thr	Ala	Gln	Ser	Phe	Met	Asp	Thr					
	1100					1105					1110								
Leu	Leu	Ser	Asn	Val	Ile	Thr	Tyr	Gly	Glu	Ser	Asp	Glu	Asn	Ile					
	1115					1120					1125								
Ser	Ile	Glu	Gly	Tyr	His	Leu	Leu	Cys	Ser	Tyr	Thr	Val	Thr	Leu					
	1130					1135					1140								

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Phe Ala Met Gly Leu Phe Asp	Leu Lys Ile Asn Asn	Glu Lys Arg
1145	1150	1155
Gln Ile Leu Leu Asp Ile Thr	Val Lys Phe Met Lys	Val Arg Ser
1160	1165	1170
His Leu Ala Gly Ile Ala Glu	Ala Ser His His Met	Glu Tyr Ile
1175	1180	1185
Ser Asp Ser Glu Lys Leu Thr	Phe Pro Leu Ile Met	Gly Thr Val
1190	1195	1200
Gly Arg Ala Leu Phe Val Ser	Leu Tyr Ser Ser Gln	Gln Lys Ile
1205	1210	1215
Glu Lys Thr Leu Lys Ile Ala	Tyr Thr Glu Tyr Leu	Ser Ala Ile
1220	1225	1230
Asn Phe His Glu Arg Asn Ile	Asp Asp Ala Asp Lys	Thr Trp Val
1235	1240	1245
His Asn Ile Glu Phe Val Glu	Ala Met Cys His Asp	Asn Tyr Thr
1250	1255	1260
Thr Ser Gly Ser Ile Ala Phe	Gln Arg Arg Thr Arg	Asn Asn Ile
1265	1270	1275
Leu Arg Phe Ala Thr Ile Pro	Asn Ala Ile Leu Leu	Asp Ser Met
1280	1285	1290
Arg Met Ile Tyr Lys Lys Trp	His Thr Tyr Thr His	Ser Lys Ser
1295	1300	1305
Leu Glu Lys Gln Glu Arg Asn	Asp Phe Arg Asn Phe	Ala Gly Ile
1310	1315	1320
Leu Ala Ser Leu Ser Gly Ile	Leu Phe Ile Asn Lys	Lys Ile Leu
1325	1330	1335
Gln Glu Met Tyr Pro Tyr Leu	Leu Asp Thr Val Ser	Glu Leu Lys
1340	1345	1350
Lys Asn Ile Asp Ser Phe Ile	Ser Lys Gln Cys Gln	Trp Leu Asn
1355	1360	1365
Tyr Pro Asp Leu Leu Thr Arg	Glu Asn Ser Arg Asp	Ile Leu Ser
1370	1375	1380
Val Glu Leu His Pro Leu Ser	Phe Asn Leu Leu Phe	Asn Asn Leu
1385	1390	1395
Arg Leu Lys Leu Lys Glu Leu	Ala Cys Ser Asp Leu	Ser Ile Pro
1400	1405	1410
Glu Asn Glu Ser Ser Tyr Val	Leu Leu Glu Gln Ile	Ile Lys Met
1415	1420	1425
Leu Arg Thr Ile Leu Gly Arg	Asp Asp Asp Asn Tyr	Val Met Met
1430	1435	1440
Leu Phe Ser Thr Glu Ile Val	Asp Leu Ile Asp Leu	Leu Thr Asp
1445	1450	1455
Glu Ile Lys Lys Ile Pro Ala	Tyr Cys Pro Lys Tyr	Leu Lys Ala
1460	1465	1470
Ile Ile Gln Met Thr Lys Met	Phe Ser Ala Leu Gln	His Ser Glu
1475	1480	1485
Val Asn Leu Gly Val Lys Asn	His Phe His Val Lys	Asn Lys Trp
1490	1495	1500
Leu Arg Gln Val Thr Asp Trp	Phe Gln Val Ser Ile	Ala Arg Glu
1505	1510	1515
Tyr Asp Phe Glu Asn Leu Ser	Lys Pro Leu Lys Glu	Met Asp Leu
1520	1525	1530

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Val Lys	Arg Asp	Met Asp	Ile	Leu Tyr	Ile Asp	Thr	Ala Ile	Glu	
1535			1540			1545			
Ala Ser	Thr Ala	Ile Ala	Tyr	Leu Thr	Arg His	Thr	Phe Leu	Glu	
1550			1555			1560			
Ile Pro	Pro Ala	Ala Ser	Asp	Pro Glu	Leu Ser	Arg	Ser Arg	Ser	
1565			1570			1575			
Val Ile	Phe Gly	Phe Tyr	Phe	Asn Ile	Leu Met	Lys	Gly Leu	Glu	
1580			1585			1590			
Lys Ser	Ser Asp	Arg Asp	Asn	Tyr Pro	Val Phe	Leu	Arg His	Lys	
1595			1600			1605			
Met Ser	Val Leu	Asn Asp	Asn	Val Ile	Leu Ser	Leu	Thr Asn	Leu	
1610			1615			1620			
Ser Asn	Thr Asn	Val Asp	Ala	Ser Leu	Gln Phe	Thr	Leu Pro	Met	
1625			1630			1635			
Gly Tyr	Ser Gly	Asn Arg	Asn	Ile Arg	Asn Ala	Phe	Leu Glu	Val	
1640			1645			1650			
Phe Ile	Asn Ile	Val Thr	Asn	Tyr Arg	Thr Tyr	Thr	Ala Lys	Thr	
1655			1660			1665			
Asp Leu	Gly Lys	Leu Glu	Ala	Ala Asp	Lys Phe	Leu	Arg Tyr	Thr	
1670			1675			1680			
Ile Glu	His Pro	Gln Leu	Ser	Ser Phe	Gly Ala	Ala	Val Cys	Pro	
1685			1690			1695			
Ala Ser	Asp Ile	Asp Ala	Tyr	Ala Ala	Gly Leu	Ile	Asn Ala	Phe	
1700			1705			1710			
Glu Thr	Arg Asn	Ala Thr	His	Ile Val	Val Ser	Gln	Leu Ile	Lys	
1715			1720			1725			
Asn Glu	Ile Glu	Asn Ser	Ser	Arg Pro	Thr Asp	Ile	Leu Arg	Arg	
1730			1735			1740			
Asn Ser	Cys Ala	Thr Arg	Ser	Leu Ser	Met Leu	Ala	Arg Ser	Lys	
1745			1750			1755			
Gly Ser	Glu Tyr	Leu Ile	Arg	Thr Leu	Gln Pro	Leu	Leu Lys	Lys	
1760			1765			1770			
Ile Ile	Gln Asn	Arg Asp	Phe	Phe Glu	Ile Glu	Lys	Leu Lys	Pro	
1775			1780			1785			
Glu Asp	Ser Asp	Ala Glu	Arg	Gln Ile	Glu Leu	Phe	Val Lys	Tyr	
1790			1795			1800			
Met Asn	Glu Leu	Leu Glu	Ser	Ile Ser	Asn Ser	Val	Ser Tyr	Phe	
1805			1810			1815			
Pro Pro	Pro Leu	Phe Tyr	Ile	Cys Gln	Asn Ile	Tyr	Lys Val	Ala	
1820			1825			1830			
Cys Glu	Lys Phe	Pro Asp	His	Ala Ile	Ile Ala	Ala	Gly Ser	Phe	
1835			1840			1845			
Val Phe	Leu Arg	Phe Phe	Cys	Pro Ala	Leu Val	Ser	Pro Asp	Ser	
1850			1855			1860			
Glu Asn	Ile Ile	Asp Ile	Ser	His Leu	Ser Glu	Lys	Arg Thr	Phe	
1865			1870			1875			
Ile Ser	Leu Ala	Lys Val	Ile	Gln Asn	Ile Ala	Asn	Gly Ser	Glu	
1880			1885			1890			
Asn Phe	Ser Arg	Trp Pro	Ala	Leu Cys	Ser Gln	Lys	Asp Phe	Leu	
1895			1900			1905			
Lys Glu	Cys Ser	Asp Arg	Ile	Phe Arg	Phe Leu	Ala	Glu Leu	Cys	
1910			1915			1920			
Arg Thr	Asp Arg	Thr Ile	Asp	Ile Gln	Val Arg	Thr	Asp Pro	Thr	

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1925	1930	1935
Pro Ile Ala Phe Asp Tyr Gln Phe Leu His Ser Phe Val Tyr Leu 1940 1945 1950		
Tyr Gly Leu Glu Val Arg Arg Asn Val Leu Asn Glu Ala Lys His 1955 1960 1965		
Asp Asp Gly Asp Ile Asp Gly Asp Asp Phe Tyr Lys Thr Thr Phe 1970 1975 1980		
Leu Leu Ile Asp Asp Val Leu Gly Gln Leu Gly Gln Pro Lys Met 1985 1990 1995		
Glu Phe Ser Asn Glu Ile Pro Ile Tyr Ile Arg Glu His Met Asp 2000 2005 2010		
Asp Tyr Pro Glu Leu Tyr Glu Phe Met Asn Arg His Ala Phe Arg 2015 2020 2025		
Asn Ile Glu Thr Ser Thr Ala Tyr Ser Pro Ser Val His Glu Ser 2030 2035 2040		
Thr Ser Ser Glu Gly Ile Pro Ile Ile Thr Leu Thr Met Ser Asn 2045 2050 2055		
Phe Ser Asp Arg His Val Asp Ile Asp Thr Val Ala Tyr Lys Phe 2060 2065 2070		
Leu Gln Ile Tyr Ala Arg Ile Trp Thr Thr Lys His Cys Leu Ile 2075 2080 2085		
Ile Asp Cys Thr Glu Phe Asp Glu Gly Gly Leu Asp Met Arg Lys 2090 2095 2100		
Phe Ile Ser Leu Val Met Gly Leu Leu Pro Glu Val Ala Pro Lys 2105 2110 2115		
Asn Cys Ile Gly Cys Tyr Tyr Phe Asn Val Asn Glu Thr Phe Met 2120 2125 2130		
Asp Asn Tyr Gly Lys Cys Leu Asp Lys Asp Asn Val Tyr Val Ser 2135 2140 2145		
Ser Lys Ile Pro His Tyr Phe Ile Asn Ser Asn Ser Asp Glu Gly 2150 2155 2160		
Leu Met Lys Ser Val Gly Ile Thr Gly Gln Gly Leu Lys Val Leu 2165 2170 2175		
Gln Asp Ile Arg Val Ser Leu His Asp Ile Thr Leu Tyr Asp Glu 2180 2185 2190		
Lys Arg Asn Arg Phe Thr Pro Val Ser Leu Lys Ile Gly Asp Ile 2195 2200 2205		
Tyr Phe Gln Val Leu His Glu Thr Pro Arg Gln Tyr Lys Ile Arg 2210 2215 2220		
Asp Met Gly Thr Leu Phe Asp Val Lys Phe Asn Asp Val Tyr Glu 2225 2230 2235		
Ile Ser Arg Ile Phe Glu Val His Val Ser Ser Ile Thr Gly Val 2240 2245 2250		
Ala Ala Glu Phe Thr Val Thr Phe Gln Asp Glu Arg Arg Leu Ile 2255 2260 2265		
Phe Ser Ser Pro Lys Tyr Leu Glu Ile Val Lys Met Phe Tyr Tyr 2270 2275 2280		
Ala Gln Ile Arg Leu Glu Ser Glu Tyr Glu Met Asp Asn Asn Ser 2285 2290 2295		
Ser Thr Ser Ser Pro Asn Ser Asn Asn Lys Asp Lys Gln Gln Lys 2300 2305 2310		
Glu Arg Thr Lys Leu Leu Cys His Leu Leu Leu Val Ser Leu Ile 2315 2320 2325		

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Gly Leu Phe Asp Glu Ser Lys	Lys Met Lys Asn Ser	Ser Tyr Asn
2330	2335	2340
Leu Ile Ala Ala Thr Glu Ala	Ser Phe Gly Leu Asn	Phe Gly Ser
2345	2350	2355
His Phe His Arg Ser Pro Glu	Val Tyr Val Pro Glu	Asp Thr Thr
2360	2365	2370
Thr Phe Leu Gly Val Ile Gly	Lys Ser Leu Ala Glu	Ser Asn Pro
2375	2380	2385
Glu Leu Thr Ala Tyr Met Phe	Ile Tyr Val Leu Glu	Ala Leu Lys
2390	2395	2400
Asn Asn Val Ile Pro His Val	Tyr Ile Pro His Thr	Ile Cys Gly
2405	2410	2415
Leu Ser Tyr Trp Ile Pro Asn	Leu Tyr Gln His Val	Tyr Leu Ala
2420	2425	2430
Asp Asp Glu Glu Gly Pro Glu	Asn Ile Ser His Ile	Phe Arg Ile
2435	2440	2445
Leu Ile Arg Leu Ser Val Arg	Glu Thr Asp Phe Lys	Ala Val Tyr
2450	2455	2460
Met Gln Tyr Val Trp Leu Leu	Leu Leu Asp Asp Gly	Arg Leu Thr
2465	2470	2475
Asp Ile Ile Val Asp Glu Val	Ile Asn His Ala Leu	Glu Arg Asp
2480	2485	2490
Ser Glu Asn Arg Asp Trp Lys	Lys Thr Ile Ser Leu	Leu Thr Val
2495	2500	2505
Leu Pro Thr Thr Glu Val Ala	Asn Asn Ile Ile Gln	Lys Ile Leu
2510	2515	2520
Ala Lys Ile Arg Ser Phe Leu	Pro Ser Leu Lys Leu	Glu Ala Met
2525	2530	2535
Thr Gln Ser Trp Ser Glu Leu	Thr Ile Leu Val Lys	Ile Ser Ile
2540	2545	2550
His Val Phe Phe Glu Thr Ser	Leu Leu Val Gln Met	Tyr Leu Pro
2555	2560	2565
Glu Ile Leu Phe Ile Val Ser	Leu Leu Ile Asp Val	Gly Pro Arg
2570	2575	2580
Glu Leu Arg Ser Ser Leu His	Gln Leu Leu Met Asn	Val Cys His
2585	2590	2595
Ser Leu Ala Ile Asn Ser Ala	Leu Pro Gln Asp His	Arg Asn Asn
2600	2605	2610
Leu Asp Glu Ile Ser Asp Ile	Phe Ala His Gln Lys	Val Lys Phe
2615	2620	2625
Met Phe Gly Phe Ser Glu Asp	Lys Gly Arg Ile Leu	Gln Ile Phe
2630	2635	2640
Ser Ala Ser Ser Phe Ala Ser	Lys Phe Asn Ile Leu	Asp Phe Phe
2645	2650	2655
Ile Asn Asn Ile Leu Leu Leu	Met Glu Tyr Ser Ser	Thr Tyr Glu
2660	2665	2670
Ala Asn Val Trp Lys Thr Arg	Tyr Lys Lys Tyr Val	Leu Glu Ser
2675	2680	2685
Val Phe Thr Ser Asn Ser Phe	Leu Ser Ala Arg Ser	Ile Met Ile
2690	2695	2700
Val Gly Ile Met Gly Lys Ser	Tyr Ile Thr Glu Gly	Leu Cys Lys
2705	2710	2715

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Ala Met	Leu Ile Glu Thr	Met	Lys Val Ile Ala	Glu	Pro Lys Ile
2720		2725		2730	
Thr Asp	Glu His Leu Phe	Leu	Ala Ile Ser His	Ile	Phe Thr Tyr
2735		2740		2745	
Ser Lys	Ile Val Glu Gly	Leu	Asp Pro Asn Leu	Asp	Leu Met Lys
2750		2755		2760	
His Leu	Phe Trp Phe Ser	Thr	Leu Phe Leu Glu	Ser	Arg His Pro
2765		2770		2775	
Ile Ile	Phe Glu Gly Ala	Leu	Leu Phe Val Ser	Asn	Cys Ile Arg
2780		2785		2790	
Arg Leu	Tyr Met Ala Gln	Phe	Glu Asn Glu Ser	Glu	Thr Ser Leu
2795		2800		2805	
Ile Ser	Thr Leu Leu Lys	Gly	Arg Lys Phe Ala	His	Thr Phe Leu
2810		2815		2820	
Ser Lys	Ile Glu Asn Leu	Ser	Gly Ile Val Trp	Asn	Glu Asp Asn
2825		2830		2835	
Phe Thr	His Ile Leu Ile	Phe	Ile Ile Asn Lys	Gly	Leu Ser Asn
2840		2845		2850	
Pro Phe	Ile Lys Ser Thr	Ala	Phe Asp Phe Leu	Lys	Met Met Phe
2855		2860		2865	
Arg Asn	Ser Tyr Phe Glu	His	Gln Ile Asn Gln	Lys	Ser Asp His
2870		2875		2880	
Tyr Leu	Cys Tyr Met Phe	Leu	Leu Tyr Phe Val	Leu	Asn Cys Asn
2885		2890		2895	
Gln Phe	Glu Glu Leu Leu	Gly	Asp Val Asp Phe	Glu	Gly Glu Met
2900		2905		2910	
Val Asn	Ile Glu Asn Lys	Asn	Thr Ile Pro Lys	Ile	Leu Leu Glu
2915		2920		2925	
Trp Leu	Ser Ser Asp Asn	Glu	Asn Ala Asn Ile	Thr	Leu Tyr Gln
2930		2935		2940	
Gly Ala	Ile Leu Phe Lys	Cys	Ser Val Thr Asp	Glu	Pro Ser Arg
2945		2950		2955	
Phe Arg	Phe Ala Leu Ile	Ile	Arg His Leu Leu	Thr	Lys Lys Pro
2960		2965		2970	
Ile Cys	Ala Leu Arg Phe	Tyr	Ser Val Ile Arg	Asn	Glu Ile Arg
2975		2980		2985	
Lys Ile	Ser Ala Phe Glu	Gln	Thr Ser Asp Cys	Val	Pro Leu Ala
2990		2995		3000	
Phe Asp	Ile Leu Asn Leu	Leu	Val Thr His Ser	Glu	Ser Asn Ser
3005		3010		3015	
Leu Glu	Lys Leu His Glu	Glu	Ser Ile Glu Arg	Leu	Thr Lys Arg
3020		3025		3030	
Gly Leu	Ser Ile Val Thr	Ser	Ser Gly Ile Phe	Ala	Lys Asn Ser
3035		3040		3045	
Asp Met	Met Ile Pro Leu	Asp	Val Lys Pro Glu	Asp	Ile Tyr Glu
3050		3055		3060	
Arg Lys	Arg Ile Met Thr	Met	Ile Leu Ser Arg	Met	Ser Cys Ser
3065		3070		3075	

Ala

<210> SEQ ID NO 23

<211> LENGTH: 3079

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

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<220> FEATURE:

<223> OTHER INFORMATION: Consensus IRA2 sequence

<400> SEQUENCE: 23

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Met Ser Gln Pro Thr Lys Asn Lys Lys Lys Glu His Gly Thr Asp Ser
1          5          10          15

Lys Ser Ser Arg Met Thr Arg Thr Leu Val Asn His Ile Leu Phe Glu
20          25          30

Arg Ile Leu Pro Ile Leu Pro Val Glu Ser Asn Leu Ser Thr Tyr Ser
35          40          45

Glu Val Glu Glu Tyr Ser Ser Phe Ile Ser Cys Arg Ser Val Leu Ile
50          55          60

Asn Val Thr Val Ser Arg Asp Ala Asn Ala Met Val Glu Gly Thr Leu
65          70          75          80

Glu Leu Ile Glu Ser Leu Leu Gln Gly His Glu Ile Ile Ser Asp Lys
85          90          95

Gly Ser Ser Asp Val Ile Glu Ser Ile Leu Ile Ile Leu Arg Leu Leu
100         105         110

Ser Asp Ala Leu Glu Tyr Asn Trp Gln Asn Gln Glu Ser Leu His Tyr
115         120         125

Asn Asp Ile Ser Thr His Val Glu His Asp Gln Glu Gln Lys Tyr Arg
130         135         140

Pro Lys Leu Asn Ser Ile Leu Pro Asp Tyr Ser Ser Thr His Ser Asn
145         150         155         160

Gly Asn Lys His Phe Phe His Gln Ser Lys Pro Gln Ala Leu Ile Pro
165         170         175

Glu Leu Ala Ser Lys Leu Leu Glu Ser Cys Ala Lys Leu Lys Phe Asn
180         185         190

Thr Arg Thr Leu Gln Ile Leu Gln Asn Met Ile Ser His Val His Gly
195         200         205

Asn Ile Leu Thr Thr Leu Ser Ser Ser Ile Leu Pro Arg His Lys Ser
210         215         220

Tyr Leu Thr Arg His Asn His Pro Ser His Cys Lys Met Ile Asp Ser
225         230         235         240

Thr Leu Gly His Ile Leu Arg Phe Val Ala Ala Ser Asn Pro Ser Glu
245         250         255

Tyr Phe Glu Phe Ile Arg Lys Ser Val Gln Val Pro Val Thr Gln Thr
260         265         270

His Thr His Ser His Ser His Ser His Ser Leu Pro Ser Ser Val Tyr
275         280         285

Asn Ser Ile Val Pro His Phe Asp Leu Phe Ser Phe Ile Tyr Leu Ser
290         295         300

Lys His Asn Phe Lys Lys Tyr Leu Glu Leu Ile Lys Asn Leu Ser Val
305         310         315         320

Thr Leu Arg Lys Thr Ile Tyr His Cys Leu Leu Leu His Tyr Ser Ala
325         330         335

Lys Ala Ile Met Phe Trp Ile Met Ala Arg Pro Ala Glu Tyr Tyr Glu
340         345         350

Leu Phe Asn Leu Leu Lys Asp Asn Asn Asn Glu His Ser Lys Ser Leu
355         360         365

Asn Thr Leu Asn His Thr Leu Phe Glu Glu Ile His Ser Thr Phe Asn
370         375         380

Val Asn Ser Met Ile Thr Thr Asn Gln Asn Ala His Gln Gly Ser Ser
385         390         395         400

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Ser	Pro	Ser	Ser	Ser	Ser	Pro	Ser	Ser	Pro	Pro	Ser	Ser	Ser	Ser	Ser	Ser	
				405													415
Asp	Asn	Asn	Asn	Gln	Asn	Ile	Ile	Ala	Lys	Ser	Leu	Ser	Arg	Gln	Leu		
				420					425					430			
Ser	His	His	Gln	Ser	Tyr	Ile	Gln	Gln	Gln	Ser	Glu	Arg	Lys	Leu	His		
				435					440					445			
Ser	Ser	Trp	Thr	Thr	Asn	Ser	Gln	Ser	Ser	Thr	Ser	Leu	Ser	Ser	Ser		
				450					455					460			
Thr	Ser	Asn	Ser	Thr	Thr	Thr	Asp	Phe	Ser	Thr	His	Thr	Gln	Pro	Gly		
465					470					475						480	
Glu	Tyr	Asp	Pro	Ser	Leu	Pro	Asp	Thr	Pro	Thr	Met	Ser	Asn	Ile	Thr		
				485					490						495		
Ile	Ser	Ala	Ser	Ser	Leu	Leu	Ser	Gln	Thr	Pro	Thr	Pro	Thr	Thr	Gln		
				500					505					510			
Leu	Gln	Gln	Arg	Leu	Asn	Ser	Ala	Ala	Ala	Ala	Ala	Ala	Ala	Ala	Ala		
				515					520					525			
Ser	Pro	Ser	Asn	Ser	Thr	Pro	Thr	Gly	Tyr	Thr	Ala	Glu	Gln	Gln	Ser		
				530					535					540			
Arg	Ala	Ser	Tyr	Asp	Ala	His	Lys	Thr	Gly	His	Thr	Gly	Lys	Asp	Tyr		
545					550					555						560	
Asp	Glu	His	Phe	Leu	Ser	Val	Thr	Arg	Leu	Asp	Asn	Val	Leu	Glu	Leu		
				565					570					575			
Tyr	Thr	His	Phe	Asp	Asp	Thr	Glu	Val	Leu	Pro	His	Thr	Ser	Val	Leu		
				580					585					590			
Lys	Phe	Leu	Thr	Thr	Leu	Thr	Met	Phe	Asp	Ile	Asp	Leu	Phe	Asn	Glu		
				595					600					605			
Leu	Asn	Ala	Thr	Ser	Phe	Lys	Tyr	Ile	Pro	Asp	Cys	Thr	Met	His	Arg		
				610					615					620			
Pro	Lys	Glu	Arg	Thr	Ser	Ser	Phe	Asn	Asn	Thr	Ala	His	Glu	Thr	Gly		
625					630					635						640	
Ser	Glu	Lys	Thr	Ser	Gly	Ile	Lys	His	Ile	Thr	Gln	Gly	Leu	Lys	Lys		
				645					650					655			
Leu	Thr	Ser	Leu	Pro	Ser	Ser	Thr	Lys	Lys	Thr	Val	Lys	Phe	Val	Lys		
				660					665					670			
Met	Leu	Leu	Arg	Asn	Leu	Asn	Gly	Asn	Gln	Ala	Val	Ser	Asp	Val	Ala		
				675					680					685			
Leu	Leu	Asp	Thr	Met	Arg	Ala	Leu	Leu	Ser	Phe	Phe	Thr	Met	Thr	Ser		
				690					695					700			
Ala	Val	Phe	Leu	Val	Asp	Arg	Asn	Leu	Pro	Ser	Val	Leu	Phe	Ala	Lys		
705					710					715						720	
Arg	Leu	Ile	Pro	Ile	Met	Gly	Thr	Asn	Leu	Ser	Val	Gly	Gln	Asp	Trp		
				725					730					735			
Asn	Ser	Lys	Ile	Asn	Asn	Ser	Leu	Met	Val	Cys	Leu	Lys	Lys	Asn	Ser		
				740					745					750			
Thr	Thr	Phe	Val	Gln	Leu	Gln	Leu	Ile	Phe	Phe	Ser	Ser	Ala	Ile	Gln		
				755					760					765			
Phe	Asp	His	Glu	Leu	Leu	Leu	Ala	Arg	Leu	Ser	Ile	Asp	Thr	Met	Ala		
				770					775					780			
Asn	Asn	Leu	Asn	Met	Gln	Lys	Leu	Cys	Leu	Tyr	Thr	Glu	Gly	Phe	Arg		

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Val Lys Ile Ser Lys Phe Phe Lys Thr Leu Phe Ser Ile Ile Ala Asp	820	825	830
Ile Leu Leu Gln Glu Phe Pro Tyr Phe Asp Glu Gln Ile Thr Asp Ile	835	840	845
Val Ala Ser Ile Leu Asp Gly Thr Ile Ile Asn Glu Tyr Gly Thr Lys	850	855	860
Lys His Phe Lys Gly Ser Ser Pro Ser Leu Cys Ser Thr Thr Arg Ser	865	870	875
Arg Ser Gly Ser Thr Ser Gln Ser Ser Met Thr Pro Val Ser Pro Leu	885	890	895
Gly Leu Asp Thr Asp Ile Cys Pro Met Asn Thr Leu Ser Leu Val Gly	900	905	910
Ser Ser Thr Ser Arg Asn Ser Asp Asn Val Asn Ser Leu Asn Ser Ser	915	920	925
Pro Lys Asn Leu Ser Ser Asp Pro Tyr Leu Ser His Leu Val Ala Pro	930	935	940
Arg Ala Arg His Ala Leu Gly Gly Pro Ser Ser Ile Ile Arg Asn Lys	945	950	955
Ile Pro Thr Thr Leu Thr Ser Pro Pro Gly Thr Glu Lys Ser Ser Pro	965	970	975
Val Gln Arg Pro Gln Thr Glu Ser Ile Ser Ala Thr Pro Met Ala Ile	980	985	990
Thr Asn Ser Thr Pro Leu Ser Ser Ala Ala Phe Gly Ile Arg Ser Pro	995	1000	1005
Leu Gln Lys Ile Arg Thr Arg Arg Tyr Ser Asp Glu Ser Leu Gly	1010	1015	1020
Lys Phe Met Lys Ser Thr Asn Asn Tyr Ile Gln Glu His Leu Ile	1025	1030	1035
Pro Lys Asp Leu Asn Glu Ala Thr Leu Gln Asp Ala Arg Arg Ile	1040	1045	1050
Met Ile Asn Ile Phe Ser Ile Phe Lys Arg Pro Asn Ser Tyr Phe	1055	1060	1065
Ile Ile Pro His Asn Ile Asn Ser Asn Leu Gln Trp Val Ser Gln	1070	1075	1080
Asp Phe Arg Asn Ile Met Lys Pro Ile Phe Val Ala Ile Val Ser	1085	1090	1095
Pro Asp Val Asp Leu Gln Asn Thr Ala Gln Ser Phe Met Asp Thr	1100	1105	1110
Leu Leu Ser Asn Val Ile Thr Tyr Gly Glu Ser Asp Glu Asn Ile	1115	1120	1125
Ser Ile Glu Gly Tyr His Leu Leu Cys Ser Tyr Thr Val Thr Leu	1130	1135	1140
Phe Ala Met Gly Leu Phe Asp Leu Lys Ile Asn Asn Glu Lys Arg	1145	1150	1155
Gln Ile Leu Leu Asp Ile Thr Val Lys Phe Met Lys Val Arg Ser	1160	1165	1170
His Leu Ala Gly Ile Ala Glu Ala Ser His His Met Glu Tyr Ile	1175	1180	1185
Ser Asp Ser Glu Lys Leu Thr Phe Pro Leu Ile Met Gly Thr Val	1190	1195	1200
Gly Arg Ala Leu Phe Val Ser Leu Tyr Ser Ser Gln Gln Lys Ile	1205	1210	1215
Glu Lys Thr Leu Lys Ile Ala Tyr Thr Glu Tyr Leu Ser Ala Ile			

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1220	1225	1230
Asn Phe His Glu Arg Asn Ile Asp Asp Ala Asp Lys Thr Trp Val		
1235	1240	1245
His Asn Ile Glu Phe Val Glu Ala Met Cys His Asp Asn Tyr Thr		
1250	1255	1260
Thr Ser Gly Ser Ile Ala Phe Gln Arg Arg Thr Arg Asn Asn Ile		
1265	1270	1275
Leu Arg Phe Ala Thr Ile Pro Asn Ala Ile Leu Leu Asp Ser Met		
1280	1285	1290
Arg Met Ile Tyr Lys Lys Trp His Thr Tyr Thr His Ser Lys Ser		
1295	1300	1305
Leu Glu Lys Gln Glu Arg Asn Asp Phe Arg Asn Phe Ala Gly Ile		
1310	1315	1320
Leu Ala Ser Leu Ser Gly Ile Leu Phe Ile Asn Lys Lys Ile Leu		
1325	1330	1335
Gln Glu Met Tyr Pro Tyr Leu Leu Asp Thr Val Ser Glu Leu Lys		
1340	1345	1350
Lys Asn Ile Asp Ser Phe Ile Ser Lys Gln Cys Gln Trp Leu Asn		
1355	1360	1365
Tyr Pro Asp Leu Leu Thr Arg Glu Asn Ser Arg Asp Ile Leu Ser		
1370	1375	1380
Val Glu Leu His Pro Leu Ser Phe Asn Leu Leu Phe Asn Asn Leu		
1385	1390	1395
Arg Leu Lys Leu Lys Glu Leu Ala Cys Ser Asp Leu Ser Ile Pro		
1400	1405	1410
Glu Asn Glu Ser Ser Tyr Val Leu Leu Glu Gln Ile Ile Lys Met		
1415	1420	1425
Leu Arg Thr Ile Leu Gly Arg Asp Asp Asp Asn Tyr Val Met Met		
1430	1435	1440
Leu Phe Ser Thr Glu Ile Val Asp Leu Ile Asp Leu Leu Thr Asp		
1445	1450	1455
Glu Ile Lys Lys Ile Pro Ala Tyr Cys Pro Lys Tyr Leu Lys Ala		
1460	1465	1470
Ile Ile Gln Met Thr Lys Met Phe Ser Ala Leu Gln His Ser Glu		
1475	1480	1485
Val Asn Leu Gly Val Lys Asn His Phe His Val Lys Asn Lys Trp		
1490	1495	1500
Leu Arg Gln Ile Thr Asp Trp Phe Gln Val Ser Ile Ala Arg Glu		
1505	1510	1515
Tyr Asp Phe Glu Asn Leu Ser Lys Pro Leu Lys Glu Met Asp Leu		
1520	1525	1530
Val Lys Arg Asp Met Asp Ile Leu Tyr Ile Asp Thr Ala Ile Glu		
1535	1540	1545
Ala Ser Thr Ala Ile Ala Tyr Leu Thr Arg His Thr Phe Leu Glu		
1550	1555	1560
Ile Pro Pro Ala Ala Ser Asp Pro Glu Leu Ser Arg Ser Arg Ser		
1565	1570	1575
Val Ile Phe Gly Phe Tyr Phe Asn Ile Leu Met Lys Gly Leu Glu		
1580	1585	1590
Lys Ser Ser Asp Arg Asp Asn Tyr Pro Val Phe Leu Arg His Lys		
1595	1600	1605
Met Ser Val Leu Asn Asp Asn Val Ile Leu Ser Leu Thr Asn Leu		
1610	1615	1620

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Ser	Asn	Thr	Asn	Val	Asp	Ala	Ser	Leu	Gln	Phe	Thr	Leu	Pro	Met
1625						1630					1635			
Gly	Tyr	Ser	Gly	Asn	Arg	Asn	Ile	Arg	Asn	Ala	Phe	Leu	Glu	Val
1640						1645					1650			
Phe	Ile	Asn	Ile	Val	Thr	Asn	Tyr	Arg	Thr	Tyr	Thr	Ala	Lys	Thr
1655						1660					1665			
Asp	Leu	Gly	Lys	Leu	Glu	Ala	Ala	Asp	Lys	Phe	Leu	Arg	Tyr	Thr
1670						1675					1680			
Ile	Glu	His	Pro	Gln	Leu	Ser	Ser	Phe	Gly	Ala	Ala	Val	Cys	Pro
1685						1690					1695			
Ala	Ser	Asp	Ile	Asp	Ala	Tyr	Ala	Ala	Gly	Leu	Ile	Asn	Ala	Phe
1700						1705					1710			
Glu	Thr	Arg	Asn	Ala	Thr	His	Ile	Val	Val	Ala	Gln	Leu	Ile	Lys
1715						1720					1725			
Asn	Glu	Ile	Glu	Lys	Ser	Ser	Arg	Pro	Thr	Asp	Ile	Leu	Arg	Arg
1730						1735					1740			
Asn	Ser	Cys	Ala	Thr	Arg	Ser	Leu	Ser	Met	Leu	Ala	Arg	Ser	Lys
1745						1750					1755			
Gly	Asn	Glu	Tyr	Leu	Ile	Arg	Thr	Leu	Gln	Pro	Leu	Leu	Lys	Lys
1760						1765					1770			
Ile	Ile	Gln	Asn	Arg	Asp	Phe	Phe	Glu	Ile	Glu	Lys	Leu	Lys	Pro
1775						1780					1785			
Glu	Asp	Ser	Asp	Ala	Glu	Arg	Gln	Ile	Glu	Leu	Phe	Val	Lys	Tyr
1790						1795					1800			
Met	Asn	Glu	Leu	Leu	Glu	Ser	Ile	Ser	Asn	Ser	Val	Ser	Tyr	Phe
1805						1810					1815			
Pro	Pro	Pro	Leu	Phe	Tyr	Ile	Cys	Gln	Asn	Ile	Tyr	Lys	Val	Ala
1820						1825					1830			
Cys	Glu	Lys	Phe	Pro	Asp	His	Ala	Ile	Ile	Ala	Ala	Gly	Ser	Phe
1835						1840					1845			
Val	Phe	Leu	Arg	Phe	Phe	Cys	Pro	Ala	Leu	Val	Ser	Pro	Asp	Ser
1850						1855					1860			
Glu	Asn	Ile	Ile	Asp	Ile	Ser	His	Leu	Ser	Glu	Lys	Arg	Thr	Phe
1865						1870					1875			
Ile	Ser	Leu	Ala	Lys	Val	Ile	Gln	Asn	Ile	Ala	Asn	Gly	Ser	Glu
1880						1885					1890			
Asn	Phe	Ser	Arg	Trp	Pro	Ala	Leu	Cys	Ser	Gln	Lys	Asp	Phe	Leu
1895						1900					1905			
Lys	Glu	Cys	Ser	Asp	Arg	Ile	Phe	Arg	Phe	Leu	Ala	Glu	Leu	Cys
1910						1915					1920			
Arg	Thr	Asp	Arg	Thr	Ile	Asp	Ile	Gln	Val	Arg	Thr	Asp	Pro	Thr
1925						1930					1935			
Pro	Ile	Ala	Phe	Asp	Tyr	Gln	Phe	Leu	His	Ser	Phe	Val	Tyr	Leu
1940						1945					1950			
Tyr	Gly	Leu	Glu	Val	Arg	Arg	Asn	Val	Leu	Asn	Glu	Ala	Lys	His
1955						1960					1965			
Asp	Asp	Gly	Asp	Ile	Asp	Gly	Asp	Asp	Phe	Tyr	Lys	Thr	Thr	Phe
1970						1975					1980			
Leu	Leu	Ile	Asp	Asp	Val	Leu	Gly	Gln	Leu	Gly	Gln	Pro	Lys	Met
1985						1990					1995			
Glu	Phe	Ser	Asn	Glu	Ile	Pro	Ile	Tyr	Ile	Arg	Glu	His	Met	Asp
2000						2005					2010			

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Asp	Tyr	Pro	Glu	Leu	Tyr	Glu	Phe	Met	Asn	Arg	His	Ala	Phe	Arg
2015						2020					2025			
Asn	Ile	Glu	Thr	Ser	Thr	Ala	Tyr	Ser	Pro	Ser	Val	His	Glu	Ser
2030						2035					2040			
Thr	Ser	Ser	Glu	Gly	Ile	Pro	Ile	Ile	Thr	Leu	Thr	Met	Ser	Asn
2045						2050					2055			
Phe	Ser	Asp	Arg	His	Val	Asp	Ile	Asp	Thr	Val	Ala	Tyr	Lys	Phe
2060						2065					2070			
Leu	Gln	Ile	Tyr	Ala	Arg	Ile	Trp	Thr	Thr	Lys	His	Cys	Leu	Ile
2075						2080					2085			
Ile	Asp	Cys	Thr	Glu	Phe	Asp	Glu	Gly	Gly	Leu	Asp	Met	Arg	Lys
2090						2095					2100			
Phe	Ile	Ser	Leu	Val	Met	Gly	Leu	Leu	Pro	Glu	Val	Ala	Pro	Lys
2105						2110					2115			
Asn	Cys	Ile	Gly	Cys	Tyr	Tyr	Phe	Asn	Val	Asn	Glu	Thr	Phe	Met
2120						2125					2130			
Asp	Asn	Tyr	Gly	Lys	Cys	Leu	Asp	Lys	Asp	Asn	Val	Tyr	Val	Ser
2135						2140					2145			
Ser	Lys	Ile	Pro	His	Tyr	Phe	Ile	Asn	Ser	Asn	Ser	Asp	Glu	Gly
2150						2155					2160			
Leu	Met	Lys	Ser	Val	Gly	Ile	Thr	Gly	Gln	Gly	Leu	Lys	Val	Leu
2165						2170					2175			
Gln	Asp	Ile	Arg	Val	Ser	Leu	His	Asp	Ile	Thr	Leu	Tyr	Asp	Glu
2180						2185					2190			
Lys	Arg	Asn	Arg	Phe	Thr	Pro	Val	Ser	Leu	Lys	Ile	Gly	Asp	Ile
2195						2200					2205			
Tyr	Phe	Gln	Val	Leu	His	Glu	Thr	Pro	Arg	Gln	Tyr	Lys	Ile	Arg
2210						2215					2220			
Asp	Met	Gly	Thr	Leu	Phe	Asp	Val	Lys	Phe	Asn	Asp	Val	Tyr	Glu
2225						2230					2235			
Ile	Ser	Arg	Ile	Phe	Glu	Val	His	Val	Ser	Ser	Ile	Thr	Gly	Val
2240						2245					2250			
Ala	Ala	Glu	Phe	Thr	Val	Thr	Phe	Gln	Asp	Glu	Arg	Arg	Leu	Ile
2255						2260					2265			
Phe	Ser	Ser	Pro	Lys	Tyr	Leu	Glu	Ile	Val	Lys	Met	Phe	Tyr	Tyr
2270						2275					2280			
Ala	Gln	Ile	Arg	Leu	Glu	Ser	Glu	Tyr	Glu	Met	Asp	Asn	Asn	Ser
2285						2290					2295			
Ser	Thr	Ser	Ser	Pro	Asn	Ser	Asn	Asn	Lys	Asp	Lys	Gln	Gln	Lys
2300						2305					2310			
Glu	Arg	Thr	Lys	Leu	Leu	Cys	His	Leu	Leu	Leu	Val	Ser	Leu	Ile
2315						2320					2325			
Gly	Leu	Phe	Asp	Glu	Ser	Lys	Lys	Met	Lys	Asn	Ser	Ser	Tyr	Asn
2330						2335					2340			
Leu	Ile	Ala	Ala	Thr	Glu	Ala	Ser	Phe	Gly	Leu	Asn	Phe	Gly	Ser
2345						2350					2355			
His	Phe	His	Arg	Ser	Pro	Glu	Val	Tyr	Val	Pro	Glu	Asp	Thr	Thr
2360						2365					2370			
Thr	Phe	Leu	Gly	Val	Ile	Gly	Lys	Ser	Leu	Ala	Glu	Ser	Asn	Pro
2375						2380					2385			
Glu	Leu	Thr	Ala	Tyr	Met	Phe	Ile	Tyr	Val	Leu	Glu	Ala	Leu	Lys
2390						2395					2400			
Asn	Asn	Val	Ile	Pro	His	Val	Tyr	Ile	Pro	His	Thr	Ile	Cys	Gly

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2405	2410	2415
Leu Ser Tyr Trp Ile Pro Asn	Leu Tyr Gln His Val	Tyr Leu Ala
2420	2425	2430
Asp Asp Glu Glu Gly Pro Lys	Asn Ile Ser His Ile	Phe Arg Ile
2435	2440	2445
Leu Ile Arg Leu Ser Val Arg	Glu Thr Asp Phe Lys	Ala Val Tyr
2450	2455	2460
Met Gln Tyr Val Trp Leu Leu	Leu Leu Asp Asp Gly	Arg Leu Thr
2465	2470	2475
Asp Ile Ile Val Asp Glu Val	Ile Asn His Ala Leu	Glu Arg Asp
2480	2485	2490
Ser Glu Asn Arg Asp Trp Lys	Lys Thr Ile Ser Leu	Leu Thr Val
2495	2500	2505
Leu Pro Thr Thr Glu Val Ala	Asn Asn Ile Ile Gln	Lys Ile Leu
2510	2515	2520
Ala Lys Ile Arg Ser Phe Leu	Pro Ser Leu Lys Leu	Glu Ala Met
2525	2530	2535
Thr Gln Ser Trp Ser Glu Leu	Thr Ile Leu Val Lys	Ile Ser Ile
2540	2545	2550
His Val Phe Phe Glu Thr Ser	Leu Leu Val Gln Met	Tyr Leu Pro
2555	2560	2565
Glu Ile Leu Phe Ile Val Ser	Leu Leu Ile Asp Val	Gly Pro Arg
2570	2575	2580
Glu Leu Arg Ser Ser Leu His	Gln Leu Leu Met Asn	Val Cys His
2585	2590	2595
Ser Leu Ala Ile Asn Ser Ala	Leu Pro Gln Asp His	Arg Asn Asn
2600	2605	2610
Leu Asp Glu Ile Ser Asp Ile	Phe Ala His Gln Lys	Val Lys Phe
2615	2620	2625
Met Phe Gly Phe Ser Glu Asp	Lys Gly Arg Ile Leu	Gln Ile Phe
2630	2635	2640
Ser Ala Ser Ser Phe Ala Ser	Lys Phe Asn Ile Leu	Asp Phe Phe
2645	2650	2655
Ile Asn Asn Ile Leu Leu Leu	Met Glu Tyr Ser Ser	Thr Tyr Glu
2660	2665	2670
Ala Asn Val Trp Lys Thr Arg	Tyr Lys Lys Tyr Val	Leu Glu Ser
2675	2680	2685
Val Phe Thr Ser Asn Ser Phe	Leu Ser Ala Arg Ser	Ile Met Ile
2690	2695	2700
Val Gly Ile Met Gly Lys Ser	Tyr Ile Thr Glu Gly	Leu Cys Lys
2705	2710	2715
Ala Met Leu Ile Glu Thr Met	Lys Val Ile Ala Glu	Pro Lys Ile
2720	2725	2730
Thr Asp Glu His Leu Phe Leu	Ala Ile Ser His Ile	Phe Thr Tyr
2735	2740	2745
Ser Lys Ile Val Glu Gly Leu	Asp Pro Asn Leu Asp	Leu Met Lys
2750	2755	2760
His Leu Phe Trp Phe Ser Thr	Leu Phe Leu Glu Ser	Arg His Pro
2765	2770	2775
Ile Ile Phe Glu Gly Ala Leu	Leu Phe Val Ser Asn	Cys Ile Arg
2780	2785	2790
Arg Leu Tyr Met Ala Gln Phe	Glu Asn Glu Ser Glu	Thr Ser Leu
2795	2800	2805

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Ile Ser Thr Leu Leu Lys Gly Arg Lys Phe Ala His Thr Phe Leu		
2810	2815	2820
Ser Lys Ile Glu Asn Leu Ser Gly Ile Val Trp Asn Glu Asp Asn		
2825	2830	2835
Phe Thr His Ile Leu Ile Phe Ile Ile Asn Lys Gly Leu Ser Asn		
2840	2845	2850
Pro Phe Ile Lys Ser Thr Ala Phe Asp Phe Leu Lys Met Met Phe		
2855	2860	2865
Arg Asn Ser Tyr Phe Glu His Gln Ile Asn Gln Lys Ser Asp His		
2870	2875	2880
Tyr Leu Cys Tyr Met Phe Leu Leu Tyr Phe Val Leu Asn Cys Asn		
2885	2890	2895
Gln Phe Glu Glu Leu Leu Gly Asp Val Asp Phe Glu Gly Glu Met		
2900	2905	2910
Val Asn Ile Glu Asn Lys Asn Thr Ile Pro Lys Ile Leu Leu Glu		
2915	2920	2925
Trp Leu Ser Ser Asp Asn Glu Asn Ala Asn Ile Thr Leu Tyr Gln		
2930	2935	2940
Gly Ala Ile Leu Phe Lys Cys Ser Val Thr Asp Glu Pro Ser Arg		
2945	2950	2955
Phe Arg Phe Ala Leu Ile Ile Arg His Leu Leu Thr Lys Lys Pro		
2960	2965	2970
Ile Cys Ala Leu Arg Phe Tyr Ser Val Ile Arg Asn Glu Ile Arg		
2975	2980	2985
Lys Ile Ser Ala Phe Glu Gln Asn Ser Asp Cys Val Pro Leu Ala		
2990	2995	3000
Phe Asp Ile Leu Asn Leu Leu Val Thr His Ser Glu Ser Asn Ser		
3005	3010	3015
Leu Glu Lys Leu His Glu Glu Ser Ile Glu Arg Leu Thr Lys Arg		
3020	3025	3030
Gly Leu Ser Ile Val Thr Ser Ser Gly Ile Phe Ala Lys Asn Ser		
3035	3040	3045
Asp Met Met Ile Pro Leu Asp Val Lys Pro Glu Asp Ile Tyr Glu		
3050	3055	3060
Arg Lys Arg Ile Met Thr Met Ile Leu Ser Arg Met Ser Cys Ser		
3065	3070	3075

Ala

<210> SEQ ID NO 24
 <211> LENGTH: 293
 <212> TYPE: PRT
 <213> ORGANISM: *Saccharomyces cerevisiae*

<400> SEQUENCE: 24

Met Ser Gln Pro Thr Lys Asn Lys Lys Lys Glu His Gly Thr Asp Ser		
1	5	10 15
Lys Ser Ser Arg Met Thr Arg Thr Leu Val Asn His Ile Leu Phe Glu		
	20	25 30
Arg Ile Leu Pro Ile Leu Pro Val Glu Ser Asn Leu Ser Thr Tyr Ser		
	35	40 45
Glu Val Glu Glu Tyr Ser Ser Phe Ile Ser Cys Arg Ser Val Leu Ile		
	50	55 60
Asn Val Thr Val Ser Gln Asp Ala Asn Ala Met Val Glu Gly Thr Leu		
65	70	75 80

[illegible]

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<210> SEQ ID NO 25
<211> LENGTH: 3081
<212> TYPE: PRT
<213> ORGANISM: Saccharomyces cerevisiae

<400> SEQUENCE: 25
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Met	Ser	Gln	Pro	Thr	Lys	Asn	Lys	Lys	Lys	Glu	His	Gly	Thr	Asp	Ser
1				5					10					15	
Lys	Ser	Ser	Arg	Met	Thr	Arg	Thr	Leu	Val	Asn	His	Ile	Leu	Phe	Glu
			20					25					30		
Arg	Ile	Leu	Pro	Ile	Leu	Pro	Val	Glu	Ser	Asn	Leu	Ser	Thr	Tyr	Ser
		35					40					45			
Glu	Val	Glu	Glu	Tyr	Ser	Ser	Phe	Ile	Ser	Cys	Arg	Ser	Val	Leu	Ile
		50				55					60				
Asn	Val	Thr	Val	Ser	Gln	Asp	Ala	Asn	Ala	Met	Val	Glu	Gly	Thr	Leu
65					70					75					80
Glu	Leu	Ile	Glu	Ser	Leu	Leu	Gln	Gly	His	Glu	Ile	Ile	Ser	Asp	Lys
				85					90					95	
Cys	Ser	Ser	Asp	Val	Ile	Glu	Ser	Ile	Leu	Ile	Ile	Leu	Arg	Leu	Leu
			100					105					110		
Ser	Asp	Ala	Leu	Glu	Tyr	Asn	Trp	Gln	Asn	Gln	Glu	Ser	Leu	His	Tyr
		115					120					125			
Asn	Asp	Ile	Ser	Thr	His	Val	Glu	His	Asp	Gln	Glu	Gln	Lys	Tyr	Arg
	130					135					140				
Pro	Lys	Leu	Asn	Ser	Ile	Leu	Pro	Asp	Tyr	Ser	Ser	Thr	His	Ser	Asn

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145					150						155					160
Gly	Asn	Lys	His	Phe	Phe	His	Gln	Ser	Lys	Pro	Gln	Ala	Leu	Ile	Pro	
				165					170					175		
Glu	Leu	Ala	Ser	Lys	Leu	Leu	Glu	Ser	Cys	Ala	Lys	Leu	Lys	Phe	Asn	
			180					185					190			
Thr	Arg	Thr	Leu	Gln	Ile	Leu	Gln	Asn	Met	Ile	Ser	His	Val	His	Gly	
	195						200					205				
Asn	Ile	Leu	Thr	Thr	Leu	Ser	Ser	Ser	Ile	Leu	Pro	Arg	His	Lys	Ser	
	210					215					220					
Tyr	Leu	Thr	Arg	His	Asn	His	Pro	Ser	His	Cys	Lys	Met	Ile	Asp	Ser	
225					230					235				240		
Thr	Leu	Gly	His	Ile	Leu	Arg	Phe	Val	Ala	Ala	Ser	Asn	Pro	Ser	Glu	
			245						250					255		
Tyr	Phe	Glu	Phe	Ile	Arg	Lys	Ser	Val	Gln	Val	Pro	Val	Thr	Gln	Thr	
		260						265					270			
His	Thr	His	Thr	His	Ser	His	Ser	His	Ser	His	Ser	Leu	Pro	Ser	Ser	
	275						280					285				
Val	Tyr	Asn	Ser	Ile	Val	Pro	His	Phe	Asp	Leu	Phe	Ser	Phe	Ile	Tyr	
	290					295					300					
Leu	Ser	Lys	His	Asn	Phe	Lys	Lys	Tyr	Leu	Glu	Leu	Ile	Lys	Asn	Leu	
305					310					315				320		
Ser	Val	Thr	Leu	Arg	Lys	Thr	Ile	Tyr	His	Cys	Leu	Leu	Leu	His	Tyr	
			325						330					335		
Ser	Ala	Lys	Ala	Ile	Met	Phe	Trp	Ile	Met	Ala	Arg	Pro	Ala	Glu	Tyr	
		340						345					350			
Tyr	Glu	Leu	Phe	Asn	Leu	Leu	Lys	Asp	Asn	Asn	Asn	Glu	His	Ser	Lys	
	355						360					365				
Ser	Leu	Asn	Thr	Leu	Asn	His	Thr	Leu	Phe	Glu	Glu	Ile	His	Ser	Thr	
	370					375					380					
Phe	Asn	Val	Asn	Ser	Met	Ile	Thr	Thr	Asn	Gln	Asn	Ala	His	Gln	Gly	
385					390					395				400		
Ser	Ser	Ser	Pro	Ser	Ser	Ser	Ser	Pro	Ser	Ser	Pro	Pro	Ser	Ser	Ser	
			405						410					415		
Ser	Ser	Asp	Asn	Asn	Asn	Gln	Asn	Ile	Ile	Ala	Lys	Ser	Leu	Ser	Arg	
		420						425					430			
Gln	Leu	Ser	His	His	Gln	Ser	Tyr	Ile	Gln	Gln	Gln	Ser	Glu	Arg	Lys	
	435						440						445			
Leu	His	Ser	Ser	Trp	Thr	Thr	Asn	Ser	Gln	Ser	Ser	Thr	Ser	Leu	Ser	
	450					455					460					
Ser	Ser	Thr	Ser	Asn	Ser	Thr	Thr	Thr	Asp	Phe	Ser	Thr	His	Thr	Gln	
465				470						475				480		
Pro	Gly	Glu	Tyr	Asp	Pro	Ser	Leu	Pro	Asp	Thr	Pro	Thr	Met	Ser	Asn	
			485						490					495		
Ile	Thr	Ile	Ser	Ala	Ser	Ser	Leu	Leu	Ser	Gln	Thr	Pro	Thr	Pro	Thr	
		500						505					510			
Thr	Gln	Leu	Gln	Gln	Arg	Leu	Asn	Ser	Ala	Ala	Ala	Ala	Ala	Ala	Ala	
	515						520					525				
Ala	Ala	Ser	Pro	Ser	Asn	Ser	Thr	Pro	Thr	Gly	Tyr	Thr	Ala	Glu	Gln	
	530					535					540					
Gln	Ser	Arg	Ala	Ser	Tyr	Asp	Ala	His	Lys	Thr	Gly	His	Thr	Gly	Lys	
545					550					555				560		
Asp	Tyr	Asp	Glu	His	Phe	Leu	Ser	Val	Thr	Arg	Leu	Asp	Asn	Val	Leu	
			565						570					575		

Glu 580	Leu	Tyr	Thr	His	Phe	Asp	Asp	Thr	Glu	Val	Leu	Pro	His	Thr	Ser
Val 595	Leu	Lys	Phe	Leu	Thr	Thr	Leu	Thr	Met	Phe	Asp	Ile	Asp	Leu	Phe
Asn 610	Glu	Leu	Asn	Ala	Thr	Ser	Phe	Lys	Tyr	Ile	Pro	Asp	Cys	Thr	Met
His 625	Arg	Pro	Lys	Glu	Arg	Thr	Ser	Ser	Phe	Asn	Asn	Thr	Ala	His	Glu
Thr 640	Gly	Ser	Glu	Lys	Thr	Ser	Gly	Ile	Lys	His	Ile	Thr	Gln	Gly	Leu
Lys 660	Lys	Leu	Thr	Ser	Leu	Pro	Ser	Ser	Thr	Lys	Lys	Thr	Val	Lys	Phe
Met 675	Lys	Met	Leu	Leu	Arg	Asn	Leu	Asn	Gly	Asn	Gln	Ala	Val	Ser	Asp
Val 690	Ala	Leu	Leu	Asp	Thr	Met	Arg	Ala	Leu	Leu	Ser	Phe	Phe	Thr	Met
Thr 705	Ser	Ala	Val	Phe	Leu	Val	Asp	Arg	Asn	Leu	Pro	Ser	Val	Leu	Phe
Ala 720	Lys	Arg	Leu	Ile	Pro	Ile	Met	Gly	Thr	Asn	Leu	Ser	Val	Gly	Gln
Asp 735	Trp	Asn	Ser	Lys	Ile	Asn	Asn	Ser	Leu	Met	Val	Cys	Leu	Lys	Lys
Asn 750	Ser	Thr	Thr	Phe	Val	Gln	Leu	Gln	Leu	Ile	Phe	Phe	Ser	Ser	Ala
Ile 770	Gln	Phe	Asp	His	Glu	Leu	Leu	Leu	Ala	Arg	Leu	Ser	Ile	Asp	Thr
Met 785	Ala	Asn	Asn	Leu	Asn	Met	Gln	Lys	Leu	Cys	Leu	Tyr	Thr	Glu	Gly
Phe 800	Arg	Ile	Phe	Phe	Asp	Ile	Pro	Ser	Lys	Lys	Glu	Leu	Arg	Lys	Ala
Ile 815	Ala	Val	Lys	Ile	Ser	Lys	Phe	Phe	Lys	Thr	Leu	Phe	Ser	Ile	Ile
Ala 830	Asp	Ile	Leu	Leu	Gln	Glu	Phe	Pro	Tyr	Phe	Asp	Glu	Gln	Ile	Thr
Asp 845	Ile	Val	Ala	Ser	Ile	Leu	Asp	Gly	Thr	Ile	Ile	Asn	Glu	Tyr	Gly
Thr 860	Lys	Lys	His	Phe	Lys	Gly	Ser	Ser	Pro	Ser	Leu	Cys	Ser	Thr	Thr
Arg 875	Ser	Arg	Ser	Gly	Ser	Thr	Ser	Gln	Ser	Ser	Met	Thr	Pro	Val	Ser
Pro 890	Leu	Gly	Leu	Asp	Thr	Asp	Ile	Arg	Pro	Met	Asn	Thr	Leu	Ser	Leu
Val 905	Gly	Ser	Ser	Thr	Ser	Arg	Asn	Ser	Asp	Asn	Val	Asn	Ser	Leu	Asn
Ser 920	Ser	Pro	Lys	Asn	Leu	Ser	Ser	Asp	Pro	Tyr	Leu	Ser	His	Leu	Val
Ala 935	Pro	Arg	Ala	Arg	His	Ala	Leu	Gly	Gly	Pro	Ser	Ser	Ile	Ile	Arg
Asn 950	Lys	Ile	Pro	Thr	Thr	Leu	Thr	Ser	Pro	Pro	Gly	Thr	Glu	Lys	Ser
Ser 965	Pro	Val	Gln	Arg	Pro	Gln	Thr	Glu	Ser	Ile	Ser	Ala	Thr	Pro	Met

Ala	Ile	Thr	Asn	Ser	Thr	Pro	Leu	Ser	Ser	Ala	Ala	Phe	Gly	Ile	Arg
995			1000					1005							
Ser	Pro		Leu	Gln	Lys	Ile	Arg	Thr	Arg	Arg	Tyr	Ser	Asp	Glu	Ser
	1010						1015					1020			
Leu	Gly		Lys	Phe	Met	Lys	Ser	Thr	Asn	Asn	Tyr	Ile	Gln	Glu	His
	1025						1030					1035			
Leu	Ile		Pro	Lys	Asp	Leu	Asn	Glu	Ala	Thr	Leu	Gln	Asp	Ala	Arg
	1040						1045					1050			
Arg	Ile		Met	Ile	Asn	Ile	Phe	Ser	Ile	Phe	Lys	Arg	Pro	Asn	Ser
	1055						1060					1065			
Tyr	Phe		Ile	Ile	Pro	His	Asn	Ile	Asn	Ser	Asn	Leu	Gln	Trp	Val
	1070						1075					1080			
Ser	Gln		Asp	Phe	Arg	Asn	Ile	Met	Lys	Pro	Ile	Phe	Val	Ala	Ile
	1085						1090					1095			
Val	Ser		Pro	Asp	Val	Asp	Leu	Gln	Asn	Thr	Ala	Gln	Ser	Phe	Met
	1100						1105					1110			
Asp	Thr		Leu	Leu	Ser	Asn	Val	Ile	Thr	Tyr	Gly	Glu	Ser	Asp	Glu
	1115						1120					1125			
Asn	Ile		Ser	Ile	Glu	Gly	Tyr	His	Leu	Leu	Cys	Ser	Tyr	Thr	Val
	1130						1135					1140			
Thr	Leu		Phe	Ala	Met	Gly	Leu	Phe	Asp	Leu	Lys	Ile	Asn	Asn	Glu
	1145						1150					1155			
Lys	Arg		Gln	Ile	Leu	Leu	Asp	Ile	Thr	Val	Lys	Phe	Met	Lys	Val
	1160						1165					1170			
Arg	Ser		His	Leu	Ala	Gly	Ile	Ala	Glu	Ala	Ser	His	His	Met	Glu
	1175						1180					1185			
Tyr	Ile		Ser	Asp	Ser	Glu	Lys	Leu	Thr	Phe	Pro	Leu	Ile	Met	Gly
	1190						1195					1200			
Thr	Val		Gly	Arg	Ala	Leu	Phe	Val	Ser	Leu	Tyr	Ser	Ser	Gln	Gln
	1205						1210					1215			
Lys	Ile		Glu	Lys	Thr	Leu	Asn	Ile	Ala	Tyr	Thr	Glu	Tyr	Leu	Ser
	1220						1225					1230			
Ala	Ile		Asn	Phe	His	Glu	Arg	Asn	Ile	Asp	Asp	Ala	Asp	Lys	Thr
	1235						1240					1245			
Trp	Val		His	Asn	Ile	Glu	Phe	Val	Glu	Ala	Met	Cys	His	Asp	Asn
	1250						1255					1260			
Tyr	Thr		Thr	Ser	Gly	Ser	Ile	Ala	Phe	Gln	Arg	Arg	Thr	Arg	Asn
	1265						1270					1275			
Asn	Ile		Leu	Arg	Phe	Ala	Thr	Ile	Pro	Asn	Ala	Ile	Leu	Leu	Asp
	1280						1285					1290			
Ser	Met		Arg	Met	Ile	Tyr	Lys	Lys	Trp	His	Thr	Tyr	Thr	His	Ser
	1295						1300					1305			
Lys	Ser		Leu	Glu	Lys	Gln	Glu	Arg	Asn	Asp	Phe	Arg	Asn	Phe	Ala
	1310						1315					1320			
Gly	Ile		Leu	Ala	Ser	Leu	Ser	Gly	Ile</						

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1385	1390	1395
Asn Leu Arg Leu Lys Leu Lys Glu Leu Ala Cys Ser Asp Leu Ser		
1400	1405	1410
Ile Pro Glu Asn Glu Ser Ser Tyr Val Leu Leu Glu Gln Ile Ile		
1415	1420	1425
Lys Met Leu Arg Thr Ile Leu Gly Arg Asp Asp Asp Asn Tyr Val		
1430	1435	1440
Met Met Leu Phe Ser Thr Glu Ile Val Asp Leu Ile Asp Leu Leu		
1445	1450	1455
Thr Asp Glu Ile Lys Lys Ile Pro Ala Tyr Cys Pro Lys Tyr Leu		
1460	1465	1470
Lys Ala Ile Ile Gln Met Thr Lys Met Phe Ser Ala Leu Gln His		
1475	1480	1485
Ser Glu Val Asn Leu Gly Val Lys Asn His Phe His Val Lys Asn		
1490	1495	1500
Lys Trp Leu Arg Gln Ile Thr Asp Trp Phe Gln Val Ser Ile Ala		
1505	1510	1515
Arg Glu Tyr Asp Phe Glu Asn Leu Ser Lys Pro Leu Lys Glu Met		
1520	1525	1530
Asp Leu Val Lys Arg Asp Met Asp Ile Leu Tyr Ile Asp Thr Ala		
1535	1540	1545
Ile Glu Ala Ser Thr Ala Ile Ala Tyr Leu Thr Arg His Thr Phe		
1550	1555	1560
Leu Glu Ile Pro Pro Ala Ala Ser Asp Pro Glu Leu Ser Arg Ser		
1565	1570	1575
Arg Ser Val Ile Phe Gly Phe Tyr Phe Asn Ile Leu Met Lys Gly		
1580	1585	1590
Leu Glu Lys Ser Ser Asp Arg Asp Asn Tyr Pro Val Phe Leu Arg		
1595	1600	1605
His Lys Met Ser Val Leu Asn Asp Asn Val Ile Leu Ser Leu Thr		
1610	1615	1620
Asn Leu Ser Asn Thr Asn Val Asp Ala Ser Leu Gln Phe Thr Leu		
1625	1630	1635
Pro Met Gly Tyr Ser Gly Asn Arg Asn Ile Arg Asn Ala Phe Leu		
1640	1645	1650
Glu Val Phe Ile Asn Ile Val Thr Asn Tyr Arg Thr Tyr Thr Ala		
1655	1660	1665
Lys Thr Asp Leu Gly Lys Leu Glu Ala Ala Asp Lys Phe Leu Arg		
1670	1675	1680
Tyr Thr Ile Glu His Pro Gln Leu Ser Ser Phe Gly Ala Ala Val		
1685	1690	1695
Cys Pro Ala Ser Asp Ile Asp Ala Tyr Ala Ala Gly Leu Ile Asn		
1700	1705	1710
Ala Phe Glu Thr Arg Asn Ala Thr His Ile Val Val Ala Gln Leu		
1715	1720	1725
Ile Lys Asn Glu Ile Glu Lys Ser Ser Arg Pro Thr Asp Ile Leu		
1730	1735	1740
Arg Arg Asn Ser Cys Ala Thr Arg Ser Leu Ser Met Leu Ala Arg		
1745	1750	1755
Ser Lys Gly Asn Glu Tyr Leu Ile Arg Thr Leu Gln Pro Leu Leu		
1760	1765	1770
Lys Lys Ile Ile Gln Asn Arg Asp Phe Phe Glu Ile Glu Lys Leu		
1775	1780	1785

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Lys	Pro	Glu	Asp	Ser	Asp	Ala	Glu	Arg	Gln	Ile	Glu	Leu	Phe	Val
1790						1795					1800			
Lys	Tyr	Met	Asn	Glu	Leu	Leu	Glu	Ser	Ile	Ser	Asn	Ser	Val	Ser
1805						1810					1815			
Tyr	Phe	Pro	Pro	Pro	Leu	Phe	Tyr	Ile	Cys	Gln	Asn	Ile	Tyr	Lys
1820						1825					1830			
Val	Ala	Cys	Glu	Lys	Phe	Pro	Asp	His	Ala	Ile	Ile	Ala	Ala	Gly
1835						1840					1845			
Ser	Phe	Val	Phe	Leu	Arg	Phe	Phe	Cys	Pro	Ala	Leu	Val	Ser	Pro
1850						1855					1860			
Asp	Ser	Glu	Asn	Ile	Ile	Asp	Ile	Ser	His	Leu	Ser	Glu	Lys	Arg
1865						1870					1875			
Thr	Phe	Ile	Ser	Leu	Ala	Lys	Val	Ile	Gln	Asn	Ile	Ala	Asn	Gly
1880						1885					1890			
Ser	Glu	Asn	Phe	Ser	Arg	Trp	Pro	Ala	Leu	Cys	Ser	Gln	Lys	Asp
1895						1900					1905			
Phe	Leu	Lys	Glu	Cys	Ser	Asp	Arg	Ile	Phe	Arg	Phe	Leu	Ala	Glu
1910						1915					1920			
Leu	Cys	Arg	Thr	Asp	Arg	Thr	Ile	Asp	Ile	Gln	Val	Arg	Thr	Asp
1925						1930					1935			
Pro	Thr	Pro	Ile	Ala	Phe	Asp	Tyr	Gln	Phe	Leu	His	Ser	Phe	Val
1940						1945					1950			
Tyr	Leu	Tyr	Gly	Leu	Glu	Val	Arg	Arg	Asn	Val	Leu	Asn	Glu	Ala
1955						1960					1965			
Lys	His	Asp	Asp	Gly	Asp	Ile	Asp	Gly	Asp	Asp	Phe	Tyr	Lys	Thr
1970						1975					1980			
Thr	Phe	Leu	Leu	Ile	Asp	Asp	Val	Leu	Gly	Gln	Leu	Gly	Gln	Pro
1985						1990					1995			
Lys	Met	Glu	Phe	Ser	Asn	Glu	Ile	Pro	Ile	Tyr	Ile	Arg	Glu	His
2000						2005					2010			
Met	Asp	Asp	Tyr	Pro	Glu	Leu	Tyr	Glu	Phe	Met	Asn	Arg	His	Ala
2015						2020					2025			
Phe	Arg	Asn	Ile	Glu	Thr	Ser	Thr	Ala	Tyr	Ser	Pro	Ser	Val	His
2030						2035					2040			
Glu	Ser	Thr	Ser	Ser	Glu	Gly	Ile	Pro	Ile	Ile	Thr	Leu	Thr	Met
2045						2050					2055			
Ser	Asn	Phe	Ser	Asp	Arg	His	Val	Asp	Ile	Asp	Thr	Val	Ala	Tyr
2060						2065					2070			
Lys	Phe	Leu	Gln	Ile	Tyr	Ala	Arg	Ile	Trp	Thr	Thr	Lys	His	Cys
2075						2080					2085			
Leu	Ile	Ile	Asp	Cys	Thr	Glu	Phe	Asp	Glu	Gly	Gly	Leu	Asp	Met
2090						2095					2100			
Arg	Lys	Phe	Ile	Ser	Leu	Val	Met	Gly	Leu	Leu	Pro	Glu	Val	Ala
2105						2110					2115			
Pro	Lys	Asn	Cys	Ile	Gly	Cys	Tyr	Tyr	Phe	Asn	Val	Asn	Glu	Thr
2120						2125					2130			
Phe	Met	Asp	Asn	Tyr	Gly	Lys	Cys	Leu	Asp	Lys	Asp	Asn	Val	Tyr
2135						2140					2145			
Val	Ser	Ser	Lys	Ile	Pro	His	Tyr	Phe	Ile	Asn	Ser	Asn	Ser	Asp
2150						2155					2160			
Glu	Gly	Leu	Met	Lys	Ser	Val	Gly	Ile	Thr	Gly	Gln	Gly	Leu	Lys
2165						2170					2175			

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Val	Leu	Gln	Asp	Ile	Arg	Val	Ser	Leu	His	Asp	Ile	Thr	Leu	Tyr
2180						2185					2190			
Asp	Glu	Lys	Arg	Asn	Arg	Phe	Thr	Pro	Val	Ser	Leu	Lys	Ile	Gly
2195						2200					2205			
Asp	Ile	Tyr	Phe	Gln	Val	Leu	His	Glu	Thr	Pro	Arg	Gln	Tyr	Lys
2210						2215					2220			
Ile	Arg	Asp	Met	Gly	Thr	Leu	Phe	Asp	Val	Lys	Phe	Asn	Asp	Val
2225						2230					2235			
Tyr	Glu	Ile	Ser	Arg	Ile	Phe	Glu	Val	His	Val	Ser	Ser	Ile	Thr
2240						2245					2250			
Gly	Val	Ala	Ala	Glu	Phe	Thr	Val	Thr	Phe	Gln	Asp	Glu	Arg	Arg
2255						2260					2265			
Leu	Ile	Phe	Ser	Ser	Pro	Lys	Tyr	Leu	Glu	Ile	Val	Lys	Met	Phe
2270						2275					2280			
Tyr	Tyr	Ala	Gln	Ile	Arg	Leu	Glu	Ser	Glu	Tyr	Glu	Met	Asp	Asn
2285						2290					2295			
Asn	Ser	Ser	Thr	Ser	Ser	Pro	Asn	Ser	Asn	Asn	Lys	Asp	Lys	Gln
2300						2305					2310			
Gln	Lys	Glu	Arg	Thr	Lys	Leu	Leu	Cys	His	Leu	Leu	Leu	Val	Ser
2315						2320					2325			
Leu	Ile	Gly	Leu	Phe	Asp	Glu	Ser	Lys	Lys	Met	Lys	Asn	Ser	Ser
2330						2335					2340			
Tyr	Asn	Leu	Ile	Ala	Ala	Thr	Glu	Ala	Ser	Phe	Gly	Leu	Asn	Phe
2345						2350					2355			
Gly	Ser	His	Phe	His	Arg	Ser	Pro	Glu	Val	Tyr	Val	Pro	Glu	Tyr
2360						2365					2370			
Thr	Thr	Thr	Phe	Leu	Gly	Val	Ile	Gly	Lys	Ser	Leu	Ala	Glu	Ser
2375						2380					2385			
Asn	Pro	Glu	Leu	Thr	Ala	Tyr	Met	Phe	Ile	Tyr	Val	Leu	Glu	Ala
2390						2395					2400			
Leu	Lys	Asn	Asn	Val	Ile	Pro	His	Val	Tyr	Ile	Pro	His	Thr	Ile
2405						2410					2415			
Cys	Gly	Leu	Ser	Tyr	Trp	Ile	Pro	Asn	Leu	Tyr	Gln	His	Val	Tyr
2420						2425					2430			
Leu	Ala	Asp	Asp	Glu	Glu	Gly	Pro	Glu	Asn	Ile	Ser	His	Ile	Phe
2435						2440					2445			
Arg	Ile	Leu	Ile	Arg	Leu	Ser	Val	Arg	Glu	Thr	Asp	Phe	Lys	Ala
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Val	Tyr	Met	Gln	Tyr	Val	Trp	Leu	Leu	Leu	Leu	Asp	Asp	Gly	Arg
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Leu	Thr	Asp	Ile	Ile	Val	Asp	Glu	Val	Ile	Asn	His	Ala	Leu	Glu
2480						2485					2490			
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2495						2500					2505			
Thr	Val	Leu	Pro	Thr	Thr	Glu	Val	Ala	Asn	Asn	Ile	Ile	Gln	Lys
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Ile	Leu	Ala	Lys	Ile	Arg	Ser	Phe	Leu	Pro	Ser	Leu	Lys	Leu	Glu
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Ala	Met	Thr	Gln	Ser	Trp	Ser	Glu	Leu	Thr	Ile	Leu	Val	Lys	Ile
2540						2545					2550			
Ser	Ile	His	Val	Phe	Phe	Glu	Thr	Ser	Leu	Leu	Val	Gln	Met	Tyr
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Cys His Ser Leu Ala Ile Asp Ser Ala Leu Ser Gln Asp His Arg		
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Asn Asn Leu Asp Glu Ile Ser Asp Ile Phe Ala His Gln Lys Val		
2615	2620	2625
Lys Phe Met Phe Gly Phe Ser Glu Asp Lys Gly Arg Ile Leu Gln		
2630	2635	2640
Ile Phe Ser Ala Ser Ser Phe Ala Ser Lys Phe Asn Ile Leu Asp		
2645	2650	2655
Phe Phe Ile Asn Asn Ile Leu Leu Leu Met Glu Tyr Ser Ser Thr		
2660	2665	2670
Tyr Glu Ala Asn Val Trp Lys Thr Arg Tyr Lys Lys Tyr Val Leu		
2675	2680	2685
Glu Ser Val Phe Thr Ser Asn Ser Phe Leu Ser Ala Arg Ser Ile		
2690	2695	2700
Met Ile Val Gly Ile Met Gly Lys Ser Tyr Ile Thr Glu Gly Leu		
2705	2710	2715
Cys Lys Ala Met Leu Ile Glu Thr Met Lys Val Ile Ala Glu Pro		
2720	2725	2730
Lys Ile Thr Asp Glu His Leu Phe Leu Ala Ile Ser His Ile Phe		
2735	2740	2745
Thr Tyr Ser Lys Ile Val Glu Gly Leu Asp Pro Asn Leu Asp Leu		
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Met Lys His Leu Phe Trp Phe Ser Thr Leu Phe Leu Glu Ser Arg		
2765	2770	2775
His Pro Ile Ile Phe Glu Gly Ala Leu Leu Phe Val Ser Asn Cys		
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Ile Arg Arg Leu Tyr Met Ala Gln Phe Glu Asn Glu Ser Glu Thr		
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Ser Leu Ile Ser Thr Leu Leu Lys Gly Arg Lys Phe Ala His Thr		
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Phe Leu Ser Lys Ile Glu Asn Leu Ser Gly Ile Val Trp Asn Glu		
2825	2830	2835
Asp Asn Phe Thr His Ile Leu Ile Phe Ile Ile Asn Lys Gly Leu		
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Ser Asn Pro Phe Ile Lys Ser Thr Ala Phe Asp Phe Leu Lys Met		
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 Cys Ser Ala
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<210> SEQ ID NO 26
 <211> LENGTH: 309
 <212> TYPE: PRT
 <213> ORGANISM: *Saccharomyces cerevisiae*

<400> SEQUENCE: 26

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 Phe Asp Glu Leu Leu Ser Tyr Tyr Gln Gln Ile Gln Arg Val Lys Asp
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 Ser Asp Tyr Ile Pro Val Val Val Val Gly Asn Lys Leu Asp Leu Glu
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 Asn Glu Arg Gln Val Ser Tyr Glu Asp Gly Leu Arg Leu Ala Lys Gln
 130 135 140
 Leu Asn Ala Pro Phe Leu Glu Thr Ser Ala Lys Gln Ala Ile Asn Val
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 Asp Glu Ala Phe Tyr Ser Leu Ile Arg Leu Val Arg Asp Asp Gly Gly
 165 170 175
 Lys Tyr Asn Ser Met Asn Arg Gln Leu Asp Asn Thr Asn Glu Ile Arg
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 Asp Ser Glu Leu Thr Ser Ser Ala Thr Ala Asp Arg Glu Lys Lys Asn
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 Asn Gly Ser Tyr Val Leu Asp Asn Ser Leu Thr Asn Ala Gly Thr Gly
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<212> TYPE: PRT															
<213> ORGANISM: <i>Saccharomyces cerevisiae</i>															
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Lys	Gln	Val	Val	Ile	Asp	Asp	Lys	Val	Ser	Ile	Leu	Asp	Ile	Leu	Asp
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Phe	Asp	Glu	Leu	Leu	Ser	Tyr	Tyr	Gln	Gln	Ile	Gln	Arg	Val	Lys	Asp
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Gly	Asn	Gln	Asn	Asn	Ala	Leu	Asn	Ser	Arg	Ser	Lys	Gln	Ser	Ala	Glu
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Cys Cys Ile Ile Cys
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<210> SEQ ID NO 28
<211> LENGTH: 9240
<212> TYPE: DNA
<213> ORGANISM: *Saccharomyces cerevisiae*

<400> SEQUENCE: 28

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gcgtataaca	tgcaatatgt	ttggctgcta	cttttagatg	atggccgctt	aactgacatt	7440
atcgttgatg	aagttattaa	tcatgcgtta	gaaagagact	ccgaaaaccg	cgattggaag	7500
aaaacaatat	cgttactaac	tgctctaccc	actactgagg	ttgctaataa	tattattcaa	7560
aaaatatttg	caaaaattag	atcattttta	ccgtcattga	agttagaagc	tatgacccaa	7620
agttggctcg	aactaacaat	attagttaag	ataagcatcc	atgttttttt	tgaaacttct	7680
ttgctggtac	agatgtactt	accagagatc	ctgtttatcg	tatccttatt	aattgatggt	7740
ggtccaaggg	aactcagatc	atcactacac	cagctattaa	tgaatgtatg	ccattccttg	7800
gctattaact	cagctttacc	acaagatcat	agaaataatc	tagatgaaat	aagtgatata	7860
tttgcacatc	aaaagggtgaa	gtttatgttt	gggttcagcg	aggacaaagg	acgaatttta	7920
cagattttta	gcgctctctc	ttttgcaagc	aagttaata	tcctggattt	cttcatcaat	7980
aatatattat	tgctgatgga	atattcttca	acgtacgaag	caaacgtgtg	gaagacaaga	8040
tacaagaaat	atgtcttgga	atctgtgttt	acaagtaatt	cttttctttc	ggcacgttca	8100
atcatgattg	ttggtataat	gggtaaatct	tacataactg	aagggttatg	caaggctatg	8160
ttaattgaaa	ccatgaaagt	tatgcgcgaa	ccaaagatta	ctgacgagca	tcttttctta	8220
gccatatctc	atatttttac	ttattccaaa	attgttgaag	gtttggatcc	caaccttgac	8280
ttaatgaagc	acttattttg	gttttcaaca	ctcttccttg	aatcacgtca	cccagataatt	8340
tttgagggtg	cccttctctt	tgtgtcaaac	tgtataaggc	gcctatacat	ggcccagttt	8400
gaaaatgaaa	gtgaaacatc	attgataagt	actttactta	aggggagaaa	gtttgctcat	8460
acctttttta	gcaagataga	gaatcttagt	ggtattgttt	ggaatgaaga	taattttaca	8520
cacattctga	ttttcatcat	taataaagga	ctatccaatc	ctttcattaa	gagtacggct	8580
tttgatttct	tgaagatgat	gtttagaaac	tcctactttg	agcatcaaat	caatcagaaa	8640
tctgatcatt	atttgtgcta	tatgttccca	ttgtattttg	ttttaaactg	taatcaattt	8700
gaggaaactt	taggtgacgt	tgattttgaa	ggagaaatgg	ttaacattga	aaacaagaac	8760
accattccta	aaattttggt	ggagtgggtg	agttcggata	acgaaaatgc	aaacattacc	8820
ctctatcaag	gtgcgatact	gttcaaatgt	tcagttacgg	atgaaccaag	tagatttagg	8880
tttgcgttga	ttattaggca	tctattgaca	aagaaaccca	tttgtgcatt	gcgtttttac	8940
agtgttattc	gtaacgaaat	aagaaaaata	tcagcatttg	agcaaaactc	ggatttgtgt	9000
ccacttgctt	tcgatatttt	aaacttatta	gtgacgcatt	cagagtctaa	tcccttagaa	9060
aaacttcacg	aagaatccat	tgaacgtctg	acccaaagag	gtttatcaat	tgtgacttct	9120
tctggtatat	ttgcgaagaa	ttccgacatg	atgataccgt	tagatgtaaa	acctgaagat	9180

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atctatgaac gtaagagaat aatgacaatg attttatcaa ggatgtcatg ttctgcttag 9240

<210> SEQ ID NO 29

<211> LENGTH: 475

<212> TYPE: PRT

<213> ORGANISM: Streptococcus mutans

<400> SEQUENCE: 29

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Met Thr Lys Gln Tyr Lys Asn Tyr Val Asn Gly Glu Trp Lys Leu Ser
 1           5           10           15

Glu Asn Glu Ile Lys Ile Tyr Glu Pro Ala Ser Gly Ala Glu Leu Gly
 20           25           30

Ser Val Pro Ala Met Ser Thr Glu Glu Val Asp Tyr Val Tyr Ala Ser
 35           40           45

Ala Lys Lys Ala Gln Pro Ala Trp Arg Ser Leu Ser Tyr Ile Glu Arg
 50           55           60

Ala Ala Tyr Leu His Lys Val Ala Asp Ile Leu Met Arg Asp Lys Glu
 65           70           75           80

Lys Ile Gly Ala Val Leu Ser Lys Glu Val Ala Lys Gly Tyr Lys Ser
 85           90           95

Ala Val Ser Glu Val Val Arg Thr Ala Glu Ile Ile Asn Tyr Ala Ala
100          105          110

Glu Glu Gly Leu Arg Met Glu Gly Glu Val Leu Glu Gly Gly Ser Phe
115          120          125

Glu Ala Ala Ser Lys Lys Lys Ile Ala Val Val Arg Arg Glu Pro Val
130          135          140

Gly Leu Val Leu Ala Ile Ser Pro Phe Asn Tyr Pro Val Asn Leu Ala
145          150          155          160

Gly Ser Lys Ile Ala Pro Ala Leu Ile Ala Gly Asn Val Ile Ala Phe
165          170          175

Lys Pro Pro Thr Gln Gly Ser Ile Ser Gly Leu Leu Leu Ala Glu Ala
180          185          190

Phe Ala Glu Ala Gly Leu Pro Ala Gly Val Phe Asn Thr Ile Thr Gly
195          200          205

Arg Gly Ser Glu Ile Gly Asp Tyr Ile Val Glu His Gln Ala Val Asn
210          215          220

Phe Ile Asn Phe Thr Gly Ser Thr Gly Ile Gly Glu Arg Ile Gly Lys
225          230          235          240

Met Ala Gly Met Arg Pro Ile Met Leu Glu Leu Gly Gly Lys Asp Ser
245          250          255

Ala Ile Val Leu Glu Asp Ala Asp Leu Glu Leu Thr Ala Lys Asn Ile
260          265          270

Ile Ala Gly Ala Phe Gly Tyr Ser Gly Gln Arg Cys Thr Ala Val Lys
275          280          285

Arg Val Leu Val Met Glu Ser Val Ala Asp Glu Leu Val Glu Lys Ile
290          295          300

Arg Glu Lys Val Leu Ala Leu Thr Ile Gly Asn Pro Glu Asp Asp Ala
305          310          315          320

Asp Ile Thr Pro Leu Ile Asp Thr Lys Ser Ala Asp Tyr Val Glu Gly
325          330          335

Leu Ile Asn Asp Ala Asn Asp Lys Gly Ala Ala Ala Leu Thr Glu Ile
340          345          350

Lys Arg Glu Gly Asn Leu Ile Cys Pro Ile Leu Phe Asp Lys Val Thr
355          360          365

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Thr Asp Met Arg Leu Ala Trp Glu Glu Pro Phe Gly Pro Val Leu Pro
 370 375 380
 Ile Ile Arg Val Thr Ser Val Glu Glu Ala Ile Glu Ile Ser Asn Lys
 385 390 395 400
 Ser Glu Tyr Gly Leu Gln Ala Ser Ile Phe Thr Asn Asp Phe Pro Arg
 405 410 415
 Ala Phe Gly Ile Ala Glu Gln Leu Glu Val Gly Thr Val His Ile Asn
 420 425 430
 Asn Lys Thr Gln Arg Gly Thr Asp Asn Phe Pro Phe Leu Gly Ala Lys
 435 440 445
 Lys Ser Gly Ala Gly Ile Gln Gly Val Lys Tyr Ser Ile Glu Ala Met
 450 455 460
 Thr Thr Val Lys Ser Val Val Phe Asp Ile Lys
 465 470 475

<210> SEQ ID NO 30
 <211> LENGTH: 1428
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Encoding SEQ ID NO: 29 and codon-optimized for
 expression in *Saccharomyces cerevisiae*

<400> SEQUENCE: 30

atgacaaaac aatataaaaa ttatgtcaat ggcgagtggg agctttcaga aaatgaaatt 60
 aaaatctacg aaccggccag tggagctgaa ttgggttcag ttccagcaat gagtactgaa 120
 gaagtagatt atgtttatgc ttcagccaag aaagctcaac cagcttggcg atcactttca 180
 tacatagaac gtgctgccta ccttcataag gtagcagata ttttgatgcg tgataaagaa 240
 aaaatagggt ctgttcttct caaagagggt gctaaagggt ataaatcagc agtcagcgaa 300
 gttgttcgta ctgcagaaat cattaattat gcagctgaag aaggccttcg tatggaagggt 360
 gaagtccttg aaggcgccag ttttgaagca gccagcaaga aaaaaattgc cgttgttcgt 420
 cgtgaaccag taggtcttgt attagctatt tcaccattta actaccctgt taacttggca 480
 ggttcgaaaa ttgcaccggc tcttattgcg ggaaatgtta ttgcttttaa accaccgacg 540
 caaggatcaa tctcagggtc cttacttgct gaagcatttg ctgaagctgg acttcctgca 600
 ggtgtcttta ataccattac aggtcgtggt tctgaaattg gagactatat tgtagaacat 660
 caagccgtta actttatcaa tttcactggt tcaacaggaa ttggggaacg tattggcaaa 720
 atggctggta tgcgtccgat tatgctgaa ctccgtggaa aagattcagc catcgttctt 780
 gaagatgcag acctgaatt gactgctaaa aatattattg caggtgcttt tggttattca 840
 ggtcaacgct gtacagcagt taaacgtgtt cttgtgatgg aaagtgttgc tgatgaactg 900
 gtcgaaaaaa tccgtgaaaa agttcttgca ttaacaattg gtaatccaga agacgatgca 960
 gatattacac cgttgattga taaaaatca gctgattatg tagaaggctt tattaatgat 1020
 gccaatgata aaggagccgc tgccttact gaaatcaaac gtgaaggtaa tcttatctgt 1080
 ccaatcctct ttgataagggt aacgacagat atgcgtcttg cttgggaaga accatttggt 1140
 cctgttcttc cgatcattcg tgtgacatct gtagaagaag ccattgaaat ttctaacaaa 1200
 tcggaatatg gacttcaggc ttctatcttt acaaatgatt tcccacgcgc ttttggattt 1260
 gctgagcagc ttgaagtggg tacagttcat atcaataata agacacagcg cggtagcgac 1320
 aacttcccat tcttaggggc taaaaaatca ggtgcaggta ttcaaggggt aaaatattct 1380

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 attgaagcta tgacaactgt taaatccgtc gtatttgata tcaaataa

1428

<210> SEQ ID NO 31

<211> LENGTH: 535

<212> TYPE: PRT

<213> ORGANISM: *Alternaria alternata*

<400> SEQUENCE: 31

 Met Asp Lys Ile Ser Lys Tyr Asn Val Val His Arg Tyr Glu Lys Arg
 1 5 10 15

 Ser Leu Leu Ile Ala Ile Asn Cys Ile Ala Gly Leu Ser Ile Leu Phe
 20 25 30

 Phe Gly Tyr Asp Gln Gly Met Met Gly Gly Val Asn Thr Ala Tyr Asp
 35 40 45

 Tyr Tyr Thr Leu Met Gly Phe Gly His Lys Gly Ala Asp Gly Gly Pro
 50 55 60

 Val Val Asp Asp Thr Leu Leu Gln Gly Gly Ile Val Ser Val Tyr Tyr
 65 70 75 80

 Leu Gly Thr Leu Val Gly Cys Leu Val Gly Gly Ser Ile Gly Asp Arg
 85 90 95

 Phe Gly Arg Ile Lys Thr Ile Phe Val Gly Ala Ala Val Ala Thr Val
 100 105 110

 Gly Ala Cys Leu Gln Cys Ser Ala Met Asn His Glu Trp Met Ile Cys
 115 120 125

 Ala Arg Leu Val Asn Gly Trp Gly Thr Gly Ile Leu Asn Ser Ile Ile
 130 135 140

 Pro Val Trp Ala Thr Glu Thr Ala Glu His Thr Ser Arg Gly Gln Phe
 145 150 155 160

 Ile Ala Ile Glu Phe Thr Leu Asn Ile Leu Gly Val Val Ile Ala Tyr
 165 170 175

 Trp Leu Glu Tyr Ala Cys Ser Phe Tyr Gly Asp Gly Thr Ser Ser Phe
 180 185 190

 Ile Trp Arg Phe Pro Ile Ala Phe Gln Ile Pro Met Leu Met Ile Leu
 195 200 205

 Met Ala Ala Val Met Phe Phe Pro Glu Ser Pro Arg Trp Leu Val Lys
 210 215 220

 Val Gly Arg Glu Ala Glu Gly Arg Tyr Val Leu Ser Arg Leu Arg Gly
 225 230 235 240

 Asp Ala Gly Glu Asp Arg Glu Arg Ala Glu Thr Glu Phe Gln Glu Ile
 245 250 255

 Val Ala Ser Cys Glu Leu Glu Arg Ser Asn Phe Arg Lys Gln Ser Tyr
 260 265 270

 Phe His Met Leu Phe Gly Ile Gly Ser Gly Lys Leu His Thr Gly Arg
 275 280 285

 Arg Val Gln Leu Val Ile Trp Leu Gln Ile Met Gln Glu Trp Val Gly
 290 295 300

 Ile Ala Gly Val Thr Ile Tyr Ala Pro Thr Ile Phe Gly Ile Ala Gly
 305 310 315 320

 Met Ser Pro Ala Lys Arg Gln Trp Ile Ser Gly Leu Asn Asn Ile Phe
 325 330 335

 Tyr Met Phe Ala Thr Leu Ile Cys Val Phe Thr Leu Asp Arg Ile Gly
 340 345 350

 Arg Arg Trp Thr Cys Tyr Trp Gly Ser Ala Gly Gln Gly Ile Ala Met
 355 360 365

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Ala Leu Ala Gly Gly Phe Ser Tyr Leu Gly Gln Glu Ala Thr Lys Arg
 370 375 380

Gly Asp Thr Ser Ala Ala Ser Ser Tyr Gly Asn Ala Ala Val Ser Met
 385 390 395 400

Val Phe Ile Phe Thr Ala Ile Phe Gly Ala Thr Trp Leu Thr Val Pro
 405 410 415

Trp Leu Tyr Pro Ala Glu Ile Phe Pro Leu Glu Val Arg Ala Lys Gly
 420 425 430

Asn Ala Trp Gly Val Val Gly Trp Ser Ile Gly Asn Gly Trp Leu Thr
 435 440 445

Leu Leu Cys Pro Ile Met Phe Glu Ala Leu Gly Glu Lys Thr Leu Tyr
 450 455 460

Ile Phe Ala Ala Cys Asn Ala Ile Thr Ile Pro Met Val Trp Ala Leu
 465 470 475 480

Tyr Pro Glu Thr Asn Gln Arg Thr Leu Glu Glu Ile Asp Leu Leu Phe
 485 490 495

Ala Ser Asp Ser Ile Trp Asn Trp Glu Ala Glu Lys Asn Phe Ala Ala
 500 505 510

Leu Gln Glu Thr Leu Pro Phe Glu Ala Thr Ser His Ala Ala Lys Asn
 515 520 525

Asp Ile Glu Arg Val Ser Leu
 530 535

<210> SEQ ID NO 32
 <211> LENGTH: 542
 <212> TYPE: PRT
 <213> ORGANISM: Aspergillus fumigatus
 <400> SEQUENCE: 32

Met Trp Thr Thr Thr Ser Gly Leu Ser Gly Arg Ser Leu Arg Leu Ser
 1 5 10 15

Ile Thr Phe Ala Ala Val Val Gly Phe Ser Leu Phe Gly Tyr Asn Gln
 20 25 30

Gly Met Met Ala Gly Leu Leu Asn Gly Asp Glu Phe Val Asn Ser Phe
 35 40 45

Pro Ile Leu Lys Met Pro Asp Asn Pro Thr Ala Gly Glu Lys His Tyr
 50 55 60

Ile Asp Val Ile Arg Gly Ala Val Thr Ser Cys Tyr Glu Leu Gly Cys
 65 70 75 80

Phe Phe Gly Ala Leu Phe Ser Met Phe Cys Gly Asn Arg Leu Gly Arg
 85 90 95

Thr Arg Leu Ile Phe Met Gly Ala Ser Ile Leu Ile Val Gly Ala Leu
 100 105 110

Leu Thr Thr Val Cys Tyr Thr Gly Lys Trp Glu Val Gly Gln Phe Val
 115 120 125

Ile Gly Arg Val Val Ser Gly Ile Gly Asn Gly Met Asn Thr Ala Thr
 130 135 140

Ile Pro Val Trp Gln Ser Glu Cys Ser Gly Ala His Asn Arg Gly Phe
 145 150 155 160

Leu Val Cys Phe Glu Gly Ala Met Ile Ala Gly Gly Thr Phe Ile Ala
 165 170 175

Tyr Trp Val Val Phe Gly Ile Ser His Ala Ala Asp Ser Val Gln Trp
 180 185 190

Arg Phe Pro Val Ala Leu Gln Ile Phe Phe Ala Leu Val Val Ala Thr
 195 200 205

-continued

Gly Ala Leu Met Leu Pro Asp Ser Pro Ser Trp Phe Val Ser Arg Gly
 210 215 220
 Leu Asp Asn Glu Ala Cys Glu Val Leu Gly Lys Ile Lys Gly Thr Ser
 225 230 235 240
 Pro Asp Ser Asp Gln Val Leu His Asp Phe Asn Leu Ile Lys Thr Asp
 245 250 255
 Met Glu Ser Thr Lys Ser Glu Gln Ser Asn Trp Lys Thr Val Phe Thr
 260 265 270
 Phe Gly Lys Thr Gln Glu Phe Gln Arg Leu Leu Ile Gly Cys Ser Gly
 275 280 285
 Gln Phe Phe Gln Gln Phe Thr Gly Cys Asn Ala Ala Ile Tyr Tyr Ser
 290 295 300
 Thr Leu Leu Phe Gln Glu Asn Leu His Met Glu Lys Tyr Leu Ser Leu
 305 310 315 320
 Ile Met Gly Gly Val Phe Ala Ser Val Tyr Ala Leu Ala Thr Ile Pro
 325 330 335
 Ser Phe Phe Met Ile Glu Arg Val Gly Arg Arg Lys Leu Tyr Leu Ile
 340 345 350
 Gly Phe Leu Gly Gln Gly Leu Ser Phe Val Ile Thr Phe Ala Cys Leu
 355 360 365
 Ile Lys Glu Thr Glu Glu Asn Ser Lys Gly Ala Ala Val Gly Ile Phe
 370 375 380
 Leu Phe Ile Thr Phe Phe Ala Phe Thr Leu Leu Pro Leu Pro Trp Ile
 385 390 395 400
 Tyr Pro Pro Glu Ile Asn Pro Leu Arg Thr Arg Thr Val Gly Ala Ser
 405 410 415
 Ala Ser Thr Cys Thr Asn Trp Met Cys Asn Phe Ala Val Val Met Phe
 420 425 430
 Thr Pro Leu Phe Ala Gly Gln Ser Pro Trp Gly Val Tyr Leu Phe Phe
 435 440 445
 Ala Leu Phe Asn Phe Val Gly Leu Ile Phe Gly Tyr Phe Phe Tyr Val
 450 455 460
 Glu Thr Ala Gly Arg Glu Leu Glu Glu Val Asp Ile Ile Tyr Ala Lys
 465 470 475 480
 Ala His Val Glu Gly Lys Met Pro Phe Arg Val Ala His Asp Leu Pro
 485 490 495
 Lys Leu Ser Phe Glu Glu Ile Val Gln Gln Ser Arg Glu Leu Gly Leu
 500 505 510
 Asp Thr Asn Asp His Val Met Leu Glu Lys Lys Glu Leu Gly Leu Ser
 515 520 525
 Ser Asp Ser Ala Gln Glu Thr Glu Glu Val Tyr Glu Lys Gln
 530 535 540

<210> SEQ ID NO 33
 <211> LENGTH: 539
 <212> TYPE: PRT
 <213> ORGANISM: Aspergillus terreus

<400> SEQUENCE: 33

Met Trp Thr Thr Ser Gly Thr Ser Gly Arg Leu Leu Arg Ala Ser
 1 5 10 15
 Ile Thr Phe Thr Ala Val Met Gly Phe Ser Leu Phe Gly Tyr Asn Gln
 20 25 30
 Gly Met Met Ser Gly Leu Ile Ala Gly Asp Glu Phe Thr Lys Thr Phe

-continued

35	40	45
Gly Thr Leu Ala Met Pro Asp Asn Pro Asp Asp Ala Thr Tyr Arg His		
50	55	60
Ile Asn Val Ile Arg Gly Ala Val Thr Ala Cys Tyr Glu Ile Gly Cys		
65	70	75
Phe Phe Gly Ala Leu Phe Ser Met Phe Phe Gly Glu Lys Ile Gly Arg		
	85	90
Thr Arg Ile Ile Phe Ser Gly Ala Leu Val Leu Ile Ile Gly Ala Val		
	100	105
Ile Ser Thr Ala Ala Phe Gly Pro Gln Trp Gly Leu Gly Gln Phe Val		
	115	120
Val Gly Arg Val Ile Ser Gly Leu Gly Asn Gly Met Asn Thr Ala Thr		
	130	135
Ile Pro Val Trp Gln Ser Glu Cys Ser Ser Ala His Asn Arg Gly Leu		
145	150	155
Leu Val Cys Phe Glu Gly Ala Met Ile Ala Val Gly Thr Phe Ile Ala		
	165	170
Tyr Trp Val Val Phe Gly Leu Ser Tyr Val Pro Glu Thr Val Gln Trp		
	180	185
Arg Phe Pro Val Ala Leu Gln Val Phe Phe Ala Leu Ile Val Ala Ala		
	195	200
Gly Ala Met Met Phe Pro Glu Ser Pro Arg Trp Phe Val Met Arg Gly		
	210	215
Tyr His Lys Glu Ala Cys Glu Val Ile Ala Lys Leu Lys Asn Ser Thr		
225	230	235
Pro Asp Ser Asp Glu Val Leu Thr Glu Phe Asn Phe Met Lys Ala Asp		
	245	250
Val Glu Met Thr Ser Ala Ser Gln Ala Ser Trp Lys Thr Val Phe Thr		
	260	265
Phe Gly Lys Thr Gln Glu Phe Gln Arg Leu Leu Val Gly Cys Ser Gly		
	275	280
Gln Phe Phe Gln Gln Phe Thr Gly Cys Asn Ala Ala Ile Tyr Tyr Ser		
	290	295
Thr Leu Leu Phe Glu Glu Asn Leu Asn Leu Glu His Arg Leu Ala Leu		
305	310	315
Ile Met Gly Gly Val Phe Ala Thr Val Tyr Ala Leu Ala Thr Ile Pro		
	325	330
Ser Phe Phe Met Val Glu Arg Val Gly Arg Arg Lys Leu Tyr Leu Ile		
	340	345
Gly Phe Leu Gly Gln Gly Leu Ser Phe Val Ile Thr Phe Ala Cys Leu		
	355	360
Ile Lys Pro Thr Thr Glu Asn Ser Lys Gly Ala Ile Val Gly Ile Phe		
	370	375
Leu Phe Ile Thr Phe Phe Ala Phe Thr Leu Leu Pro Leu Pro Trp Ile		
385	390	395
Tyr Pro Pro Glu Ile Asn Pro Leu Arg Thr Arg Thr Val Gly Ala Ala		
	405	410
Ala Ser Thr Cys Thr Asn Trp Ile Cys Asn Phe Ala Val Val Met Phe		
	420	425
Thr Pro Leu Phe Ala Ala Ser Ser Gly Trp Gly Val Tyr Leu Phe Phe		
	435	440
Ala Leu Ile Asn Leu Ile Gly Val Pro Phe Gly Trp Phe Phe Phe Val		
	450	455
		460

-continued

Glu Thr Ala Gly Arg Glu Leu Glu Glu Ile Asp Ile Val Tyr Ala Lys
 465 470 475 480
 Ala His Val Glu Lys Lys Trp Pro Phe Met Val Ala Lys Asp Met Pro
 485 490 495
 Lys Leu Ser Leu Glu Glu Ile Thr Gln Gln Ser Arg Glu Leu Gly Leu
 500 505 510
 Asp Leu Asn Asp Ser Ser Pro Pro Asp Gln Asn Glu Ser Gly Leu Ser
 515 520 525
 Ser Asp Asn Asp Gln Ser Val Ala Glu Lys Gln
 530 535

<210> SEQ ID NO 34
 <211> LENGTH: 572
 <212> TYPE: PRT
 <213> ORGANISM: *Cordyceps militaris*

<400> SEQUENCE: 34

Met Val Asp Phe Lys Ala Leu Thr Glu Lys His Asn Val Ala His Lys
 1 5 10 15
 Leu Tyr Lys Ser Ser Leu Leu Asn Thr Val Cys Leu Val Ala Gly Leu
 20 25 30
 Ser Ile Phe Phe Phe Gly Tyr Asp Gln Gly Leu Met Gly Gly Val Asn
 35 40 45
 Thr Thr Arg Asn Tyr Ala Glu Thr Met Gly Phe Gly His Trp Asp Glu
 50 55 60
 Glu Gln Gly Ile Val Val Val Asp Lys Pro Leu Leu Gln Gly Gly Ile
 65 70 75 80
 Val Ala Val Tyr Tyr Leu Pro Gly Thr Leu Ala Gly Cys Leu Leu Gly
 85 90 95
 Gly Trp Met Gly Asp Arg Tyr Gly Arg Ile Thr Thr Ile Gly Leu Ala
 100 105 110
 Cys Ile Trp Cys Ile Phe Ala Ala Ala Leu Gln Ser Ala Ala Gln Asn
 115 120 125
 Ala Ser Trp Met Phe Cys Ala Arg Val Leu Asn Gly Ile Gly Thr Gly
 130 135 140
 Ile Leu Asn Ala Ile Thr Pro Val Trp Ala Thr Glu Thr Ala Ser His
 145 150 155 160
 Thr Ser Arg Gly Gln Phe Val Ala Phe Glu Phe Thr Leu Asn Ile Phe
 165 170 175
 Gly Val Val Val Ala Tyr Trp Leu Glu Phe Gly Thr Ser Lys Tyr His
 180 185 190
 Asp Pro Glu Ser Ser Phe Ile Trp Arg Phe Pro Val Ala Phe Gln Ile
 195 200 205
 Leu Pro Leu Ile Val Leu Leu Ala Val Ile Trp Phe Met Pro Glu Ser
 210 215 220
 Pro Arg Trp Leu Val Lys Val Gly Arg Glu Glu Glu Ala Arg Phe Ile
 225 230 235 240
 Leu Gly Arg Leu Arg Gly Asn Glu Gly Glu Glu Gly Glu Ala Ala Glu
 245 250 255
 Ala Glu Leu Gln Asp Ile Ile Ser Ile Lys Asn Leu Glu Asn Asp Thr
 260 265 270
 Ser Glu Gln Gln Ser Tyr Phe His Met Phe Phe Gly Ile Gly Ser Gly
 275 280 285
 Lys Leu His Thr Gly Arg Arg Val Gln Leu Val Ile Trp Leu Gln Ile

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290	295	300
Leu Gln Glu Trp Ile Gly Ile Ala Gly Ile Thr Ile Tyr Gly Pro Gln		
305	310	315 320
Ile Phe Thr Ile Ala Gly Ile Gly Ala Ser Asp Arg Leu Trp Val Ser		
	325 330	335
Gly Val Asn Asn Ile Thr Tyr Met Phe Ala Thr Leu Ile Cys Val Phe		
	340 345	350
Thr Leu Asp Arg Ile Gly Arg Arg Trp Thr Leu Tyr Trp Gly Ala Val		
	355 360	365
Gly Gln Gly Ile Cys Met Phe Ala Gly Gly Gly Leu Ala Arg Ala Thr		
	370 375	380
Ile Asn Ala Ser Asp Ser Asn Arg Gly Gln Ile Gly Gly Ala Ala Thr		
385	390	395 400
Phe Phe Val Phe Leu Tyr Thr Ala Ile Phe Gly Ala Thr Trp Leu Thr		
	405 410	415
Val Pro Trp Leu Tyr Pro Ala Glu Ile Phe Pro Leu Gln Val Arg Ala		
	420 425	430
Lys Gly Asn Ala Trp Gly Val Val Gly Trp Ser Ile Gly Asn Gly Trp		
	435 440	445
Cys Val Leu Leu Leu Pro Thr Ile Phe Asp Lys Leu Lys Glu Lys Thr		
	450 455	460
Leu Tyr Ile Phe Gly Ala Val Asn Val Val Ser Ile Val Ala Val Trp		
465	470 475	480
Ala Leu Tyr Pro Glu Ser Asn Gln Arg Thr Leu Glu Glu Met Asp Leu		
	485 490	495
Val Phe Ala Ser Asp Ser Ile Trp Thr Trp Asp Ala Glu Arg Asn Phe		
	500 505	510
Ala Lys Leu Lys Ala Glu Asn Pro Asp Leu Ile Lys Ser His Lys Asp		
	515 520	525
Thr Arg Gly Ala Asn Asp Leu Glu Arg Gly Ile Phe Ala Ser Arg Arg		
	530 535	540
Ala Ser Lys Ala Ser Ala Thr Pro Pro Ser Val Asp Lys Glu Thr Thr		
545	550 555	560
Val Glu Asn Val Glu Asn Ser Ser Arg Pro His Pro		
	565 570	

<210> SEQ ID NO 35

<211> LENGTH: 524

<212> TYPE: PRT

<213> ORGANISM: Diplodia corticola

<400> SEQUENCE: 35

Met Gly Ala Lys Gly Leu Phe Gly Cys Thr Gly His Ala Ile Glu Arg		
1	5 10	15
Leu Gln Leu Ala Leu Ile Val Ala Pro Ser Phe Ile Leu Phe Gly Tyr		
	20 25	30
Asn Gln Ala Gly Leu Gly Gly Leu Leu Thr Glu Ala Asp Trp Val Lys		
	35 40	45
Thr Phe Pro Glu Ile Asp Thr Val Asn Thr Glu Gly Ala Glu Lys Ser		
	50 55	60
His Asn Ser Thr Ile Gln Gly Leu Val Val Ala Thr Phe Val Ile Gly		
65	70 75	80
Ala Leu Ile Gly Ser Leu Ser Cys Ser Tyr Thr Gly Asp Lys Tyr Gly		
	85 90	95

Arg	Arg	Ala	Val	Val	Met	Ala	Gly	Ala	Ile	Cys	Thr	Leu	Ile	Gly	Glu
			100					105					110		
Val	Leu	Glu	Cys	Ser	Ser	Phe	Gly	Leu	Pro	Gln	Phe	Ile	Val	Gly	Arg
		115					120					125			
Ile	Ile	Val	Gly	Leu	Gly	Ile	Gly	Gln	Leu	Ser	Ala	Thr	Val	Pro	Val
		130				135					140				
Trp	Gln	Ser	Glu	Thr	Ser	Gly	Ala	Lys	Asn	Arg	Gly	Gln	His	Val	Val
145				150					155						160
Leu	Asp	Gly	Leu	Phe	Ile	Cys	Val	Gly	Tyr	Val	Leu	Glu	Ser	Trp	Ile
				165					170					175	
Asp	Leu	Gly	Phe	Phe	Glu	Leu	Pro	Glu	Gly	Gln	Val	Thr	Trp	Arg	Pro
			180					185					190		
Pro	Ile	Ala	Ile	Ala	Ile	Val	Phe	Ser	Leu	Ile	Leu	Ile	Gly	Ser	Val
		195					200					205			
Tyr	Leu	Phe	Pro	Glu	Ser	Pro	Arg	Trp	Leu	Val	Arg	Lys	Ala	Arg	Asn
		210				215					220				
Asp	Glu	Ala	Arg	Ser	Ala	Ile	Ala	Leu	Phe	Arg	Gly	His	Glu	Glu	Asp
225				230					235						240
Ala	Ile	Glu	Val	Gln	Ala	Glu	Leu	Ala	Gly	Ile	Glu	Leu	Ser	Leu	Glu
				245					250					255	
Glu	Thr	Ser	Lys	Ser	Gly	Ala	Lys	Leu	Lys	Asp	Met	Leu	Lys	Met	Gly
			260					265					270		
Glu	Asp	Lys	Leu	Leu	Tyr	Arg	Phe	Gly	Leu	Cys	Ile	Leu	Leu	Gln	Phe
		275					280					285			
Tyr	Gln	Gln	Met	Ser	Gly	Ser	Asn	Leu	Ile	Ser	Val	Tyr	Thr	Ser	Ile
		290				295					300				
Leu	Phe	Gln	Gln	Asn	Leu	Gly	Leu	Ser	Pro	Glu	Leu	Ser	Arg	Val	Leu
305				310					315						320
Ser	Gly	Gly	Ala	Leu	Thr	Trp	Lys	Phe	Leu	Ser	Ser	Phe	Val	Ala	Phe
				325					330					335	
Phe	Thr	Ile	Asp	Arg	Phe	Gly	Arg	Arg	Ala	Val	Phe	Met	Phe	Ser	Gly
			340					345					350		
Ala	Gly	Met	Ser	Leu	Cys	Met	Ile	Ala	Leu	Ala	Ile	Thr	Thr	Ser	Met
		355					360					365			
Thr	Asp	Ser	His	Gly	Ala	Gln	Val	Ala	Ala	Gly	Cys	Phe	Ile	Tyr	Met
		370				375					380				
Phe	Asn	Phe	Phe	Val	Pro	Ile	Gly	Phe	Leu	Gly	Ala	Asn	Phe	Leu	Tyr
385				390					395						400
Cys	Thr	Glu	Val	Ala	Pro	Leu	Arg	Leu	Arg	Val	Ala	Met	Ser	Ser	Ile
				405					410					415	
Ser	Thr	Ala	Asn	His	Trp	Leu	Trp	Asn	Phe	Val	Val	Ile	Met	Val	Thr
			420					425					430		
Pro	Val	Ala	Leu	Ala	Asn	Ile	Ala	Trp	Arg	Tyr	Tyr	Ile	Val	Tyr	Ala
		435					440					445			
Ala	Val	Ala</													

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515	520
<210> SEQ ID NO 36	
<211> LENGTH: 558	
<212> TYPE: PRT	
<213> ORGANISM: Eremothecium gossypii	
<400> SEQUENCE: 36	
Met Ile Lys Glu Ser Phe Ser Arg Thr Ala Thr Gly Gly Phe Ser Gly	
1 5 10 15	
Arg Ala Leu Arg Leu Ala Val Thr Ile Thr Ala Val Thr Gly Phe Ser	
20 25 30	
Leu Phe Gly Tyr Asp Gln Gly Leu Leu Ser Gly Leu Ile Asn Gly Glu	
35 40 45	
Gln Phe Asn His Glu Phe Pro Ala Thr Leu Glu Gln Gly Asp Asn Asp	
50 55 60	
Arg His Ala Thr Val Val Gln Gly Ala Val Thr Ser Cys Tyr Glu Leu	
65 70 75 80	
Gly Cys Phe Phe Gly Ser Leu Tyr Val Met Val Asp Gly Glu Arg Arg	
85 90 95	
Gly Arg Lys Pro Leu Ile Ile Met Gly Ser Leu Leu Thr Ile Leu Gly	
100 105 110	
Thr Val Val Ser Val Ala Ala Phe Arg Glu His Trp Gly Leu Gly Gln	
115 120 125	
Phe Val Ile Gly Arg Val Val Thr Gly Leu Gly Thr Gly Leu Asn Thr	
130 135 140	
Ser Thr Ile Pro Val Trp Gln Ser Glu Met Ser Lys Pro Lys Asn Arg	
145 150 155 160	
Gly Leu Leu Val Asn Leu Glu Gly Ser Val Ile Ala Val Gly Thr Met	
165 170 175	
Ile Ala Tyr Trp Ile Asp Phe Gly Leu Ser Tyr Val Asp Ser Ser Val	
180 185 190	
Gln Trp Arg Phe Pro Val Ala Met Gln Ile Val Phe Ala Val Leu Leu	
195 200 205	
Leu Val Gly Ile Val Gln Leu Pro Glu Ser Pro Arg Trp Leu Met Ala	
210 215 220	
His Gly Arg Thr Ala Glu Ser Lys Tyr Val Leu Gly Lys Leu Asp Asn	
225 230 235 240	
Leu Asp Pro Ser Asp Asp Ser Val Leu Ala Glu Ala Ala Ala Ile Glu	
245 250 255	
Asp Ala Val Asn Arg Phe Lys His Glu Lys Arg Ser Leu Lys Asp Ala	
260 265 270	
Leu Thr Gly Gly Arg Gly Gln Asn Leu Gln Arg Thr Leu Val Ala Cys	
275 280 285	
Ser Thr Gln Phe Phe Gln Gln Phe Thr Gly Cys Asn Ala Ala Ile Tyr	
290 295 300	
Tyr Ser Thr Val Leu Phe His Lys Thr Ile Asn Leu Glu Tyr Arg Leu	
305 310 315 320	
Ser Leu Ile Leu Gly Gly Val Phe Ser Thr Val Tyr Thr Leu Ala Thr	
325 330 335	
Ile Pro Ser Phe Phe Leu Ile Glu Thr Val Gly Arg Arg Lys Leu Phe	
340 345 350	
Leu Ile Gly Ala Leu Gly Gln Gly Phe Ser Phe Thr Ile Thr Phe Ala	
355 360 365	

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Cys Leu Ile Asn Asp Thr Lys Asn Asn Ala Lys Gly Ala Ala Val Gly
 370 375 380
 Leu Phe Leu Phe Ile Ile Phe Phe Gly Met Thr Ile Leu Ser Leu Pro
 385 390 395 400
 Trp Ile Tyr Pro Pro Glu Ile Ser Ser Met Lys Val Arg Ser Ile Thr
 405 410 415
 Asn Ala Met Ser Thr Cys Thr Asn Trp Leu Cys Asn Phe Ala Val Val
 420 425 430
 Met Phe Thr Pro Ile Phe Ile His Glu Ala Gly Trp Gly Cys Tyr Leu
 435 440 445
 Phe Phe Ala Val Met Asn Phe Leu Tyr Val Pro Ile Ile Phe Phe Phe
 450 455 460
 Tyr Pro Glu Thr Ala Gly Arg Ser Leu Glu Glu Ile Asp Ile Ile Tyr
 465 470 475 480
 Ala Lys Ser Phe Val Asp Gly Thr Gln Ala Trp Arg Val Ala Ala Asn
 485 490 495
 Leu Pro Lys Leu Ser Leu Lys Glu Val Glu Glu Tyr Gly Ile Gln Leu
 500 505 510
 Gly Leu Tyr Asn Asn Glu His Asp Phe Lys Asp Glu Met Lys Val Ile
 515 520 525
 Thr Glu Ala Glu Gly Gln Val Pro Arg Ser Ser Ile Ser Asp Ser Asn
 530 535 540
 Met Glu Ser Ser Ser Thr Gly Asn Gly Ile Met Lys Lys Pro
 545 550 555

<210> SEQ ID NO 37
 <211> LENGTH: 557
 <212> TYPE: PRT
 <213> ORGANISM: Eremothecium sincaudum

<400> SEQUENCE: 37

Met Glu Ser Thr Val Arg Lys Trp Phe Ser Arg Thr Ala Thr Ala Ser
 1 5 10 15
 Leu Ser Gly Arg Ala Leu Arg Leu Ala Ile Thr Ile Thr Ala Val Thr
 20 25 30
 Gly Phe Ser Leu Phe Gly Tyr Asp Gln Gly Leu Leu Ser Gly Leu Ile
 35 40 45
 Asn Gly Asp Lys Phe Asn His Glu Phe Pro Ala Thr Leu Glu Lys His
 50 55 60
 Asp Asn Asp Thr His Ala Thr Val Val Gln Gly Ala Val Thr Ser Cys
 65 70 75 80
 Tyr Glu Leu Gly Cys Phe Phe Gly Ser Leu Phe Val Val Met Lys Gly
 85 90 95
 Glu Lys Met Gly Arg Lys Pro Leu Ile Ile Cys Gly Ala Leu Leu Thr
 100 105 110
 Ile Val Gly Thr Ile Ile Ser Thr Leu Ala Phe Arg Glu His Trp Gly
 115 120 125
 Leu Gly Gln Phe Val Val Gly Arg Val Ile Thr Gly Leu Gly Thr Gly
 130 135 140
 Phe Asn Thr Ser Thr Ile Pro Val Trp Gln Ser Glu Met Ser Lys Pro
 145 150 155 160
 Glu Asn Arg Gly Leu Leu Val Asn Leu Glu Gly Ser Val Ile Ala Val
 165 170 175
 Gly Thr Met Ile Ala Tyr Trp Ile Asp Phe Gly Leu Ser Tyr Val Asp
 180 185 190

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Ser Ser Ala Gln Trp Arg Phe Pro Val Ala Met Gln Ile Val Phe Ala
 195 200 205
 Val Phe Leu Leu Ile Gly Cys Thr Gln Leu Pro Glu Ser Pro Arg Trp
 210 215 220
 Leu Met Ala His Gly Arg Gly Glu Glu Ala Arg Tyr Ile Leu Ser Gln
 225 230 235 240
 Leu Asp Gly Val Pro Ile Asp Asp Pro Thr Val Leu Ala Glu Ala Ala
 245 250 255
 Ala Ile Glu Glu Val Val Asp Lys Phe Lys Asn Gln Arg Trp Ser Leu
 260 265 270
 Lys Asp Ala Phe Thr Gly Gly Arg Gly Gln Asn Leu Gln Arg Thr Leu
 275 280 285
 Val Ala Cys Ser Thr Gln Phe Phe Gln Gln Phe Thr Gly Cys Asn Ala
 290 295 300
 Ala Ile Tyr Tyr Ser Thr Val Leu Phe His Lys Thr Ile Lys Leu Glu
 305 310 315 320
 Tyr Arg Met Ser Leu Ile Leu Gly Gly Val Phe Ala Thr Val Tyr Thr
 325 330 335
 Leu Ser Thr Ile Pro Ser His Phe Leu Ile Glu Thr Val Gly Arg Arg
 340 345 350
 Lys Leu Phe Leu Val Gly Ala Leu Gly Gln Gly Val Ala Phe Thr Ile
 355 360 365
 Thr Phe Ala Cys Leu Ile His Asp Thr Lys Gln Asn Ala Lys Gly Ala
 370 375 380
 Ala Phe Gly Leu Phe Leu Phe Ile Val Phe Phe Gly Met Thr Ile Leu
 385 390 395 400
 Ser Leu Pro Trp Ile Tyr Pro Pro Glu Ile Ser Ser Leu Arg Val Arg
 405 410 415
 Ser Leu Thr Asn Ser Leu Ser Thr Cys Thr Asn Trp Leu Ser Asn Phe
 420 425 430
 Ala Val Val Met Phe Thr Pro Ile Phe Ile Gln Lys Ser Ser Trp Gly
 435 440 445
 Cys Tyr Met Phe Phe Ala Ile Val Asn Phe Ser Tyr Leu Pro Ile Ile
 450 455 460
 Phe Phe Phe Tyr Pro Glu Thr Ser Gly Arg Ser Leu Glu Glu Ile Asp
 465 470 475 480
 Ile Ile Tyr Ala Lys Ser Asn Lys Asp Gly Ile Ala Ser Trp Arg Val
 485 490 495
 Ala Ala His Leu Pro Lys Leu Ser Leu Lys Glu Ile Glu Gln Tyr Ala
 500 505 510
 Ile Glu Tyr Asp Leu Tyr Asp Gly Pro Val Ser Glu Ala Ser Thr Glu
 515 520 525
 Ser Asp Asn Glu Arg Asn Gln Ser Glu Leu Asp Leu Lys Gln Asp Ser
 530 535 540
 Pro Gln Asn Leu Ala Asp Lys Ser Lys Asp Gly Ser Leu
 545 550 555

<210> SEQ ID NO 38

<211> LENGTH: 525

<212> TYPE: PRT

<213> ORGANISM: *Gaeumannomyces tritici*

<400> SEQUENCE: 38

Met Thr Val Ala Thr Thr Gly Val Ser Lys Ser Tyr Phe Gly Gly Leu

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1	5	10	15
Thr Gly His Arg Leu Ala Val Leu Gln Thr Ala Leu Ile Thr Ala Pro	20	25	30
Ser Phe Ile Leu Phe Gly Tyr Asn Gln Ala Gly Leu Gly Gly Leu Ile	35	40	45
Gly Leu Lys Asp Trp Glu Arg Thr Phe Pro Glu Ile Asp Thr Thr His	50	55	60
His Lys Glu Ser Ser Val Ser Thr Leu Lys Gly Phe Val Val Ala Thr	65	70	75
Phe Val Ile Gly Ala Leu Phe Gly Ser Leu Leu Cys Ser Tyr Thr Gly	85	90	95
Asp Lys Phe Gly Arg Arg His Val Ile Phe Phe Ala Gly Leu Cys Thr	100	105	110
Leu Val Gly Glu Val Leu Glu Cys Ala Ala Phe Gly Leu Ala Gln Leu	115	120	125
Ile Val Gly Arg Val Ile Val Gly Phe Gly Ile Gly Gln Leu Ser Ser	130	135	140
Ile Val Pro Val Trp Gln Ser Glu Thr Ser Gly Ala Lys Asn Arg Gly	145	150	155
Arg His Val Val Leu Asp Gly Val Phe Ile Cys Leu Gly Phe Val Leu	165	170	175
Glu Ser Trp Ile Asn Leu Gly Phe Phe Gln Phe Gly Asp Ser Pro Ile	180	185	190
Thr Trp Arg Pro Pro Ile Ala Ile Ala Ile Leu Phe Ser Leu Ile Leu	195	200	205
Ser Gly Ser Val Tyr Leu Phe Pro Glu Ser Pro Arg Trp Leu Met Ala	210	215	220
Lys Asn Arg Lys Ser Asp Ala Val Ala Val Leu Ser Val Leu Arg Gly	225	230	235
Leu Pro Gln Asp Ser Ile Glu Val Gln Ala Glu Ile Ser Gly Ile Glu	245	250	255
Leu Ser Leu Glu Asp Val Thr Gly Lys Asp Ala Lys Leu Ser Glu Met	260	265	270
Leu Ser Pro Lys Asn Glu Asp Lys Leu Leu Tyr Arg Phe Gly Leu Cys	275	280	285
Ile Leu Leu Gln Phe Phe Gln Gln Met Ser Gly Thr Asn Leu Val Ser	290	295	300
Val Tyr Ala Thr Ile Leu Phe Gln Asp Asn Leu Gly Met Ser Pro Gln	305	310	315
Leu Ala Arg Val Leu Thr Gly Gly Ala Leu Thr Trp Lys Phe Leu Ser	325	330	335
Ser Phe Leu Ala Phe Phe Cys Ile Asp Arg Phe Gly Arg Arg Val Cys	340	345	350
Phe Met Ile Ser Gly Gly Gly Met Ala Ala Cys Met Val Ala Leu Ala	355	360	365
Val Ala Thr Ser Phe Pro Glu Asp Asn Arg Gly Ala Gln Ile Ser Ala	370	375	380
Ala Cys Phe Leu Tyr Leu Phe Asn Thr Phe Val Pro Ile Gly Phe Leu	385	390	395
Gly Ala Asn Phe Leu Tyr Cys Thr Glu Val Ala Pro Val Arg Leu Arg	405	410	415
Met Val Met Thr Ser Ile Ser Thr Ala Asn His Trp Leu Trp Asn Phe	420	425	430

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Val	Ile	Val	Met	Ile	Thr	Pro	Val	Ala	Ile	Ser	Thr	Ile	Gly	Ser	Arg
		435						440					445		
Tyr	Tyr	Ile	Met	Tyr	Ala	Val	Ile	Ala	Ala	Cys	Ile	Pro	Leu	Ser	Val
	450					455					460				
Tyr	Phe	Leu	Phe	Pro	Glu	Thr	Met	Gly	Arg	Asn	Leu	Glu	Glu	Ile	Asn
465					470					475					480
Leu	Met	Phe	Arg	Asp	Ser	Pro	Ser	Val	Trp	Ala	Thr	Val	Lys	Tyr	Ala
			485						490					495	
Arg	Asn	Arg	Pro	Ile	Gly	Met	Pro	Gln	Glu	Phe	Asp	Arg	Lys	Gln	Lys
			500					505					510		
Thr	Asp	His	Ile	Glu	Gly	Asn	Gly	Asp	Glu	Ser	Ser	Lys			
		515				520						525			

<210> SEQ ID NO 39

<211> LENGTH: 550

<212> TYPE: PRT

<213> ORGANISM: Isaria fumosorosea

<400> SEQUENCE: 39

Met	Thr	Arg	Lys	Thr	Tyr	Met	Gly	Met	Ser	Gly	Ser	Asn	Leu	Asn	Trp
1				5					10					15	
Ala	Ile	Ser	Thr	Ile	Ala	Gly	Cys	Asp	Phe	Leu	Leu	Phe	Gly	Tyr	Asp
			20					25					30		
Gln	Gly	Val	Met	Gly	Gly	Ile	Leu	Thr	Leu	Pro	Ile	Phe	Leu	Asp	Gln
		35					40					45			
Phe	Pro	Gln	Ile	Asn	Asp	Leu	Ala	Asp	Ala	Arg	Asn	Met	Ala	Gln	Leu
	50					55					60				
Pro	Gly	Thr	Arg	Ser	Gln	Arg	Ser	Leu	Asn	Gln	Gly	Ile	Ala	Ile	Ala
65					70					75					80
Ser	Tyr	Asn	Leu	Gly	Cys	Phe	Leu	Gly	Ala	Val	Ile	Thr	Ile	Phe	Ile
			85						90					95	
Gly	Asn	Pro	Leu	Gly	Arg	Arg	Arg	Met	Ile	Phe	Leu	Gly	Thr	Ala	Ile
		100						105					110		
Met	Thr	Val	Gly	Ala	Ala	Leu	Gln	Ala	Ser	Ala	Phe	Thr	Ile	Glu	His
		115					120					125			
Phe	Ile	Ile	Gly	Arg	Ile	Ile	Thr	Gly	Ile	Gly	Asn	Gly	Gly	Asn	Thr
	130					135					140				
Ser	Thr	Val	Pro	Met	Trp	Gln	Ser	Glu	Thr	Cys	Ser	Ser	His	Lys	Arg
145					150					155					160
Gly	Lys	Leu	Val	Met	Ile	Glu	Gly	Ala	Leu	Ile	Thr	Phe	Gly	Ile	Met
			165						170					175	
Val	Ser	Tyr	Trp	Ile	Asp	Leu	Gly	Leu	Ser	Phe	Thr	Asp	Ser	Ser	Val
		180						185					190		
Ser	Trp	Arg	Phe	Pro	Leu	Ala	Phe	Gln	Leu	Val	Phe	Cys	Ile	Phe	Ile
		195					200					205			
Leu	Ala	Phe	Val	Leu	Gly	Leu	Pro	Glu	Ser	Pro	Arg	Trp	Leu	Ile	Leu
	210					215					220				
Lys	Gly	Gln	Glu	Asp	Glu	Ala	Arg	Ala	Val	Ile	Ala	Ala	Ile	Ala	Asp
225					230					235					240
Lys	Glu	Ile	Glu	Asp	Pro	Phe	Val	Ala	Asn	Glu	Phe	Arg	Ala	Ile	Lys
			245						250					255	
Asp	Thr	Ala	Val	Lys	Ser	Ala	Gly	Gly	Tyr	Gly	Glu	Val	Phe	His	Met
		260						265						270	
Asp	Glu	Asn	Arg	Thr	Leu	His	Arg	Thr	Ile	Leu	Gly	Tyr	Val	Asn	Gln

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275	280	285
Met Phe Gln Gln Ile Ser Gly Ile Asn Leu Ile Thr Tyr Tyr Ala Ala		
290	295	300
Lys Ile Tyr Ala Asp Leu Gly Met Ser Pro Phe Leu Ala Arg Leu Leu		
305	310	315 320
Ala Ala Leu Asn Gly Thr Glu Tyr Phe Leu Ala Ser Trp Pro Ala Val		
	325	330 335
Phe Leu Val Glu Arg Val Gly Arg Arg Lys Leu Met Leu Phe Gly Ala		
	340	345 350
Val Gly Gln Ala Cys Thr Met Ala Ile Leu Ala Gly Val Asn Ser His		
	355	360 365
Lys Thr Asp Ala Ser Arg Ile Ala Gly Val Val Phe Leu Phe Val Phe		
	370	375 380
Asn Ser Phe Phe Ala Val Gly Trp Leu Gly Met Thr Trp Leu Tyr Pro		
385	390	395 400
Ala Glu Ile Thr Pro Leu Arg Thr Arg Ala Pro Ala Asn Ala Leu Ser		
	405	410 415
Thr Ser Ala Asn Trp Ile Phe Asn Phe Met Val Val Met Ile Thr Pro		
	420	425 430
Val Ser Phe Thr Asn Ile Asp Tyr His Thr Tyr Thr Ile Phe Ala Val		
	435	440 445
Ile Asn Ala Ile Met Val Pro Ser Val Tyr Phe Phe Phe Pro Glu Thr		
	450	455 460
Ala Tyr Arg Ser Leu Glu Glu Met Asp Ser Ile Phe Arg Lys Val Thr		
465	470	475 480
Gly Leu Arg Gly Ala Leu Asp Val Val Lys Val Ala Arg Glu Met Pro		
	485	490 495
His Arg Tyr Gly Lys Asn Gly Glu Leu Leu Ile Ala Phe Asp Glu Ser		
	500	505 510
Thr Glu Lys Val Gln Ala Glu His Ala Ser Gly Ser Val Ser Asp Gly		
	515	520 525
Ser Gly Asn Asn Asn Asn Thr Met Phe Thr Lys Thr Asp Glu Glu Thr		
	530	535 540
Ser Arg Pro Arg Glu Lys		
545	550	

<210> SEQ ID NO 40

<211> LENGTH: 498

<212> TYPE: PRT

<213> ORGANISM: Metarhizium majus

<400> SEQUENCE: 40

Met Ala Val Pro Glu Leu Glu Gly Arg Ala Leu Leu Leu Thr Val Thr		
1	5	10 15
Thr Leu Thr Ser Leu Gly Phe Met Leu Ile Gly Tyr Asp Asn Gly Leu		
	20	25 30
Met Gly Gly Leu Val Asn Ser Pro Ala Phe Gly Ser Ser Phe Asn Tyr		
	35	40 45
Pro Gly Ser Lys Met Ile Gly Val Ile Val Ala Ile Phe Glu Val Gly		
	50	55 60
Cys Phe Leu Gly Ser Ile Leu Ser Ala Ile Phe Gly Glu Arg Leu Gly		
65	70	75 80
Arg Arg Ser Thr Ile Gly His Gly Cys Trp Ile Met Ile Val Gly Ala		
	85	90 95

Val	Leu	Gln	Ala	Ser	Ser	Tyr	Gly	Arg	Ala	Gln	Leu	Ile	Val	Gly	Arg
			100					105					110		
Ile	Val	Ser	Gly	Leu	Gly	Leu	Gly	Ile	Ile	Asn	Ser	Thr	Val	Pro	Val
		115					120					125			
Leu	Gln	Ala	Glu	Phe	Ser	Pro	Lys	Ala	Thr	Arg	Gly	Val	Tyr	Val	Cys
		130				135					140				
Ala	Gln	Leu	Ser	Thr	Leu	Asn	Phe	Gly	Ile	Phe	Leu	Val	Tyr	Trp	Ile
					150					155					160
Asp	Tyr	Ala	Phe	Ser	Ser	His	Thr	Ser	Ser	Tyr	Ala	Trp	Arg	Val	Pro
				165					170					175	
Val	Ile	Leu	Gln	Cys	Val	Cys	Ile	Leu	Pro	Met	Met	Gly	Ile	Leu	Met
			180					185					190		
Leu	Ile	Pro	Glu	Thr	Pro	Arg	Trp	Leu	Ala	Ser	His	Asp	Arg	Pro	Asp
		195					200					205			
Asp	Cys	Leu	Arg	Val	Leu	Ala	Arg	Met	Lys	Ser	Ser	Ser	Ile	Asn	Asp
		210				215					220				
Pro	Glu	Val	Gln	Arg	Gln	His	His	Asn	Ile	Leu	Gln	Val	Val	Ala	Phe
					230					235					240
Glu	Ala	Ser	Ile	Gly	Thr	Gly	Ser	Trp	Lys	Asp	Leu	Leu	Ser	Asn	Asp
				245					250					255	
Arg	Val	Lys	Ser	Gln	Thr	Arg	Leu	Leu	Ile	Ala	Cys	Ser	Ile	Gln	Ala
				260				265					270		
Phe	Gln	Gln	Leu	Gly	Gly	Ile	Asn	Ala	Val	Ile	Tyr	Tyr	Thr	Asn	Thr
		275				280						285			
Leu	Phe	Ser	Lys	Ser	Ile	Gly	Phe	Asp	Glu	Arg	Met	Ser	Ala	Leu	Met
		290				295					300				
Ser	Gly	Phe	Leu	Gln	Thr	Trp	Phe	Phe	Val	Ala	Ser	Phe	Ile	Pro	Trp
				310					315						320
Phe	Leu	Ile	Asp	Arg	Ile	Gly	Arg	Lys	Pro	Leu	Phe	Val	Ser	Met	Ile
				325					330					335	
Ser	Leu	Met	Ala	Ala	Ala	Met	Ala	Val	Gln	Ala	Gly	Leu	Ile	Tyr	Gln
			340					345					350		
Val	Gln	Asn	Glu	Thr	Asn	Ile	Ser	His	Ser	Ala	Gly	Ile	Gly	Ala	Ala
		355				360						365			
Val	Met	Leu	Phe	Val	Phe	Gln	Gly	Ala	Phe	Thr	Ile	Gly	Phe	Gln	Ala
		370				375					380				
Thr	Val	Trp	Val	Tyr	Pro	Ser	Glu	Ile	Leu	Pro	Leu	Arg	Leu	Arg	Gln
				390						395					400
Lys	Gly	Ser	Ser	Ile	Ser	Thr	Ala	Ala	Asn	Trp	Ile	Phe	Asn	Tyr	Met
				405					410					415	
Val	Val	Gln	Ile	Thr	Pro	Ile	Ala	Ile	Asp	Asn	Ile	Gly	Trp	Lys	Thr
			420					425					430		
Tyr	Ile	Ile	Phe	Ala	Ile	Leu	Asn	Ala	Thr	Trp	Val	Pro	Ile	Ile	Phe
		435					440					445			
Phe	Phe	Phe	Pro	Glu	Thr										

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<211> LENGTH: 498

<212> TYPE: PRT

<213> ORGANISM: Metarhizium robertsii

<400> SEQUENCE: 41

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Met Ala Val Pro Glu Leu Glu Gly Arg Ala Leu Leu Leu Thr Val Thr
 1              5              10              15

Ala Leu Thr Ser Leu Gly Phe Met Leu Ile Gly Tyr Asp Asn Gly Leu
      20              25              30

Met Gly Gly Leu Val Asn Ser Pro Ala Phe Gly Ser Ser Phe Asp Tyr
      35              40              45

Pro Asp Ser Lys Met Ile Gly Val Ile Val Ala Ile Phe Glu Val Gly
      50              55              60

Cys Phe Leu Gly Ser Ile Leu Ser Ala Ile Phe Gly Glu Arg Leu Gly
      65              70              75              80

Arg Arg Ser Thr Ile Gly His Gly Cys Trp Ile Met Ile Val Gly Ala
      85              90              95

Val Leu Gln Ala Ser Ser Tyr Gly Arg Ala Gln Leu Ile Val Gly Arg
      100             105             110

Ile Val Ser Gly Leu Gly Leu Gly Ile Ile Asn Ser Thr Val Pro Val
      115             120             125

Leu Gln Ala Glu Phe Ser Pro Lys Ala Thr Arg Gly Val Tyr Val Cys
      130             135             140

Ala Gln Leu Ser Thr Leu Asn Phe Gly Ile Phe Leu Val Tyr Trp Ile
      145             150             155             160

Asp Tyr Ala Phe Ser Ser His Thr Ser Ser Tyr Ala Trp Arg Val Pro
      165             170             175

Val Ile Leu Gln Cys Val Cys Ile Leu Pro Met Met Gly Ile Leu Met
      180             185             190

Leu Ile Pro Glu Thr Pro Arg Trp Leu Ala Ser His Asp Arg Pro Asp
      195             200             205

Asp Cys Leu Arg Val Leu Ala Arg Met Lys Ser Ser Ser Ile Asn Asp
      210             215             220

Pro Glu Val Gln Arg Gln His His Asn Ile Leu Gln Val Val Ala Phe
      225             230             235             240

Glu Ala Ser Ile Gly Thr Gly Ser Trp Lys Asp Leu Leu Ser Asn Asp
      245             250             255

Arg Val Lys Ser Gln Thr Arg Leu Leu Ile Ala Cys Ser Ile Gln Ala
      260             265             270

Phe Gln Gln Leu Gly Gly Ile Asn Ala Val Ile Tyr Tyr Thr Asn Thr
      275             280             285

Leu Phe Ser Lys Ser Ile Gly Phe Asp Glu Arg Met Ser Ala Leu Met
      290             295             300

Ser Gly Phe Leu Gln Thr Trp Phe Phe Val Ala Ser Phe Ile Pro Trp
      305             310             315             320

Phe Leu Ile Asp Arg Ile Gly Arg Lys Pro Leu Phe Val Ser Met Ile
      325             330             335

Ser Leu Met Ala Ala Ala Met Ala Val Gln Ala Gly Leu Ile Tyr Gln
      340             345             350

Val Gln Asn Glu Thr Asn Ile Ser His Ser Ala Gly Ile Gly Ala Ala
      355             360             365

Val Met Leu Phe Val Phe Gln Gly Ala Phe Thr Ile Gly Phe Gln Ala
      370             375             380

Thr Val Trp Val Tyr Pro Ser Glu Ile Leu Pro Leu Arg Leu Arg Gln

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385	390	395	400
Lys Gly Ser Ser Ile Ser Thr Ala Ala Asn Trp Ile Phe Asn Tyr Met	405	410	415
Val Val Gln Ile Thr Pro Ile Ala Ile Glu Asn Ile Gly Trp Lys Thr	420	425	430
Tyr Ile Ile Phe Ala Ile Leu Asn Ala Thr Trp Val Pro Ile Ile Phe	435	440	445
Phe Phe Phe Pro Glu Thr Lys Gly Leu Glu Leu Glu Asp Val Asp Arg	450	455	460
Leu Phe Ala Gly Asp Asp Val Ile Glu Ser Val Gly Glu Lys Ser Met	465	470	475
Ala Leu Ala Thr Val Glu His Val Gly Arg Thr Asp Gly Thr Asn Glu	485	490	495
Arg Val			
<210> SEQ ID NO 42			
<211> LENGTH: 551			
<212> TYPE: PRT			
<213> ORGANISM: Nannizzia gypsea			
<400> SEQUENCE: 42			
Met Gly Val Gly Arg Gln Tyr Phe Gly Leu Arg Gly Thr Lys Leu Asn	5	10	15
Ile Ala Ile Gly Ile Ile Ala Gly Leu Asp Phe Leu Gly Val Met Gly	20	25	30
Gly Leu Leu Thr Leu Pro Ser Phe Glu Lys Val Phe Pro Glu Ile Ala	35	40	45
Thr Ser Lys Glu Ala Val Val Gly Leu Thr Gln Ala Glu Lys Asn His	50	55	60
Arg Ser Thr Ile Gln Gly Ile Ser Val Ala Ser Tyr Asn Leu Gly Cys	65	70	75
Phe Val Gly Ala Ile Ala Cys Ile Trp Val Gly Asp Arg Leu Gly Arg	85	90	95
Arg Lys Thr Ile Trp Leu Gly Ala Ala Val Met Val Val Gly Ala Ala	100	105	110
Leu Gln Ala Ser Ala Phe Gly Leu Pro His Phe Ile Val Gly Arg Leu	115	120	125
Val Thr Gly Phe Gly Asn Gly Leu Asn Thr Ser Thr Val Pro Thr Trp	130	135	140
Gln Ser Glu Cys Ser Lys Ser His Arg Arg Gly Gln Leu Val Met Val	145	150	155
Glu Gly Ala Leu Ile Thr Gly Gly Ile Cys Ile Ser Tyr Trp Leu Asp	165	170	175
Phe Gly Phe Ser Phe Leu Glu Pro Ser Ser Val Thr Trp Arg Phe Pro	180	185	190
Ile Ala Phe Gln Ile Val Phe Ala Leu Ile Ile Met Leu Val Val Met	195	200	205
Gly Leu Pro Glu Ser Pro Arg Trp Leu Val Leu Lys Gly Gln Glu Asp	210	215	220
Glu Ala Met Asn Val Leu Ala Ala Leu Ser Asp Leu Asp Arg Glu Asp	225	230	235
Arg Phe Val His Ala Glu Phe Ser Ala Ile Lys Asp Thr Val Ile Glu	245	250	255
Met Gln Lys Gly Gly Phe Arg Asp Leu Phe Thr Met Asp Lys Asp Arg			

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260				265				270							
His	Leu	His	Arg	Val	Ile	Leu	Ala	Tyr	Val	Asn	Gln	Met	Phe	Gln	Gln
	275						280						285		
Ile	Ser	Gly	Ile	Asn	Leu	Ile	Thr	Tyr	Tyr	Ala	Ala	Thr	Ile	Tyr	Glu
	290				295					300					
Gly	Ser	Ile	Gly	Leu	Ser	Pro	Phe	Leu	Ser	Arg	Val	Leu	Ala	Ala	Cys
305					310					315					320
Asn	Gly	Thr	Glu	Tyr	Phe	Ile	Ala	Ser	Trp	Ile	Ala	Val	Phe	Val	Val
			325						330					335	
Glu	Lys	Ile	Gly	Arg	Arg	Ile	Leu	Met	Leu	Phe	Gly	Ala	Val	Gly	Met
		340							345				350		
Ser	Leu	Ser	Met	Ala	Val	Leu	Ala	Ile	Ala	Thr	Ser	Phe	Lys	Gly	Gln
	355					360						365			
Thr	Glu	Ala	Gly	Ile	Val	Ala	Ala	Val	Phe	Leu	Phe	Val	Phe	Asn	Thr
	370				375						380				
Phe	Phe	Ala	Ile	Gly	Trp	Leu	Gly	Met	Thr	Trp	Leu	Tyr	Pro	Ala	Glu
385					390					395					400
Ile	Val	Pro	Leu	Arg	Ile	Arg	Ala	Pro	Ala	Asn	Ala	Leu	Ala	Thr	Ser
			405						410						415
Gly	Asn	Trp	Ile	Phe	Asn	Phe	Met	Val	Val	Met	Ile	Thr	Pro	Val	Ser
	420						425						430		
Phe	Ser	Ser	Ile	Glu	Tyr	Lys	Thr	Tyr	Ile	Ile	Phe	Ala	Val	Ile	Asn
	435					440						445			
Ala	Phe	Ile	Val	Pro	Val	Val	Tyr	Phe	Phe	Tyr	Pro	Glu	Thr	Ala	Tyr
	450				455					460					
Arg	Ser	Leu	Glu	Glu	Met	Asp	Ser	Ile	Phe	Arg	Lys	Thr	Lys	Ser	Ile
465					470					475					480
Phe	Thr	Val	Val	Lys	Ile	Ala	His	Glu	Thr	Pro	Arg	Arg	Tyr	Gly	Lys
			485						490						495
Asn	Gly	Glu	Val	Leu	Ile	Asp	Tyr	Asp	Glu	Thr	Asp	Glu	His	Arg	Ala
		500							505				510		
Arg	Ala	Gly	Ile	Thr	Gln	Glu	Glu	Thr	Thr	Thr	Ser	Phe	Pro	Glu	Lys
	515					520						525			
Ser	His	Ala	Asn	Pro	Asp	His	Asp	Ala	Glu	Thr	Gly	Asn	Ser	Asn	Ser
	530				535						540				
Pro	Ser	Asn	Gln	Ser	Thr	Ala									
545					550										

<210> SEQ ID NO 43

<211> LENGTH: 634

<212> TYPE: PRT

<213> ORGANISM: Pachysolen tannophilus

<400> SEQUENCE: 43

Met	Asp	Ser	Asn	Ile	Asp	Asp	Thr	Ala	Ile	Pro	Pro	Ser	Gly	Tyr	Leu
1				5					10					15	
Val	Gly	Lys	Pro	Leu	Leu	Tyr	Phe	Thr	Ser	Val	Phe	Val	Ser	Leu	Gly
		20						25					30		
Val	Phe	Leu	Phe	Gly	Tyr	Asp	Gln	Gly	Val	Met	Ser	Gly	Ile	Ile	Thr
		35					40					45			
Gly	Pro	Tyr	Phe	Lys	Tyr	Tyr	Phe	Asn	Asp	Pro	Ser	Ser	Thr	Thr	Ile
	50					55					60				
Gly	Thr	Met	Val	Ala	Ile	Leu	Glu	Ile	Gly	Ala	Leu	Ile	Ser	Ser	Leu
65					70					75					80

Leu	Val	Gly	Arg	Met	Gly	Asp	Ile	Ile	Gly	Arg	Arg	Arg	Thr	Ile	Arg
				85					90					95	
Tyr	Gly	Ala	Phe	Ile	Phe	Val	Val	Gly	Gly	Ser	Ile	Gln	Thr	Phe	Ala
			100					105					110		
Thr	Asp	Met	His	His	Leu	Ile	Leu	Gly	Arg	Ile	Val	Ser	Gly	Val	Ala
		115					120					125			
Val	Gly	Leu	Leu	Ser	Ala	Thr	Val	Pro	Val	Tyr	Gln	Ser	Glu	Ile	Ser
		130				135					140				
Gln	Pro	His	Asn	Arg	Gly	Gln	Leu	Ser	Cys	Val	Gln	Phe	Thr	Gly	Asn
					150				155						160
Ile	Phe	Gly	Tyr	Ala	Thr	Ser	Val	Trp	Thr	Asp	Tyr	Gly	Cys	Ser	Phe
				165					170					175	
Phe	Glu	Ser	Asn	Leu	Ser	Trp	Arg	Phe	Pro	Leu	Phe	Val	Gln	Cys	Val
			180					185					190		
Ile	Gly	Leu	Leu	Leu	Phe	Leu	Gly	Thr	Phe	Val	Ile	Val	Glu	Thr	Pro
		195					200					205			
Arg	Trp	Leu	Leu	Asn	Asn	Asp	His	Asp	Ala	Glu	Gly	Ile	Val	Val	Leu
		210				215						220			
Ala	Asp	Leu	Tyr	Ser	Asn	Gly	Asp	Val	His	Asp	Ile	Arg	Ala	Arg	Asn
					230				235						240
Glu	Phe	Arg	Asn	Ile	Lys	Glu	Asp	Val	Leu	Met	Ser	Arg	Leu	Glu	Asp
			245						250					255	
Thr	Gly	Thr	Ser	Tyr	Ser	Tyr	Met	Trp	Lys	Arg	Tyr	Lys	Thr	Arg	Ile
			260					265					270		
Leu	Ile	Ala	Met	Ser	Ser	Gln	Met	Phe	Ala	Gln	Phe	Asn	Gly	Ile	Asn
		275					280					285			
Val	Ile	Ser	Tyr	Tyr	Ala	Pro	Leu	Val	Phe	Glu	Gln	Ala	Gly	Trp	Phe
		290				295					300				
Gly	Arg	Asp	Ala	Ile	Leu	Met	Thr	Gly	Ile	Asn	Ser	Ile	Ile	Tyr	Phe
				310						315					320
Leu	Ser	Ser	Ile	Pro	Pro	Trp	Tyr	Leu	Val	Asp	Arg	Trp	Gly	Arg	Lys
				325					330					335	
Pro	Ile	Leu	Ile	Ile	Gly	Gly	Ile	Ile	Met	Ala	Ile	Ser	Leu	Phe	Ser
			340				345						350		
Ile	Ser	Phe	Val	Leu	Phe	Ile	Asn	Val	Pro	Lys	Thr	Pro	Val	Tyr	Val
		355					360					365			
Val	Ile	Leu	Val	Ile	Ile	Tyr	Asn	Ala	Leu	Phe	Gly	Phe	Ser	Trp	Gly
		370				375					380				
Pro	Ile	Pro	Trp	Leu	Leu	Pro	Pro	Glu	Ile	Leu	Pro	Leu	Ser	Ile	Arg
				390						395					400
Ser	Lys	Gly	Ala	Ser	Leu	Ser	Thr	Ala	Thr	Asn	Trp	Phe	Cys	Asn	Phe
			405						410					415	
Leu	Val	Gly	Glu	Leu	Thr	Pro	Val	Leu	Gln	Glu	Thr	Ile	Lys	Trp	Arg
			420					425					430		
Leu	Tyr	Leu	Ile	His	Gly	Thr									

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500	505	510
Gly Leu Pro Val Gln Tyr Gln Ser Pro Ala Val Leu Ala Arg His Pro		
515	520	525
Ile Val Ala Ala Gln Gln Leu Gln Asn Leu Lys Ser Ser Thr Pro Ser		
530	535	540
Leu Arg Ser Thr Ser Asn Tyr Met Asn Asn Val Ser Pro Leu Ile Gln		
545	550	555
Pro His Glu Leu Glu Pro Pro Asn Ile Glu Glu Ile Arg Ala Tyr Lys		
565	570	575
Leu Ser Asp Asn Asn Ser Ile Lys Gly Asn Ile Arg Arg Ser Ser Glu		
580	585	590
Asn Ile Gly Ser Met Phe His Lys Val Phe Asn Asn Asn Ser Leu Lys		
595	600	605
Arg Ser Asp Ser Thr Ser Glu Phe Thr Asn Asp Ser Glu Asp Glu Glu		
610	615	620
Ala Asn Leu Thr Ser Asn Ser Gly Arg His		
625	630	
<210> SEQ ID NO 44		
<211> LENGTH: 484		
<212> TYPE: PRT		
<213> ORGANISM: Paracoccidioides lutzii		
<400> SEQUENCE: 44		
Met Ser Gln Lys Ile Arg Pro Ala Leu Ala Leu Trp Gly Thr Arg Met		
1	5	10
Ala Phe Met Ser Tyr Gly Trp Asp Ala Gly Val Leu Gly Gly Val Leu		
20	25	30
Glu Thr Ala Pro Phe Gln Asp Ala Met Lys His Pro Ser Thr Thr Thr		
35	40	45
Met Ser Met Ile Val Ala Ala Phe Leu Leu Ala Ser Trp Leu Gly Cys		
50	55	60
Cys Ile Val Ala Ser Pro Trp Ser Asp Arg Val Gly Arg Arg Met Trp		
65	70	75
Val Ile Ser Gly Ala Ala Ile Gln Ile Ile Gly Thr Ile Ile Ser Thr		
85	90	95
Ala Ser Tyr Ser Ser Gly Gln Met Ile Ala Gly Arg Thr Ile Ile Gly		
100	105	110
Ile Gly Asn Gly Ile Val Val Ala Ser Ala Pro Val Tyr Ile Ala Glu		
115	120	125
Ile Thr Pro Thr Thr Ser Met Arg Gly Pro Leu Ile Gly Ile Leu Met		
130	135	140
Gly Phe Ala Cys Thr Gly Thr Thr Leu Ala Tyr Trp Val Asp Phe Ala		
145	150	155
Phe Thr His Ala Arg Gly Gln Val Val Trp Arg Val Pro Val Gly Leu		
165	170	175
Gln Ile Ile Trp Ser Leu Leu Thr Ile Ile Leu Thr Leu Pro Asn Met		
180	185	190
Asp Ser Pro Arg Trp Tyr Tyr Leu Arg Asn Arg Asp His Glu Gly Leu		
195	200	205
Asn Val Leu Gln Gln Leu His Pro Asp Gln Glu Val Ala Leu Arg Val		
210	215	220
Gln Gly Glu Ile Leu Lys Glu Leu Arg Glu Glu Lys Glu Glu Lys Leu		
225	230	235
		240

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Gly	Val	Trp	Gln	Ala	Glu	Thr	Ser	Lys	Met	Ser	Ser	Arg	Gly	Lys	Leu
130						135				140					
Val	Ile	Ile	Gln	Met	Ala	Asn	Cys	Ile	Thr	Gly	Phe	Ser	Ile	Ser	Asn
145					150					155					160
Trp	Leu	Thr	Leu	Ala	Phe	Ser	Phe	Ala	Pro	Gly	Asp	Val	Ala	Trp	Arg
			165						170					175	
Phe	Pro	Leu	Ala	Phe	Gln	Leu	Phe	Phe	Thr	Phe	Cys	Ile	Tyr	Ala	Leu
		180					185						190		
Cys	Pro	Phe	Leu	Pro	Asp	Ser	Pro	Arg	Leu	Leu	Ile	Arg	Lys	Gly	Lys
		195					200					205			
Pro	Asp	Glu	Ala	Leu	Glu	Val	Leu	Ala	Ala	Leu	Glu	Gly	His	Gly	Ala
	210					215					220				
Thr	Pro	Glu	Ser	Ala	Ser	Val	Arg	Thr	Gln	Tyr	Asn	Ile	Ile	Lys	Asp
225					230					235					240
Ile	Leu	Asp	Arg	Glu	His	Met	Asn	Thr	Tyr	Thr	Trp	Trp	Gln	Leu	Leu
			245						250					255	
Ser	Gly	Lys	Gly	Pro	Ser	Gly	Val	Leu	Arg	Arg	Met	Ile	Leu	Gly	Ala
			260					265					270		
Trp	Met	Gln	Ala	Met	Asn	Gln	Ile	Ser	Gly	Ile	Asn	Val	Thr	Ser	Tyr
		275					280					285			
Tyr	Met	Ser	Tyr	Ile	Phe	Ile	Asn	Ala	Leu	Gly	Ile	Ser	Glu	Leu	Leu
	290					295					300				
Ser	Arg	Ile	Leu	Ala	Ala	Ala	Gly	Ser	Val	Asp	Tyr	Leu	Val	Phe	Ala
305				310						315					320
Cys	Leu	Ala	Phe	Phe	Val	Ile	Glu	Arg	Tyr	Gly	Arg	Arg	Lys	Val	Met
			325						330					335	
Met	Val	Ser	Ala	Ala	Ala	Cys	Ser	Thr	Cys	Trp	Ile	Val	Ile	Ala	Ile
			340					345					350		
Ala	Leu	Gly	Leu	Ser	Ala	Asn	Gly	Gly	Asp	Ser	Tyr	Lys	Leu	Gly	Ile
		355					360					365			
Val	Ala	Val	Ser	Phe	Phe	Phe	Val	Phe	Phe	Ala	Ser	Phe	Gly	Met	Gly
	370					375					380				
Val	Leu	Gly	Val	Pro	Trp	Leu	Tyr	Pro	Thr	Glu	Ile	Asn	Ala	Leu	Glu
385					390					395					400
Met	Arg	Thr	Lys	Gly	Ala	Ser	Leu	Ala	Met	Ser	Thr	Asn	Trp	Ile	Met
			405						410					415	
Asn	Tyr	Ala	Val	Val	Gln	Val	Thr	Leu	Pro	Gly	Ile	Gln	Asn	Ile	Gly
			420					425					430		
Trp	Lys	Phe	Trp	Ile	Ile	Trp	Ala	Val	Ile	Cys	Phe	Ser	Phe	Ile	Pro
		435					440					445			
Ile	Thr	Tyr	Phe	Phe	Tyr	Pro	Glu	Thr	Ala	Asn	Arg	Thr	Leu	Glu	Asp
	450					455					460				
Ile	Asp	Arg	Phe	Phe	Glu	Thr	Asn	Pro	Gly	Leu	Phe	Val	His	Arg	Asn
465					470					475					480
Lys	Leu	Ala	Ile	Gln	Leu	His	Arg	Pro	Val	Glu	Phe	Ile	Glu	Ala	Asp
			485						490					495	
Glu	Arg	Ile	Ala	Thr	Ala	Gln	Ala	Glu	Glu	Glu	Lys	Asn	Leu	Gly	Glu
			500					505					510		
Lys	Thr	Asp	Phe	Val	Glu	Ile	Lys	Glu	Ala	Val					
		515					520								

<210> SEQ ID NO 46

<211> LENGTH: 198

<212> TYPE: PRT

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<213> ORGANISM: *Penicillium digitatum*

<400> SEQUENCE: 46

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Met Glu Lys Tyr Leu Ser Leu Ile Met Val Gly Ala Phe Ala Thr Val
 1           5           10           15

Tyr Thr Phe Ala Thr Ile Pro Ser Ile Phe Leu Ile Glu Arg Glu Asn
          20           25           30

Glu Asp Asn Ala Lys Gly Ala Ala Val Val Phe Pro Phe Ile Thr Phe
          35           40           45

Phe Ala Phe Thr Leu Leu Pro Leu Leu Trp Ile Tyr Pro Pro Glu Ile
          50           55           60

Asn Pro Leu Ser Thr Arg Thr Leu Ile Ala Ser Thr Cys Ala Asn Trp
65           70           75           80

Ile Cys Asn Phe Ala Leu Val Leu Phe Thr Pro Leu Val Ala Asp His
          85           90           95

Ser Pro Arg Ser Val Asp Leu Phe Phe Ala Leu Phe Lys Phe Ile Gly
          100          105          110

Leu Ile Phe Gly Val Phe Phe Tyr Val Glu Thr Ala Gly Arg Gln Leu
          115          120          125

Gly Glu Val Asp Arg Ile Tyr Ala Lys Ala His Ile Glu Gly Lys Met
          130          135          140

Ala Trp Arg Val Ala Gln Asp Met Pro Lys Leu Asn Phe Glu Glu Ile
145          150          155          160

Val Gln Gln Phe Arg Gly Leu Gly Leu Asp Thr Asn Glu Leu Ala Ala
          165          170          175

His Glu Lys Ile Glu Leu Gly Leu Asn Ser Asn Ser Gly Gln Glu Leu
          180          185          190

Glu Glu Val Arg Glu Lys
          195

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<210> SEQ ID NO 47

<211> LENGTH: 533

<212> TYPE: PRT

<213> ORGANISM: *Phialophora attinorum*

<400> SEQUENCE: 47

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Met Gly Arg Arg Tyr Leu Gly Gly Ser Gly Gln Arg Leu Thr Val Trp
 1           5           10           15

Ile Ser Ile Ala Ser Ser Thr Val Leu Ile Phe Tyr Gly Tyr Asp Gln
          20           25           30

Gly Val Phe Gly Asn Val Leu Val Ser Glu Asp Phe Leu Arg Thr Val
          35           40           45

Gly Tyr Pro Ser Val Thr Ala Gln Gly Thr Met Thr Ser Val Tyr Asn
          50           55           60

Leu Gly Cys Phe Ala Gly Ala Leu Ser Thr Leu Tyr Thr Gly Asp Lys
65           70           75           80

Leu Gly Arg Pro Arg Thr Leu Ile Leu Gly Ser Cys Thr Ile Ala Val
          85           90           95

Gly Ala Ile Val Gln Ala Ala Cys Met Asn Ala Ala Met Gln Tyr Ala
          100          105          110

Gly Arg Val Ile Ala Gly Met Gly Thr Gly Met Asn Thr Ala Thr Ala
          115          120          125

Gly Val Trp Gln Ser Glu Thr Ser Lys Met Arg Ser Arg Gly Lys Leu
          130          135          140

Ile Ile Ile Gln Met Ala Asn Cys Ile Thr Gly Phe Ala Ile Ser Asn

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145	150	155	160
Trp Leu Thr	Leu Gly Phe Ser Phe Ala	Pro Gly Ser Val Ser Trp Arg	
	165	170	175
Phe Pro Leu	Ala Phe Gln Met Phe Phe Thr	Ile Leu Ile Cys Leu Met	
	180	185	190
Cys Pro Phe	Leu Pro Asp Ser Pro Arg Leu	Leu Ile Arg Lys Gly Lys	
	195	200	205
Tyr Asp Glu	Ala Tyr Glu Val Leu Ala Ala	Leu Glu Gly Asn Gly Ala	
	210	215	220
Thr Val Asn	Ser Pro Val Val Arg Thr Gln Phe	Ala Ile Ile Lys Gln	
225	230	235	240
Val Leu Asp	Glu Glu Tyr Ala Val Lys Tyr	Thr Trp Trp Gln Ile Leu	
	245	250	255
Thr Gly Lys	Gly Pro Ser Gly Val Leu Arg	Arg Met Val Leu Gly Ala	
	260	265	270
Trp Met Gln	Ala Ser Asn Gln Ile Ser Gly	Ile Asn Val Thr Ser Tyr	
	275	280	285
Tyr Met Thr	Tyr Val Phe Ile Asn Ala	Ile Asn Phe Ser Gln Leu Thr	
	290	295	300
Ala Arg Ile	Leu Ala Ala Ala Gly Ala Met	Asp Tyr Leu Phe Phe Ser	
305	310	315	320
Phe Met Ala	Tyr Phe Val Ile Glu Arg Phe	Gly Arg Arg Ser Val Met	
	325	330	335
Met Thr Ser	Ala Ala Ala Cys Ser Ile Cys	Trp Thr Val Ile Ala Ile	
	340	345	350
Ser Leu Gly	Leu Ser Glu Thr Gly Arg Ala	Asp Ser Tyr Thr Met Gly	
	355	360	365
Ala Val Ala	Val Ser Phe Phe Phe Leu Phe	Phe Ala Ser Phe Ala Met	
	370	375	380
Gly Val Leu	Gly Val Pro Trp Leu Tyr Pro	Thr Glu Val Asn Ala Leu	
385	390	395	400
Ala Phe Arg	Ala Lys Gly Ala Ser Leu Ala	Met Ser Thr Asn Trp Ile	
	405	410	415
Met Asn Tyr	Met Val Ala Gln Ile Thr Pro	Pro Gly Ile Asp Asn Leu	
	420	425	430
Gly Tyr Lys	Phe Trp Ile Ile Trp Ala Val	Ile Cys Ala Ala Phe Val	
	435	440	445
Pro Ile Thr	Tyr Leu Phe Tyr Pro Glu Thr	Ala Asn Arg Ser Leu Glu	
	450	455	460
Asp Ile Asp	Arg Phe Phe His Ser Asn His	Gly Ile Leu Val Phe Asn	
465	470	475	480
Asn Lys Val	Ala Thr Gln Leu Lys Arg Pro	Glu Ile Tyr Glu Glu Ala	
	485	490	495
Asp Arg Arg	Val Ala Ala Ala His Glu Lys	Val Gly Ala Gly Ala Asp	
	500	505	510
Gln Ala Asp	Glu Gly Lys Glu Ala Gly Ala	Thr Leu Val Glu Glu Asn	
	515	520	525
Gly Ala Ala	Arg Ala		
	530		

<210> SEQ ID NO 48

<211> LENGTH: 213

<212> TYPE: PRT

<213> ORGANISM: Pochonia chlamydosporia

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<400> SEQUENCE: 48

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Met Gly Leu Ile Gly Arg Pro Leu Asn Trp Ala Ile Thr Ala Thr Ala
 1           5           10           15
Gly Ala Gly Phe Leu Leu Phe Gly Tyr Asp Gln Gly Val Met Ser Gly
 20           25           30
Leu Leu Ala Gly Asp Ala Phe Thr Arg Thr Phe Pro Glu Met Asp Thr
 35           40           45
Thr Glu Ser Gly His Gly Ser Ala Ser Leu Gln Gly Thr Val Val Ala
 50           55           60
Ile Tyr Glu Ile Gly Cys Phe Phe Gly Ala Leu Ile Ala Phe Val Phe
 65           70           75           80
Ala Glu Arg Leu Gly Arg Arg Arg Thr Ile Met Leu Gly Cys Val Ile
 85           90           95
Leu Ser Ile Gly Gly Ala Leu Gln Ala Cys Ala Ser Thr Ile Pro His
 100          105          110
Met Ile Ala Gly Arg Ile Val Ala Gly Leu Gly Asn Gly Leu Asn Thr
 115          120          125
Ser Thr Ile Pro Val Cys His Ser Glu Leu Met Val Ala Ser Lys Arg
 130          135          140
Gly Lys Gly Leu Cys Ile Glu Leu Ser Ile Thr Val Phe Gly Val Met
 145          150          155          160
Ile Ala Tyr Trp Val Asp Cys Gly Met Ser Tyr Val Pro Asn Asp Ala
 165          170          175
Gln Phe Arg Phe Pro Leu Ala Leu Gln Cys Leu Phe Ala Ile Ile Thr
 180          185          190
Val Ile Gly Ile Leu Phe Leu Pro Glu Ser Pro Arg Trp Leu Val Ala
 195          200          205
His Asp Arg His Asp
 210

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<210> SEQ ID NO 49

<211> LENGTH: 498

<212> TYPE: PRT

<213> ORGANISM: Pyrenophora tritici-repentis

<400> SEQUENCE: 49

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Met Pro Glu Leu Arg Gly Arg Ala Leu Met Leu Thr Ile Ser Val Leu
 1           5           10           15
Thr Ser Leu Gly Phe Met Leu Ile Gly Tyr Asp Asn Gly Leu Met Gly
 20           25           30
Gly Leu Val Gly Ala Pro Ala Phe Asn Lys Thr Phe Asp His Pro Ser
 35           40           45
Ser Asp Met Ile Gly Thr Ile Val Ala Ile Phe Glu Ile Gly Cys Phe
 50           55           60
Phe Gly Ala Met Ala Thr Ala Val Ile Gly Glu Lys Leu Gly Arg Arg
 65           70           75           80
Lys Ser Val Ala Ile Gly Ala Val Ile Ser Ile Leu Gly Ala Leu Leu
 85           90           95
Gln Ala Thr Ala Tyr Gly Arg Ala His Leu Ile Val Gly Arg Ile Val
 100          105          110
Ser Gly Val Gly Leu Gly Ile Ile Asn Ser Thr Val Pro Val Met Gln
 115          120          125
Ala Glu Phe Ser Pro Lys Ala Ser Arg Gly Ile Tyr Val Cys Ala Gln
 130          135          140

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Leu Ser Thr Leu Asn Phe Gly Ile Phe Leu Val Tyr Trp Ile Asp Tyr
145          150          155          160

Ala Phe Val Ser His Thr Ser Asp Tyr Ala Trp Arg Ile Pro Thr Ile
          165          170          175

Leu Gln Cys Ile Ile Val Leu Ala Ile Leu Gly Leu Leu Thr Val Ile
          180          185          190

Pro Glu Thr Pro Arg Trp Leu Ala Ala His Asp Arg Pro Asp Glu Cys
          195          200          205

Leu Arg Val Leu Ala Arg Val Ala Asp Val Pro Glu Thr Asp Pro Glu
          210          215          220

Val Gln Arg Leu His Thr Val Ile Thr Glu Thr Val Ala Phe Glu Gln
225          230          235          240

Ser Arg Gln Ala Gly Trp Lys Asp Ile Val Arg Ser Asp Pro Ile Lys
          245          250          255

Ser Arg Arg Arg Phe Leu Ile Ala Cys Gly Ile Gln Met Phe Gln Gln
          260          265          270

Leu Gly Gly Ile Asn Ala Ile Ile Tyr Tyr Ser Gly Thr Leu Phe Gln
          275          280          285

Lys Ser Ile Gly Phe Asp Thr His Met Ser Ala Leu Met Ser Gly Phe
          290          295          300

Leu Gln Thr Trp Phe Phe Val Ala Ser Phe Ile Pro Trp Phe Leu Ile
305          310          315          320

Asp Arg Val Gly Arg Arg Pro Leu Leu Leu Ser Met Ile Ser Leu Met
          325          330          335

Ala Ala Val Met Ala Val Gln Ser Gly Leu Ile Tyr Gln Val Gln Tyr
          340          345          350

Lys Thr Ala Ser Ala Lys Gly Ala Gly Ile Ala Ala Ala Ala Met Leu
          355          360          365

Phe Ile Phe Gln Gly Ala Phe Thr Ile Gly Phe Gln Ala Thr Val Trp
          370          375          380

Val Tyr Pro Ser Glu Ile Leu Pro Leu Arg Leu Arg Gln Arg Gly Ser
385          390          395          400

Ala Ile Ser Thr Ala Ala Asn Trp Ile Cys Asn Tyr Ile Ile Val Gln
          405          410          415

Ile Thr Pro Arg Ala Ile Ser Asn Ile Gly Trp Lys Thr Tyr Ile Ile
          420          425          430

Phe Ala Val Leu Asn Gly Leu Trp Val Pro Ile Ile Phe Phe Phe Phe
          435          440          445

Pro Glu Thr Lys Gly Leu Glu Leu Glu Asp Val Asp Arg Leu Phe Ser
          450          455          460

Gly Glu Ala Ser Arg Thr Asp Leu Leu Asp Lys Asp Leu Asp Asp Glu
465          470          475          480

Arg Val Glu Ser Val Val Val Ala Lys Thr Val Glu Cys Ala Gly His
          485          490          495

Val Ser

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<210> SEQ ID NO 50

<211> LENGTH: 533

<212> TYPE: PRT

<213> ORGANISM: Pyricularia oryzae

<400> SEQUENCE: 50

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Met Trp Leu Thr Asp Lys Phe Tyr Gly Leu Arg Gly Lys Ser Leu Asn
1          5          10          15

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-continued

Tyr Ala Ile Val Cys Ile Ala Gly Val Asp Phe Leu Leu Phe Gly Tyr
 20 25 30
 Asp Gln Gly Val Met Gly Gly Ile Leu Thr Leu Pro Val Phe Leu Ser
 35 40 45
 Gln Phe Pro Thr Ile Asn Pro Glu Ala Asp Gly Leu Thr Ser Ile Glu
 50 55 60
 Ser Ala Gln Arg Ala Thr Asn Gln Gly Ile Ala Val Ala Ser Tyr Asn
 65 70 75 80
 Leu Gly Cys Phe Val Gly Ala Val Ile Ala Ile Trp Ile Ser Asn Pro
 85 90 95
 Leu Gly Arg Lys Arg Met Ile Ile Leu Gly Thr Ser Ile Met Val Val
 100 105 110
 Gly Thr Ile Ile Lys Ile Thr Ala Phe Ser Leu Val His Leu Val Ile
 115 120 125
 Gly Arg Ile Ile Met Gly Leu Gly Asn Gly Met Asn Thr Ser Thr Val
 130 135 140
 Pro Thr Trp Gln Ser Glu Thr Ser Ser Ser His Lys Arg Gly Lys Met
 145 150 155 160
 Val Met Ile Glu Gly Ala Leu Ile Thr Cys Gly Ile Met Ile Ser Tyr
 165 170 175
 Trp Ile Asp Leu Gly Leu Ser Phe Ala Pro Gly Ser Val Ala Trp Arg
 180 185 190
 Phe Pro Ile Ala Phe Gln Leu Val Phe Cys Phe Phe Ile Leu Ala Phe
 195 200 205
 Val Trp Asn Leu Pro Glu Ser Pro Arg Trp Leu Ile Leu Lys Gly His
 210 215 220
 His Glu Glu Ala Lys Leu Val Ile Ala Ala Ile Ala Asp Leu Glu Val
 225 230 235 240
 Glu Asp Asn Phe Val Gln Asn Glu Phe Leu Ala Ile Lys Glu Thr Val
 245 250 255
 Glu Glu Met Ser Lys Gly Ser Phe Arg Asp Leu Phe Ala Thr Asn Glu
 260 265 270
 Asn Arg Asn Leu His Arg Thr Phe Leu Ala Tyr Leu Asn Gln Val Phe
 275 280 285
 Gln Gln Ile Ser Gly Ile Asn Leu Ile Thr Tyr Tyr Ala Ala Val Ile
 290 295 300
 Tyr Ser Gly Leu Gly Met Ser Asp Phe Met Ser Arg Leu Leu Ala Ala
 305 310 315 320
 Leu Asn Gly Thr Glu Tyr Phe Ile Ala Ser Trp Pro Ala Val Trp Leu
 325 330 335
 Val Glu Arg Val Gly Arg Arg Lys Leu Met Leu Phe Gly Ala Ala Gly
 340 345 350
 Gln Ala Leu Thr Met Ala Ala Ser Ala Gly Val Thr Ser Arg Ser Glu
 355 360 365
 Glu Gly Phe Gln Ile Ala Gly Val Val Leu Leu Phe Ile Phe Asn Thr
 370 375 380
 Phe Phe Ala Ile Gly Trp Leu Gly Met Thr Trp Leu Tyr Pro Ala Glu
 385 390 395 400
 Ile Val Pro Leu Arg Ile Arg Ala Pro Ala Asn Ala Leu Ser Thr Ser
 405 410 415
 Ala Asn Trp Ile Phe Asn Phe Met Val Val Met Ile Thr Pro Val Ala
 420 425 430

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Phe Asn Thr Ile Lys His His Thr Tyr Thr Ile Phe Ala Ile Ile Asn
 435 440 445
 Ala Ile Met Val Pro Ser Val Tyr Phe Leu Phe Pro Glu Thr Ala Tyr
 450 455 460
 Arg Ser Leu Glu Glu Met Asp Thr Ile Phe Gln Lys Val His Gly Trp
 465 470 475 480
 Lys Gly Leu Phe Thr Val Val Arg Gln Ala Glu Ile Glu Pro Arg Arg
 485 490 495
 Tyr Gly Lys Asn Gly Glu Leu Leu Leu Asp Val Asp Ala Val Arg Ala
 500 505 510
 Glu Lys Gly Asp Ser Ser Ser Ala Gly Arg Glu Asn Glu His Arg Glu
 515 520 525
 Val Arg Thr Glu Pro
 530

<210> SEQ ID NO 51
 <211> LENGTH: 540
 <212> TYPE: PRT
 <213> ORGANISM: *Scedosporium apiospermum*

<400> SEQUENCE: 51

Met Ser Val Gln Glu Asn Gly Val Cys Ile Pro Thr Phe Trp Gly Thr
 1 5 10 15
 Ser Gly Arg Lys Leu Gln Met Leu Val Thr Ala Val Ala Thr Ala Asp
 20 25 30
 Phe Leu Leu Phe Gly Tyr Asp Gln Gly Val Met Ser Gly Ile Ile Ser
 35 40 45
 Ala Asp Ala Phe Thr Glu Asp Phe Pro Glu Val Val Thr Gly Gly Ser
 50 55 60
 Ala Tyr Glu Gly Phe Val Thr Ser Ile Tyr Ala Val Gly Cys Phe Leu
 65 70 75 80
 Gly Ala Val Phe Ile Leu Leu Phe Gly Asp His Leu Gly Arg Arg Met
 85 90 95
 Ser Ile Tyr Leu Gly Ala Thr Thr Met Ile Val Gly Val Ile Ile Gln
 100 105 110
 Val Ser Cys Val Pro Val Ser Gly Gly Thr Thr Ala Gln Phe Ile Ile
 115 120 125
 Gly Arg Cys Ile Thr Gly Val Gly Asn Gly Ile Asn Thr Ser Thr Ile
 130 135 140
 Pro Thr Tyr Gln Ala Glu Cys Ser His Ser His Asn Arg Gly Lys Leu
 145 150 155 160
 Ile Cys Ile Glu Gly Gly Asn Val Ala Ile Gly Thr Leu Ile Ala Tyr
 165 170 175
 Trp Ile Asp Tyr Gly Ala Ile Tyr Gly Pro His Asp Phe Thr Trp Arg
 180 185 190
 Phe Pro Ile Ala Phe Gln Cys Val Phe Ala Ile Thr Val Leu Ile Leu
 195 200 205
 Asn Thr Arg Leu Pro Glu Ser Pro Arg Trp Leu Leu Thr Lys Asp Lys
 210 215 220
 His Glu Glu Ala Ala Met Val Leu Ala Ala Leu Ala Gly Lys Pro Thr
 225 230 235 240
 Asp Asp Tyr Glu Val Arg Ser Gln Met Thr Ala Ile Val Glu Ser Ile
 245 250 255
 Lys Ala Ser Gly His Ser Gly Gly Val Thr Pro Met Ser Ala Leu Phe
 260 265 270

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Thr Asn Gly Lys Thr Gln His Phe Arg Arg Met Ile Leu Gly Phe Ser
 275 280 285
 Ser Gln Met Met Gln Gln Leu Ser Gly Cys Asn Ala Val Ile Tyr Tyr
 290 295 300
 Phe Pro Ile Leu Phe Gln Thr Ser Ile Gly Val Ser His Asn Met Ala
 305 310 315 320
 Leu Leu Leu Gly Gly Val Asn Met Ile Val Tyr Ser Ile Phe Ala Thr
 325 330 335
 Thr Ser Trp Phe Ala Val Glu Arg Ile Gly Arg Arg Lys Leu Phe Leu
 340 345 350
 Ile Gly Thr Val Gly Gln Cys Leu Ser Met Val Leu Ala Phe Gly Ala
 355 360 365
 Leu Ile Pro Gly Thr Glu Ala Ala Ala Arg Gly Ala Ala Val Gly Leu
 370 375 380
 Phe Thr Tyr Ile Ala Phe Phe Gly Ala Thr Trp Leu Pro Leu Pro Trp
 385 390 395 400
 Leu Tyr Pro Ala Glu Ile Asn Pro Leu Lys Thr Arg Thr Lys Ala Asn
 405 410 415
 Ala Val Ser Thr Val Ser Asn Trp Leu Trp Asn Phe Phe Ile Val Met
 420 425 430
 Ile Thr Pro Val Leu Val Asp Asn Ile Gly Trp Gly Thr Tyr Leu Phe
 435 440 445
 Phe Ala Val Leu Asn Ala Ile Phe Phe Pro Ile Ile Tyr Phe Phe Tyr
 450 455 460
 Pro Glu Thr Ser Gln Arg Ser Leu Glu Glu Ile Asp His Ile Phe Ala
 465 470 475 480
 Lys Gly Tyr Thr Glu Asn Met Ser Tyr Val Arg Ala Ala Lys Glu Leu
 485 490 495
 Pro Arg Leu Ser Gly Glu Ile Asn Ala Gln Thr Val Gly Arg Asp Asp
 500 505 510
 Val Asp Val Glu Lys Ser Gly Glu Val Ser Asp Glu Met Asp Ser Arg
 515 520 525
 Ser Val Glu Asn Thr Glu Lys Leu Ser His Leu Asp
 530 535 540

<210> SEQ ID NO 52

<211> LENGTH: 512

<212> TYPE: PRT

<213> ORGANISM: Sphaerulina musiva

<400> SEQUENCE: 52

Met Ala Phe Glu Leu Arg Gly Ser Lys Leu Ile Ala Val Ile Leu Leu
 1 5 10 15
 Ala Ser Gly Leu Asp Phe Leu Leu Phe Gly Tyr Asp Gln Gly Leu Phe
 20 25 30
 Gly Gly Ile Leu Gly Gly Lys Arg Phe Lys Ala Met Leu Gly Glu Pro
 35 40 45
 Gly Pro Thr Met Thr Gly Phe Val Thr Gly Val Tyr Asp Ile Gly Cys
 50 55 60
 Ala Ile Gly Ala Val Ala Ala Phe Phe Phe Gly Glu Ala Ile Gly Arg
 65 70 75 80
 Lys Lys Ser Ile Ile Tyr Ala Asn Val Ile Val Ile Ile Gly Ala Thr
 85 90 95
 Ile Gln Thr Ala Cys Tyr Ser Tyr Ala Gln Met Ala Val Ala Arg Val

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100							105					110				
Ile	Ala	Gly	Val	Gly	Val	Gly	Leu	Ser	Thr	Val	Ala	Val	Pro	Ile	Leu	
		115					120					125				
Gln	Ser	Glu	Thr	Leu	Pro	Ser	His	Asn	Arg	Gly	Ala	Leu	Leu	Val	Val	
		130			135						140					
Gln	Ser	Ala	Leu	Ile	Ile	Ile	Gly	Val	Ala	Ile	Ala	Ser	Trp	Leu	Cys	
		145			150				155						160	
Phe	Ala	Thr	Leu	Phe	Ala	Asn	Ser	Ser	Leu	Gln	Trp	Arg	Phe	Pro	Ile	
				165					170					175		
Ala	Cys	Gln	Ile	Leu	Phe	Ser	Leu	Leu	Val	Leu	Cys	Cys	Cys	Pro	Phe	
				180					185			190				
Leu	Pro	Glu	Thr	Pro	Arg	Trp	Leu	Cys	Lys	His	Asn	Arg	Ile	Asp	Glu	
		195				200						205				
Ala	Arg	Tyr	Thr	Ile	Ser	Arg	Leu	Leu	Asp	Lys	Pro	Glu	Asp	Asp	Ala	
		210			215						220					
Glu	Val	Lys	Gly	Gln	Leu	His	Glu	Ile	Leu	Ala	Asn	Ile	Glu	Ala	Glu	
		225			230				235						240	
Asn	Glu	Asp	Gly	Glu	Pro	Ser	Trp	Ser	Glu	Val	Phe	Ser	Asn	Ala	Thr	
				245					250					255		
Lys	Ala	Arg	Asn	Leu	Gln	Arg	Val	Leu	Leu	Gly	Met	Gly	Pro	Tyr	Met	
				260					265			270				
Met	Asn	Gln	Trp	Ser	Gly	Ile	Asn	Ala	Leu	Cys	Tyr	Tyr	Leu	Ala	Tyr	
		275				280						285				
Ile	Leu	Glu	Thr	Tyr	Leu	Asp	Phe	Ser	Gln	Asn	Met	Ser	Leu	Ile	Leu	
		290			295						300					
Ala	Ser	Val	Ala	Phe	Thr	Gln	Tyr	Ala	Ile	Phe	Ser	Trp	Pro	Pro	Tyr	
		305			310				315						320	
Phe	Tyr	Ile	Asp	Lys	Ile	Gly	Arg	Arg	Trp	Thr	Val	Met	Leu	Ser	Ser	
				325					330					335		
Ile	Gly	Cys	Ala	Ile	Cys	Met	Ala	Val	Ile	Ala	Gly	Cys	Leu	Ile	Lys	
		340						345					350			
Gln	Ser	Tyr	Ser	Ser	Ala	Ala	Ala	Ala	Val	Ala	Phe	Met	Phe	Leu	Tyr	
		355				360						365				
Leu	Asp	Cys	Phe	Thr	Leu	Gly	Ile	Leu	Pro	Val	Ser	Trp	Ser	Tyr	Ser	
		370				375						380				
Ala	Glu	Ile	Gln	Pro	Leu	Arg	Val	Arg	Asn	Lys	Ala	Thr	Ala	Val	Gly	
		385			390				395						400	
Val	Phe	Ser	His	Trp	Thr	Ser	Asn	Phe	Val	Val	Val	Met	Val	Thr	Pro	
				405					410					415		
Ile	Gly	Leu	Asn	His	Ile	Gly	Gly	Asn	Tyr	Phe	Trp	Ile	Trp	Ala	Ile	
		420						425					430			
Val	Cys	Ala	Ser	Phe	Val	Pro	Leu	Thr	Tyr	Phe	Phe	Gly	Val	Glu	Thr	
		435				440						445				
Ser	Gly	Arg	Thr	Leu	Glu	Gln	Ile	Asp	Glu	Ser	Phe	Phe	Glu	Asn	Pro	
		450				455						460				
Arg	Ile	Leu	Met	Gly	Leu	Asp	Lys	Lys	Asn	Thr	Val	Val	Ile	Lys	Ala	
		465			470				475						480	
Ser	Lys	Gln	Asp	Glu	Glu	Ser	Arg	Phe	Arg	Ala	Leu	Ala	Lys	Glu	Glu	
				485					490					495		
Glu	Lys	His	Pro	Glu	Arg	Val	Ser	Val	Glu	Gln	Val	Glu	Glu	Lys	Ser	
		500						505					510			

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<211> LENGTH: 534

<212> TYPE: PRT

<213> ORGANISM: *Talaromyces atrovirens*

<400> SEQUENCE: 53

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Met Trp Thr Thr Thr Ser Gly Leu Arg Gly Lys Lys Leu His Leu Ala
 1          5          10          15

Ile Thr Phe Thr Ser Val Ile Gly Phe Ser Leu Phe Gly Tyr Asp Gln
 20          25          30

Gly Leu Met Ser Gly Leu Ile Ser Gly Asp Gln Phe Val Lys Glu Phe
 35          40          45

Pro Val Leu Asp Gly Asp Ser Leu His Val Ser Val Leu Gln Gly Ala
 50          55          60

Val Thr Ser Cys Tyr Glu Ile Gly Cys Phe Phe Gly Ala Ile Phe Thr
 65          70          75          80

Leu Leu Phe Gly Gln Arg Ile Gly Arg Thr Pro Leu Leu Val Gly Gly
 85          90          95

Gly Ala Leu Met Val Val Gly Thr Val Ile Ser Thr Ala Ser Phe Gly
100          105          110

Pro His Trp Gly Leu Gly Gln Phe Val Ile Gly Arg Val Ile Ser Gly
115          120          125

Val Gly Asn Gly Met Asp Thr Ala Thr Ile Pro Val Trp Gln Ser Glu
130          135          140

Cys Ser Arg Ala His Asn Arg Gly Phe Leu Val Cys Phe Glu Gly Ala
145          150          155          160

Ile Ile Ala Val Gly Thr Phe Ile Ala Tyr Trp Ile Asp Phe Gly Leu
165          170          175

Ser Tyr Val Glu Ser Ser Val Gln Trp Arg Phe Pro Val Ala Phe Gln
180          185          190

Ile Leu Phe Ala Leu Leu Val Val Ile Gly Ala Leu Met Leu Pro Glu
195          200          205

Ser Pro Arg Trp Phe Ile Met Ser Gly Lys Thr Gln Glu Ala Leu His
210          215          220

Val Leu Ala Gln Leu Asn Asp Ser Ser Glu Asp Ala Asp Asp Val Leu
225          230          235          240

Arg Asp Phe Asn Leu Met Gln Ala Asp Leu Lys Ser Leu Glu Asn Ala
245          250          255

Glu Ala Ser Ser Trp Lys Thr Leu Phe Thr Phe Gly Lys Thr Gln Glu
260          265          270

Phe Gln Arg Met Met Ile Gly Cys Ser Gly Gln Phe Phe Gln Gln Phe
275          280          285

Thr Gly Cys Asn Ala Ala Ile Tyr Tyr Ser Thr Leu Leu Phe Lys Asn
290          295          300

Asn Leu His Met Thr Gly Lys Leu Pro Leu Val Leu Gly Gly Val Phe
305          310          315          320

Ala Thr Val Tyr Ala Leu Ala Thr Ile Pro Ser Phe Phe Met Val Glu
325          330          335

Lys Val Gly Arg Arg Asn Leu Phe Leu Ile Gly Phe Leu Gly Gln Gly
340          345          350

Leu Ser Phe Ile Ile Thr Met Gly Cys Leu Val Asp Asp Thr Thr Gln
355          360          365

Asn Ala Lys Gly Ala Ala Val Gly Ile Phe Leu Phe Ile Thr Phe Phe
370          375          380

Ala Phe Thr Thr Leu Pro Leu Pro Trp Ile Tyr Pro Pro Glu Ile Asn

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385	390	395	400
Pro Leu Arg Thr Arg Thr Met Ala Ala Ser Ala Ser Thr Cys Val Asn			
	405	410	415
Trp Ile Cys Asn Phe Ala Val Val Met Phe Thr Pro Val Phe Ser Asn			
	420	425	430
Lys Ser Ala Trp Gly Ile Tyr Leu Phe Phe Ala Leu Val Asn Phe Ile			
	435	440	445
Ala Ile Pro Phe Ala Trp Phe Phe Tyr Val Glu Thr Ala Gly Arg Asp			
	450	455	460
Leu Glu Glu Val Asp Ile Ile Phe Ala Lys Ala His Val Glu Asn Lys			
	465	470	475
Trp Pro Phe Gln Ile Ala Gln Gln Met Pro Lys Leu Thr His Glu Glu			
	485	490	495
Ile Ser Gln Gln Leu Phe Asp Leu Gly Leu Ser Val Ser Asp His Ser			
	500	505	510
Ala Ser Ser Glu Ser Glu Lys Val Glu Ile Ala Ala Thr Asn Asn Glu			
	515	520	525
His Arg Thr Ala Ser Asp			
	530		

<210> SEQ ID NO 54

<211> LENGTH: 488

<212> TYPE: PRT

<213> ORGANISM: Trichophyton benhamiae

<400> SEQUENCE: 54

Met Ala Gly Ile Ile Ser Ala Met Pro Phe Asn Thr Val Phe Pro Glu			
1	5	10	15
Thr Lys Asp Asn Pro Thr Asn Gln Gly Phe Val Thr Ala Ile Tyr Glu			
	20	25	30
Ile Gly Cys Leu Leu Gly Ala Val Ser Ile Ile Trp Gly Gly Asp Met			
	35	40	45
Leu Gly Arg Arg Lys Ser Ile Val Thr Gly Ala Ile Ile Met Ala Ile			
	50	55	60
Gly Ala Ile Ile Gln Val Thr Ser Phe Val Gly His Gln Pro Tyr Ala			
	65	70	75
Gln Phe Ile Ile Gly Arg Ile Ile Thr Gly Val Gly Asn Gly Ile Asn			
	85	90	95
Thr Ser Thr Ile Pro Thr Tyr Gln Ala Glu Cys Ser His Ala Ser Asn			
	100	105	110
Arg Gly Leu Leu Ile Cys Ile Glu Gly Ala Thr Ile Ala Phe Gly Thr			
	115	120	125
Leu Ile Ala Tyr Trp Ile Asp Tyr Gly Ala Ser Tyr Gly Pro Asp Ser			
	130	135	140
Phe Ser Trp Arg Phe Pro Ile Ala Phe Gln Ile Ala Phe Ser Ile Val			
	145	150	155
Met Val Thr Gly Met Ile Trp Leu Pro Glu Ser Pro Arg Trp Leu Cys			
	165	170	175
Met Arg Asp Arg Ser Asp Glu Gly Glu Arg Val Ile Ala Ala Leu His			
	180	185	190
Gly Val Pro Ile Thr Asp Pro Leu Val Gln Ala Glu Lys Asn Ala Val			
	195	200	205
Met Glu Ser Ile Arg Ala Ser Gly Glu Val Gly Lys Pro Thr Pro Leu			
	210	215	220

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Ser Val Val Phe Thr Gly Gly Lys Thr Gln His Arg Arg Arg Met Phe
225                230                235                240

Leu Gly Val Phe Gly Gln Phe Ala Gln Gln Leu Ser Gly Cys Asn Ala
                245                250                255

Ile Ile Tyr Phe Phe Pro Val Leu Phe Glu Lys Ser Ile Gly Val Asp
                260                265                270

His Asn Met Ala Thr Leu Leu Gly Gly Val Asn Met Ile Val Tyr Ser
                275                280                285

Ile Phe Ala Thr Thr Ser Trp Phe Leu Ile Glu Arg Ala Gly Arg Arg
290                295                300

Lys Leu Phe Leu Tyr Gly Ala Ala Gly Gln Ala Ile Ser Met Thr Ile
305                310                315                320

Thr Phe Ala Cys Leu Ile Pro Asn Thr Pro Ala Thr Ala Lys Gly Ala
                325                330                335

Ala Val Gly Leu Phe Thr Tyr Ile Ala Ser Phe Gly Ala Thr Trp Leu
                340                345                350

Pro Leu Pro Trp Leu Tyr Ala Ala Glu Ile Ser Pro Ile Lys Thr Arg
355                360                365

Ala Lys Ala Asn Ala Leu Ser Thr Cys Ser Asn Trp Leu Phe Asn Phe
370                375                380

Phe Ile Val Met Ile Thr Pro Val Met Leu Ala Gly Ile Gly Trp Gly
385                390                395                400

Thr Tyr Leu Phe Phe Ala Ile Ile Asn Val Cys Phe Leu Pro Ile Ile
                405                410                415

Tyr Phe Phe Tyr Pro Glu Thr Ala Lys Arg Ser Leu Glu Glu Ile Asp
420                425                430

Ile Ile Phe Ala Lys Gly Tyr Cys Glu Asn Lys Ser Tyr Val Gln Ala
435                440                445

Ala Arg Glu Leu Pro Tyr Leu Thr Glu Glu Glu Ile Ser Arg Met Asp
450                455                460

Ala Glu Tyr Gly His Gly Lys Pro Ser Glu Thr Ala Ser Pro Val Asn
465                470                475                480

Glu Lys Glu Ser Asp Ser Glu Gln
485

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<210> SEQ ID NO 55

<211> LENGTH: 472

<212> TYPE: PRT

<213> ORGANISM: Verticillium alfalfae

<400> SEQUENCE: 55

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Met Ser Glu Lys Ser Asp Arg Ala Asp Thr Ile Asn Asn His Gln Gly
1          5          10          15

Thr Val Asp Ser Thr Pro Lys Ser Gly Phe Ser Arg Phe Cys Ser Lys
20        25        30

Met Gly Asp Leu Pro Gln Trp Lys Val Asn Gly Lys Leu Leu Arg Gly
35        40        45

Ala Ala Leu Asn Trp Gly Ile Gly Val Ile Ala Ser Cys Gly Phe Leu
50        55        60

Met Phe Gly Tyr Asp Gln Gly Val Leu Ser Gly Leu Leu Thr Leu Asp
65        70        75        80

Asp Phe Gln Lys Asn Gln Ala Leu Met Thr Pro Leu Asp Ala Ser Asn
85        90        95

Pro Leu Cys Trp Asn Asp Asp Gly Ser Arg Asp Glu Arg Tyr Cys His
100       105       110

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Gly Asp Ala Asn Thr Gln Ala Ala Gly Val Ala Met Tyr Gln Ile Gly
 115 120 125
 Cys Phe Leu Gly Ala Val Leu Ile Leu Phe Tyr Gly Glu Ser Trp Gly
 130 135 140
 Arg Arg Ser Ser Thr Phe Trp Gly Ser Leu Ile Met Ile Ile Gly Gly
 145 150 155 160
 Ile Met Gln Ala Ala Ser Leu Glu Tyr Gly Leu Phe Val Ser Gly Arg
 165 170 175
 Val Ile Gly Gly Arg Phe Lys Arg Leu Leu Thr Asn Arg Pro Ser Gln
 180 185 190
 Lys Leu Arg Arg Thr Leu Leu Gly Ile Ala Ala Gln Phe Phe Gln Gln
 195 200 205
 Ile Cys Gly Ile Thr Leu Ile Thr Tyr Tyr Ala Thr Phe Val Phe Glu
 210 215 220
 Asn Ser Leu Gly Phe Gly Pro Gln Leu Ser Arg Leu Leu Ala Ala Leu
 225 230 235 240
 Asn Gly Thr Glu Tyr Phe Leu Ala Ser Leu Val Ala Leu Pro Leu Ile
 245 250 255
 Glu Arg Val Gly Arg Arg Lys Leu Met Leu Phe Gly Ala Phe Gly Met
 260 265 270
 Met Gly Ser Met Ala Ile Leu Ala Gly Thr Thr Ser Thr Gly Thr Thr
 275 280 285
 Asn Glu Asp Gly Ala Pro Gln Leu Ser Thr Ala Tyr Gly Val Thr Ala
 290 295 300
 Val Val Phe Leu Phe Ala Phe Asn Ser Phe Phe Ala Val Gly Trp Leu
 305 310 315 320
 Gly Met Thr Trp Leu Tyr Pro Ala Glu Val Thr Gly Leu Asn Ile Arg
 325 330 335
 Ile Gln Ala Asn Ala Leu Ser Thr Cys Ser Asn Trp Ile Ser Asn Phe
 340 345 350
 Leu Ile Val Met Ile Thr Pro Pro Ala Phe Ala Asn Leu Gln Trp Lys
 355 360 365
 Thr Tyr Val Met Phe Ala Val Phe Asn Ala Ala Leu Ile Pro Cys Val
 370 375 380
 Tyr Leu Tyr Phe Pro Glu Thr Ser Lys Arg Ser Leu Glu Glu Ile Asp
 385 390 395 400
 Leu Tyr Phe Ala Arg Ala Trp Ser Glu Gly Val Ser Pro Val Lys Met
 405 410 415
 Ala Lys Thr Met Pro Arg Tyr Thr Ala Thr Glu Leu Asp Asn Glu Leu
 420 425 430
 Ala Lys Tyr Phe Ser Ala Ala Asp Ile Glu Gln Arg Arg Gly Ser Met
 435 440 445
 Leu Gln Ser Arg Arg Pro Ser Ala Met Thr Gln Ala Pro Asp Glu Pro
 450 455 460
 Ser Val Asn Lys Asn Ala Thr Gln
 465 470

<210> SEQ ID NO 56

<211> LENGTH: 533

<212> TYPE: PRT

<213> ORGANISM: Verticillium dahliae

<400> SEQUENCE: 56

Met Gly Tyr Thr Thr Trp Trp Lys Arg Leu Ser Pro Arg Gln Leu Asn

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1	5	10	15
Ile Ala Ile Gln Thr Phe Ser Val Ile Ser Ile Phe Phe Glu Gly Tyr	20	25	30
Asp Gln Gly Val Met Gly Gly Val Asn Ala Ser Pro Arg Tyr Val Glu	35	40	45
Glu Val Gly Ile Gly Leu Pro Asp Gly Thr Val Thr Asp Thr Leu His	50	55	60
Gln Gly Gly Ile Val Ser Val Tyr Tyr Leu Gly Cys Ile Ala Gly Cys	65	70	75
Phe Ala Gly Gly Trp Leu Ala Asp Arg Ile Gly Arg Ile Asn Gly Leu	85	90	95
Phe Ile Gly Cys Ile Phe Ala Ile Ile Gly Gly Ala Leu Gln Ala Ala	100	105	110
Ala Gln Ser Ser Asn Phe Ile Ile Val Ala Arg Val Ile Thr Gly Ile	115	120	125
Gly Thr Gly Ala Leu Thr Gly Ile Thr Pro Val Met Val Ser Glu Thr	130	135	140
Ser Thr Ala Glu His Arg Gly Gly Phe Leu Gly Tyr Val Phe Ile Ala	145	150	155
Asn Tyr Leu Gly Ile Ser Ile Ala Tyr Trp Leu Ser Phe Gly Leu Ala	165	170	175
Phe Ile Asp Gly Gly Tyr Ser Asp Ile Arg Trp Arg Phe Gln Leu Ala	180	185	190
Phe Gln Cys Leu Pro Ala Leu Leu Leu Phe Leu Gly Ile Lys Ile Leu	195	200	205
Pro Asp Thr Pro Arg Phe Leu Ala Ser Val Gly Arg Tyr Asp Glu Ala	210	215	220
Arg Glu Val Ile Glu His Val Arg Gly Asn Phe Gly Pro Leu Val Glu	225	230	235
Arg Glu Phe Leu Glu Ile Arg Thr Val Ala Glu Glu Ser Thr Lys Ser	245	250	255
Ser Pro Ile Glu Phe Ile Lys Ile Leu Phe Gly Arg Gly Pro Lys Pro	260	265	270
Gly Tyr Asn Leu Gly Gln Arg Ala Trp Leu Cys Leu Phe Leu Gln Ile	275	280	285
Met Ala Ser Trp Thr Gly Ile Thr Ala Val Thr Ala Tyr Ser Pro Ile	290	295	300
Leu Leu Ser Gln Ala Gly Tyr Thr Glu Leu Thr Gln Asn Gly Leu Ala	305	310	315
Gly Gly Leu Asn Thr Val Gly Ile Val Gly Thr Ile Ile Ser Ala Gln	325	330	335
Ile Val Asp Arg Leu Gly Arg Arg Thr Cys Leu Met Gly Gly Ala Leu	340	345	350
Ala Leu Ser Ala Val Asn Leu Ile Ala Gly Ala Leu Tyr Glu Gly Ser	355	360	365
Arg Ala His Pro Asp Arg Ala Ser Gln Phe Ala Pro Val Ala Val Ala	370	375	380
Met Leu Phe Leu Phe Asn Leu Ser Tyr Ala Ala Thr Trp Gly Thr Val	385	390	395
Ala Phe Leu Ile Pro Thr Glu Ile Trp Ser Ser Asp Leu Arg Ala Gln	405	410	415
Gly Asn Gly Phe Gly Ile Thr Gly Trp Ala Val Gly Val Gly Met Thr	420	425	430

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Thr Leu Val Asn Pro Ile Met Phe Gly Val Leu Lys Asn Trp Thr Tyr
435 440 445

Phe Leu Phe Ala Gly Leu Asn Leu Leu Trp Val Pro Val Val Phe Leu
450 455 460

Phe Tyr Pro Glu Thr Ser Gly Arg Ser Leu Glu Ser Ile Asp Ala Leu
465 470 475 480

Phe Ala Ala Asn Ser Ile Phe Asn Thr Lys Met Glu Arg Ser Tyr Met
485 490 495

Ala His Gly Asp Val Leu Ala Glu Arg Gly Asn His Asp Asn Gln Val
500 505 510

Leu Ser Ala Ser Asp Ser Gly Ser Lys Pro Gly Pro Pro Gly Ser Val
515 520 525

Glu Lys Leu Gln Val
530

<210> SEQ ID NO 57

<211> LENGTH: 573

<212> TYPE: PRT

<213> ORGANISM: *Torulaspora delbrueckii*

<400> SEQUENCE: 57

Met Ile Glu Lys Lys Ser Leu Lys Ser Arg Phe Phe Ser Arg Thr Ser
1 5 10 15

His Phe Gly Leu Thr Gly Lys Thr Leu Arg Tyr Val Ile Thr Leu Cys
20 25 30

Ala Met Thr Gly Phe Ser Leu Phe Gly Tyr Asp Gln Gly Leu Met Ala
35 40 45

Ser Leu Ile Thr Gly Thr Gln Phe Asn Tyr Glu Phe Pro Ala Thr Lys
50 55 60

Ser Lys Ser Asp Asn Asp Thr His Ala Ser Thr Val Gln Gly Ala Val
65 70 75 80

Thr Ser Cys Tyr Glu Ile Gly Cys Phe Phe Gly Ser Leu Phe Val Met
85 90 95

Phe Tyr Gly Glu Lys Ile Gly Arg Lys Pro Leu Ile Val Ile Gly Ser
100 105 110

Val Ile Thr Ile Val Gly Ala Val Ile Ser Thr Thr Ala Phe Arg Asp
115 120 125

Tyr Trp Ala Leu Gly Gln Phe Val Val Gly Arg Val Ile Thr Gly Val
130 135 140

Gly Thr Gly Leu Asn Thr Ser Thr Ile Pro Val Trp Gln Ser Glu Met
145 150 155 160

Ser Asp Pro Ser Ile Arg Gly Ile Leu Val Asn Leu Glu Gly Ser Thr
165 170 175

Ile Ala Ile Gly Thr Met Leu Ala Tyr Trp Ile Asp Phe Gly Phe Ser
180 185 190

Phe Ile Asp Ser Ser Val Gln Trp Arg Phe Pro Val Ser Met Gln Ile
195 200 205

Leu Phe Ala Leu Ile Leu Cys Phe Met Ile Val Asn Leu Pro Glu Ser
210 215 220

Pro Arg Trp Leu Ile Ser Gln Ser Arg Thr Glu Glu Ala Arg Tyr Leu
225 230 235 240

Leu Gly Gln Leu Asp Asp Val Asp Pro Asn Asp Asp Arg Ile Val Ala
245 250 255

Glu Val Ala Met Ile His Asp Ala Val Asn Arg Ser Lys Gln Glu Lys

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260				265				270							
Asn	Ser	Met	Ser	Val	Leu	Phe	Ser	Gly	Gly	Lys	Ser	Gln	Asn	Met	Gln
		275					280						285		
Arg	Ala	Leu	Val	Ala	Ala	Ser	Thr	Gln	Phe	Phe	Gln	Gln	Phe	Thr	Gly
		290				295					300				
Cys	Asn	Ala	Ala	Ile	Tyr	Tyr	Ser	Thr	Val	Leu	Phe	His	Glu	Thr	Ile
		305			310						315				320
Gln	Leu	Ser	Pro	Arg	Leu	Ser	Met	Ile	Leu	Gly	Ala	Val	Phe	Ser	Thr
				325					330					335	
Val	Tyr	Ala	Leu	Ser	Thr	Ile	Pro	Ser	Phe	Phe	Leu	Ile	Glu	Arg	Leu
			340						345					350	
Gly	Arg	Arg	Lys	Leu	Phe	Leu	Leu	Gly	Ala	Thr	Gly	Gln	Ala	Ile	Ser
		355					360						365		
Phe	Thr	Ile	Thr	Phe	Ala	Cys	Leu	Val	Arg	Gln	Thr	Glu	Glu	Asn	Ala
		370				375					380				
Lys	Gly	Ala	Ala	Val	Gly	Leu	Phe	Leu	Phe	Ile	Val	Phe	Phe	Gly	Cys
		385			390					395					400
Ser	Met	Leu	Ser	Leu	Pro	Trp	Ile	Tyr	Pro	Pro	Glu	Ile	Ala	Ser	Met
			405						410					415	
Lys	Val	Arg	Ala	Ser	Thr	Asn	Ala	Phe	Ser	Thr	Cys	Thr	Asn	Trp	Leu
			420						425					430	
Cys	Asn	Phe	Ala	Val	Val	Met	Phe	Thr	Pro	Ile	Phe	Ile	Asn	Lys	Ser
		435					440						445		
Gly	Trp	Gly	Cys	Tyr	Leu	Phe	Phe	Ala	Cys	Ile	Asn	Tyr	Leu	Tyr	Ile
	450				455						460				
Pro	Val	Ile	Phe	Phe	Phe	Tyr	Pro	Glu	Thr	Ala	Gly	Arg	Ser	Leu	Glu
	465				470					475					480
Glu	Ile	Asp	Ile	Ile	Tyr	Ala	Lys	Ser	His	Glu	Glu	Gly	Thr	Gln	Ala
			485						490					495	
Trp	Arg	Val	Ala	Ala	His	Leu	Pro	Lys	Leu	Ser	Leu	Gln	Glu	Val	Asp
			500						505					510	
Asp	His	Ala	Asn	Ala	Leu	Gly	Leu	Tyr	Glu	Asp	Asp	Leu	Glu	Lys	Glu
		515					520						525		
Asp	Phe	Ala	Ala	Glu	Glu	Glu	Gly	Lys	Glu	Ala	Gly	Ala	His	Gly	Tyr
		530				535					540				
Ala	Leu	Phe	Ala	Arg	Asn	Thr	Ser	Thr	Ser	Asn	Asn	Glu	Asp	Gly	Ser
			545		550					555					560
Ser	Glu	Ser	Glu	Lys	Asp	Gln	Asn	Ala	Thr	Pro	Arg	Ala			
			565						570						

<210> SEQ ID NO 58

<211> LENGTH: 532

<212> TYPE: PRT

<213> ORGANISM: Penicillium rubens

<400> SEQUENCE: 58

Met	Gly	Tyr	Thr	Thr	Leu	Trp	Lys	Arg	Leu	Ser	Pro	Arg	Gln	Leu	Asn
1				5					10					15	
Val	Ala	Ile	Gln	Ile	Phe	Ser	Leu	Ile	Ser	Ile	Phe	Phe	Glu	Gly	Tyr
		20					25						30		
Asp	Gln	Gly	Val	Met	Gly	Gly	Val	Asn	Asn	Ser	Pro	Arg	Tyr	Val	Glu
		35					40					45			
Glu	Val	Gly	Ile	Gly	Lys	Pro	Asp	Gly	Thr	Val	Thr	Asp	Thr	Thr	His
	50					55					60				

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Gln	Gly	Gly	Ile	Val	Ser	Ile	Tyr	Tyr	Leu	Gly	Ala	Ile	Phe	Gly	Cys	65	70	75	80
Phe	Ala	Gly	Gly	Trp	Leu	Ala	Asp	Arg	Val	Gly	Arg	Ile	Asn	Gly	Leu	85	90	95	
Leu	Ala	Gly	Ser	Leu	Phe	Ala	Leu	Val	Gly	Gly	Ala	Leu	Gln	Ala	Gly	100	105	110	
Ala	Gln	Asn	Ser	Asp	Phe	Met	Leu	Cys	Ala	Arg	Val	Ile	Thr	Gly	Ile	115	120	125	
Gly	Thr	Gly	Ala	Leu	Thr	Gly	Ile	Thr	Pro	Val	Leu	Val	Ser	Glu	Thr	130	135	140	
Ser	Ser	Ala	Asn	His	Arg	Gly	Gly	Phe	Leu	Gly	Tyr	Val	Phe	Ile	Ala	145	150	155	160
Asn	Tyr	Leu	Gly	Ile	Ser	Val	Ala	Tyr	Trp	Ile	Ser	Phe	Gly	Leu	Ala	165	170	175	
Phe	Val	Asp	Asn	Gly	Tyr	Ser	Asp	Val	Arg	Trp	Arg	Phe	Leu	Leu	Ala	180	185	190	
Phe	Gln	Cys	Phe	Pro	Ala	Leu	Leu	Leu	Ala	Ala	Phe	Ile	Lys	Met	Leu	195	200	205	
Pro	Asp	Ser	Pro	Arg	Phe	Leu	Ala	Ser	Val	Gly	Arg	Asn	Asp	Glu	Ala	210	215	220	
Arg	Asp	Leu	Leu	Asn	Arg	Ile	Arg	Lys	Asp	Arg	Ala	Ser	Gln	Asp	Asp	225	230	235	240
Ile	Asp	Arg	Glu	Tyr	Leu	Glu	Ile	Ile	Val	Thr	Ala	Lys	Gly	Ser	Lys	245	250	255	
Phe	Ser	Ser	Pro	Ile	Glu	Phe	Val	Lys	Ile	Leu	Phe	Gly	Lys	Gly	Gly	260	265	270	
Arg	Pro	Gly	Met	Asn	Leu	Gly	Arg	Arg	Ala	Trp	Leu	Cys	Val	Trp	Leu	275	280	285	
Gln	Ile	Met	Ala	Ser	Trp	Thr	Gly	Ile	Thr	Ala	Val	Thr	Ala	Tyr	Ser	290	295	300	
Pro	Val	Leu	Leu	Ala	Gln	Ala	Gly	Tyr	Ser	Asp	Ile	Lys	Gln	Asn	Gly	305	310	315	320
Leu	Ala	Gly	Gly	Ile	Asn	Thr	Ile	Gly	Ile	Ile	Gly	Thr	Ile	Ile	Ser	325	330	335	
Ala	Ile	Ile	Ile	Asp	Arg	Leu	Gly	Arg	Arg	Val	Cys	Leu	Met	Gly	Gly	340	345	350	
Ala	Ala	Val	Leu	Phe	Ala	Val	Asn	Leu	Ile	Ala	Gly	Ala	Val	Tyr	Glu	355	360	365	
Gly	Ser	Leu	His	Asn	Pro	Glu	Lys	Ala	Ser	Gln	Tyr	Ala	Pro	Gly	Ala	370	375	380	
Val	Thr	Met	Leu	Phe	Leu	Phe	Asn	Leu	Gly	Tyr	Ala	Ala	Thr	Trp	Gly	385	390	395	400
Thr	Val	Ala	Phe	Leu	Val	Pro	Thr	Glu	Ile	Phe	Pro	Ser	Asp	Leu	Arg	405	410	415	
Ala	Gln	Gly	Asn	Gly	Phe	Gly	Ile	Thr	Gly	Trp	Ala	Ile	Gly	Val	Gly	420	425	430	
Met	Thr	Thr	Leu	Val	Asn	Pro	Ile	Met	Phe	Asp	Val	Met	Thr	Ser	Arg	435	440	445	
Thr	Tyr	Phe	Leu	Phe	Ala	Gly	Leu	Asn	Leu	Ile	Trp	Ile	Pro	Ile	Val	450	455	460	
Tyr	Leu	Phe	Tyr	Pro	Glu	Thr	Arg	Asn	Arg	Ser	Leu	Glu	Ser	Ile	Asp	465	470	475	480
Ala	Leu	Phe	Ser	Thr	Pro	Ser	Pro	Phe	His	Trp	Lys	Met	Glu	Gln	Ala				

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	485		490		495										
Tyr	Lys	Leu	His	Gly	Asp	Val	Leu	Ala	Glu	His	Gly	Val	Asn	Arg	Asn
		500						505					510		
Glu	Ala	Leu	Gly	Asp	Gly	Lys	Ser	Glu	Leu	Thr	Thr	Ser	Pro	Thr	Glu
		515					520					525			
Leu	Gly	Thr	Val												
		530													

<210> SEQ ID NO 59
 <211> LENGTH: 237
 <212> TYPE: PRT
 <213> ORGANISM: Trichophyton rubrum

<400> SEQUENCE: 59

Met	Ala	Lys	Lys	Tyr	Leu	Gly	Gly	Ser	Gly	Asp	Lys	Leu	Thr	Ile	Trp
1				5					10					15	
Ile	Ser	Ile	Ala	Ala	Ser	Thr	Val	Leu	Ile	Phe	Tyr	Gly	Tyr	Asp	Gln
			20					25					30		
Gly	Val	Phe	Gly	Asn	Val	Leu	Ile	Gly	Glu	Asp	Phe	Leu	Gln	Thr	Met
		35				40						45			
Gly	Tyr	Pro	Ser	Thr	Asn	Leu	Gln	Gly	Thr	Met	Thr	Ser	Val	Tyr	Asn
	50					55					60				
Ile	Gly	Cys	Phe	Val	Gly	Ala	Met	Ser	Thr	Val	Trp	Thr	Gly	Asp	Tyr
65					70					75				80	
Phe	Gly	Arg	Pro	Arg	Gln	Ile	Ile	Val	Gly	Ser	Thr	Ile	Ile	Ala	Ile
				85				90						95	
Gly	Gly	Ile	Ile	Gln	Ala	Ser	Ala	Tyr	Gly	Val	Pro	Gln	Met	Met	Val
			100					105					110		
Gly	Arg	Val	Val	Ala	Gly	Leu	Gly	Thr	Gly	Met	Asn	Thr	Ser	Thr	Ala
		115				120						125			
Gly	Val	Trp	Gln	Ser	Glu	Thr	Ser	Lys	Met	Ser	Ser	Arg	Gly	Lys	Leu
	130					135					140				
Val	Ile	Ile	Gln	Met	Val	Phe	Phe	Thr	Leu	Cys	Ile	Tyr	Ala	Met	Cys
145					150					155				160	
Pro	Phe	Leu	Pro	Asp	Ser	Pro	Arg	Leu	Leu	Ile	Arg	Lys	Gly	Glu	Tyr
			165					170						175	
Ser	Glu	Ala	Leu	Glu	Val	Leu	Ala	Ala	Leu	Glu	Gly	Asn	Gly	Ala	Thr
			180				185						190		
Ser	Asn	Ser	His	Ser	Val	Lys	Thr	Gln	Phe	Asn	Val	Ile	Lys	Asp	Val
	195					200						205			
Leu	Asp	Arg	Glu	Asn	Leu	Asn	Ser	Tyr	Thr	Trp	Phe	Lys	Leu	Leu	Met
	210					215					220				
Gly	Lys	Gly	Glu	Ser	Ser	Arg	Phe	Pro	Ser	Val	Tyr	Thr			
225					230					235					

<210> SEQ ID NO 60
 <211> LENGTH: 568
 <212> TYPE: PRT
 <213> ORGANISM: Lachancea thermotolerans

<400> SEQUENCE: 60

Met	Pro	Ser	Arg	Leu	Ser	Val	Asn	Arg	Thr	Ser	Thr	Leu	Gly	Leu	Asn
1				5					10					15	
Gly	Arg	Ser	Leu	Arg	Leu	Ala	Ile	Thr	Ile	Thr	Ser	Val	Ile	Gly	Phe
			20				25						30		
Ser	Leu	Phe	Gly	Tyr	Asp	Gln	Gly	Leu	Met	Ser	Gly	Leu	Ile	Thr	Gly

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35					40					45					
Lys	Glu	Phe	Asn	Ser	Glu	Phe	Pro	Ala	Thr	Gly	Gly	Asp	Asp	Arg	Arg
50					55					60					
Thr	Lys	Leu	Val	Gln	Gly	Ala	Val	Thr	Ala	Cys	Tyr	Glu	Ile	Gly	Cys
65				70					75					80	
Phe	Phe	Gly	Ser	Leu	Phe	Val	Met	Phe	Arg	Gly	Glu	Gln	Ile	Gly	Arg
			85					90						95	
Lys	Pro	Leu	Val	Ile	Leu	Gly	Ala	Cys	Leu	Thr	Ile	Val	Gly	Thr	Val
		100						105					110		
Ile	Ser	Thr	Ala	Ala	Phe	Gly	Pro	His	Trp	Gly	Leu	Gly	Gln	Phe	Val
	115						120					125			
Val	Gly	Arg	Val	Cys	Thr	Gly	Val	Gly	Thr	Gly	Leu	Asn	Thr	Ser	Thr
	130					135					140				
Ile	Pro	Val	Trp	Gln	Ser	Glu	Met	Ser	Lys	Pro	Glu	Asn	Arg	Gly	Ile
145				150					155					160	
Leu	Val	Asn	Leu	Glu	Gly	Ser	Met	Ile	Ala	Val	Gly	Thr	Met	Ile	Ala
		165						170						175	
Tyr	Trp	Ile	Asp	Phe	Gly	Leu	Ser	Tyr	Val	Asp	Ser	Ser	Val	Gln	Trp
		180						185					190		
Arg	Phe	Pro	Val	Ala	Met	Gln	Ile	Phe	Phe	Ala	Leu	Leu	Leu	Leu	Ala
	195					200						205			
Gly	Ile	Trp	Glu	Leu	Pro	Asp	Ser	Pro	Arg	Trp	Leu	Met	Ser	Arg	Gly
	210				215						220				
Arg	Arg	Asp	Asp	Ala	Leu	His	Val	Leu	Ala	Lys	Leu	Asp	Asn	Leu	Pro
225				230						235				240	
Glu	Asp	Asp	Asp	Ala	Ile	Ile	Ala	Glu	Ala	Thr	Val	Ile	Gln	Asp	Ala
			245					250						255	
Val	Asn	Arg	Phe	Arg	His	Glu	Lys	Arg	Ser	Val	Lys	Asp	Leu	Phe	Thr
		260						265					270		
Gly	Gly	Lys	Thr	Gln	Asn	Leu	Gln	Arg	Ala	Leu	Val	Ala	Ser	Ser	Thr
	275					280						285			
Gln	Phe	Phe	Gln	Gln	Phe	Thr	Gly	Cys	Asn	Ala	Ala	Ile	Tyr	Tyr	Ser
	290				295					300					
Thr	Val	Leu	Phe	Gln	Glu	Ser	Ile	Gly	Leu	Thr	Gly	Lys	Leu	Pro	Leu
305				310						315				320	
Ile	Leu	Gly	Gly	Val	Phe	Ala	Thr	Ile	Tyr	Ala	Cys	Phe	Thr	Ile	Pro
		325						330						335	
Ser	Phe	Phe	Leu	Val	Glu	Thr	Leu	Gly	Arg	Arg	Lys	Leu	Phe	Met	Leu
		340						345					350		
Gly	Ala	Ala	Gly	Gln	Ala	Ile	Ser	Phe	Thr	Ile	Thr	Phe	Gly	Cys	Leu
	355					360						365			
Thr	Lys	Asp	Asp	Thr	Glu	Val	Ala	Lys	Gly	Ala	Ala	Val	Gly	Leu	Phe
	370				375					380					
Leu	Phe	Ile	Cys	Phe	Phe	Gly	Met	Ser	Met	Leu	Ser	Leu	Pro	Trp	Ile
385				390						395				400	
Tyr	Pro	Pro	Glu	Ile	Ala	Ser	Met	Arg	Val	Arg	Ser	Ala	Thr	Asn	Ala
			405					410						415	
Leu	Ser	Thr	Cys	Thr	Asn	Trp	Leu	Cys	Asn	Phe	Ala	Val	Val	Met	Phe
		420						425					430		
Thr	Pro	Ile	Phe	Ile	Met	Asp	Thr	Gly	Tyr	Gly	Cys	Tyr	Leu	Phe	Phe
	435					440						445			
Ala	Val	Met	Asn	Tyr	Leu	Tyr	Leu	Pro	Val	Ile	Phe	Phe	Phe	Tyr	Pro
	450				455					460					

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Glu Thr Ala Gly Arg Ser Leu Glu Glu Ile Asp Ile Ile Phe Ala Lys
 465 470 475 480

Ala His Val Asp Gly Thr Met Pro Trp Arg Val Ala Ala Asn Leu Pro
 485 490 495

Lys Leu Ser Phe Ser Glu Ile Glu Asp Gln Ala Asn Ala Leu Gly Leu
 500 505 510

Tyr Glu Asp Asp Asn Glu Lys Gln Asp Leu Asp Leu Asp Glu Ala Ala
 515 520 525

Asp Arg Glu Asp Ala Ile Arg Gln Asn Ser Gly Val Gln Gly Tyr Thr
 530 535 540

Leu Phe Glu Lys Arg Glu Asn Gly Ala Glu Ser Gly Gln Pro Asp Gly
 545 550 555 560

Ser Ser Thr Ser Gln Lys Ser Asp
 565

<210> SEQ ID NO 61
 <211> LENGTH: 542
 <212> TYPE: PRT
 <213> ORGANISM: Aspergillus oryzae

<400> SEQUENCE: 61

Met Trp Thr Thr Thr Ser Gly Leu Ser Gly Arg Ser Leu Arg Leu Ser
 1 5 10 15

Ile Thr Phe Ala Ala Val Val Gly Phe Ser Leu Phe Gly Tyr Asn Gln
 20 25 30

Gly Met Met Ala Gly Leu Leu Asn Gly Asp Glu Phe Val Asp Ser Phe
 35 40 45

Pro Ile Leu Lys Met Pro Asp Asn Pro Thr Ala Gly Glu Lys His Tyr
 50 55 60

Ile Asp Val Ile Arg Gly Ala Val Thr Ser Cys Tyr Glu Leu Gly Cys
 65 70 75 80

Phe Phe Gly Ala Leu Phe Ser Met Phe Leu Gly Asp Lys Leu Gly Arg
 85 90 95

Thr Arg Leu Ile Phe Met Gly Ala Ser Ile Leu Ile Ile Gly Ala Leu
 100 105 110

Leu Thr Thr Val Cys Phe Thr Gly His Trp Glu Val Gly Gln Phe Val
 115 120 125

Ile Gly Arg Val Val Ser Gly Ile Gly Asn Gly Met Asn Thr Ala Thr
 130 135 140

Ile Pro Val Trp Gln Ser Glu Cys Ser Gly Ala His Asn Arg Gly Phe
 145 150 155 160

Leu Val Cys Phe Glu Gly Ala Met Ile Ala Gly Gly Thr Phe Ile Ala
 165 170 175

Tyr Trp Val Val Phe Gly Met Ser His Ala Ala Asp Ser Val Gln Trp
 180 185 190

Arg Phe Pro Val Ala Leu Gln Ile Phe Phe Ala Leu Val Val Ala Ala
 195 200 205

Gly Ala Met Met Leu Pro Asp Ser Pro Ser Trp Phe Val Met Arg Gly
 210 215 220

Leu Asp Lys Glu Ala Cys Glu Val Leu Gly Lys Leu Lys Gly Thr Ser
 225 230 235 240

Pro Asp Ser Asp Gln Val Leu His Asp Phe Asn Phe Leu Lys Gln Asp
 245 250 255

Met Glu Ser Ser Lys Asn Thr Gln Ser Asn Trp Lys Thr Val Phe Thr

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260				265				270							
Phe	Gly	Lys	Thr	Gln	Glu	Phe	Gln	Arg	Leu	Leu	Ile	Gly	Cys	Ser	Gly
	275						280						285		
Gln	Phe	Phe	Gln	Gln	Phe	Thr	Gly	Cys	Asn	Ala	Ala	Ile	Tyr	Tyr	Ser
	290					295					300				
Thr	Leu	Leu	Phe	Gln	Glu	Asn	Leu	Gly	Met	Glu	Lys	Tyr	Leu	Ser	Leu
305					310					315					320
Ile	Met	Gly	Gly	Val	Phe	Ala	Thr	Val	Tyr	Val	Leu	Ala	Thr	Ile	Pro
			325						330					335	
Ser	Phe	Phe	Met	Ile	Glu	Lys	Val	Gly	Arg	Arg	Asn	Leu	Tyr	Leu	Val
			340						345					350	
Gly	Phe	Leu	Gly	Gln	Gly	Leu	Ser	Phe	Val	Ile	Thr	Phe	Ala	Cys	Leu
		355					360						365		
Ile	Lys	Glu	Thr	Glu	Glu	Asn	Ser	Lys	Gly	Ala	Ala	Val	Gly	Ile	Phe
	370					375					380				
Leu	Phe	Ile	Thr	Phe	Phe	Ala	Phe	Thr	Leu	Leu	Pro	Leu	Pro	Trp	Ile
385					390					395					400
Tyr	Pro	Pro	Glu	Ile	Asn	Pro	Leu	Arg	Thr	Arg	Thr	Val	Gly	Ala	Ser
			405						410					415	
Ala	Ser	Thr	Cys	Thr	Asn	Trp	Ile	Cys	Asn	Phe	Ala	Val	Val	Met	Phe
			420						425					430	
Thr	Pro	Leu	Phe	Ala	Gly	Gln	Ser	Pro	Trp	Gly	Val	Tyr	Leu	Phe	Phe
		435				440							445		
Ala	Leu	Phe	Asn	Phe	Leu	Gly	Leu	Ile	Phe	Gly	Phe	Phe	Phe	Tyr	Val
	450				455						460				
Glu	Thr	Ala	Gly	Arg	Glu	Leu	Glu	Glu	Val	Asp	Ile	Ile	Tyr	Ala	Lys
465					470					475					480
Ala	His	Val	Glu	Gly	Lys	Met	Ala	Trp	Arg	Val	Ala	Asn	Thr	Met	Pro
			485						490					495	
Lys	Leu	Ser	Phe	Glu	Glu	Ile	Thr	Gln	Gln	Ser	Arg	Glu	Leu	Gly	Leu
		500							505					510	
Asp	Thr	Asn	Asp	His	Gly	Val	His	Glu	Lys	Thr	Glu	Leu	Gly	Leu	Ser
		515				520							525		
Ser	Asp	Ser	Gly	Gln	Glu	Thr	Glu	Glu	Val	His	Glu	Lys	His		
	530				535						540				

<210> SEQ ID NO 62

<211> LENGTH: 488

<212> TYPE: PRT

<213> ORGANISM: Trichophyton verrucosum

<400> SEQUENCE: 62

Met	Ala	Gly	Ile	Ile	Ser	Ala	Met	Pro	Phe	Asn	Thr	Val	Phe	Pro	Glu
1				5						10				15	
Thr	Lys	Asp	Asn	Pro	Thr	Asn	Gln	Gly	Phe	Val	Thr	Ala	Ile	Tyr	Glu
		20						25					30		
Ile	Gly	Cys	Leu	Leu	Gly	Ala	Val	Ser	Ile	Ile	Trp	Ser	Gly	Asp	Met
		35				40						45			
Leu	Gly	Arg	Arg	Lys	Ser	Ile	Val	Thr	Gly	Ala	Ile	Ile	Met	Ala	Ile
		50				55					60				
Gly	Ala	Ile	Ile	Gln	Val	Thr	Ser	Phe	Val	Gly	His	Gln	Pro	Tyr	Ala
65				70						75				80	
Gln	Phe	Ile	Ile	Gly	Arg	Ile	Ile	Thr	Gly	Val	Gly	Asn	Gly	Ile	Asn
			85					90						95	

Thr	Ser	Thr	Ile	Pro	Thr	Tyr	Gln	Ala	Glu	Cys	Ser	His	Ala	Ser	Asn
			100					105					110		
Arg	Gly	Leu	Leu	Ile	Cys	Ile	Glu	Gly	Ala	Thr	Ile	Ala	Phe	Gly	Thr
		115					120					125			
Leu	Ile	Ala	Tyr	Trp	Ile	Asp	Tyr	Gly	Ala	Ser	Tyr	Gly	Ala	Asp	Ser
		130				135					140				
Phe	Ser	Trp	Arg	Phe	Pro	Ile	Ala	Phe	Gln	Ile	Ala	Phe	Ser	Ile	Val
145					150					155					160
Met	Val	Thr	Gly	Met	Ile	Trp	Leu	Pro	Glu	Ser	Pro	Arg	Trp	Leu	Cys
				165					170					175	
Met	Arg	Asp	Arg	Ser	Asp	Glu	Gly	Glu	Arg	Val	Ile	Ala	Ala	Leu	His
			180					185					190		
Gly	Val	Pro	Val	Thr	Asp	Pro	Leu	Val	Gln	Ala	Glu	Lys	Asn	Ala	Val
		195					200					205			
Met	Glu	Ser	Ile	Arg	Ala	Ser	Gly	Glu	Val	Gly	Lys	Pro	Thr	Pro	Leu
		210				215					220				
Ser	Val	Val	Phe	Thr	Gly	Gly	Lys	Thr	Gln	His	Arg	Arg	Arg	Met	Phe
225					230					235					240
Leu	Gly	Val	Phe	Gly	Gln	Phe	Ala	Gln	Gln	Leu	Ser	Gly	Cys	Asn	Ala
				245				250						255	
Ile	Ile	Tyr	Phe	Phe	Pro	Val	Leu	Phe	Glu	Lys	Ser	Ile	Gly	Val	Asp
			260					265					270		
His	Asn	Met	Ala	Thr	Leu	Leu	Gly	Gly	Val	Asn	Met	Ile	Val	Tyr	Ser
		275					280					285			
Ile	Phe	Ala	Thr	Thr	Ser	Trp	Phe	Leu	Ile	Glu	Arg	Ala	Gly	Arg	Arg
		290				295					300				
Lys	Leu	Phe	Leu	Tyr	Gly	Ala	Ala	Gly	Gln	Ala	Ile	Ser	Met	Thr	Ile
305					310					315					320
Thr	Phe	Ala	Cys	Leu	Ile	Pro	Asn	Thr	Pro	Ala	Thr	Ala	Lys	Gly	Ala
				325					330					335	
Ala	Val	Gly	Leu	Phe	Thr	Tyr	Ile	Ala	Ser	Phe	Gly	Ala	Thr	Trp	Leu
			340					345					350		
Pro	Leu	Pro	Trp	Leu	Tyr	Ala	Ala	Glu	Ile	Ser	Pro	Ile	Lys	Thr	Arg
		355					360					365			
Ala	Lys	Ala	Asn	Ala	Leu	Ser	Thr	Cys	Ser	Asn	Trp	Leu	Phe	Asn	Phe
		370				375					380				
Phe	Ile	Val	Met	Ile	Thr	Pro	Val	Met	Leu	Ala	Gly	Ile	Gly	Trp	Gly
385					390					395					400
Thr	Tyr	Leu	Phe	Phe	Ala	Ile	Ile	Asn	Val	Cys	Phe	Leu	Pro	Ile	Ile
				405				410						415	
Tyr	Phe	Phe	Tyr	Pro	Glu	Thr	Ala	Lys	Arg	Ser	Leu	Glu	Glu	Ile	Asp
			420					425					430		
Ile	Ile	Phe	Ala	Lys	Gly	Tyr	Cys	Glu	Asn	Lys	Ser	Tyr	Val	Gln	Ala
		435					440					445			
Ala	Arg	Glu	Leu												

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<210> SEQ ID NO 63
<211> LENGTH: 488
<212> TYPE: PRT
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<213> ORGANISM: *Neofusicoccum parvum*

<400> SEQUENCE: 63

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Met Ser Gly Leu Leu Thr Gly Ser Ala Phe Thr Lys Val Phe Pro Glu
 1          5          10          15

Ile Asp Thr Gln Asn Gly Gly Ser Ser Ser Leu Gln Gly Thr Val Val
 20          25          30

Ala Ile Tyr Glu Ile Gly Cys Phe Ala Gly Ala Leu Ile Thr Phe Ala
 35          40          45

Phe Gly Glu Gln Leu Gly Arg Arg Lys Cys Ile Met Ala Gly Cys Thr
 50          55          60

Ile Leu Thr Ile Gly Ala Thr Ile Gln Cys Ala Ser Tyr Gly Ile Pro
 65          70          75          80

Gln Leu Ile Val Gly Arg Ile Val Ala Gly Ile Gly Asn Gly Leu Asn
 85          90          95

Thr Ser Thr Ile Pro Val Trp His Ala Glu Leu Met Gln Ala His Asp
100          105          110

Arg Gly Lys Gly Leu Ala Ile Glu Phe Ile Leu Asn Ile Phe Gly Val
115          120          125

Ala Leu Ala Tyr Trp Val Asp Tyr Ala Phe Ser Phe Val Asp Asn Glu
130          135          140

Ser Gln Phe Arg Phe Pro Ile Ala Phe Gln Ile Ala Phe Ala Leu Val
145          150          155          160

Thr Leu Ala Ser Ile Ile Phe Leu Pro Glu Ser Pro Arg Trp Leu Leu
165          170          175

Asn His Asp Arg Glu Ala Glu Ala Arg Asn Ile Leu Trp Arg Leu Gln
180          185          190

Pro Asn Ala Lys Glu Ile Ala Glu Asp Ser Asp Val Val Asn Asn Glu
195          200          205

Met Ala Ile Ile Gln His Ala Leu Tyr Glu Glu Lys Glu Val Ala Gly
210          215          220

Gly Thr Thr Phe Lys Ala Ile Phe Lys Asp Gly Pro Gln Arg Phe Arg
225          230          235          240

Tyr Arg Thr Leu Leu Gly Ile Gly Gly Gln Phe Met Gln Gln Leu Ser
245          250          255

Gly Ile Asn Leu Ile Thr Tyr Tyr Ala Ala Val Ile Phe Glu Thr Ser
260          265          270

Ile Gly Met Ser His Asn Thr Ala Leu Leu Val Ala Gly Ala Asn Gly
275          280          285

Ile Ala Tyr Phe Leu Ser Thr Phe Pro Val Val Trp Val Leu Asp Arg
290          295          300

Leu Gly Arg Arg Lys Leu Met Leu Phe Ala Val Ile Gly Gln Ser Cys
305          310          315          320

Cys Met Ala Ile Leu Ala Gly Thr Val Ser Asn Gly Gly Lys Ser Ala
325          330          335

Gly Ile Val Ala Ala Val Met Leu Phe Leu Phe Asn Phe Phe Phe Ala
340          345          350

Ile Gly Leu Leu Ala Ile Pro Trp Leu Leu Pro Ala Glu Tyr Ala Pro
355          360          365

Leu Ala Ile Arg Thr Lys Ala Ala Ser Leu Ala Thr Ala Ser Asn Trp
370          375          380

Ile Phe Thr Phe Leu Val Val Glu Ile Val Pro Val Ser Ile Asn Asn
385          390          395          400

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Ile Ala Trp Arg Thr Tyr Ile Tyr Phe Ala Val Phe Asn Ala Phe Phe
      405                      410                      415

Val Pro Ile Ile Tyr Phe Phe Tyr Pro Glu Thr Lys Asn Leu Ser Leu
      420                      425                      430

Glu Glu Ile Asp Met Leu Phe Thr Gly Asp Lys Val Leu Met His Leu
      435                      440                      445

Pro Asp Ser Met Arg Ile Pro Ala Asn Asp Asn Ala Ala Val Ala Ser
      450                      455                      460

Ala Ile Arg Gly Glu Lys Asp Asn Val Ala Val Val Glu Asp Val Asn
      465                      470                      475                      480

Asn Gly Asn Asn Ser Glu Lys Ser
      485

<210> SEQ ID NO 64
<211> LENGTH: 516
<212> TYPE: PRT
<213> ORGANISM: Eutypa lata

<400> SEQUENCE: 64

Met Ser Ser Pro Thr Thr Phe Leu Gly Phe Gln Gly Ser Ser Leu Thr
 1          5          10          15

Leu Ala Gln Leu Phe Leu Val Val Cys Pro Ala Phe Val Leu Phe Gly
 20          25          30

Tyr Asn Gln Ser Gly Leu Gly Gly Leu Val Gly Leu Gln Asp Trp Ser
 35          40          45

Gln Thr Phe Pro Arg Ile Asp Thr Leu Asn Thr Glu Gly Ala Gln Lys
 50          55          60

Asp Asn Asn Ala Thr Ile Gln Gly Leu Val Val Ala Thr Phe Thr Leu
 65          70          75          80

Gly Ala Leu Pro Gly Cys Leu Ser Cys Ala Tyr Thr Ala Asp Arg Phe
 85          90          95

Gly Arg Arg Thr Val Ile Phe Val Gly Ala Leu Leu Thr Leu Ile Gly
100          105          110

Glu Val Leu Glu Ala Ser Ala Phe His Leu Ala Gln Met Ile Val Gly
115          120          125

Arg Val Ile Leu Gly Ala Gly Val Gly Met Leu Ser Gly Val Val Pro
130          135          140

Thr Trp Gln Ser Glu Cys Ser Asn Ser Lys Asn Arg Gly Lys His Val
145          150          155          160

Val Leu Glu Gly Leu Phe Ile Ser Met Gly Tyr Val Leu Gln Ala Trp
165          170          175

Ile Asn Leu Gly Phe Tyr Gln Phe Glu Thr Gly Pro Val Thr Trp Arg
180          185          190

Pro Pro Ile Ala Ile Pro Ile Phe Phe Ser Leu Val Leu Met Ser Phe
195          200          205

Ile Tyr Leu Met Pro Glu Ser Pro Arg Trp Leu Ile Arg Gln Gly Arg
210          215          220

Val Ser Glu Ala Arg Ala Ala Met Ser Ala Leu Lys Gly Leu Ala Asp
225          230          235          240

Asp Ala Gln Glu Ile His Ala Glu Val Ala Ala Val Glu Leu Ser Leu
245          250          255

Glu Glu Thr Gly Gln Lys Lys Ala Ala Leu Ala Asp Leu Leu Arg Met
260          265          270

Asp Glu Asp Lys Leu Leu Tyr Arg Phe Gly Ile Cys Ile Leu Leu Gln
275          280          285

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Phe Phe Gln Gln Met Ser Gly Gly Asn Leu Ile Ser Val Tyr Ser Thr
 290 295 300
 Ile Ile Phe Gln Arg Gly Leu Asn Leu Glu Ala Glu Thr Ser Arg Ile
 305 310 315 320
 Leu Ser Gly Gly Thr Leu Thr Trp Lys Phe Leu Ser Cys Phe Val Ser
 325 330 335
 Phe Phe Thr Ile Asp Arg Phe Gly Arg Arg Val Ala Leu Met Val Ser
 340 345 350
 Gly Thr Gly Met Ala Val Cys Met Met Ser Leu Ala Ile Ala Thr Ser
 355 360 365
 Phe Pro Thr Ser Asn Leu Ala Ala Gln Ile Val Ser Val Leu Phe Val
 370 375 380
 Phe Leu Phe Asn Phe Phe Ile Pro Ile Gly Phe Leu Gly Ala Asn Phe
 385 390 395 400
 Leu Tyr Cys Thr Glu Val Ala Pro Leu Arg Leu Arg Val Ala Met Ser
 405 410 415
 Ser Ile Ser Thr Ala Asn His Trp Leu Trp Asn Phe Val Val Thr Met
 420 425 430
 Ile Thr Pro Val Ala Ile Glu Ser Ile Gly Tyr Lys Tyr Tyr Ile Val
 435 440 445
 Tyr Thr Val Val Gly Phe Cys Ile Pro Leu Thr Val Tyr Phe Leu Tyr
 450 455 460
 Pro Glu Thr Met Gly Met Arg Leu Glu Asp Ile Asp Leu Val Phe Arg
 465 470 475 480
 Glu Ser Pro Ser Val Leu Ala Thr Val Lys Tyr Ala Arg Ser Arg Ser
 485 490 495
 Gln Arg Ser Asn Glu Glu Val Leu Ala Asp Lys Lys Lys Val Glu Tyr
 500 505 510
 Ala Glu Lys Ile
 515

<210> SEQ ID NO 65
 <211> LENGTH: 517
 <212> TYPE: PRT
 <213> ORGANISM: Phaeoacremonium minimum

<400> SEQUENCE: 65

Met Gly Phe Lys Thr Ala Phe Gly Leu Thr Gly His Ala Leu Ser Ile
 1 5 10 15
 Leu Gln Ile Ala Leu Ile Val Ala Pro Ser Phe Val Leu Phe Gly Tyr
 20 25 30
 Asn Gln Ala Gly Ile Gly Gly Leu Leu Ser Glu Glu Asp Trp Val Lys
 35 40 45
 Thr Phe Pro Glu Ile Asp Thr Val His Ala Thr Gly Thr Thr Lys Ser
 50 55 60
 Ser Lys Ser Thr Leu Gln Gly Phe Val Val Ala Thr Phe Val Ile Gly
 65 70 75 80
 Ala Leu Ile Gly Ala Leu Ser Cys Ser Tyr Thr Gly Asp Ile Phe Gly
 85 90 95
 Arg Arg Asn Val Ile Phe Ala Gly Ala Val Phe Thr Leu Val Gly Glu
 100 105 110
 Val Leu Glu Ala Ser Ser Phe Ser Leu Ala Gln Phe Ile Val Gly Arg
 115 120 125
 Val Leu Ile Gly Ala Gly Val Gly Gln Leu Ser Ser Ile Val Pro Val

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130	135	140
Trp Gln Ser Glu Thr Ser Gly Ala Lys Asn Arg Gly Arg Ser Val Val		
145	150	155 160
Val Thr Gly Leu Phe Ile Cys Leu Gly Tyr Val Leu Glu Ser Trp Ile		
	165	170 175
Asp Leu Gly Phe Phe Glu Phe Lys Thr Gly Pro Leu Thr Trp Arg Pro		
	180	185 190
Pro Ile Ala Ile Ala Val Ala Phe Ser Leu Val Leu Met Ala Ser Val		
	195	200 205
Tyr Val Phe Pro Glu Ser Pro Arg Trp Leu Leu Met Lys Asn Arg Val		
	210	215 220
Gln Glu Ala Arg Glu Ser Leu Ser Val Leu Arg Gly His Ala Glu Asp		
225	230	235 240
Ser Leu Glu Val Gln Ala Glu Leu Ala Gly Ile Glu Leu Ser Leu Glu		
	245	250 255
Glu Thr Ser Gly Asn Ala Ala Lys Leu Gly Asp Met Leu Lys Met Gly		
	260	265 270
Glu Glu Lys Leu Leu Tyr Arg Phe Phe Leu Cys Met Leu Leu Gln Phe		
	275	280 285
Tyr Gln Gln Met Ser Gly Ser Asn Leu Val Ser Val Tyr Ala Thr Thr		
	290	295 300
Leu Phe Gln Thr Asn Leu Gly Leu Ser Ser Glu Leu Ser Arg Val Leu		
305	310	315 320
Thr Gly Gly Ala Leu Thr Trp Lys Phe Leu Ser Ser Phe Ile Ala Phe		
	325	330 335
Val Thr Ile Asp Arg Phe Gly Arg Arg Ala Val Phe Ile Leu Ser Gly		
	340	345 350
Ile Gly Met Ser Cys Cys Met Ile Ala Leu Ala Val Ser Thr Ser Phe		
	355	360 365
Gly Lys Glu Asn Arg Ala Ala Gln Ile Ala Ala Gly Cys Phe Ile Tyr		
	370	375 380
Leu Tyr Asn Thr Phe Val Pro Ile Gly Phe Leu Gly Ala Asn Phe Leu		
385	390	395 400
Tyr Cys Thr Glu Val Ala Pro Ile Arg Leu Arg Met Ala Met Ser Ser		
	405	410 415
Ile Ser Thr Ala Asn His Trp Leu Trp Asn Phe Val Val Val Met Val		
	420	425 430
Thr Pro Val Ala Ile Glu Thr Ile Gly Trp Gln Phe Tyr Ile Val Phe		
	435	440 445
Ala Val Ile Ala Ala Cys Val Pro Val Ser Val Tyr Phe Leu Phe Pro		
	450	455 460
Glu Thr Met Gly Arg Asn Leu Glu Glu Ile Asp Met Val Phe Arg Glu		
465	470	475 480
Ser Pro Ser Val Trp Ala Thr Val Arg Phe Ala Arg Ser Arg Pro Ala		
	485	490 495
Leu Thr Ala Val Glu Tyr Ala Glu Lys His Asp Asn Val Asp His Leu		
	500	505 510
Glu Lys Thr Ala Glu		
515		

<210> SEQ ID NO 66

<211> LENGTH: 517

<212> TYPE: PRT

<213> ORGANISM: Aureobasidium namibiae

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<400> SEQUENCE: 66

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Met Val Gly Tyr Leu Asp Gly Ile Gln Gly Lys Thr Leu Tyr Lys Ile
1          5          10          15

Met Ser Ala Ala Cys Gly Ser Ala Phe Met Leu Tyr Gly Trp Asp Ala
20          25          30

Gly Val Leu Gly Gly Ile Gln Glu Thr Lys Glu Phe Arg Ala Ala Ile
35          40          45

Gly Asp Pro Gln Gly Ala Phe Ile Ile Pro Ile Ile Ala Ala Ile Tyr
50          55          60

Asn Leu Ala Ala Gly Val Met Ser Leu Cys Val Ser Phe Tyr Gly Met
65          70          75          80

Gln Ile Gly Arg Lys Gly Thr Ile Leu Leu Gly Cys Leu Leu Ile Cys
85          90          95

Ile Gly Ala Leu Leu Gln Ala Ser Thr Tyr Ser Val Gly Gln Ile Ile
100         105         110

Val Gly Arg Ile Val Thr Gly Ala Gly Ile Gly Asn Ile Ala Ala Ala
115         120         125

Val Pro Thr Tyr Met Ala Glu Met Ser Leu Glu Ala Lys Glu Arg Gly
130         135         140

Pro Glu Val Ser Tyr Gln Leu Ala Leu Leu Ile Thr Gly Val Ala Leu
145         150         155         160

Ala Tyr Trp Ile Asp Leu Gly Phe Val Gln Gly Leu Asp Arg His Pro
165         170         175

Trp Leu Trp Arg Ile Pro Leu Ala Leu Gln Ser Cys Phe Ala Ile Phe
180         185         190

Ser Ala Val Leu Leu Phe Met Leu Pro Asp Thr Pro Arg Trp Tyr Tyr
195         200         205

Ala Arg Gly Lys Glu Ala Lys Gly Asp Arg Val Leu Ala Arg Leu His
210         215         220

Gly Leu Pro Val Glu His Gln Asn Val Gln Ala Val Lys Ala Asp Ile
225         230         235         240

Met Ala Ser Met Glu Glu Glu Asp Glu Thr Gly Lys Ile Ser Ile Val
245         250         255

Ser Leu Phe Trp Asp Asn Thr Glu Leu Gln Phe Gly Arg Arg Leu Arg
260         265         270

Thr Ser Phe Leu Ile Asn Trp Ala Gln Gln Phe Leu Gly Ile Asn Met
275         280         285

Leu Val Tyr Phe Ser Thr Gln Ile Phe Ser Asn Leu Asn Tyr Ser Pro
290         295         300

Leu Leu Ser Gly Ile Leu Ala Gly Val Leu Asn Thr Ala Phe Ala Ile
305         310         315         320

Ala Ser Tyr Pro Pro Ile Trp Tyr Ile Glu Lys Val Gly Arg Arg Ala
325         330         335

Met Met Ile Trp Ser Ala Leu Gly Cys Gly Val Cys Met Leu Ile Tyr
340         345         350

Val Val Leu Thr Thr Leu Pro Ala His Met Gln Ser Ala Gly Thr Asn
355         360         365

Trp Gly Ala Val Ala Ile Ile Leu Tyr Glu Ile Val Phe Ala Phe
370         375         380

Gly Trp Leu Gly Thr Cys Trp Ile Tyr Gly Pro Glu Ile Ala Pro Leu
385         390         395         400

Lys Tyr Arg His Val Ala Gly Ser Leu Gly Ala Ala Gly Glu Trp Phe

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405	410	415
Ser Thr Phe Val Met Val Phe Gly Gly Gly Thr Gly Ile Asn Ala Val 420 425 430		
Gly Pro Lys Ile Phe Ile Trp Pro Leu Leu Cys Cys Phe Leu Ala Ala 435 440 445		
Ala Tyr Val Tyr Phe Leu Cys Pro Glu Thr Thr Gly Lys Thr Leu Glu 450 455 460		
Glu Ile Asp Ala Leu Phe Ala Arg Ser Pro Glu Val Arg Glu Arg Leu 465 470 475 480		
Glu Arg Asp Ile Ala Ala Arg Arg Ala Gly Val Leu Pro Gly Asn Glu 485 490 495		
Lys Ser Met Ser Arg Asp Ser Ser Asp Met Ser Lys Met Glu Val Ser 500 505 510		
Asn Ile Glu Lys Ile 515		
<210> SEQ ID NO 67		
<211> LENGTH: 677		
<212> TYPE: PRT		
<213> ORGANISM: Kluyveromyces lactis		
<400> SEQUENCE: 67		
Met Ser Asp Ser Thr Asp Lys Asp Arg His Asn Lys Asn Ser Ser Arg 1 5 10 15		
Gly Ile Val Asn Thr Asp Ile Glu Asp Asp Asn Ser Pro Ser Pro Leu 20 25 30		
Asp Thr Thr Asp Lys Lys Gly Ala Pro Glu Leu Gln Lys Pro Gly Asn 35 40 45		
Pro Ser Leu Phe Lys Glu Ser Ala Leu Lys Asn Ser Asp Gln Leu Gln 50 55 60		
Phe Arg Asn Ser Phe Asn Ile Pro Asn Ala Val Gly Gly Thr Asn Gly 65 70 75 80		
Ile Val Ile Pro Pro Asn Val Gln Pro Leu Asp Gln Gln Arg Ile Pro 85 90 95		
Pro Ser Tyr Thr Asp His Leu Gln Val Lys Asp Thr Tyr Leu Thr Gly 100 105 110		
Arg Thr Leu Leu Tyr Phe Thr Ser Ile Phe Val Ser Leu Gly Val Phe 115 120 125		
Leu Phe Gly Tyr Asp Gln Gly Val Met Ser Gly Ile Ile Thr Gly Pro 130 135 140		
Tyr Phe Lys Thr Tyr Phe Asn Asn Pro Thr Ala Ala Thr Ile Gly Thr 145 150 155 160		
Met Val Ser Ile Leu Glu Ile Gly Ala Leu Val Ser Ser Leu Leu Val 165 170 175		
Ser Asn Ile Gly Glu Lys Phe Gly Arg Arg Phe Thr Ile Lys Tyr Gly 180 185 190		
Ser Leu Ile Phe Ile Leu Gly Gly Leu Val Gln Thr Phe Ser Trp Glu 195 200 205		
Met Gly His Met Ile Phe Gly Arg Ile Ile Ser Gly Ile Gly Val Gly 210 215 220		
Leu Leu Ser Thr Ile Val Pro Ile Tyr Gln Ser Glu Ile Ser Pro Pro 225 230 235 240		
His Asn Arg Gly Lys Leu Ala Cys Ile Glu Phe Thr Gly Asn Ile Val 245 250 255		

Gly 260	Tyr 260	Ala 260	Ser 260	Ser 260	Val 260	Trp 260	Val 265	Asp 265	Tyr 265	Ala 265	Cys 265	Ser 270	Tyr 270	Ile 270	Glu 270
Ser 275	Asp 275	Thr 275	Ser 275	Trp 275	Arg 275	Leu 280	Pro 280	Leu 280	Phe 280	Ile 280	Gln 285	Cys 285	Val 285	Met 285	Gly 285
Leu 290	Leu 290	Leu 290	Phe 290	Leu 295	Gly 295	Ser 295	Phe 295	Val 295	Ile 295	Val 300	Glu 300	Thr 300	Pro 300	Arg 300	Trp 300
Leu 305	Leu 305	Asn 310	His 310	Asp 310	His 310	Asp 310	Ile 315	Glu 315	Gly 315	Leu 315	Val 315	Val 320	Ile 320	Ala 320	Asp 320
Leu 325	His 325	Ser 330	Asp 330	Gly 330	Asp 335	Val 335	Leu 335	His 335	Ser 335	Lys 335	Ala 340	His 340	Glu 340	Glu 345	Tyr 345
Lys 350	Leu 350	Ile 350	Lys 350	Glu 350	Thr 350	Val 350	Leu 350	Ile 350	Ser 350	Arg 350	Leu 350	Glu 350	Gly 350	Glu 355	Lys 355
Lys 355	Ser 355	Leu 355	Arg 355	Phe 355	Ala 355	Phe 360	Lys 360	Arg 360	Tyr 360	Arg 360	Thr 360	Arg 365	Met 365	Leu 365	Ile 365
Ala 370	Met 370	Ser 370	Ser 370	Gln 370	Met 370	Phe 375	Ala 375	Gln 375	Leu 375	Asn 375	Gly 380	Ile 380	Asn 380	Val 380	Ile 380
Ser 385	Tyr 385	Tyr 385	Ala 385	Pro 385	Leu 390	Val 390	Phe 390	Glu 390	Gln 390	Ala 395	Gly 395	Trp 395	Val 395	Gly 400	Arg 400
Glu 405	Ala 405	Leu 405	Leu 405	Met 405	Thr 405	Gly 405	Ile 410	Asn 410	Ser 410	Ile 410	Ile 410	Tyr 410	Ile 415	Leu 415	Ser 415
Thr 420	Ile 420	Leu 420	Pro 420	Trp 420	Lys 420	Leu 420	Val 420	Asp 425	Lys 425	Trp 425	Gly 425	Arg 425	Lys 430	Pro 430	Ile 430
Leu 435	Leu 435	Ser 435	Gly 435	Ala 435	Leu 435	Val 440	Met 440	Gly 440	Thr 440	Ser 440	Leu 440	Leu 445	Ala 445	Ile 445	Ala 445
Met 450	Ser 450	Leu 450	Trp 450	Ala 450	Asn 450	Val 455	Ala 455	Ala 455	Thr 455	Pro 455	Arg 455	Leu 460	Val 460	Val 460	Val 460
Phe 465	Val 465	Ile 465	Ile 465	Phe 465	Asn 470	Ala 470	Phe 470	Phe 470	Gly 470	Tyr 475	Ser 475	Trp 475	Gly 475	Pro 480	Ile 480
Pro 485	Trp 485	Leu 485	Tyr 485	Pro 485	Val 485	Glu 485	Ile 485	Ala 485	Pro 490	Ala 490	Met 490	Ala 490	Arg 490	Ser 495	Ala 495
Met 500	Ala 500	Ser 500	Ala 500	Ser 500	Thr 500	Ala 500	Thr 500	Asn 505	Trp 505	Leu 505	Phe 505	Asn 505	Trp 510	Leu 510	Val 510
Gly 515	Ile 515	Met 515	Thr 515	Pro 515	Ile 515	Leu 515	Gln 520	Glu 520	Lys 520	Ile 520	His 520	Trp 525	Arg 525	Met 525	Tyr 525
Leu 530	Ile 530	His 530	Thr 530	Val 530	Ser 530	Cys 535	Tyr 535	Leu 535	Ser 535	Phe 535	Trp 540	Cys 540	Val 540	Leu 540	Lys 540
Val 545	Tyr 545	Pro 545	Glu 545	Thr 545	Ala 550	Gly 550	Leu 550	Arg 550	Leu 550	Glu 555	Asp 555	Met 555	Asp 555	Ser 555	Val 560
Phe 565	Asp 565	Asp 565	Arg 565	Ser 565	Ser 565	Thr 565	Phe 565	Ser 565	Phe 570	Gln 570	Ser 570	Gly 570	Thr 570	Ser 575	Ala 575
Glu 580	Ile 580	Glu 580	Gln 580	Gln 580	Ser 580	His 580	Leu 580	Val 585	Ser 585	Gly 585	Gly 585	Gly 585	Glu 590	Val 590	Ala 590
Pro 595	Ser 595	Thr 595	Arg 595	Ser 595	Arg 595	Lys 595	Ser 600	Val 600	Tyr 600	Ser 600	Asn 600	Ala 605	Gln 605	Ser 605	Met 605
Phe 610	Asn 610	Lys 610	Asp 610	Glu 610	Ile 610	Gln 615	Pro 615	Pro 615	Thr 615	Leu 615	Thr 615	Gln 620	Val 620	Leu 620	Gln 620
Trp 625	Lys 625	Glu 625	Glu 625	Arg 625	Thr 625	Gln 630	Thr 630	Lys 630	Pro 630	Leu 635	Lys 635	Lys 635	Phe 635	Ile 640	Arg 640
Arg 645	Gly 645	Ser 645	Glu 645	Thr 645</											

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<210> SEQ ID NO 68
 <211> LENGTH: 163
 <212> TYPE: PRT
 <213> ORGANISM: *Rasamsonia emersonii*

<400> SEQUENCE: 68

```
Met Thr Ser Ile Gly Gly Pro Lys Cys Gly Ile Val Ala Ala Thr Phe
 1             5             10             15
Leu Phe Val Phe Asn Thr Phe Phe Ala Leu Gly Trp Leu Ser Ile Pro
                20             25             30
Trp Leu Tyr Pro Ala Glu Leu Val Pro Leu Glu Ile Arg Ala Gln Ala
            35             40             45
Asn Ala Leu Ser Thr Ser Ala Asn Trp Ile Phe Asn Phe Met Val Val
            50             55             60
Met Ile Thr Pro Val Ala Phe Ser Ser Ile Gly Trp Arg Thr Tyr Ile
65             70             75             80
Ile Phe Ala Val Phe Asn Ala Ala Ser Ile Pro Ile Leu Tyr Trp Cys
            85             90             95
Tyr Pro Glu Thr Ala Tyr Arg Ser Leu Glu Glu Met Asp Ile Ile Phe
            100            105            110
Ala Lys Ala Thr Gly Val Val Asp Ala Val Arg Val Ala Arg Thr Glu
            115            120            125
Pro Arg His Phe Gly Lys His Gly Glu Val Leu Arg Glu Met Val Pro
            130            135            140
Asp Val Lys Asp Thr Thr Gly Ile Gly Ser Glu Lys Gly Gly Val Glu
145            150            155            160
His Val Gln
```

<210> SEQ ID NO 69
 <211> LENGTH: 575
 <212> TYPE: PRT
 <213> ORGANISM: *Pachysolen tannophilus*

<400> SEQUENCE: 69

```
Met Phe Lys Lys Ile Asp Lys Ile Asp Lys Ser Asp Tyr Val Ala Ser
 1             5             10             15
Ser Ser Lys Lys Tyr Leu Gly Met Arg Gly Ala Pro Leu His Lys Ala
            20             25             30
Ile Ala Thr Ile Ala Gly Leu Gly Phe Leu Leu Phe Gly Tyr Asp Gln
            35             40             45
Gly Val Met Gly Ser Leu Leu Thr Leu Asp Ser Phe Leu Glu Thr Phe
            50             55             60
Pro Gln Ile Asn Asp Ser Val Asp Thr Ser Lys Ser Thr Leu Lys Gly
65             70             75             80
Phe Val Ile Ala Val Tyr Glu Leu Gly Cys Met Thr Gly Ala Phe Phe
            85             90             95
Thr Met Trp Lys Gly Asp Ile Phe Gly Arg Arg Lys Met Ile Phe Tyr
            100            105            110
Gly Ser Ile Ile Met Thr Ile Gly Gly Ile Leu Gln Cys Thr Ser Tyr
            115            120            125
Ser Val Ala Gln Leu Ala Val Ala Arg Val Val Ser Gly Val Gly Asn
            130            135            140
Gly Phe Ile Thr Ser Thr Ile Pro Thr Leu Gln Ser Glu Cys Ala Lys
145            150            155            160
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Pro	His	Arg	Arg	Gly 165	Ala	Leu	Ile	Met	Met 170	Ser	Gly	Ala	Leu	Ile	Ser
Phe	Gly	Ile	Cys 180	Phe	Ser	Tyr	Trp	Val 185	Asp	Phe	Gly	Leu	Tyr 190	Phe	Ala
Thr	Gly	Asp	Val 195	Gln	Trp	Arg	Phe 200	Pro	Ile	Ala	Phe	Gln 205	Ile	Val	Phe
Ser	Leu	Leu	Leu	Thr	Ser	Leu 215	Ile	Phe	Glu	Leu	Pro 220	Glu	Ser	Pro	Arg
Trp	Leu	Val	Lys	Ile	His 230	Glu	Ile	Glu	Arg	Ala 235	Arg	Glu	Thr	Phe	Ala 240
Ala	Leu	Asp	Asp	Val 245	Ser	Val	Asp	Asp	Pro 250	Leu	Ile	Asp	Asp	Glu 255	Ile
Lys	Asp	Ile	Gln 260	Ala	Val	Leu	Lys	Arg 265	Asp	Leu	Asp	Leu	Gly 270	Ala	Asp
Lys	Phe	Ser	Phe	Ser	Val	Val	Phe 280	Lys	Phe	Asp	Glu	Lys 285	Lys	Thr	Phe
His	Arg	Thr	Met	Leu	Ala	Tyr 295	Phe	Val	Gln	Val	Met 300	Gln	Gln	Ile	Ser
Gly	Ile	Asn	Leu	Ile	Thr 310	Tyr	Tyr	Ala	Gly	Thr 315	Ile	Tyr	Glu	Thr	Tyr 320
Ile	Gly	Met	Asn	Ala 325	Leu	Asp	Ser	Arg	Ile 330	Leu	Ala	Ala	Cys	Asn 335	Gly
Thr	Glu	Tyr	Phe 340	Leu	Ala	Ser	Leu	Ile 345	Pro	Phe	Tyr	Thr	Val 350	Glu	Arg
Phe	Gly	Arg	Arg	Ser	Leu	Phe 360	Leu	Phe	Gly	Thr	Ala	Gly 365	Gln	Ala	Ile
Thr	Met	Ala	Ile	Leu	Thr	Gly 375	Val	Gln	Trp	Ala	Ser 380	Glu	Tyr	Lys	Gly
Asp	Gln	Gly	Ala	Ala 390	Ile	Ala	Cys	Ala	Val	Phe 395	Leu	Phe	Val	Phe	Asn 400
Thr	Phe	Phe	Ala	Ile 405	Gly	Met	Leu	Gly	Met 410	Thr	Trp	Leu	Leu	Pro 415	Pro
Glu	Leu	Val	Thr 420	Leu	Glu	Ser	Arg	Ala 425	Ser	Val	Thr	Gly	Leu 430	Ser	Thr
Ser	Ala	Asn	Trp 435	Leu	Phe	Asn	Phe 440	Val	Val	Val	Met	Ile 445	Thr	Pro	Val
Cys	Phe	Thr	His	Ile	Gly	Pro 455	Tyr	Thr	Tyr	Thr	Ile 460	Phe	Ala	Val	Val
Asn	Ala	Ile	Met	Val	Pro 470	Cys	Ile	Phe	Phe	Phe 475	Tyr	Pro	Glu	Thr	Lys 480
Gly	Arg	Ser	Leu 485	Glu	Glu	Met	Asp	Arg	Ile 490	Phe	Glu	Gln	Ser	Asn 495	Pro
Lys	Thr	Pro	Trp 500	Asp	Val	Val	Arg	Ile 505	Ala	Arg	Glu	Met 510	Pro	Phe	Glu
Asn	Arg	Asp	Ile 515	Asp	Asn	Glu	Asp 520	Glu	Glu	Asp	Lys	Ile 525	Asn	Leu	Asp
Arg	Ser	Ser	Glu 530	Thr	Ser	Ser 535	Val	Ser	Asn	Glu	Lys 540	Gly	Ser	Ala	Ser
Phe	Thr	Leu	Asp 545	Ser	Val 550	Asn	Asp	Thr	Gly	Phe 555	Phe	Val	Lys	Asn	Glu 560
Glu	Asn	Lys	Asn 565	Glu	Gln	Glu	Thr	Ser	Gln 570	Pro	Glu	Lys	Lys	Glu 575	

What is claimed is:

1. A recombinant yeast host cell expressing a heterologous sugar transporter protein (STL1) and/or having, when compared to a parental cell, a decreased expression of a native NAD-dependent glycerol-3-phosphate dehydrogenase gene,

wherein the heterologous sugar transporter protein comprises the amino acid sequence of any one of SEQ ID NO: 8 and 31 to 69, is a variant having at least 70% identity to the amino acid sequence of any one of SEQ ID NO: 8 and 31 to 69 and glycerol transport activity, or is a fragment having at least 70% identity to the amino acid sequence of any one of SEQ ID NO: 8 and 31 to 69 and glycerol transport activity; and

wherein the recombinant yeast host cell has at least one of phenotypic trait providing persistence of the recombinant yeast host cell in a plurality of fermentation cycles, and wherein the at least one phenotypic trait is triploidy.

2. The recombinant yeast host cell of claim 1, wherein the recombinant yeast host cell expresses the heterologous sugar transporter protein (STL1).

3. The recombinant yeast host cell of claim 1, wherein the recombinant yeast host cell has decreased expression of the native NAD-dependent glycerol-3-phosphate dehydrogenase gene.

4. The recombinant yeast host cell of claim 3, wherein the native NAD-dependent glycerol-3-phosphate dehydrogenase gene is a native glycerol-3-phosphate dehydrogenase 1 (GPD1) gene and/or a native glycerol-3-phosphate dehydrogenase 2 (GPD2) gene.

5. The recombinant yeast host cell of claim 1 further exhibiting a fast settling phenotype.

6. The recombinant yeast host cell of claim 5, wherein:

(i) at least 5% of a population consisting essentially of the recombinant yeast host cells is able to sediment by gravity after 5 minutes and/or

(ii) the population consisting essentially of the recombinant yeast host cells is able to sediment by gravity in 5 minutes in a proportion equal to or higher than a control population consisting essentially of control yeast cells lacking the fast settling phenotype, and wherein the control yeast cells are from a *Saccharomyces cerevisiae* PE-2 strain.

7. The recombinant yeast host cell of claim 1 further exhibiting a rugose phenotype.

8. The recombinant yeast host cell of claim 7, wherein at least 90% of a population consisting essentially of the recombinant yeast host cells, after exponential growth in a medium inoculated at low recombinant yeast host cell density, has at least two daughter cells attached.

9. The recombinant yeast host cell of claim 7 being capable of:

reducing the transcription factor activity of an Activator of CUP1 Expression (ACE2) polypeptide; and/or expressing a mutated ACE2 polypeptide, wherein the mutated ACE2 polypeptide has decreased activity when compared to a wild type ACE2 polypeptide.

10. The recombinant yeast host cell of claim 1 further exhibiting improved invertase activity.

11. The recombinant yeast host cell of claim 10, wherein a population consisting essentially of the recombinant yeast host cells is able to hydrolyze more than 0.05 gram of sucrose per gram of dry cell weight per minute and/or exhibits more than 1.0 times invertase activity than a control population consisting essentially of control yeast cells lacking the improved invertase activity phenotypic trait, wherein

the control yeast cells are from a *Saccharomyces cerevisiae* PE-2 strain, and wherein invertase activity is measured after exponential growth of the population diluted to a concentration of 9 mg/mL on a wet cell weight in a buffer and wherein the buffer comprises 40 g/L of sucrose, is at pH 5 and at a temperature of 35° C.

12. The recombinant yeast host cell of claim 1 being capable of increasing the enzymatic activity of at least one polypeptide having invertase activity.

13. The recombinant yeast host cell of claim 12, wherein the at least one polypeptide having invertase activity comprises SUC1, SUC2, SUC3, SUC4, SUC5, SUC6, SUC7, SUC8 or SUC9.

14. The recombinant yeast host cell of claim 1 further exhibiting increased signaling in the RAS/cAMP/PKA pathway.

15. The recombinant yeast host cell of claim 14, wherein a population consisting essentially of the recombinant yeast host cells is able to exhibit a fold increase in the production of cAMP of equal to or less than 1.7 and/or a fold increase in the production of cAMP production of less than 70% when compared a control population consisting essentially of control yeast cells lacking the increased signaling in the RAS/cAMP/PKA pathway phenotypic trait, wherein the control yeast cells are from a *Saccharomyces cerevisiae* PE-2 strain, and wherein the production of cAMP is measured in the population having been glucose depleted and at 5 minutes after a glucose spike.

16. The recombinant yeast host cell of claim 14 being capable of expressing a mutated polypeptide involved in the RAS/cAMP/PKA pathway.

17. The recombinant yeast host cell of claim 16, wherein the mutated polypeptide involved in the RAS/cAMP/PKA pathway comprises a mutated RAS2 polypeptide having increased activity when compared to a wild-type RAS2 polypeptide, a mutated IRA2 polypeptide having a reduced inhibitory activity towards a wild-type RAS1 and/or a wild-type RAS2 polypeptide when compared to a wild-type IRA2 polypeptide.

18. The recombinant yeast host cell of claim 1 being from the genus *Saccharomyces* sp. or from the species *Saccharomyces cerevisiae*.

19. The recombinant yeast host cell of claim 2 comprising a heterologous nucleic acid encoding a sugar transporter protein (STL1).

20. The recombinant yeast host cell of claim 4 having a deletion in the native glycerol-3-phosphate dehydrogenase 1 gene (GPD1).

21. The recombinant yeast host cell of claim 4 having a deletion in the native glycerol-3-phosphate dehydrogenase 2 gene (GPD2).

22. The recombinant yeast host cell of claim 1, wherein the heterologous sugar transporter protein (STL1) comprises the amino acid sequence of SEQ ID NO: 8, is the variant having at least 70% identity to the amino acid sequence of SEQ ID NO: 8 and glycerol transport activity, or is the fragment having at least 70% identity to the amino acid sequence of SEQ ID NO: 8 and glycerol transport activity.

23. The recombinant yeast host cell of claim 1, wherein the heterologous sugar transporter protein (STL1) comprises the amino acid sequence of any one of SEQ ID NO: 31 to 56, 68, or 69 is the variant having at least 70% identity to the amino acid sequence of any one of SEQ ID NO: 31 to 56, 68, and 69 and glycerol transport activity, or is the fragment

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having at least 70% identity to the amino acid sequence of any one of SEQ ID NO: 31 to 56, 68, and 69 and glycerol transport activity.

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